



Master Thesis

IU University of Applied Sciences

Study program: Msc Computer Science

Low Earth orbit satellite broadband services:
Challenges and Opportunities

Wycliffe Bahati

Enrolment number: 9051234

Nairobi 54041

00200 Nairobi

Supervisor: DR. Frank Müller

Date of submission: 27th May 2025

Acknowledgement

I sincerely thank Dr. Frank Müller, my supervisor, for his invaluable guidance and continued support throughout my research journey. His expertise and insightful feedback played an essential role in shaping my thesis's direction and quality. I also extend my heartfelt appreciation to all the interview participants who generously shared their time and knowledge for this study. Your perspectives have been essential in helping me understand the complex dynamics of low Earth orbit satellite broadband services. A special thank you goes to Henrich Nothnagel, whose detailed insights into the satellite industry added a practical depth to my research. Your contributions are truly appreciated.

Abstract

The rapid deployment of low-Earth orbit satellites for telecommunication disrupts internet broadband services worldwide. This research investigates the challenges and opportunities LEO broadband will face in a market dominated by traditional broadband providers. We also examine how prepared regulators are to manage the integration of LEO satellite internet into the existing broadband ecosystem, as well as the perceptions of stakeholders, including consumers, providers, and policymakers, regarding LEO broadband as a sustainable and competitive alternative. The study applies a qualitative research design guided by Mayring's Qualitative Content Analysis framework.

There are opportunities and challenges in the areas of reaching new markets and conquering the existing broadband market, technical opportunities and challenges that LEO broadband is facing to deploy services globally, long-term sustainability and economics of LEO broadband, Regulatory hurdles that LEO broadband faces in dealing with regulators all over the world and consumer adoption factors. The focus is also on future trends, and industry expectations are predictions. The growth of LEO broadband has implications, including increased market competition and disruption, as well as new challenges in spectrum management. This development also significantly impacts regulators, who must transform their current regulatory frameworks to accommodate the new LEO broadband business models.

Keywords: LEO satellite, broadband, ISPs, MNOs, regulatory readiness, qualitative research, Mayring

Table of contents

Contents

Abstract ii

Table of contents iii

List of tables vi

List of Figures..... vii

List of abbreviations..... viii

1. Introduction 1

 1.1 Background and Context1

 1.2 Problem Statement.....3

 1.3 Objectives of the Study3

 1.4 Thesis Outline.....4

2. Literature Review 4

 2.1 Overview of Broadband Technologies.....4

 2.2 Development of Low Earth Orbit Satellite Constellations10

 2.3 Traditional Broadband Service Challenges.....13

 2.4 Potential Advantages of LEO Satellite Internet Mobility.....16

 2.5 Challenges Facing LEO Satellites.....19

3 Research Methodology 21

 3.1 Introduction21

 3.2 Research Design and Data Collection Methods22

 3.3 Methodological Approach: Mayring’s Qualitative Content Analysis22

 3.4 Ethical Considerations23

 3.5 Emergent Themes and Interpretation23

4 Findings..... 24

 4.1 Marketing Opportunities and Challenges24

 4.1.1 Bridging the broadband access gap24

 4.1.2 Remote Locations and Isolated Islands27

 4.1.3 Support for IoT28

 4.1.4 Airline and Maritime Connectivity.....29

 4.1.5 Backhaul to Traditional Broadband Services31

 4.1.6 Disaster recovery to terrestrial infrastructure failures.....32

 4.1.7 Market viability and pricing challenges.....33

 4.1.8 Competition.....35

 4.1.9 Collaboration between LEO and traditional providers.....37

 4.1.10 Autonomous and Unmanned Vehicles38

4.2	Performance Technical Opportunities and Challenges:	39
4.2.1	Weather Conditions and Natural Disasters	39
4.2.2	Decentralised content	40
4.2.3	Internet Routing Protocols	43
4.2.4	Spectrum allocation and interference	45
4.2.5	Latency and Jitter	47
4.2.6	Network Standardisation	49
4.2.7	Scalability and Bandwidth Limitations	50
4.2.8	Environmental concerns	51
4.3	Sustainability and Economic Considerations	52
4.3.1	Deployment costs	52
4.3.2	Return on investment	55
4.3.3	Ground Station/Gateway Deployment	57
4.3.4	Operational and support challenges	58
4.4	Regulatory Hurdles	59
4.4.1	Resources	59
4.4.2	Regulatory and Licensing Barriers	61
4.4.3	Regulated internet	62
4.4.4	Geopolitical restrictions	63
4.4.5	Data privacy	64
4.4.6	National security and critical infrastructure	65
4.5	Consumer Adoption and Accessibility Factors:	66
4.5.1	Affordability	66
4.5.2	Technology availability	68
4.5.3	Digital literacy	68
4.5.4	Easy to Deploy	69
4.5.5	Enhanced services	69
4.5.6	Customer support challenges	70
4.5.7	Bandwidth Considerations	71
4.6	Future Trends & Industry Predictions	72
4.6.1	Geopolitical issues	72
4.6.2	New Market Entrants	72
4.6.3	Increased Access and Digital Growth	72
4.6.4	Satellite Replacement and Long-Term Sustainability	73
4.6.5	Evolution of Spectrum and Wireless Speeds	73
5	Discussion	73
5.1	Summary of Key Findings	74

5.2 Implications for the Broadband Industry	81
5.3 Implications for Policy and Regulation.....	83
5.4 Limitations of the Study.....	83
5.5 Suggestions for Future Research.....	84
6 Conclusion	85
7 Recommendations for the Broadband Industry	86
7.1 Collaboration.....	86
7.2 Address Regulatory Challenges	87
7.3 Improve digital literacy	87
7.4 Service Scalability and Affordability	87
7.5 Address Environmental Factors.....	87
7.6 Network Standardisation.....	87
References	88
Appendix A: Interview Questions	97
Appendix B: Interview Transcripts	98
Participant 1 MM.....	98
Participant 2 BR.....	144
Participant 3 BM	180
Participant 4 BL	211
Participant 5 CK.....	258
Participant 6 TM	299
Participant 7 HN	342
Participant 8 AK.....	380

List of tables

Table 1 Fixed broadband speeds (in Mbps), 2018–2023, 2018)4

Table 2 Comparison between GEO and LEO satellites10

Table 3 Projects and the projected number of satellites to be launched.....10

Table 4 Launch vehicle estimated payload cost per kg17

Table 5 Revenue models in last-mile retail sales34

Table 6 Starlink Costs34

Table 7 Safaricom broadband pricing35

Table 8 ISL Link Types for LEO45

Table 9 Launch Vehicle Costs for LEO Deployment.....54

List of Figures

Figure 1 Global mobile device and connection growth5

Figure 2 Common HFC architecture7

Figure 3 Global footprint Intelsat IS-209

Figure 4 Number of satellites in orbit launched over the years11

Figure 5 Starlink Satellite Orbit12

Figure 6 Kenya Population Density Map14

Figure 7 Kenya LTE Broadband Coverage14

Figure 8 ITU Broadband Map.....26

Figure 9 Starlink Coverage - Africa27

Figure 10 Map of metros where at least one edge node (GGC) is present42

Figure 11 Global internet exchange point map43

Figure 12 LEO satellite with ISL Links44

Figure 13 Starlink medium Downlink and Uplink in Europe47

Figure 14 Comparison between Starlink latency and other service providers48

Figure 15 Population density (people per square kilometre) by country in 202555

List of abbreviations

3GPP - 3rd Generation Partnership Project.
5G - Fifth Generation
6G - Sixth Generation
ADSL - Asymmetric Digital Subscriber Line.
AON – Active Optical Network
ARPANET - Advanced Research Projects Agency Network.
CATV – Cable Television
CDN – Content Delivery Network
DOCSIS - Data Over Cable Service Interface Specification
DSL – Digital Subscriber Line
DSLAM - Digital Subscriber Line Access Multiplexer
DSLAMS - Digital Subscriber Line Access Multiplexers.
FTTx – Fiber To The x
GEO - Geostationary Orbit
GGC – Google Global Cache
HDSL - High-bit-rate Digital Subscriber Line
HFC - Hybrid Fibre Coaxial
ICANN - Internet Corporation for Assigned Names and Numbers
IOT – Internet Of Things
ITU-T Telecommunication Standardisation Sector
LPWAN - Low Power Wide Area Network
MODEM – Modulator - Demodulator
PON – Passive Optical Network
POTS – Plain Old Telephony System
POTS – Plain Old Telephony System
PSTN – Public Switched Telephone Network
SDSL - Symmetric Digital Subscriber Line
WCDMA - Wide Code-Division Multi Access
XDSL – Digital Subscriber Line Technology

1. Introduction

The Internet has become the driving force of economic growth, operating within a globally complex and interconnected world that utilises undersea cables to facilitate Internet connectivity. Distributing connectivity to the last mile, where users are located, is necessary. This is where broadband plays a significant role. Broadband has become an enabler for innovation in various industries such as fintech, IOT, and healthcare. There is a high expectation for the quality of broadband that consumers need, putting a strain on broadband providers to deliver better solutions. The growth in the demand for broadband has attracted investors to invest more, with the expectation of recouping investments. This has led to exponential growth in the available technology to provide broadband services. The technology comes in various forms, primarily fixed and wireless broadband. Originally, broadband services were primarily provided using copper wires to the last mile, mainly through DSL technology(Švedek et al., 2011). Fibre was mainly used to provide capacity at the backbone of the Internet. This has changed dramatically with the deployment of FTTx technology, a general term for broadband network architectures that utilise fibre optic cables to deliver high-speed internet (Fahim & Sinha, 2023). This has enabled fibre to deliver broadband internet directly to end-users, enhancing speeds and enabling previously limited services to reach the network core, thereby providing faster and more efficient access. Apart from fixed internet, there has been tremendous progress with wireless broadband. Initially provided using WiMAX (Worldwide Interoperability for Microwave Access) radios, it quickly evolved to 2G, 3G, 4G, and 5G networks, with discussions about 6G in progress (Sharma, n.d.). The COVID-19 pandemic highlighted the importance of reliable broadband networks. Governments globally have come up with national broadband plans(ITU - DatHub, n.d.) Having realised the importance of broadband internet to the Gross Domestic Product.

Besides broadband access, cost is a significant factor in broadband growth. Most providers are quick to maximise profit to recover the return on investment in the infrastructure deployed. The entrance of LEO broadband changed the dynamics of the future of broadband deployments and accessibility. During the COVID-19 pandemic, when employees opted to work remotely, it was noted that despite Federal funding to improve broadband reach, large parts of the United States, particularly rural and remote areas, still lacked broadband access. Broadband offered via GEO satellite technology was insufficient; hence, LEO potential was identified as a solution to reach this remote area. Some of the factors that led to low broadband penetration were highlighted as low population density and topography(Congressional Research Service, 2021). LEO broadband deployment methods alter the dynamics of broadband infrastructure deployment in remote areas, thereby changing how these areas can access broadband services.

1.1 Background and Context

The Internet has constantly evolved since the first computers began communicating over the ARPANET network in 1969 (Kaur Bakshi, 2023). The development of Transmission Control Protocol and Internet Protocol (TCP/IP) standards enhanced communication between different types of networks. TCP/IP enhanced internetwork connectivity, resulting in increased growth and demand for more bandwidth to interconnect the Internet.

Initially, the internet was primarily composed of text, referred to as Web 1.0, which dominated the 90s. Content quickly evolved, and the internet started to be used by organisations for productivity. What was now constricted to Local area networks went global (Kaur Bakshi, 2023, p. 2) During this time, internet connectivity technologies struggled to meet consumers' demands. This brief overview reviews some of these technologies and provides a background for the discussions in this paper.

Initial connectivity to the internet was via dial-up systems. This was made possible by the existing PSTN (Public Switched Telephone Network), a traditional circuit-switched telephone network used for voice communication worldwide. PSTN lines, which had been laid across countries primarily for voice communications. These were primarily copper lines used for analogue communications. Dial-up systems consisted of two devices directly connected via a central switch. Dial-up systems use modems, devices that transmit computer data over a standard switched telephone line not designed for data transmission. A modem, which stands for modulator-demodulator, takes data that computers can understand and converts it into audible wavelengths that can be transmitted over a Public Switched Telephone Network (POTS) ("Modem," 2024).

There were tremendous changes in the modem industry, remarkably in the late 90s, when most telecommunication companies had switched to digital exchanges. This meant only the local loop connections remained analogue. This led to the adoption of 56kbps/s for dial-up. However, there were challenges, especially with the quality of local loop cables, which made 56 kbps challenging for most consumers ("Modem," 2024).

It seemed that the use of old copper lines was ending, and alternative solutions were needed to meet the high bandwidth demand required to keep pace with a world that was becoming increasingly interconnected. Telecommunications operators began to provide dedicated lines for customers, particularly corporate customers, with direct lines, also known as leased lines. This marked the start of the market's migration to broadband technology. The adoption of various high-speed internet technologies, such as ISDN (Integrated Services Digital Network), a set of communication standards for digital transmission of voice, video, and data over traditional PSTN lines, emerged (Dorros, 1981). Before delving into broadband technology, it is worth noting that Internet delivery over satellite has been available for some time. In 1996, Hughes (2022) Hughes Network Systems launched Directpc, the world's first satellite internet service for consumers and small businesses. Hughes Technology developed technologies that delivered high download speeds via VSAT antennas. This was mainly via shared inbound routes via time division multiplexing and shared broadcast on the downlink.

Broadband technology was initially used to refer to any connection that is not a dial-up connection. However, today, it has assumed different delivery modes, including wired and wireless. The world now utilises broadband technology to deliver high-speed internet to end users and organisations.

This paper will discuss the dominant available broadband technologies and the upcoming broadband delivered via low-Earth orbiting satellites.

1.2 Problem Statement

The global internet landscape is transforming rapidly, driven by the emergence of Low Earth Orbit (LEO) satellite broadband networks. These networks promise high-speed, low-latency internet access, particularly in remote and underserved areas where traditional broadband services provided by Mobile Network Operators (MNOs) and Internet Service Providers (ISPs) have been either slow, expensive or entirely unavailable. While LEO satellite technology holds the potential to revolutionise global connectivity, it also encounters significant challenges and opportunities as compared to existing broadband providers.

The introduction of LEO satellite services by companies like SpaceX's Starlink and Amazon's Project Kuiper is expected to disrupt the broadband market. European Commission (2024) has initiated its own program under the Infrastructure for Resilience, Interconnectivity, and Security by Satellite (IRIS²) to provide connectivity via Low-Earth Orbit (LEO) and Medium-Earth Orbit (MEO) satellites. As more consumers access affordable satellite internet, questions arise about the sustainability, competitiveness, and market positioning of LEO satellite services in relation to MNOs and ISPs, especially in rural and less profitable regions. Additionally, concerns about regulatory, operational, and infrastructure issues may impact the coexistence of satellite and terrestrial broadband technologies. According to Ericsson (2023) One of the largest producers of mobile broadband connectivity technology released a report in 2023 indicating that over 9.1 billion new mobile subscribers will be added by 2028, and 88 per cent of them are likely to use mobile broadband. The emergence of LEO broadband changes the dynamics of broadband delivery.

This thesis investigates the challenges and opportunities LEO broadband faces in relation to traditional internet service providers. It aims to analyse market trends, technical challenges and opportunities, competitive responses, and the long-term implications for both LEO broadband services and existing broadband services. This thesis will also investigate the regulatory challenges LEO broadband providers face.

1.3 Objectives of the Study

The evolving nature of broadband internet is interesting as various technologies try to maximise reach and minimise costs. The entry of LEO broadband has changed the dynamics and disrupted the standard deployment methods of broadband internet. This study explores the challenges and opportunities that low-Earth orbit (LEO) satellite broadband presents to the global broadband landscape. The research also examines how regulators are prepared to adapt to the evolving broadband landscape. The primary objective of this study is to investigate the challenges and opportunities that Low Earth Orbit (LEO) satellite broadband faces in reshaping market dynamics, service quality, and customer preferences compared to traditional broadband providers such as MNOs and ISPs. The secondary objective is to examine the preparedness of national and international regulators to integrate and support the deployment of LEO broadband internet services. The study aims to explore the core operational, technical, and economic challenges facing LEO broadband deployment.

What challenges and opportunities will LEO broadband face in a market dominated by MNOs and ISPs?

How prepared are regulators to manage the integration of LEO satellite internet into the existing broadband ecosystem, and what are stakeholders' perceptions, including consumers, providers, and policymakers, regarding LEO broadband as a sustainable and competitive alternative?

1.4 Thesis Outline

The outline of this thesis is as follows: Section 2 reviews the relevant literature, Section 3 explains the research methodology and design, Section 4 presents the findings, Section 5 discusses research findings, and Section 6 offers a conclusion and recommendations based on our findings.

2. Literature Review

2.1 Overview of Broadband Technologies

Long-distance and short-distance communication has grown over the decades. The primary factors driving this growth are increased traffic, the introduction of new services, and technological advancements. Stallings (2014) noted that the increased use of images, videos, social media, and the need for remote access for business transactions have increased the need for reliable connectivity(p. 33). The mobility and decentralisation of services, primarily relocating to the cloud, have also fueled the growth of broadband internet services. Broadband services need to provide connectivity and consistently guarantee the availability and quality of service. Speed improvement has differed with different technologies as they have evolved. According to Cisco's annual general report (CAGR) (2023) Fixed broadband speeds were projected to reach 110.4 Mbps in 2023, compared to 45.9 Mbps in 2018.

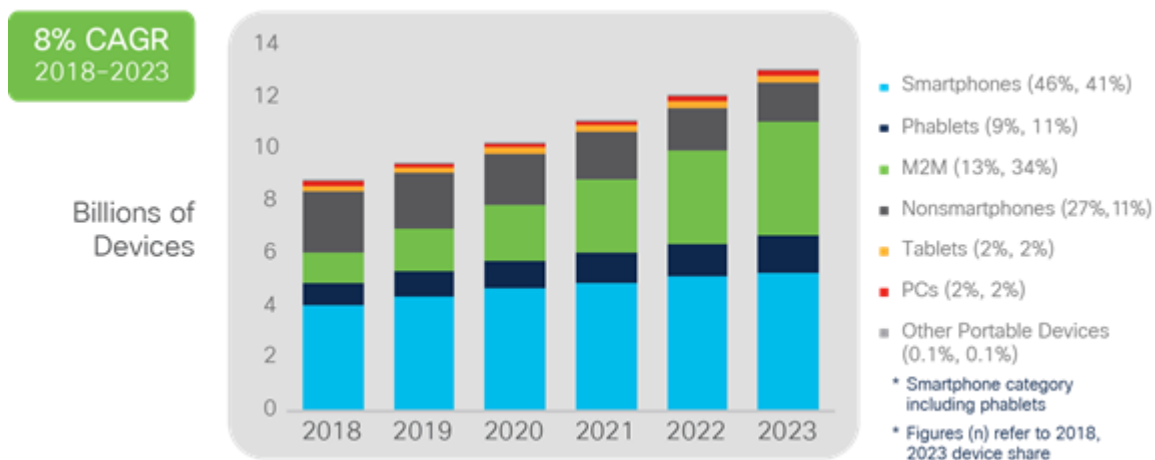
Table 1 Fixed broadband speeds (in Mbps), 2018–2023, 2018)

Region	2018	2019	2020	2021	2022	2023	CAGR (2018–2023)
Global	45.9	52.9	61.2	77.4	97.8	110.4	20%
Asia Pacific	62.8	74.9	91.8	117.1	137.4	157.1	20%
Latin America	15.7	19.7	34.5	41.2	51.5	59.3	30%
North America	56.6	70.1	92.7	106.8	126.0	141.8	20%
Western Europe	45.6	53.2	72.3	87.4	105.6	123.0	22%
Central and Eastern Europe	35.0	37.2	57.0	65.5	77.8	87.7	20%
Middle East and Africa	9.7	11.7	25.0	29.0	34.9	41.2	33%

Source: Cisco Annual Internet Report, 2018–2023

The number of devices connected to the Internet is also experiencing tremendous growth, with most of these connections expected to be made via broadband Internet, according to the Cisco Annual Internet Report (2023), as illustrated in Figure 1.

Figure 1 Global mobile device and connection growth



Source: Cisco Annual Internet Report, 2018–2023

From early dial-up systems that were unreliable and costly, the markets evolved to internet connectivity that was always on. The definition of broadband was based on connectivity that was not dial-up. Later, the definition of broadband was measured in terms of bandwidth. In an article published in 2000, Rath (2021) describes broadband networks as “connections at data rates of 250 Kbps.” Of course, this has changed as more technologies have evolved. Haidine & Aqqal (2018) This definition gives us a more consistent approach to defining broadband networks. It is “any network/access technologies used by Internet Service Providers (ISP) to provide broadband Internet access for multimedia content delivery/distribution according to technical considerations and requirements such as guaranteed Quality of Service.”

Significant improvements have been made in broadband technology over the years, enhancing both the speed and quality of broadband services. We will now examine how broadband technology has changed over the years.

In the mid-1990s, corporate networks began to deploy T-carrier technologies, which AT&T had originally developed to transmit multiple phone calls over a single link. T-carrier technologies use PCM/TDM systems. PCM (Pulse Code Modulation) is used to represent analogue signals digitally. TDM (Time Division Multiplexing) transmits multiple signals over a single channel by dividing the channel into time slots. Many PCM signals were transmitted over a common channel by time division multiplexing. In this way, multiple signals could be carried over telephone lines. This was then transmitted through a PCM transmitter, which performed an analogue-to-digital conversion. Along the way, the signal passed through several repeaters until it reached the PCM receiver, which was converted back to an analogue signal. The signal is sampled at a rate of 8,000 samples per second. The transmission rate of a T1 line is achieved at 1.544 Mbits/s (Wachtel, n.d.).

With T1, data could be transferred faster than with the dial-up system. This was the first transformation toward broadband communication. Other technologies became available to end consumers. As fast as T1 carriers were, they were restricted to the backbone, and only corporations could afford them. The good news is that there was at least a higher capacity to interconnect various exchanges and transport multiple

T1 lines.

The use of twisted pair cables to transmit more than just voice, by utilising a different frequency than that used for voice calls, led to the invention of the Digital Subscriber Line (DSL). DSL was first introduced by Bellcore in 1989, and was mainly deployed in North America (Savino, 2000a, p. 453). Unlike the modems used for dial-up systems, DSL had its modems at the subscriber end and DSLAMs (Digital Subscriber Line Access Multiplexers) at the service provider. This DSLAM aggregated connections from multiple DSL subscribers and established permanent connections over long lines (Gill, 2012). This means the connection will be established automatically when the modem is turned on and will remain active until the modem is powered off. This significant development enabled Internet providers to offer initial broadband services to end users.

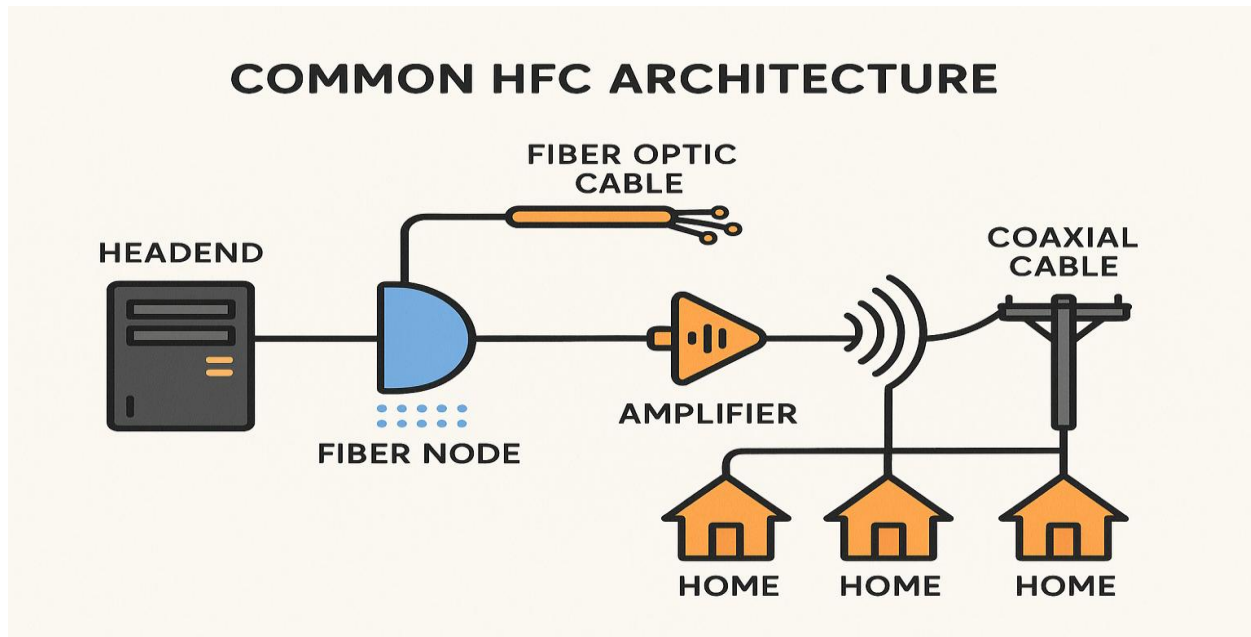
Different versions of DSL were introduced to the market. We will discuss the most common types of DSL, specifically SDSL and ADSL, which were the most popular.

ADSL stands for Asymmetric Digital Subscriber Line, which means it offers different upload and download speeds (Maxwell, 1996). In the early days, most internet connections by subscribers were asymmetric. ADSL uses the same pair of copper lines to transmit data while maintaining uninterrupted voice capabilities. Normal POTS utilises a 0 to 4 kHz frequency range with voice sampling from 300 Hz to 3.4 kHz, while ADSL starts at 25 kHz and extends up to approximately 1.1 MHz in frequency. The separation of frequency is achieved through multi-carrier modulation. (Kernan, 2002) The most used technique in ADSL is discrete multi-tone modulation, which ANSI and ITU have adopted as a standard technique. The frequencies are separated into 4.3125 kHz bands. ADSL offers asymmetric bandwidth; however, some applications, like video conferencing, require symmetric bandwidth. Another type of popular DSL is the SDSL, which stands for Symmetric Digital Subscriber Line. SDSL technology is basically the same as ADSL, except that the upload and download speeds are the same. SDSL did not allow for the sharing of lines by voice. SDSL was primarily used by businesses that hosted servers that required sending data to clients. Both businesses and home users more commonly use SDSL. SDSL utilised high-bit-rate digital subscriber line (HDSL) technology. HDSL is a scheme that utilises two pairs to provide two-way transmission at 1.544 Mb/sec or 2048 Mb/sec, depending on the technology (Savino, 2000b, p. 3).

Apart from the xDSL technologies, we also have cable broadband. Early cable television was primarily relayed via wireless technology; however, in the 1940s, the first cable television standards were developed. In 1984, the Cable Communications Policy was passed in the United States. The deregulation of the cable television business in the United States led to a significant boom in growth and adoption ("Cable Communications Policy Act of 1984," 2024). The cable television infrastructure grew nearly the same size as the PSTN network. Cable companies started utilising coaxial cable to relay internet signals in frequency bands that did not interfere with television signals. Cable TV networks were initially designed to operate in the 47-862 MHz frequency band to broadcast time-division multiplexed TV and radio signals. To provide additional services such as the internet, the frequency range between 5 and 30 MHz was allocated for upstream transmission, and the return channel could have a bandwidth of 5 to 55/65 MHz, starting from 73

or 83 MHz (Kos et al., 2003, p. 829). The cable television design features a Hybrid Fibre Coaxial (HFC) architecture. With HFC, fibre is laid to strategic locations on the distribution network, where the signal is converted back to copper for distribution to end users. As illustrated in Figure 2, the HFC design increased the available bandwidth, improving quality and reliability.

Figure 2 Common HFC architecture



Adopted From Wikipedia ("Hybrid Fiber-Coaxial," 2024)

The transition from analog to digitally enhanced capacity is available for broadband services. To transmit high-speed data, cable broadband utilises a cable modem that transmits data sent from the end user in digital format through the HFC network to the head end, where a cable modem termination system (CMTS) modulates and demodulates the signal, routing traffic to the internet. DOCSIS standardisation helped to accelerate cable modem growth. Cable modems from multiple vendors can work with cable modem terminal systems from different vendors. DOCSIS enables higher bandwidth transfer, interoperability, quality of service, and security over existing coaxial cables. Modern DOCSIS versions, such as DOCSIS 3.0 and DOCSIS 3.1, enable gigabit speeds by employing advanced modulation techniques and channel bonding, which combines multiple frequency channels to increase throughput. ("DOCSIS," 2024)

Cable broadband provided last-mile connectivity to end users. The demand from end users has been increasing over the years, and there is a need to improve last-mile connectivity to meet these demands. Previous technologies have required dedicated channels to accommodate data. Most providers had converged services, such as voice and media, into IP-deliverable products. For some time, fibre has been utilised in the core networks of internet service providers. Over the years, fibre has been used to deliver connectivity closer and closer to the end-user. These technologies have been grouped together under the banner Fiber to the x (FTTx). FTTx could be used to define any technology that involves fibre to deliver connectivity to the end user. Standard deployment methods include Fiber to the Home (FTTH), Fiber to the Neighborhood (FTTN), Fiber to the Cabinet (FTTC), and Fiber to the Building (FTTB) (Freudenrich &

Pollette, 2022). The focus will be on Fiber to the Home (FTTH), which falls within the broadband category of cable modems. There are two types of FTTH deployment: Passive Optical Network (PON) and Active Optical Network (AON). Most FTTH deployments utilise PON. PON is a point-to-multi-point optical network with no active elements in the signals path from source to destination. The only elements in the path are splitters and splices. Transmissions in a PON network are done by the optical line terminal (OLT) at the provider's head end and optical network units at the end user location (Kramer & Pesavento, 2002, p. 68). PON is a point-to-multipoint architecture. A single fiber-optic strand from an optical line terminal (OLT) can be replicated across many branches using a splitter. Depending on the network requirements, this can be designed as a Gigabit PON or Ethernet PON, with data transfer speeds ranging from 1 Gbps to 10 Gbps. The other deployment for FTTH is the Active Optical Network (AON), which features active equipment along the path (Butt et al., 2015).

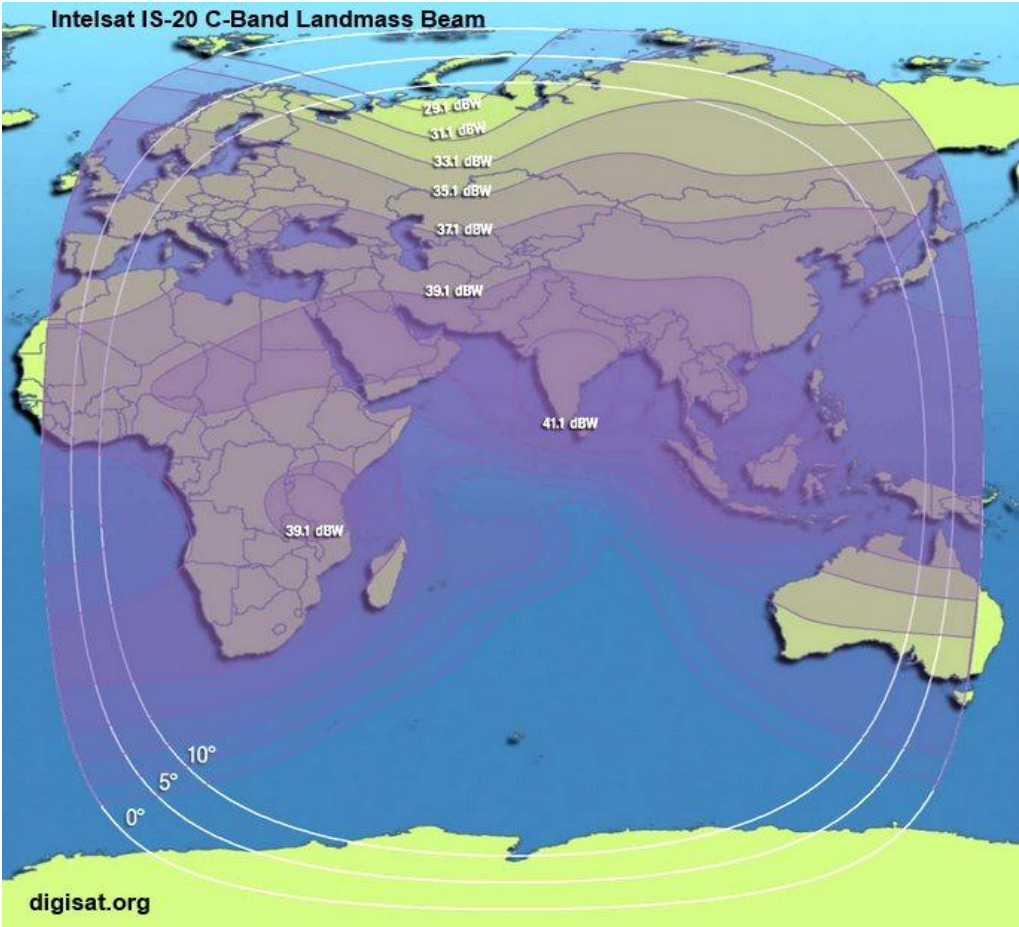
So far, discussions have centred on fixed broadband services, which require the installation of physical media to provide customers with broadband services.

According to the ITU-T (2023c) report, fixed broadband services are expected to account for 80 per cent of global internet traffic in 2022. Just as there have been enhancements in the capacity and quality of fixed broadband services, mobile broadband has experienced significant growth and made notable strides over the past 15 years. The entry of mobile broadband was enabled by the Third Generation Partnership Project (3GPP) wide code-division multi-access high-speed packet access (WCDMA-HSPA) standard (Hadden, 2009). Mobile broadband involves providing broadband services without physically wiring the end customer. The first WCDMA was launched in Japan in October 2001 and quickly spread worldwide. Initial speeds were up to 2 Mb/s; however, this would soon increase to 7.2 Mb/s with HSPA. HSPA is a mobile broadband technology that enhances the performance of 3G networks, making data transmission faster and more efficient (Furht & Ahson, 2010). The growth and penetration of smartphone users are driving demand, and recently, we have seen the deployment of Long-Term Evolution (LTE) and 5G technologies. 5G technology utilises millimetre waves with wavelengths ranging from 1 to 10 millimetres and 30 to 300 gigahertz frequencies. Low wavelengths are vital for high-bandwidth data transmission, increased throughput, enhanced capacity, and reduced latency (Ana-Maria et al., 2021). Just as 5G technology is being deployed, the focus is now on 6G, with China launching the world's first 6G experimental satellite (Tomás, 2024). Advancements in mobile broadband technology are transforming the infrastructure landscape with the emergence of mobile edge computing. Mobile edge computing is expected to reduce latencies and improve user experience in the near future (Loutfi et al., 2024, p. 2). Mobile broadband services are making inroads in internet usage. According to an ITU report in 2022, the mobile broadband share of global internet traffic increased from 3.4% to 4.2%; however, this is expected to change as mobile broadband continues to increase its penetration.

Mobile broadband is not the first technology to provide broadband services over a wireless medium. Satellite internet has been utilised to deliver internet services in remote areas for decades. Satellites are categorised into three main types based on their orbital height. Low earth orbit (LEO) deployed between

100 to 2000 kilometers above the earth's surface, medium earth orbit (MEO) deployed between 2000 to 35000 kilometers above the earth's surface and geostationary earth orbit (GEO) deployed at 35,786 kilometers above the earth's surface (Pereira & Ribeiro, 2011, p. 5). To communicate, satellites use special frequencies. The main frequencies are as follows: C-band technology operates in the 4–8 GHz range and requires antennas of 6 to 12 feet in length, while Ku-band operates on the 12–18 GHz range and requires smaller dish antennas (usually 2 to 4 feet) in diameter. Recently, other frequency allocations have been added to the bands, including Ka-band (26.5–40 GHz), V-band (40–75 GHz), and E-band (60–90 GHz). The provision of broadband internet is primarily achieved through GEO and LEO satellites, although a few services are also available on MEO satellites. GEO satellites are used because they almost appear stationary from a single point on Earth. This makes it easier to point antennas and establish continuous connectivity. Below, as illustrated in Figure 3, is an example of a footprint by the Geo satellite Intelsat IS-20. These GEO satellites have a large footprint.

Figure 3 Global footprint Intelsat IS-20



Source: (digisat.org,)

The satellite acts as a relay between two Earth stations, providing internet access. The central stations act as the internet source that is distributed over the satellite coverage area. GEO and LEO satellites have different characteristics, as summarised in Table 2. Notably, the difference in latency between the two orbits has a significant impact on broadband services. The GEO satellite has an advantage as it covers a large footprint compared to LEO, which would require thousands of satellites to cover the same area

Table 2 Comparison between GEO and LEO satellites

	GEO	LEO
Latency	~700ms	< ~50ms
Network size for global coverage	3 satellites (99%)	Thousands
Data gateways required	Few, fixed	Numerous local
Cost of deployment	\$1 – 1.5b	\$5- 15b
Satellite life cycle	15 years	5-7 years

Source: Own representation

Satellite capacity is limited by available power, transmission bandwidth and noise(Reudink et al., 1981). Historically, geostationary satellites have primarily provided backbone connections for regional computer networks. Very small aperture terminal (VSAT) technology made satellites more economical for connecting individual end-user stations to enterprise networks and linking local networks. However, the high cost of network interfaces made this technology less suitable for direct-to-home systems (Clausen et al., 1999, p. 1).

2.2 Development of Low Earth Orbit Satellite Constellations

Since the 1990s, efforts to use low-Earth orbit satellites to provide broadband connectivity have been viewed as a solution to the elusive issue of remote connectivity. Kohn (1997) discusses how LEO would address the lack of last-mile technology to distribute broadband to remote locations. The space sector development has evolved critically over the past decade as a result of many new players entering the market due to commercial interests and new technologies, applications, and markets that span across geographical regions (P.-H. Chen & Cho, 2024). GEO satellites provide communication and the Internet; however, due to their large power terminals and significant signal delays resulting from their high altitude, they are unsuitable for large-scale broadband internet deployments. On the other hand, with a low Earth orbit, a satellite can only cover a portion of the Earth's surface; however, it provides low latency for real-time applications.

To cover a large surface of the earth with LEO, you need several satellites, also known as a constellation of satellites. Wood (2003) describes a constellation of satellites as several similar satellites of a similar type and function, designed to be in similar, complementary orbits for a shared purpose under shared control. Several constellations are currently being launched to LEO. Table 3 presents some of the commonly known constellations that are to be launched.

Table 3 Projects and the projected number of satellites to be launched

Project	Number of satellites to be launched
Starlink	42,000
Amazon Kuiper	3,236
Telesat	298
OneWeb	650

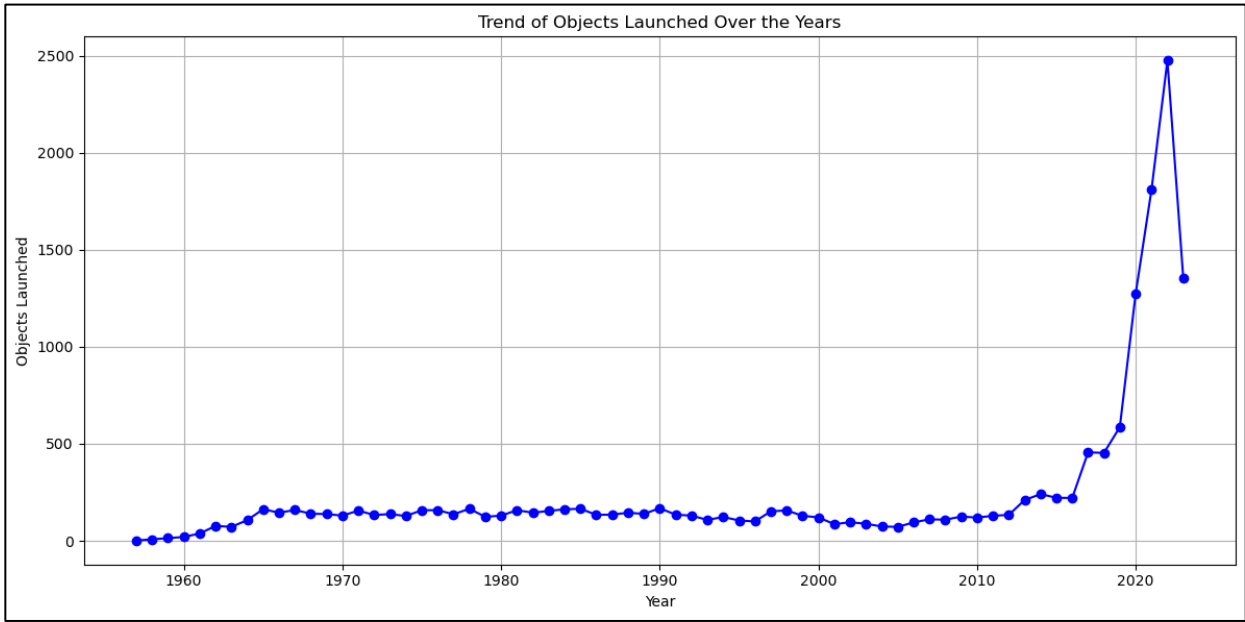
Source: Own representation

According to Pereira & Ribeiro (2011), the five key features we look at when provisioning satellite internet are global coverage, low latency, throughput, extensibility, and vertical application (p. 6). The following section provides a brief overview of past attempts to deploy constellations in Low Earth Orbit (LEO) for commercial purposes.

The Motorola Iridium project was the first attempt to establish space communication via a LEO satellite. The Iridium network comprised 66 cross-linked satellites (plus 6 in-orbit spares) at an altitude of 778 km in 6 polar planes, with 11 satellites in each. The system was designed primarily for voice and pager communication, costing over \$5 billion. This project failed due to low subscription numbers, with only 20,000 subscribers in a year, compared to the projected 500,000 subscribers (Beesemyer et al., 2012, p. 5). The other pioneering project, Teledesic in the 1990s, aimed to provide global broadband internet coverage using a constellation of low-Earth-orbit (LEO) satellites. It was one of the first major attempts to develop a satellite-based internet service that could deliver high-speed, low-latency internet globally, targeting underserved and rural areas. Initially planned to deploy 840 LEO satellites at a 700 km altitude, the project never took off due to the prohibitive cost of launching hundreds of satellites, and technological advancements in satellite communications were not yet sufficient to achieve the desired speed and latency (Beesemyer et al., 2012, p. 8).

Teledesic's vision lives on in current projects, such as SpaceX's Starlink, Amazon's Project Kuiper, and OneWeb, all working toward providing LEO-based broadband internet to underserved areas worldwide, now with more advanced technology and greater financial backing. A decade ago, the number of satellites in orbit was approximately 1,000; today, it exceeds 10,000. Figure 4 illustrates the rate of launches over the years. There has been a significant increase in 2017, and most of the satellites were to LEO orbit.

Figure 4 Number of satellites in orbit launched over the years



Source: Generated from data (Andy, 2023)

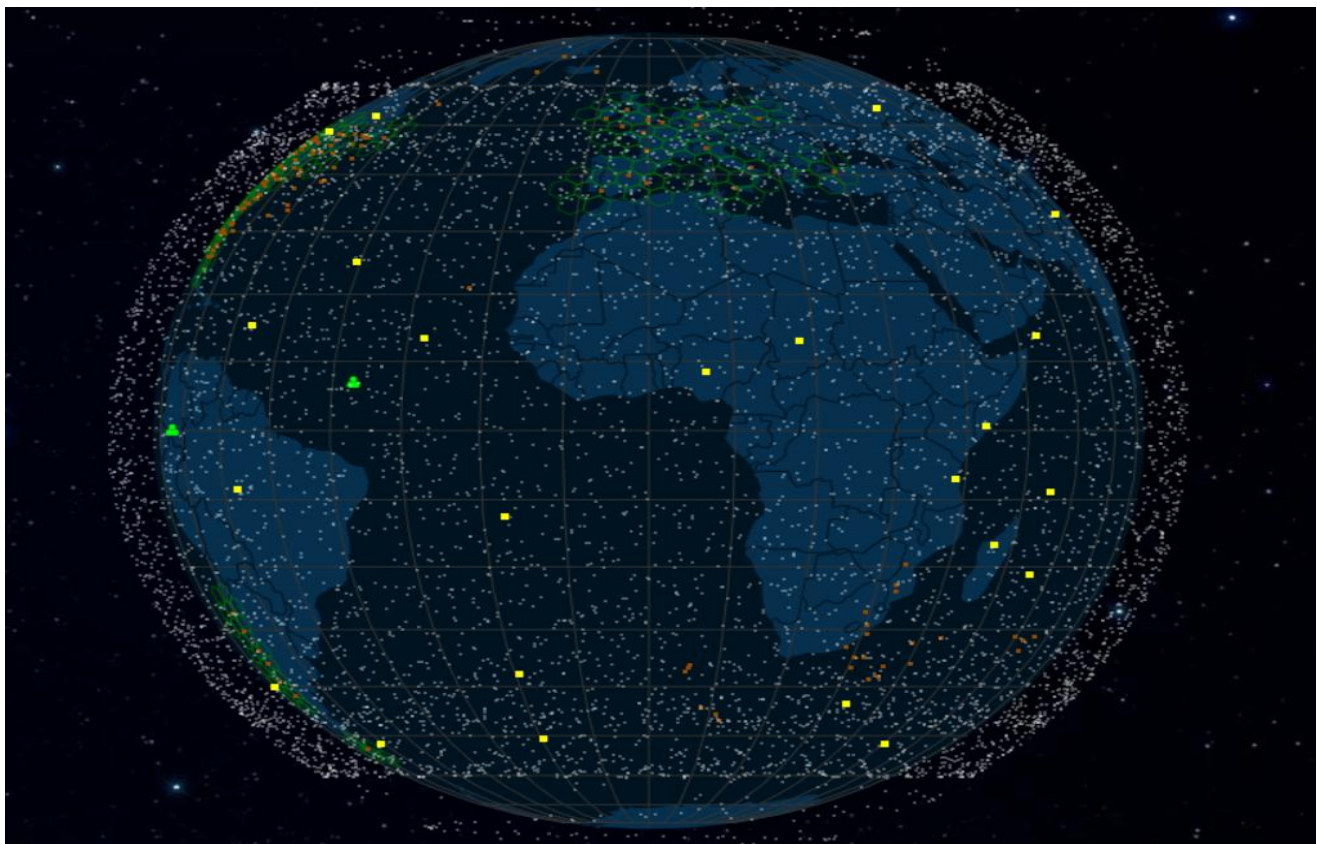
Chronopoulos (2024) notes that two factors have enabled the rapid deployment of LEO satellites. First is

the new satellite technology that has enabled the production of smaller satellites, reducing their weight from thousands of kilograms to tens of kilograms per satellite. Second, there has been a significant increase in venture capitalists investing in space businesses over the last decade. There has been significant progress in LEO broadband technology, and further discussions are underway among telecommunications and internet standards bodies regarding the future of LEO and its potential integration into existing terrestrial infrastructure. According to the World Radiocommunication Conference (2025), 80% of 2027 agenda items in the ITU-R workgroups will be focused on satellite and space communications. There has also been an upsurge in the number of spectrum requests filed with the ITU, indicating a surge in interest in space communications, including inter-satellite communications, traffic routing schemes, and space–ground communications (ITU - Space Connect, 2025).

So far, the largest satellite constellation used to provide broadband internet is the Starlink network. As of 2025, approximately 7,000 satellites have already been launched, with 6,146 operational as of mid-2024. Geespace, a Chinese company, plans to form a mega constellation it describes as China's equivalent of U.S. firm SpaceX's Starlink. Geespace plans to build a constellation of nearly 6,000 LEO satellites that would provide global broadband (Baptista, 2024)

Figure 5 shows the Starlink constellation with approximately 7000 satellites in low earth orbit.

Figure 5 Starlink Satellite Orbit



Source: (Satellite Tracker, 2025)

The design of LEO broadband consists of user terminals, satellites, and ground stations. User terminals are small phased-array antennas with a motor to track fast-moving satellites. The terminal only connects to one

satellite at a time. Broadband user traffic is transmitted from user terminals via this satellite connection (Tanveer et al., 2023, p. 38). Businesses, residences, marine traffic, and IoT sensors will use user terminals to connect to the LEO satellite. The user terminals can also connect end-user devices using WIFI or Ethernet cables to distribute the connections. The user terminals operate in Ku band (10.7-12.7 GHz downlink and 13.75-14.5 GHz uplink) and Ka band (17.7-20.2 GHz downlink, 27.5-30 GHz uplink) (ITU - Space Connect, 2025) Due to the high speed at which LEO satellites move, the user terminals may send data to different satellites. LEO terminals must be able to track and connect to multiple satellites within the LEO constellation.

A satellite or spacecraft internet constellation aims to provide global broadband internet access. It utilises low Earth orbit (LEO) satellites positioned at altitudes between 340 km and 1,110 km, significantly closer than traditional geostationary satellites, thereby reducing latency. Operating at an altitude of 550 km, the spacecraft is capable of "covering" with its signal a spot on the Earth's surface with a radius of about 950 km (i.e., a diameter of about 1900 km), provided that the elevation angle for the user terminal is at least 25 °(Dvornikov, 2023, p. 36).

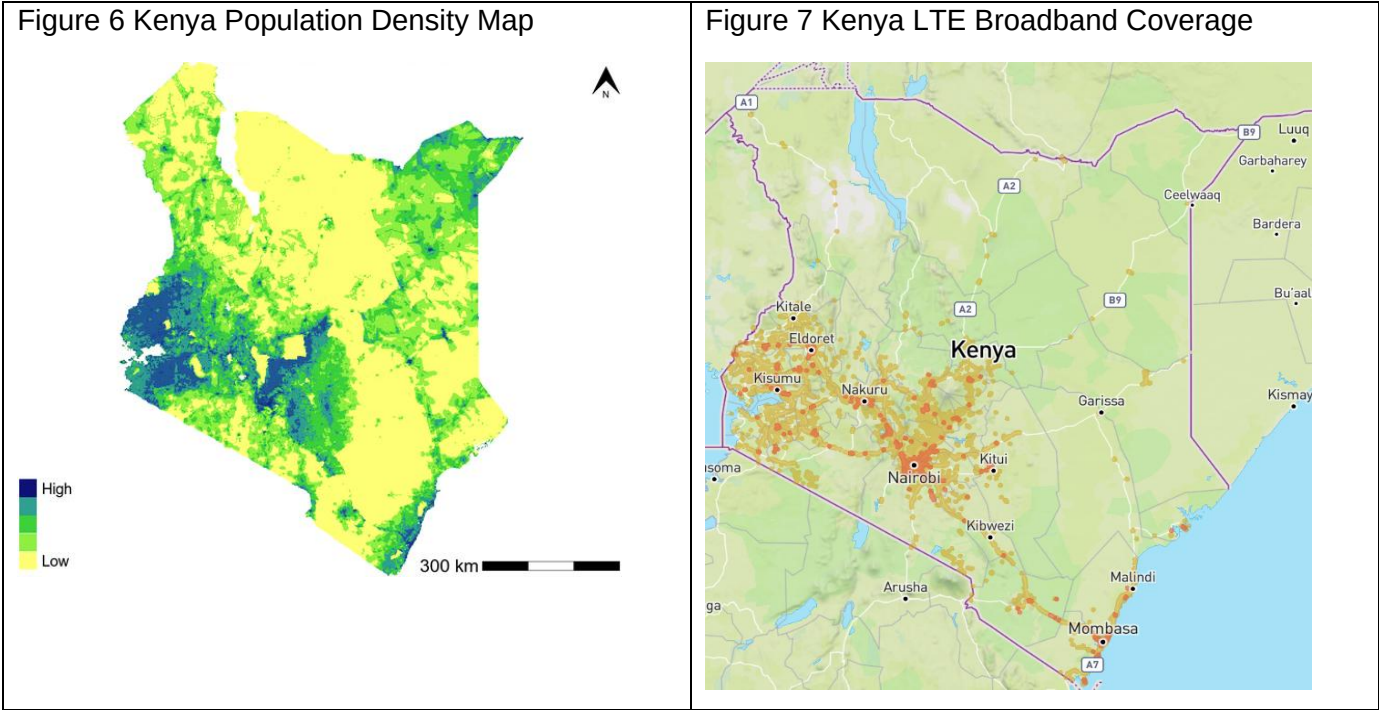
A ground station/gateway receives traffic from the spacecraft. It sends it via a terrestrial link to a gateway pop, which is, in turn, connected to the internet, cloud services and internet exchange points. A typical ground station has eight antennas, each capable of transmitting information to its own spacecraft (Dvornikov, 2023, p. 37). LEO gateways primarily operate in the Ka band, with a downlink of 17.7-20.2 GHz and an uplink of 27.5-30 GHz. There is also an extension to enable ground stations to operate in E-band (71-76 GHz downlink, 81-86 GHz uplink) and Q/V-band (37.45-42 GHz downlink, 47.2-50.2 GHz and 50.4-51.4 GHz uplink) (ITU - Space Connect, 2025).

2.3 Traditional Broadband Service Challenges

Having discussed the technology behind fixed and wireless broadband, Broadband service providers face challenges when deploying this service.

Traditional broadband providers face uphill tasks in infrastructure development. Covering large geographical areas entails deploying lengthy physical infrastructure. Building, running and maintaining such infrastructure can be very costly. Rokkas (2015) conducted a comprehensive study on the costs of deploying FTTTH and found that the cost of deploying fibre in low-density areas is relatively high. A lack of network infrastructure availability is one of the reasons why, in Africa and the Asia-Pacific region, less than half of the population is online (29 per cent and 45 per cent, respectively). Garrity and Aminata (2020) found fibre connectivity only serves approximately 2 billion people within a ten-kilometre radius of current fibre-optic cable networks, suggesting that most of the world's population still lacks access to fibre networks due to geographic distance. According to GSMA report (2024) 350 million people (4% of the global population) live in largely remote areas without mobile internet networks (the coverage gap) and 3.1 billion people (39% of the global population) live within mobile internet coverage but do not use it (the usage gap). Most of the high costs associated with deploying fixed and mobile broadband services are incurred in non-core components of the infrastructure, such as civil works and regulatory costs. The Department of Transportation has compiled statistics in the United States that put the average cost of laying fibre at

\$27,000 per mile (taoti_admin, 2017). This makes it very expensive to deploy fibre in remote areas. In developing countries, the rapid construction of infrastructure and buildings damages many last-mile connectivity infrastructures, leading to high maintenance costs. Numerous reported cases of vandalism have been documented in areas where copper is still used to bridge the last-mile connectivity (NCTA, 2024). The distribution of customers affected by various factors, such as geographical conditions, is challenging when rolling out our traditional broadband internet. The more densely populated areas tend to be favored, as the average rate of connecting a user is lower than in the geographically out-of-reach areas. Figures 6 and 7 illustrate the map of Kenya with population density and current LTE coverage zones. This illustrates one of the weaknesses of terrestrial infrastructure deployed by traditional broadband providers/



Source:(Statskenya, n.d.)

Population density can be influenced by factors other than broadband availability. However, broadband availability seems to follow population density patterns.

The distribution of broadband access infrastructure follows a private-sector model. In such cases, internet service providers assess the estimated returns before deploying broadband to neighbourhoods. Research by the Pew Charitable Trust in the United States found that broadband infrastructure was provided in areas with low-income communities. In these areas, the infrastructure is either missing or not maintained. In cases where the internet is available, it is often dominated by a few internet service providers, resulting in increased costs per household due to the lack of competition that is typically enjoyed in other areas with more competitive pricing (Chikwendiu, 2023). According to the Federal Communications Commission (FCC) , (2021) rural and tribal areas lag in broadband deployment due to low population density.

The affordability of broadband internet is a challenge facing Internet service providers. Given the input costs, providing broadband services at a cost affordable to the mass population and managing to generate good returns remains challenging. In research conducted by the National Telecommunications and Information

Administration (NTIA) in 2022 on subsidised housing households that do not use the Internet, it was found that the mean price considered acceptable was \$10 per month. However, most indicated they could only pay zero (Cao & Rafi, 2022). The broadband costs to the end user depend on several factors, including monopoly, government regulation, and the costs of delivering the broadband. According to the World Population Review (2025) conducted in 2024, residents in the United Arab Emirates pay an average of \$98 per month for broadband internet, while in Burundi, the costs paid by end users are an average of \$304.57.

The fast-evolving technology changes also affect broadband deployment, including bandwidth-hungry applications that put a strain on existing last-mile technologies. While companies deployed massive copper with the hope that technology such as CATV and XDSL would dominate the broadband market for a while, this was not the case. Most Telecom companies gradually evolved from copper-based networks to more current technologies such as FTTx (Butt et al., 2015). The rapid change and improvement in broadband technology, along with the associated costs, prompt traditional broadband providers to invest more in a short period. Fixed wireless broadband, while cost-effective, is limited by line-of-sight requirements and spectrum availability. Both fixed and wireless broadband are impacted by rapidly changing technology. Once deployed, broadband providers strive to recoup their investment before undertaking a network upgrade that may render their current communication equipment unusable.

Originally, telecom and, by extension, broadband services were provided by government-owned operators. Regulatory policy and licensing were mainly developed as part of privatisation. In the US and Canada, where private firms conducted telecom operations, licenses were not required; however, enforcement was carried out through government regulatory authorities' decisions, orders, or tariff approval processes. The WTO reference paper, or the Agreement on Basic Telecommunications, which guides member countries, contains guidelines that promote liberalisation and competition in telecom markets by discouraging anticompetitive behaviours (Gruber, 2007). This is a step towards enabling equal access to broadband services globally. Regulations sometimes impose substantial investments to enter the broadband market. This has created monopolies that come with market domination and poor pricing for end consumers. In Ethiopia, a state-owned monopoly has controlled broadband access until 2023, when the regulator allowed the second operator (Shaban, 2017).

The World Broadband Association (2024) classifies the costs of building broadband fibre into costs per building passed and per building connected. The costs per building passed are the costs of connecting a building, even though there are no subscribers. The costs per building connected are where connectivity is driven by customer subscription/demand. These costs can vary from thousands of dollars in rural areas to hundreds of dollars in urban areas. These differences make it challenging to plan for profitability in broadband fibre rollout. (p. 7).

The ever-changing bandwidth requirements of broadband users also challenge traditional broadband services. Most devices are now IP-enabled and require broadband services. Services that were traditionally provided locally, such as word processors, are rapidly being relocated to the cloud. This has put a strain on outdated technologies used to provide broadband to the end user. Some factors previously not considered

sensitive, such as latency and throughput, are now integral to the basics of broadband services.

Due to their mostly terrestrial deployment, traditional broadband services are subject to environmental factors. Excessive rainfall, hurricanes, and high temperatures can affect or destroy broadband services' performance. Vandalism and sabotage raise the costs of operating terrestrial infrastructure.

2.4 Potential Advantages of LEO Satellite Internet Mobility

The provision of satellite broadband has mainly been via GEO satellites. A long propagation delay has characterised this. LEO can provide seamless global coverage compared to fixed and wireless broadband (Y. Chen et al., 2024). LEO satellites operate at much lower altitudes (typically between 500 and 2,000 kilometres). This results in low latency that enables real-time applications to work more efficiently (Internet Society, 2022, p. 2) This places LEO satellites within acceptable latency, which terrestrial networks currently provide to deliver broadband services.

Radio waves travel at a speed of 300,000 kilometres per second (Smith-Rose, 1950). To calculate the latency, we need the distance from the Earth to the satellite.

$$\text{Round trip time} = \frac{2 \times \text{orbit height}}{\text{Speed of light}}$$

For GEO satellites that are approximately at ~ 36000 Km the total latency is

$$\text{Round trip time} = \frac{2 \times 35,786}{29792} = \sim 0.24 \text{ Sec} = 240 \text{ ms}$$

For LEO satellites that are approximately at 500 Km

$$\text{Round trip time} = \frac{2 \times 500}{29792} = \sim 0.034 \text{ Sec} = 3.4 \text{ ms}$$

This does not include processing delays at the terminal, satellite, and ground stations. Due to additional factors, such as propagation delay, inter-satellite links, and atmospheric and hardware delays, Starlink achieves a latency of approximately 25 milliseconds, rather than the theoretical minimum of 3.3 milliseconds, depending on the location of the ground stations and user terminals (Starlink | Technology, n.d.) LEO satellites have a latency of 25 milliseconds, the same as fibre optic cables, and a light speed of about 200,000 kilometres per second. They also have low-cost launchers, which makes it cheaper to access space. Another factor driving LEO satellite deployment is that the costs per kilogram have drastically reduced from \$1 million to \$1,500 per kilogram. Table 4 illustrates the cost reduction over the years.

Table 4 Launch vehicle estimated payload cost per kg

Launch Vehicle	Payload cost per kg
Vanguard	\$1,000,000
Space Shuttle	\$54,500
Electron	\$19,039
Ariane 5G	\$9,167
Long March 3B	\$4,412
Proton	\$4,320
Falcon 9	\$2,720
Falcon Heavy	\$1,500

Source: ("Space Launch Market Competition," 2025)

a significant change has been brought about by the ability to reuse the launch vehicle that promises to slash launch costs by a factor of 10 (Space Economics, 2023).

LEO broadband satellites have a lifespan of between 5 and 7 years. This makes using less sophisticated technologies easier than their GEO counterparts. This means that upgrades can be easily implemented, and newer technologies can be deployed more quickly.

Global coverage

Geographic barriers, economic viability and regulatory barriers constrain drivers of broadband coverage. One advantage of LEO satellites is their ability to provide global coverage by building constellations. Draim (1991) describes a satellite constellation is a group of satellites orbiting around a central mass as a set of n-tuple groups in different orbital planes. These satellites are placed in different orbital planes, enabling them to cover the entire Earth's surface as they orbit collectively (Draim, 1991). LEO satellites have lower orbits and cannot cover a large area like GEO satellites. While it takes a few GEO satellites to cover a larger area of the Earth, it takes a constellation of LEO satellites travelling at high speed to cover the same area (Nadoushan & Novinzadeh, 2014). Using inter-satellite links, LEO satellites can forward data from areas where ground-based base stations cannot be deployed and where there are no feeder networks (Xiao et al., 2024, p. 90).

Traditional broadband services have mainly focused on delivering services in areas where there are economic returns. The case of LEO satellites is different. An orbiting satellite must pass through remote areas and always maintain its orbital plane. This makes it ideal to provide internet access to remote locations where deploying terrestrial infrastructure is not economically feasible. LEO satellite constellations can provide Internet access to remote and underserved areas where traditional ground-based infrastructure is lacking. This can help bridge the digital divide, ensuring that rural and isolated communities can access connectivity (Internet Society, 2022, p. 1).

With reduced launch costs and low latency, LEO satellite broadband has excellent potential to provide high-quality internet to underserved communities worldwide at a comparable cost, regardless of demographics or infrastructure costs.

6G intends to provide massive user access and high capacity on the Internet, which will be challenging to

cover with only terrestrial networks. LEO has an opportunity to provide a satellite access network that can be used as 6G base stations to complement ground-based cellular networks. Apart from satellite access stations, LEO satellites can connect base stations and deliver high-speed broadband to end users. (Xiao et al., 2024)

Mobility and Flexibility

According to the World Science School (2019) approximately 71% of the Earth's surface is covered by water. Traditional broadband services are primarily concentrated in densely populated areas of the Earth's surface. Because LEO broadband is delivered via satellite, it has flexibility that is not available with fixed infrastructure. Users in remote locations can access services via satellite where terrestrial infrastructure is unavailable. Roving users can also take advantage of satellite internet at convenient locations. Ships and planes also utilise LEO satellites to provide customers with the first broadband internet.

In some cases of disaster or war, the LEO satellites can provide broadband internet where terrestrial infrastructure is destroyed or unserviceable. The Ukraine-Russia war demonstrated how Starlink could deliver high-speed internet support to a population under siege (Nahm et al., 2023).

Rapid deployment

Deploying traditional broadband infrastructure requires considerable time and planning. In developing countries, despite the presence of high-tech broadband, including 5G and fibre, broadband infrastructure has failed to support the surge in demand for coverage during national events, such as elections.

In Kenya, there was insufficient coverage of the polling stations during the elections in 2022 (Ngetich, n.d.). The compact nature of Starlink satellites allows them to be launched efficiently into orbit. A single Falcon 9 rocket can carry up to 60 Starlink satellites into orbit, significantly reducing the cost per satellite and accelerating the deployment of the constellation (Starlink, 2024b). This enables the deployment of satellites more quickly in areas where the service is needed. Going forward, more countries in developing countries will consider using LEO broadband over traditional broadband, given that they can be deployed faster when needed. LEO broadband can complement terrestrial networks, forming space and ground-integrated networks (Y. Chen et al., 2024, p. 2)

Support for IOT Applications

The use of IOT devices in E-health, home automation, smart cities, environment monitoring, and Geological disaster forecasting has made it part of the future of the internet. According to Machina Research (2016), Global Internet of Things (IoT) connections are expected to reach 27 billion by 2025, with 71% of devices now connected using short-range technologies such as Wi-Fi, ZigBee, Cellular, or in-building PLC. The IOT device's expansion into environmental monitoring and Geologic disaster forecasting places these devices in remote areas, such as deserts, large water bodies, and forests, where traditional broadband services have a minimal presence. Furthermore, terrestrial infrastructure in these remote locations is vulnerable and can be destroyed during natural disasters. The industry has primarily relied on LPWAN due to energy conservation and the long distances covered (Qu et al., 2017). The combination of IOT use and the

increasing affordability of renewable energy, specifically solar energy, will enable IOT devices to have enhanced capabilities, resulting in more data being relayed by these devices. LEO broadband provides a promising future for connectivity to future IOT devices. LEO satellites have advantages that make them better suited for IOT connectivity, such as the low propagation delay and the small antenna size required for operation.

Redundancy

Due to their large constellations, LEO satellites can also provide redundancy, making them an attractive investment. The failure of one satellite does not result in service outages, as the constellation can be reconfigured to continue providing services in areas affected by the failure.

2.5 Challenges Facing LEO Satellites

LEO satellites are generally small and orbit at latitudes less than 2000 Km. Given the limited coverage per satellite, a large satellite constellation is needed to cover the earth(Y. Chen et al., 2024, p. 5). Although the launch costs for one satellite might be low compared to GEO satellites, the need to launch thousands of satellites makes LEO satellite constellations more costly(*Digital Regulation Platform*, 2024).

Due to their low orbit, LEO satellites move at the rate of 5-10km/s, which causes ground terminals to switch from one satellite to another in just a few minutes, causing frequency degradation. The high speeds also cause severe Doppler frequency effects, creating challenges for modulation and demodulation. (Xiao et al., 2024). This frequency degradation affects broadband internet services(Su et al., 2019, p. 56).

The weather has a significant impact on satellite communication. Rain can absorb or scatter satellite radio waves, resulting in degraded signals received by ground-based antennas. Rain attenuation affects the availability of the satellite signal, thereby impacting the delivery of broadband services provided by LEO satellites. Besides signal degradation, Rain is a factor considered during link budget calculations. Link budget calculations determine the power needed to maintain acceptable service levels. This can lead to either more powerful radios or larger antennas, which will determine the costs and affordability of the broadband service to the end user as compared to traditional broadband services over terrestrial links (Cakaj, 2009). The choice of frequency to be used in extreme weather conditions can also affect the antenna size. C-band frequencies perform better in extreme weather conditions; however, the downside is the large antenna size required, which translates to higher costs for the end user. Ku-Band and Ka-Band frequencies are affected by weather conditions; however, they can work with small antennas, which is why they are the preferred choice by LEO satellite providers for delivering broadband services to ground stations.

In a study published by Khodairy (2020) the impact of solar activity on LEO satellites includes orbital decay due to the atmospheric density resulting from increased solar activity. The high atmospheric temperature and density increase the drag force on LEO satellites, reducing their orbital lifetime. This explains the relatively short operational lifespan of 5-7 years for LEO satellites. In 2022, SpaceX reported a loss of over 40 satellites to a solar storm (SpaceX, 2022).

Three satellite beams are needed to operate LEO satellites. The feeder link beam connects to the ground stations or gateways, the service beam for maintenance operations, and the signal beam delivers services

to terminals worldwide. The feeder link beam is essential as it provides the total throughput capacity delivered by the signal link beam to the end users for broadband access. The location of the feeder link may not meet broadband requirements across the entire coverage area of the LEO satellite constellation due to latency requirements for broadband services. In some cases where inter-satellite links are used, high latencies are likely to be observed. Furthermore, rerouting in the event of a satellite link failure may result in loss of connectivity or service interference. Tiwari et al. (2023) found that although Starlink provides high throughput at low latencies, the tail-end latencies were as high as 11-16 times the normal latencies.

There have been efforts to decentralise and localise content closer to the end user. This has led to the deployment of Content Delivery Networks (CDNs) worldwide. The deployment of CDN caches near end users provides low-latency, low-loss paths between the caches and the users, increasing the performance (Hasan et al., 2014). Besides serving global content, it now hosts local content, improving the broadband experience and lowering connectivity costs. The location of feeder beams will be crucial to maintaining an acceptable broadband experience. The nature of internet design, where peering and interconnections tend to keep traffic local to regions, is a challenge to the design of LEO broadband. This will require deploying multiple ground stations and complex routing algorithms to adapt regional traffic in multiple regions.

LEO broadband services are provided over a limited frequency spectrum. This spectrum is shared with already existing GEO satellites and terrestrial microwave infrastructure. The ITU classifies LEO as non-geostationary and shall not cause unacceptable interference to, nor claim protection from, geostationary-satellite networks in the fixed-satellite service and the broadcasting-satellite service (*Digital Regulation Platform*, 2024). GEO satellites have the upper hand regarding spectrum use and interference disputes. As the number of users increases, congestion will likely occur on the available spectrum.

The limited number of objects that can be accommodated in space raises concerns about equitable access. The absence of specific sharing formulas may render the sharing of finite orbital space unfeasible. In 2021, Rwanda submitted an ITU filing for 337,320 satellites on behalf of a startup, and in 2023, France filed for a 116,640-satellite constellation called E-Space. This raised questions about the availability of future orbital and spectrum resources for late entrants.

Space debris poses a significant challenge to the maintenance of LEO broadband services. NASA defines space junk, orbital debris, as “any human-made object in orbit that no longer serves a useful purpose, including spacecraft fragments and retired satellites.” (*Space Debris - NASA*, n.d.). Approximately 85 per cent of this debris resides within low-Earth orbit, which is defined as being below 2,000 kilometres in altitude. Between December 2022 and May 2023, SpaceX’s Starlink satellite constellation alone carried out more than 25,000 collision-avoidance manoeuvres. (Torres, 2024). With the race to provide service via LEO, this will continue to be a challenge for broadband services provided via LEO satellites. Apart from space debris, there are cases where debris from satellite launches is designed to burn up as it re-enters the earth’s atmosphere or falls over unoccupied areas, such as oceans and in unsafe areas. In Kenya, debris weighing about 500 kilograms fell off a rocket launch and did not burn up before entering the atmosphere (Nzioka,

n.d., p. 2025)

The capacity provided by each satellite in the LEO constellation is limited. There can also be a finite number of satellites in a coverage area. This might work well in areas where there is a low-density population; however, in areas with high density, the LEO providers may be forced to raise prices to decrease demand, limit service availability, lower speeds due to saturation, implement data caps, or ignore the saturation, leading to the degradation of services (FCC, 2021). The high concentration of users in densely populated areas like Lagos and Accra has posed challenges, and Starlink was forced to limit services until more ground stations can be deployed (AI, 2024)

The other challenges facing LEO are Regulatory considerations. LEO satellites need to deploy gateways in different jurisdictions, which require licensing. This makes it complicated for the operator. The Selection of sites to install the gateway is also limited, as the sites must have good, affordable internet connectivity and favorable weather conditions. The more gateways that are needed, the more additional fees are required for capital expenditures (CAPEX) and licensing for the gateways. Some countries require a local gateway for operators, which increases costs and prevents the utilisation of intersatellite links and cloud processing. The other challenge has been the licensing regime used to license user terminals. Under the old regulatory regime, this was licensed on a per-terminal basis. For LEO constellations, this can be very expensive (ITU - Space Connect, 2025).

3 Research Methodology

3.1 Introduction

This chapter outlines the research and methodology used to investigate how LEO satellite broadband internet is emerging and influencing market dynamics, service quality, and customer preferences for broadband internet services traditionally provided by MNOs and ISPs, as well as how regulators are prepared to accommodate the new broadband services offered by LEO. This study is grounded in the principles of Qualitative Content Analysis (QCA) formulated by Philipp Mayring. QCA is a systematic, rule-governed method of analysing qualitative data, especially textual material, with the aim of preserving the richness and complexity of the content while enabling structured interpretation (Mayring, 2014). The methodology includes expert interviews and systematic coding. NVivo software was used for data interpretation, which enables the systematic analysis of expert interviews and documents, allowing for the extraction of categories and themes relevant to broadband market dynamics, service quality, customer preferences, and regulatory preparedness. The research methodology involved in-depth interviews and specifically formulated questions in (Appendix A) to gain experts' views in the broadband industry. Interviews were in-person and via Zoom, adapting to the expert's time preference. Participants' consent was obtained for the interviews and recordings. The interviews were transcribed for further analysis. For this research, 8-10 participants were selected based on the following criteria: expertise in the broadband industry, knowledge of LEO broadband, and key decision-making positions in medium to large internet service providers in Africa. By employing an exploratory and qualitative framework, the study examines LEO broadband technology and its challenges and opportunities in markets dominated by traditional internet

service providers (ISPs). The section covers the research design, data collection methods, data analysis approach, and ethical considerations to ensure a rigorous and systematic analysis

3.2 Research Design and Data Collection Methods

In this study 8-10 experts from traditional broadband services and satellite internet are interviewed for this research. Interviewees had a background in satellite and terrestrial networks.

Primary data

Interviews: Structured interviews with industry experts, including ISP executives, technology providers, and regulators. These interviews provided firsthand insights into industry challenges and opportunities.

Secondary Data

- Industry reports and white papers from credible organisations.
- Academic research papers on satellite broadband and market trends.
- News articles and case studies focusing on key players in the LEO broadband sector, such as SpaceX (Starlink), OneWeb, and Amazon (Project Kuiper).

The study employs a qualitative, exploratory design to gain insight into expert perspectives and emerging trends. Given the novelty of LEO broadband, a qualitative approach was deemed most suitable for capturing complex, contextual, and evolving viewpoints that may not yet be quantifiable.

In this paper, we analyse the various factors that affect LEO broadband deployment, adoption, and regulatory challenges. To understand these factors, we examine the factors that affect broadband's acceptability to consumers and providers. We focus on market and technical opportunities and challenges, economic considerations, regulatory hurdles, and Consumer adoption factors, and finally, we look at competition.

3.3 Methodological Approach: Mayring's Qualitative Content Analysis

Mayring's (2014) qualitative content analysis provides a systematic, rule-based technique for analysing textual data. Mayring (2014) notes that the central feature of Qualitative Content Analysis is its step-by-step procedure, which enables the development of categories either inductively from the data itself or deductively from theoretical models. These categories serve as the primary units of analysis and are continuously refined throughout the coding process.

Given the dynamic nature of LEO broadband information, deductive category application (using predefined themes) and inductive category development (allowing new themes to emerge) guide and provide a foundation for the research. The themes were extracted using the following steps

1) The research questions were formed

This was to explore the challenges and opportunities LEO broadband providers face in a market dominated by traditional broadband providers.

2) Definition of themes

Based on the research objectives, a set of predefined themes was established, including:

- Cost and deployment challenges
- Ground station infrastructure
- Market entry implications
- Service quality and coverage
- Regulatory readiness
- Return on investment
- End-user adoption and accessibility

3) Selection of interview materials

To identify and recruit experts, a purposive sampling method was used. Ten in-depth interviews, each lasting 60 to 90 minutes, were conducted. The interviews were audio recorded, transcribed verbatim, and imported into NVivo for coding.

4) Coding and analysis

Using NVivo, transcripts were coded according to the predefined categories. Flexibility was allowed for emergent sub-themes. The coding was both thematic and comparative, enabling pattern recognition across respondents.

5) Interpretation and summarisation of the results

Key findings were extracted from the coded data and interpreted within the broader context of the broadband industry. Particular attention was given to expert views' frequency, emphasis, and contrast.

3.4 Ethical Considerations

All participants were informed of the study's purpose and rights, including the option to participate voluntarily and the assurance of confidentiality. Consent was obtained before recording, and participants were asked if their identity would be disclosed during this interview.

3.5 Emergent Themes and Interpretation

The final themes were refined through multiple iterations to ensure coherence and depth. Five major themes emerged from the analysis:

One of the themes in the findings is marketing opportunities and challenges. Participants discussed the growing demand for LEO broadband services, highlighting opportunities for expansion into underserved regions. However, challenges such as competition from terrestrial networks and gaps in public awareness were noted. The second theme is technical opportunities and challenges. The advancements in satellite communication technology were identified as key enablers of LEO broadband expansion. However, participants also emphasised technical barriers such as signal interference, latency concerns, and infrastructure limitations. Economic considerations are another theme. The financial viability of LEO broadband was a recurring theme, with discussions centred on the high costs of satellite deployment, maintenance, and the affordability of services for end-users. Cost-sharing models and public-private partnerships were explored as potential solutions. The regulatory hurdles that LEO is facing are also challenging. The complexity of obtaining licensing and spectrum allocation was a key concern. Participants

emphasised the need for standardised global policies to streamline authorisation processes and foster cross-border collaboration. Additionally, consumer adoption factors affect LEO broadband uptake. The success of LEO broadband depends on consumer perception, affordability, and accessibility. Participants highlighted factors influencing adoption, including pricing models, service reliability, and competition with existing broadband solutions.

4 Findings

This findings section provides an in-depth exploration of the emergent themes identified in the analysis. Each theme is examined with supporting evidence from participant responses, highlighting key insights into LEO broadband deployment. The discussion focuses on market opportunities and challenges, technical considerations, economic factors, regulatory hurdles, and consumer adoption dynamics. This section ensures a nuanced understanding of the perspectives shared during the interviews by presenting direct quotations from participants.

This analysis has identified key marketing, technical, economic, regulatory, and consumer-related challenges that impact the deployment of LEO broadband. The following section presents a detailed discussion of these themes, supported by direct participant quotations, to illustrate their implications for industry stakeholders and policymakers.

4.1 Marketing Opportunities and Challenges

As society increasingly relies on digital platforms for daily operations, such as government services, telemedicine, online learning, and e-commerce, ensuring universal access to the internet has become essential. Broadband internet technologies play a critical role in bridging this digital divide. While significant efforts by both private and government entities have expanded broadband access, challenges persist. Remote locations, often due to factors such as challenging terrain, security concerns, and environmental barriers, remain underserved despite these efforts.

4.1.1 Bridging the broadband access gap

There is still a market opportunity to bridge the digital divide between the connected and unconnected. Participants highlighted the significant broadband gaps in regions such as Africa, where traditional infrastructure investments have been insufficient to ensure universal access. There is sufficient broadband coverage in cities like Lagos, Nigeria and Nairobi, Kenya; however, much of the population remains uncovered for various reasons. One participant noted, *"Nigeria has a huge broadband gap... a study by the Broadband Commission estimated the capital required for universal broadband in Africa by 2030"* (Ben Roberts). This highlights the investment required to expand connectivity and bridge the digital divide.

LEO broadband has an advantage over traditional broadband services due to its large footprint. This enables LEO to serve areas where traditional broadband networks are unavailable or cannot be invested in due to a lack of business returns. As another participant explained, *"One of the advantages of LEO is that you have a huge coverage footprint"* (Bhavesh Mistry). The participant added that *"Starlink is now able to*

deliver in unconnected and underconnected areas with a service comparable to traditional ISPs and mobile networks" (Bhavesh Mistry).

Despite this potential, some participants expressed concerns that LEO broadband is currently being adopted more by urban elites rather than truly bridging the rural connectivity gap. One of the interviewees noted that this had occurred with local broadband providers in Malawi. *"Starlink was positioned as a network meant to alleviate rural connectivity issues, but in Africa, it is being used predominantly in capital cities by the urban elite"* (Brian Longwe). This raises questions about the affordability and accessibility of LEO services for rural populations, which would serve to connect the unconnected.

Another challenge is the power infrastructure. While many homes without broadband have electricity, others lack basic power access. One participant noted, *"There are 7 million homes that don't have a broadband connection but have power, while another 6 to 8 million homes don't even have power"* (Ben Roberts). Addressing this issue would require parallel investments in energy infrastructure.

Nonetheless, LEO broadband presents a clear opportunity for rural market growth. One interviewee stated, *"For Africa, LEO providers have a good chance of capturing the rural market, which currently lacks fast broadband"* (Ben Roberts). Similarly, another highlighted that *"These services will extend broadband uptake in unconnected areas"* (Tom Makau), reinforcing the potential of LEO satellites to expand internet penetration beyond existing networks.

Mobile broadband has played a significant role in extending internet services to remote locations; however, the cost of cellular infrastructure is estimated to be between USD 200,000 and 250,000 per site, which requires proper business purposes to install this site. This restricts traditional mobile broadband providers from installing cell towers where there is a return on investment. This, unfortunately, is the densely populated areas such as urban areas (Garritty & Aminata, 2020).

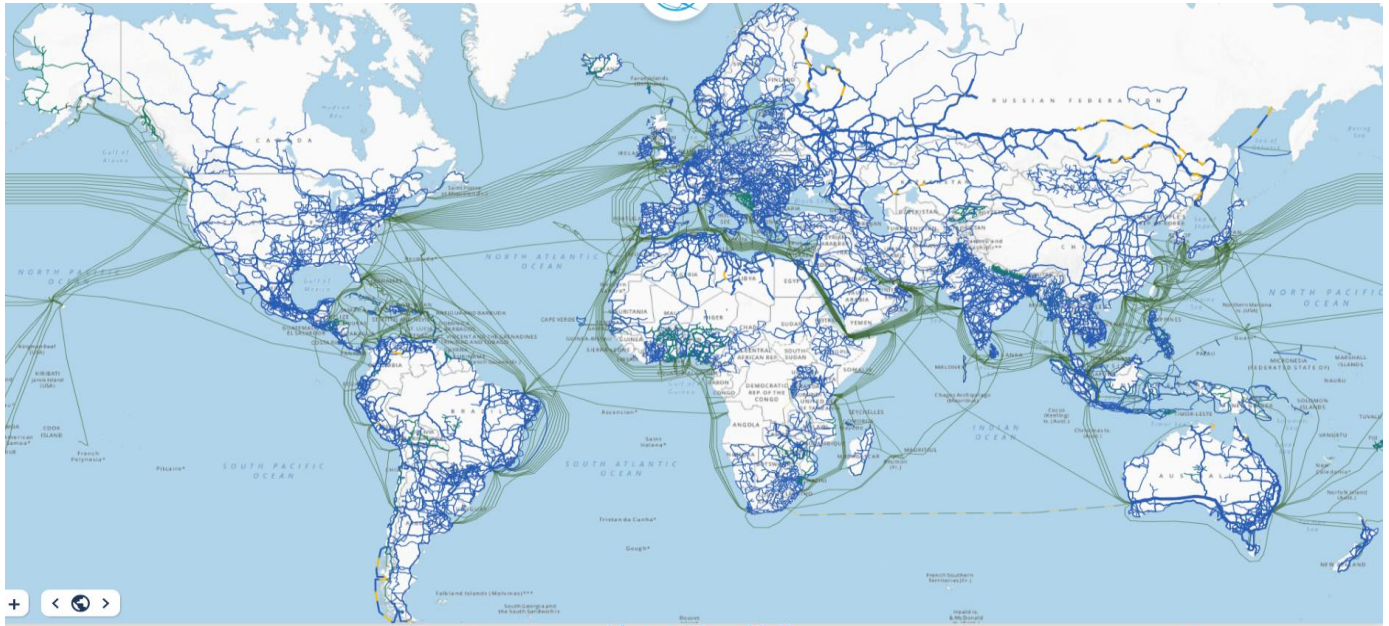
In most developing countries, internet coverage is mostly by using 3G technology. This places 3G as one of the technologies that enable bridging the digital divide. The latest GSMA report indicates that up to 750 million people live in areas with a coverage gap, with no signal from any mobile broadband operator. Another 3.28 billion people live in areas where there is a usage gap, meaning they are not using broadband internet for either affordability or lack of digital skills (ITU, 2023b). According to the ITU Facts and Figures & Broadband Commission (2023a) 33% of the world's population remains offline, (2024) while 38% live in areas with broadband availability but do not use it for some reason.

ITU (2014) classifies regions that require intervention based on current network reach and access, market efficiency and expansion gaps, smart subsidy zones, and true access gaps.

The current network reach and access are areas where the private sector deploys broadband infrastructure and may not require government or policy intervention. Market efficiency and expansion areas require policy and regulatory adjustments to allow the feasibility of deploying commercial broadband infrastructure. The smart subsidy zones would need to be subsidised to mitigate the risks of investing in these areas. Finally, the true access gap comprises areas that not only require one-time subsidies or support but also necessitate

ongoing support and are uneconomically feasible for broadband network deployment. Despite this, the need for relevant and timely information, literacy skills, and reliable, factual content has created market opportunities for broadband internet beyond urban areas (Broadband Commission: Sustainable Development Knowledge Platform, 2022). There are opportunities for telecom companies to expand their services, particularly in regions such as sub-Saharan Africa and Oceania, where coverage remains limited. The differences in broadband availability can also be reflected in the world's undersea cable and terrestrial infrastructure deployment, as illustrated in Figures 8 and 9

Figure 8 ITU Broadband Map



Source: (<https://itu.int/go/Maps>)

Figure 8 illustrates the normal distribution of undersea cable infrastructure in different regions worldwide. With a few exceptions, countries that are close to Oceans where undersea cables have landing stations tend to invest heavily in the distribution of terrestrial infrastructure. According to ITU (2023b), only 2 billion people are within ten km reach of the current fiber optic networks (p. 28). With a sufficiently large beam, LEO satellites can provide coverage of 3% to 12% of the Earth's surface, orbiting at altitudes ranging from 400 to 2000 km. In remote areas with limited connectivity capacity via GEO satellites, LEO broadband can upgrade available capacity and improve the quality of broadband services (Darwish et al., 2022). This provides opportunities to cover rural and remote areas where terrestrial networks do not ensure a consistent and fair user experience (Al Homssi et al., 2022). As of January 2025, Starlink coverage is available in areas where terrestrial infrastructure has struggled for decades.

AVAILABILITY

- AVAILABLE
- WAITLIST
- COMING SOON

Despite the capabilities to deliver capacity in areas marked as coming soon, Government approvals are still pending in most countries. However, comparing Figures 8 and 9, we can see that countries like South Sudan, which had little to no terrestrial infrastructure, immediately benefit from satellite constellations' broadband internet.

Remote Islands and landlocked countries will likely be significant beneficiaries of LEO broadband. Some of these areas have had little to no broadband access due to low population density, which does not encourage large-scale terrestrial or undersea infrastructure investments, hilly terrain, and sometimes war activities. These areas have primarily geographic isolation from most infrastructure services.

27

As seen in Ascension Island, broadband connectivity in landlocked countries and islands is further complicated by the presence of monopolies. Ascension islanders relied on geostationary Intelsat satellites to connect to the outside world, which was very expensive. The monopolies tend to fight back as they perceive LEO as threatening their revenue and services (openfalklands, 2024). Even though LEO broadband services will still rely on gateways installed in different regions, LEO's last-mile access technology bypasses the expensive tariffs and unreliable terrestrial infrastructure in some regions. LEO broadband is an innovative technology that opens new markets in underserved regions. The experts interviewed agree that LEO broadband creates new markets and equalises access to the internet backbone for landlocked countries and remote islands that lack services from undersea cables in Africa.

4.1.3 Support for IoT

The Internet of Things is one area that is growing rapidly, and LEO broadband can play a crucial role in providing much-needed connectivity to IOT devices, especially in areas where terrestrial broadband services cannot provide optimum coverage. Sometimes, coverage for IOT devices requires consistent coverage across borders and various terrains. One participant described how LEO broadband enables real-time tracking of goods across international supply chains: *"It can give global coverage around IoT devices. It means you can track a bag or a container from the port in London, across the high seas, to Mombasa, and then to its final destination in Nairobi—all on a single platform."* (Ben Roberts). This demonstrates how LEO satellites provide seamless tracking across land and sea and can be used in logistics.

Another key advantage of LEO IoT connectivity is its applicability in low-density and hard-to-reach locations. *"The real benefit is IoT coverage in areas with low network density—on the high seas, in deserts, or on Mount Kenya. You can track things globally with LEO satellites."* (Ben Roberts). This capability is particularly useful for industries such as agriculture and conservation, where IoT sensors must operate in remote regions.

IoT applications powered by LEO broadband also enable machine automation. One participant noted: *"LEO connectivity supports automation, whether it's small agricultural sensors or large-scale drone operations."* (Brian Longwe). Apart from providing coverage in logistics through satellite-based IoT, they can expand to include precision agriculture and automated industrial processes.

However, participants also pointed out challenges related to IoT over satellite networks, particularly in comparison to terrestrial solutions. *"The challenge that will come with satellite is not really the bandwidth but the number of devices one satellite can support. Unlike 5G, which can handle a million devices per square kilometer, a single LEO satellite has capacity limitations."* (Tom Makau). This highlights a potential scalability issue that LEO IoT networks may need to address large number of IOT devices in a concentrated area.

Another consideration is cost. One participant explained: *"It will be cheaper for me to use private wireless radio or shortwave to send small amounts of IoT data rather than subscribing to satellite. The price point has to be attractive to consumers."* (Tom Makau). This suggests that affordability will be a key factor in determining the success of LEO-powered IoT applications.

The observation was that LEO broadband may provide IOT as an added service. *“They can provide IoT services from their low-Earth orbit satellite...” “...without having a specific IoT constellation.”* (Jeremy Rose). This will likely change the landscape for IOT supplies that lack network infrastructure to support their devices.

Expanding IoT services through LEO broadband presents a significant opportunity, particularly for industries that require global connectivity and real-time tracking solutions.

In “Industrial 4.0,” artificial intelligence, robotics, augmented reality, additive manufacturing, the Internet of Things (IoT), big data, and cybersecurity will be significantly integrated into manufacturing processes.(Castro-Carrera, 2025). The total installed base of IoT-connected devices worldwide is projected to amount to 30.9 billion units by 2025 (Sujay Vailshery, 2022). IoT devices play a key role in transmitting real and passive data for decision-making. Low-power wide area networks have been largely successful in addressing connectivity via terrestrial broadband; however, limitations to providing IOT coverage arise in some areas due to several factors, including extreme topographies, costs to construct terrestrial broadband infrastructure to cover these areas, and damage to terrestrial broadband infrastructure in cases of disaster. A good example of how LEO has been used in a challenging environment is the Kineis & Argos System for IoT in extreme areas. Kinesis deploy IoT sensors in the polar regions to monitor sea ice movement, glacier changes, and wildlife (Anna, 2024). Because of the limitations of traditional broadband infrastructure, IoT has mainly been offered via SIM cards with data subscriptions that enable roaming.

Apart from roaming restrictions enforced in some countries, coverage is never guaranteed, as there are many areas where 4G/5G services are limited. This has particularly affected logistics companies tracking the movement of goods and services. LEO broadband provides easy coverage to these areas due to its low orbit; it can provide connectivity with low latency and high throughput. LEO broadband antennas are small and have low power consumption, making them suitable for connectivity on IOT devices, replacing low-power WAN and cellular connectivity.(Qu et al., 2017). While LEO provides universal coverage of remote locations, several challenges need to be addressed to provide IOT with broadband services. Popular IOT protocols rely on random access techniques rather than always-on access. This can be a challenge in satellite spectrum planning in large areas. Secondly, most IOT devices used in remote regions are power-limited, which requires compensation for uplink designs for satellite installations. With LEO, this can be a challenging factor for budget design (Al Homssi et al., 2022).

4.1.4 Airline and Maritime Connectivity

Maritime and aviation are areas of interest to LEO broadband. A notable example is the adoption of broadband in the airline industry. The shipping industry is also not being left behind in leveraging LEO broadband to reduce costs. A participant highlighted the wide range of use cases for in-flight and maritime connectivity: A participant highlighted the wide range of use cases for in-flight and maritime connectivity: *“In-flight connectivity and anything that is moving that requires satellite connectivity—whether it's aircraft, ships, cruise liners, or superyachts—these all depend on satellite internet”* (Carol Kamaitha). This highlights the importance of LEO broadband as a vital enabler of connectivity in these sectors.

LEO broadband is transforming crew welfare and operational efficiency in the maritime industry. One participant explained that *"Starlink is now used primarily for crew welfare, allowing seafarers to stay connected while at sea. When it works, it works well, making communication much easier in remote maritime environments"* (Carol Kamaitha). This development has reshaped how maritime companies structure their connectivity services, as they now segregate traffic for different purposes, such as operational and personal use. Also, note the limited space available for satellite equipment on vessels. *"On a cruise ship, real estate is a big deal—you don't have the luxury of space like in a data farm, where you can place multiple antennas."* (Carol Kamaitha). This constraint requires innovative hardware solutions to maximise efficiency while maintaining connectivity. LEO broadband deploys small dishes that can be easily fitted on this moving vessel as compared to the GEO counterparts

Similarly, aviation has witnessed a rise in demand for in-flight internet, driven by passenger expectations for continuous connectivity. One interviewee noted, *"More airlines are now providing high-speed broadband connectivity for as little as \$5 per flight, a service that was previously not possible."* (Michuki Mwangi). The affordability and scalability of LEO broadband have made it a viable solution for airline operators looking to enhance the passenger experience.

Michuki noted, *"Offshore drilling operations, shipping industries, and airlines have long depended on GEO satellites for connectivity, but LEO solutions are now emerging as strong alternatives."* The shift toward LEO broadband is driven by its lower latency and improved service reliability compared to traditional satellite solutions.

As demand for in-flight and maritime broadband grows, LEO satellite operators are well-positioned to capitalise on these emerging markets by addressing the unique connectivity challenges airlines and maritime industries face.

Airline and maritime connectivity will benefit from LEO broadband connectivity. In-flight connectivity is now available on some commercial flights. In maritime, the deployment of LEO has improved the quality and availability of broadband internet to ships and offshore platforms. Deploying traditional broadband in these areas has been limited to broadband services provided by GEO satellites. The main challenge is that GEO satellite networks were expensive to deploy and imposed a notable delay in communication since the path that the radio waves have to travel is fairly long (Aslam et al., 2020). Most often, GEO broadband has been expensive, making the service not affordable for commercial services in the maritime and aviation industries (Expereo, 2024). United Airlines announced the acceleration of its plan to add Starlink-enabled Wi-Fi to its planes, with the first Starlink internet-enabled flights to take off in 2025 with streaming, gaming, and even videoconferencing support (Hawkins, 2025). For decades, the US Navy has relied on a network of ageing satellites to furnish service members at sea with decidedly slow Internet access. The Navy is now testing the availability of LEO broadband to provide internet to the crew (Keller, 2024).

Despite much progress, the regulatory framework supporting internet services across jurisdictions remains challenging. Airlines and marine vessels must also overhaul their communications hardware to install new LEO-compliant ground terminals.

4.1.5 Backhaul to Traditional Broadband Services

The potential solution for backhauling terrestrial networks is becoming viable for traditional providers seeking to develop infrastructure in otherwise unreachable and underserved areas. One participant pointed out a significant industry partnership exemplifying this trend: *"They've announced a strategic partnership with Africa Mobile Networks, which operates in West Africa. AMN is utilising them to connect towers for telecommunications services. They had been using traditional VSATs."* (Ben Roberts). This transition from VSATs to LEO-based backhaul solutions illustrates a growing preference for lower latency and higher capacity connectivity in mobile network expansion.

Cost reduction was also a central theme in discussions on LEO satellite backhaul. *"They are looking for the lowering of the cost of backhaul because, of course, the further they go out to rural areas, the more expensive it becomes."* (Heinrich Nothnagel). The high cost of deploying traditional backhaul infrastructure in remote regions has long been a barrier to broadband expansion, making LEO satellites a viable alternative.

Additionally, LEO connectivity enables telecom operators to deploy base stations in remote locations where traditional backhaul is infeasible. One participant explained, *"They've started leveraging LEO connectivity to expand elsewhere. If it's a small base station that needs to be deployed, they won't subscribe to more than 200 or 400 megabits—they are using LEO for backhaul."* (Michuki Mwangi). This suggests that LEO satellites can be a flexible and scalable solution for expanding terrestrial networks.

Another participant underscored the advantage of LEO broadband's low latency for backhaul applications: *"Thanks to LEO and the low latency it provides, it is becoming a very good option or alternative for backhaul traffic."* (Tom Makau). This marks a significant departure from traditional GEO satellite backhaul solutions, which suffer from high latency and limited throughput.

By providing cost-effective and scalable backhaul solutions, LEO broadband is helping to bridge the connectivity gap in rural and underserved regions, supporting the expansion of traditional mobile and broadband networks.

Extending traditional broadband services to remote locations by providing backhaul to mobile or fixed base stations is a gap that needs addressing. Traditional ISPs heavily invest in building a transmission backbone to connect these remote areas. At times, constructing such a backhaul becomes impossible unless services are along the route to utilise, enhancing the network's cost-effectiveness. These areas may include remote islands within lakes. LEO broadband can seamlessly integrate with existing terrestrial infrastructure to offer backhaul to remote base stations. In partnership with Eutelsat OneWeb, Telstra has launched the world's largest rollout of OneWeb LEO cellular backhaul in Australia (Telstra, 2024). Due to lower latency, these services will not compare to backhaul via GEO satellite links, which can also be expensive.

Collaboration between traditional networks and LEO can extend coverage to remote areas by providing backhaul connections to core networks. Various experiments aim to integrate LEO with 5G and the forthcoming 6G networks. In these scenarios, LEO satellites serve as backhaul transmission lines between

base stations and the core network, overcoming costs and topographical challenges to access remote locations. The 3rd Generation Partnership Project (3GPP) collaborates with organisations that create technical specifications for mobile telecommunications. Established in 1998, 3GPP was formed to develop 3G standards and has since expanded to encompass 4G and 5G technologies. From Release 14 to Release 18, 3GPP has concentrated on standardising 5G satellite integration.

In addition to backhaul services, LEO broadband can be used to resolve congestion on the terrestrial backbone on a demand basis(Akhlaghpasand & Shah-Mansouri, 2023)Dynamic routing and Artificial intelligence can steer traffic away when there is congestion or a bad signal from the terrestrial backhaul. This can enhance overall network performance and user experience.

Artificial intelligence has increased the amount of bandwidth needed for backhaul capacity. inasmuch as it is an opportunity for LEO to provide this capacity, this might be short-lived as more web searches include AI. AI is reported to need 10 times the typical web search requirement. In the near future, the need for higher capacities on the backhaul might keep LEO broadband in the end-user market(World Broadband Association, 2024, p. 6).

4.1.6 Disaster recovery to terrestrial infrastructure failures

LEO broadband has emerged as a crucial alternative for disaster recovery when terrestrial infrastructure fails due to fibre cuts, natural disasters, or geopolitical conflicts. Several participants emphasized how recent fibre disruptions in Africa and other regions have highlighted the need for resilient backup solutions. One participant described the impact of major fiber cuts: *"Last year, six countries on the West Coast of Africa went offline due to fiber cuts. We also saw issues in Yemen that affected East Africa, and further disruptions impacted Southern Africa. These incidents forced governments to reconsider their stance on satellite broadband, with some countries quickly adopting Starlink as an alternative."* (Carol Kamaitha). This demonstrates how sudden infrastructure failures can shift regulatory attitudes toward LEO broadband.

LEO broadband also plays a role in global disaster recovery scenarios. *"In North America, wildfires burned down terrestrial fiber cables, leaving entire areas offline. In such situations, satellite becomes the immediate fallback option, providing a rapid response alternative."* (Michuki Mwangi). This highlights the crucial role of satellite broadband in maintaining connectivity during emergencies.

Another participant discussed how LEO broadband providers see this as a business opportunity: *"If Elon Musk or Starlink sees an opportunity, they will pursue it. Instead of waiting for fiber restoration, they can deliver connectivity directly to users, bypassing traditional backhaul entirely."* (Heinrich Nothnagel). This perspective highlights the commercial incentives for LEO operators to position themselves as a long-term solution rather than just a temporary backup.

By offering an alternative to vulnerable terrestrial infrastructure, LEO broadband is emerging as a key player in disaster recovery efforts, ensuring connectivity during crises.

Undersea fibre cables mainly connect the world's internet backbone. Several total outages have been experienced in recent years due to undersea cable cuts. Undersea cable provides high capacity, which cannot be transmitted via LEO broadband. However, in the cases where there is a total outage, like the outage in East and West Africa cable cuts in 2024, which left the continent disconnected from the rest of the world (Internet Society-1, 2024). In February 2025, More than 9,000 users of the broadband provider in London were left without broadband due to a fibre outage (The Sun, 2025). LEO broadband satellites are less susceptible to physical and natural disaster attacks and can keep vital services running to sustain the economy and communications.

4.1.7 Market viability and pricing challenges

A significant challenge for LEO broadband is market viability and pricing, particularly in emerging markets where affordability and currency fluctuations impact adoption. Participants highlighted concerns about the sustainability of pricing models and the financial accessibility of LEO services for consumers.

One participant noted that LEO broadband providers often use aggressive pricing strategies to capture market share: *"Probably selling below cost. But they've got a whole business plan where they add up all the revenues from their customers in the USA, maritime, aviation, and defense markets, then discount African ones to gain market share."* (Ben Roberts). This suggests that subsidies from high-income markets are being used to offset lower prices in developing regions.

Currency fluctuations pose a significant challenge. In some markets, pricing instability has led to significant increases in consumer costs. *"They are also susceptible to currency fluctuations. They launched at very low prices in Nigeria, but later doubled prices due to economic conditions."* (Ben Roberts). This unpredictability makes it difficult for customers in volatile economies to rely on LEO broadband as a long-term solution.

Financial infrastructure and payment systems also play a role in limiting adoption. *"Access to dollars and affordability are major issues. Some countries, like Malawi, are not as developed as Kenya or Nigeria in terms of online payments. The financial services sector is still behind."* (Carol Kamaitha). This means that even when LEO services are available, consumers in specific markets face challenges with payment accessibility.

Another challenge is the shift away from residential services in specific markets due to economic constraints. *"They initially offered heavily subsidized tariffs, but due to foreign exchange devaluations in Malawi, they've adjusted their pricing. Now, instead of offering residential subscriptions, they are focusing only on higher-value business subscriptions."* (Brian Longwe). These fluctuations affect affordability in developing countries.

LEO broadband presents significant potential. Its market viability depends on achieving sustainable pricing, navigating currency challenges, and improving financial accessibility for consumers in developing regions.

To remain sustainable and competitive in the common market, LEO broadband must adopt pricing models similar to those used by traditional providers, despite having different funding models. Traditional broadband

services are primarily categorised into four main groups (Garrity & Aminata, 2020).

Table 5 Revenue models in last-mile retail sales

Revenue model	Description
Usage-based (prepaid)	This is the standard pricing system for consumer connectivity services in emerging markets, where the consumer pays for traditional data services through a prepaid (also known as pay-as-you-go) model. If customer churn is high, prepaid service revenue can be more lumpy, irregular, and volatile than its subscription-based counterpart.
Usage-based (postpaid/ subscription)	Like the prepaid model, postpaid/subscription models are based on consumer (or corporate) connectivity services.
Value-added services	Operating expenses are covered by services other than data usage, such as paid advertising or other value-added services (mobile money, agricultural information services, education, etc.) that cross-subsidise data provision; not widespread, though seen in last-mile projects with clear specific end-user benefits.
Limited revenue, non-profit, free access	In this model, operating expenses tend to be covered by in-kind contributions (such as community management of the network) or an ongoing subsidy

Source: (Garrity & Aminata, 2020)

By January 2025, Starlink had priced the basic package at \$139 per month and the Starlink mini at \$599, with availability only limited to a few countries outside the United States of America.

Table 6 Starlink Costs

Provider	Initial Hardware costs	Monthly fee	Data Rate
Starlink	\$599	\$139	Unlimited 50-200Mbps
Starlink Premium	\$2500	\$500	Enhanced speeds

Source: Own table

Pricing provided by LEO broadband services appears to lack the flexibility typically offered by traditional broadband providers. This pricing still does not align with the competition offered by traditional broadband services. Table 7 illustrates an example of a location that remains underserved with broadband internet, as

evident in the current offering of one of the largest broadband providers in East Africa, Safaricom.

Table 7 Safaricom broadband pricing

PACKAGE	Monthly fee	Fiber Speeds
Bronze	\$23	15Mbps
Silver	\$32	30Mbps
Gold	\$48	80Mbps
Diamond	\$94	500Mbps

Source: (Safaricom, 2025)

LEO broadband providers have adopted different pricing models to compete in different markets. In Kenya, Starlink has attempted to enter the market by offering a 50GB package at \$11, a limited residential package at \$32 and a residential package priced at \$57 in an attempt to adapt to the pricing models for traditional broadband services (Starlink-P, 2025). With traditional networks having a longer end-of-life for equipment durability, costs will remain a key factor in the success of LEO broadband services.

4.1.8 Competition

The competitive landscape for LEO broadband is evolving rapidly, with both traditional broadband providers and new satellite operators vying for market share. Participants discussed how LEO providers have influenced competition, forcing traditional ISPs and mobile operators to adapt their offerings.

One significant response from fiber and mobile operators has been increased bandwidth offerings. *"Fiber providers, despite having the technology to deliver one gigabit, were limiting their plans to 100 Mbps. But when Starlink entered the market, it became more attractive to consumers, prompting fiber providers to upgrade their packages."* (Ben Roberts). This demonstrates how LEO broadband has pushed existing ISPs to enhance their service offerings.

Similarly, competition has led to price adjustments and the introduction of new service plans. *"Kenya saw Safaricom introduce a 100 Mbps product, something that was unheard of before. Other providers followed suit, increasing bandwidth availability."* (Brian Longwe). This indicates that LEO broadband has acted as a catalyst for market improvements.

One strategy discussed was for traditional broadband providers to increase bandwidth availability within existing packages. *"They've added more bandwidth to the packages. We've seen significant reductions in internet transit bandwidth over the years, but not much disruption in the bandwidth that providers offer."* (Ben Roberts). This suggests that continuous improvements in bandwidth allocation could enhance service quality and competitiveness.

Another approach is to implement fair usage policies (FUPS) that allow for high data consumption while maintaining cost efficiency. *"Safaricom has a fair usage policy of one terabyte. If you exceed that, they limit your bandwidth. Liquid, on the other hand, allows for eight terabytes—much bigger, but with a slightly lower*

top-line rate." (Ben Roberts). These differences in FUP highlight the potential for LEO broadband providers to differentiate themselves by offering more generous data caps.

Community-based models also present an opportunity for expanding LEO broadband adoption in rural areas. *"Is it possible to use the same technology to land it as a hub? Meaning, invest in a 100 or 200 Mbps plan, spend \$500–\$1,000 for terminal equipment, and then redistribute that connection to more users who only pay a lower entry cost."* (Michuki Mwangi). This model mirrors communal infrastructure approaches, such as shared water boreholes, making connectivity more affordable for rural communities.

Participants also discussed community-driven network models where local organizations take charge of broadband distribution. *"A community-based organization sets up the network using LEO satellites for backhaul, then redistributes the connection via fixed wireless or hotspots to every homestead."* (Michuki Mwangi). This suggests that alternative business models could make LEO broadband more financially viable in underserved areas. Not all LEO providers currently support this model. *"One provider said, 'No, we don't have that because our business model is based on every user buying a terminal.'"* (Michuki Mwangi). This highlights a challenge in balancing profitability with affordability, as requiring individual users to purchase expensive equipment can be a barrier to adoption.

However, some participants argued that the perception of Starlink having a technological advantage over fiber was misleading. *"Consumers believed Starlink had a technology advantage, but in reality, fiber still holds the advantage."* (Ben Roberts). This suggests that marketing strategies and consumer perceptions are crucial in shaping competition.

The entrance of LEO broadband has also created tensions between satellite operators and traditional ISPs. *"Starlink's direct-to-consumer model has led to conflicts with mobile network operators (MNOs), as seen in Kenya with Safaricom."* (Carol Kamaitha). This underscores the challenge of integrating new satellite services into existing telecom ecosystems.

Additionally, competition is intensifying with new satellite operators entering the market. *"Amazon is set to launch its Kuiper constellation, adding to the competition. We are already seeing a land grab in Africa, with Starlink placing their stations while others prepare to enter."* (Ben Roberts). The growing presence of multiple LEO providers is expected to reshape the industry further.

A significant concern raised was market dominance. *"One company dominating everything without involving local partners is problematic. Something will change in that regard."* (Ben Roberts). Some participants expressed skepticism about Starlink's approach of bypassing local ISPs, arguing that a more collaborative model could be beneficial.

Regulatory interventions may also influence competition. *"Some markets have taken a different approach by going to regulators and asking them to require LEO operators to have a local license."* (Michuki Mwangi). This regulatory push suggests that competition dynamics may be shaped by technological advancements and policy decisions.

Market dynamics are likely to affect both traditional and LEO broadband services. Both would take

advantage of the absence of competition to determine service pricing. The consumer will likely benefit from the competition; in some cases, traditional providers are doubling capacity to make the service more attractive. In Kenya, the dominant broadband provider, Safaricom, increased capacity delivered to consumers by 33 per cent (Fixed-data, 2024). Increasing competition among LEO providers (e.g., SpaceX's Starlink, OneWeb, Amazon's Project Kuiper) will likely lead to price wars. According to Gareth Owen, Associate Director at Counterpoint Technology Market Research (2025), the satellite communications market is a very competitive and challenging to be profitable in, and there are simply too many systems chasing a limited market opportunity. As a result, every satellite operator will do everything possible to impede or delay the entry of a competitor.

In as much as LEO, broadband providers provide unlimited plans. There are limitations in the capacity provided to the end user. According to Starlink's fair use policy (2024a) "If bandwidth patterns consistently exceed what is allocated to a typical residential user, Starlink may take network management measures, such as temporarily reducing a customer's speeds, to prevent or mitigate congestion of the Services. Bandwidth-intensive applications, such as streaming videos, gaming, or downloading large files, are most likely to be impacted by such actions". Fixed broadband provides capacity without the limitations in bandwidth consumed per month.

Service options on LEO broadband make them more attractive for individuals' use than for corporate businesses. One web fixed land plan in Australia starts from a package of 100GB at \$495, and unlimited plans begin at \$1325. The costs for unlimited plans are significantly beyond the affordability of most corporations. Traditional broadband providers can adopt localised service offerings that suit local communities rather than one plan that fits all. One of the providers able to harness customised local product offerings is Taara, which uses beams of light to deliver high-speed, high-capacity connectivity over long distances. In the same way, traditional fiber uses light to carry data through cables on the ground (Taara, n.d.) Taara harnesses the power of local entrepreneurs and community networks to bridge the digital divide, providing access to previously unconnected communities. Taara had to change the model to provide internet after they realised that each user connected would connect tens of more users to the same connection. To accommodate this business model, Taara developed an application that would enable bandwidth to resell, hence allowing the internet resale to be done in an organised manner and be able to get a revenue share (*Taara in Africa*, n.d.).

4.1.9 Collaboration between LEO and traditional providers

Collaboration between LEO broadband providers and traditional ISPs is emerging as a key strategy to enhance connectivity and expand market reach. Participants highlighted various partnerships and cooperative models developed in response to the growing demand for reliable broadband.

One example of collaboration is the integration of LEO broadband into mobile network infrastructure. *"Some mobile operators have converted a few hundred of their sites to Starlink to provide extra bandwidth for 4G in those locations."* (Ben Roberts). This demonstrates how LEO broadband is being used to complement traditional mobile networks, particularly in areas with limited terrestrial backhaul options.

Additionally, ISPs are leveraging LEO broadband as an upstream customer connection. *"Several ISPs are using Starlink connections at their PoPs and then distributing internet to customers via Wi-Fi or other low-cost solutions, making satellite broadband more affordable for end users."* (Bhavesh Mistry). This model allows ISPs to offer cost-effective connectivity without requiring individual customers to purchase expensive satellite terminals.

Strategic partnerships between LEO providers and telecom operators are also becoming more common. *"We've seen Starlink collaborating with companies like Paratus in Africa to sell services to enterprise customers. If Starlink wants to reach millions of subscribers in Kenya, it will likely have to pursue this approach rather than going it alone."* (Ben Roberts). This suggests that collaboration with established ISPs could help LEO providers achieve scale more efficiently.

Another participant highlighted the importance of such partnerships in global markets: *"In the US, LEO operators have partnered with mobile carriers to offer direct-to-device connectivity. These collaborations ensure seamless service integration."* (Michuki Mwangi). This approach enables mobile users to access satellite-based connectivity without requiring specialized equipment.

Despite these cooperative efforts, challenges remain. Some industry players feel that LEO providers have not fully embraced local partnerships. *"They could have done far better if they had worked more closely with local partners who could share revenue and help them navigate market dynamics."* (Ben Roberts). This suggests that while collaboration is increasing, there is still room for improvement in how LEO providers interact with existing telecom ecosystems.

Overall, integrating LEO broadband with traditional broadband services presents significant opportunities for expanding internet access. One of the larger internet service providers is partnering with Amazon's Project Kuiper satellites to provide LEO broadband to end users (Mungadze, 2025). By leveraging strategic partnerships, ISPs and LEO providers can enhance service delivery, reduce costs, and extend connectivity to underserved regions.

4.1.10 Autonomous and Unmanned Vehicles

LEO broadband presents significant opportunities for supporting autonomous and unmanned vehicles, including aerial, terrestrial, and maritime systems. One key advantage is controlling and monitoring unmanned aerial vehicles (UAVs) and other autonomous systems in real time. *"If you think about this from the perspective of being able to control drones and other types of unmanned vehicles—whether aerial, submarine, or terrestrial—LEO broadband provides a network that can actually enable them."* (Brian Longwe). This highlights how LEO satellites support the growing demand for autonomous systems.

Brian went on to emphasise the role of LEO broadband in mission-critical applications. *"An unmanned aerial vehicle can transit multiple borders and countries, traveling thousands of kilometers to carry out its mission—whether it's a strike, surveillance, delivery, or data collection."* (Brian Longwe). This demonstrates the strategic importance of LEO connectivity for applications requiring uninterrupted, long-range operations.

The use of Unmanned Aerial Vehicles (UAV) or drones for critical functions such as military operations,

logistics and delivery, environmental monitoring and evaluation, disaster response and emergency services, and agriculture and precision farming requires the use of flexible, reliable and low-cost communication broadband networks with high availability and low latency that can cover and deliver high-speed data to remote and terrain challenging areas (Earley, 2021). Currently, commercial drones are mostly restricted to operating in areas where terrestrial networks are operational. LEO can provide scalable networks that extend beyond current terrestrial network coverage by utilising lower orbits to offer high throughput and low latency with global coverage. This will enable commercial drones to extend their service coverage to remote locations while still delivering high-quality data throughput.

Beyond military and security applications, LEO broadband can also support autonomous commercial and industrial systems. Reliable satellite connectivity enables automated logistics, offshore operations, and environmental monitoring.

As the adoption of autonomous and unmanned vehicles increases, LEO broadband is expected to play a crucial role in ensuring their seamless operation. The combination of global coverage, low latency, and high-speed connectivity makes LEO networks an essential infrastructure for the future of automation.

4.2 Performance Technical Opportunities and Challenges:

4.2.1 Weather Conditions and Natural Disasters

Despite LEO broadband's ability to deliver services in areas affected by natural disasters, it faces technical challenges due to its susceptibility to weather and natural disasters. Participants highlighted how adverse weather events, such as heavy rainfall, storms, and solar activity, can affect signal quality and overall service reliability.

There are concerns about the impact of severe weather on LEO performance. *"Heavy rainfall and thick cloud cover can degrade satellite signals, leading to increased latency or temporary service disruptions."* (Bhavesh Mistry). This issue is particularly relevant for regions with frequent tropical storms or monsoon seasons, where maintaining stable connectivity remains challenging.

Another challenge is space weather, such as solar flares, which can interfere with satellite communication. *"Solar activity can ionize the Earth's atmosphere, affecting satellite signals and even causing disruptions in satellite electronics."* (Ben Roberts). As LEO constellations grow, operators must develop advanced mitigation techniques to address these potential vulnerabilities.

The performance of a satellite link depends on the transmission path between the source and the destination. Raindrops absorb and scatter radio waves. The Ku-band (12–18 GHz) and Ka-band (26.5–40 GHz) frequencies are widely used in satellite communications, including for Low Earth Orbit (LEO) broadband systems. While these bands offer high data rates and improved capacity, they also have several challenges. Both Ku and Ka bands are highly susceptible to signal attenuation caused by rain, snow, and atmospheric moisture, resulting in signal degradation and potential service interruptions, particularly in tropical and high-rainfall regions. During the interference caused by weather conditions, broadband users will experience packet drops or total service outage (Singh et al., 2018).

Alam et al.. (2019) did research on estimates of rain-induced signal attenuation across different regions of Nepal using the recommended ITU-R P.837-7 method, revealing significant temporal and spatial variations. Areas experiencing the highest attenuation necessitate fade margins of 24 db for the Ku-band and 80 db for the Ka-band. Conversely, the other regions show the least attenuation, with required fade margins of 12 db for the Ku-band and 40 db for the Ka-band. Alam et al.'s 2019 study highlights the significant impact of rain attenuation on link reliability, particularly in higher frequency bands. Another study by Osoro (n.d.) analysed uplink performance and availability at ground stations with high rainfall intensity based on ITU-R P.837-7 rain intensity results. The results reveal that the candidate stations in low-rain-intensity regions (Cairo, Praia, and Port Louis) are the most suitable terminals at 0.001%, 0.01%, and 0.5% unavailability times. In comparison, the Longonot and Hartbeesthoek ground stations performed poorly and could not achieve the threshold unavailability time of below 0.01% due to high rainfall in tropical regions.

Geomagnetic storms are another weather factor that can severely impact the operations of LEO satellites, making LEO broadband unavailable. A geomagnetic storm is a major disturbance of Earth's magnetosphere that occurs when there is a very efficient exchange of energy from the solar wind into the space environment surrounding Earth (Baruah et al., 2024). In July 2000, the Japanese ASKA satellite, which was orbiting at an altitude of 440 kilometres, lost its attitude control and reentered the Earth's atmosphere due to a geomagnetic storm. In February 2022, Starlink lost many satellites (38 out of 49) launched due to moderate magnetic storms (Kataoka et al., 2022). These storms led to a 50-125% increase in atmospheric density at altitudes between 200 and 400 km, significantly enhancing drag on the satellites (Fang et al., 2022).

Despite weather being one of the challenges to LEO broadband, there is an advantage that LEO satellites are not affected by disasters that result in weather conditions on Earth's surface, such as storms, floods, and high winds. In addition to weather conditions, LEO satellites are unaffected by other natural disasters such as earthquakes and fires. In September 2020, Starlink provided fire emergency responders and residents with internet connection during the Washington fire outbreaks by loaning seven terminals to access its satellites to the Washington Emergency Management Division (WEMD) (Kim, 2020).

Economies' reliance on connectivity has made broadband infrastructure critical to the day-to-day functioning of world economies. Significant outages have occurred because of major fiber outages that are sometimes accidental. Emerging trends of sabotage have led countries to devote resources to securing this key infrastructure. Because of the coverage and diversity, these cables become very difficult to protect (Danaher, 2025). LEO broadband offers alternative connectivity unaffected by the challenges of traditional broadband infrastructure.

4.2.2 Decentralised content

Participants discussed how the establishment of local Points of Presence (PoPs), Internet Exchange Points (IXPs), and partnerships with Content Delivery Networks (CDNs) are enhancing content accessibility and performance. LEO broadband network design has to take into consideration the decentralised nature of internet content

A participant highlighted the strategic placement of ground stations to optimize content delivery: *"Starlink has started putting earth stations in specific countries in Africa and is beginning to cross-connect with content providers and CDNs."* (Bhavesh Mistry). This enables faster content access and reduces reliance on long-haul international internet routes.

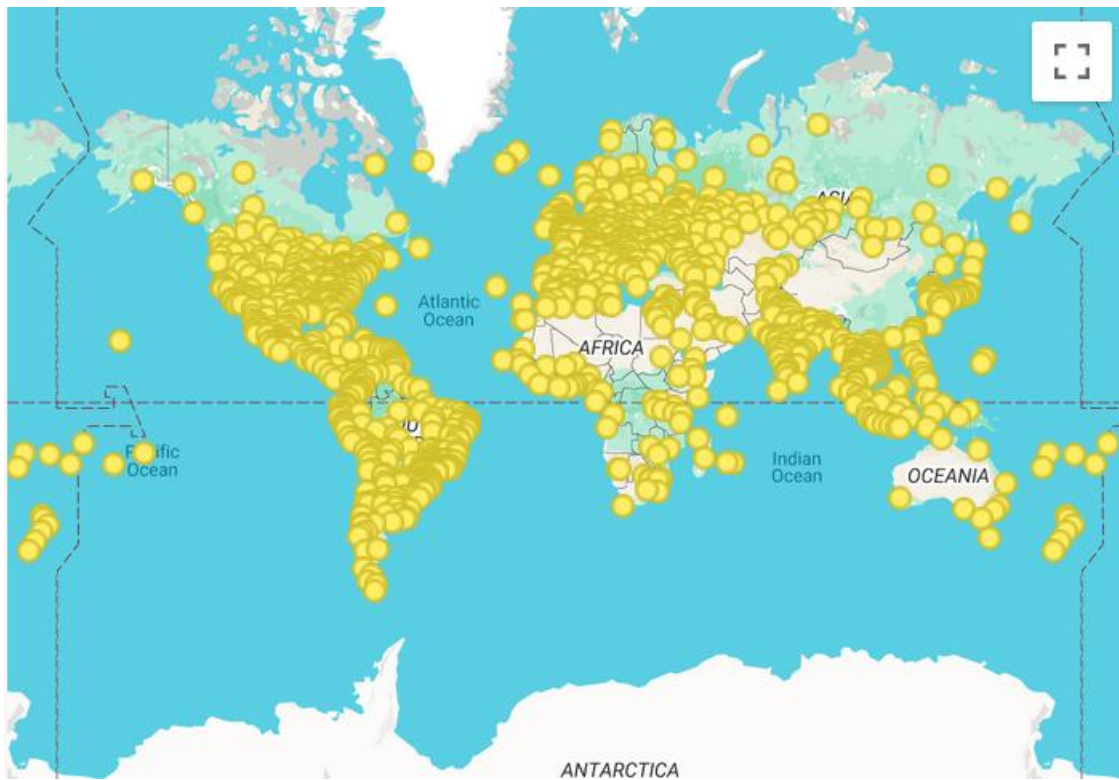
Another interviewee explained how regional PoPs enhance local content accessibility: *"If you're in Kenya, your Starlink dish will connect to a satellite and then downlink directly to a Starlink PoP in Kenya, where it can access local content like Netflix, Google, and e-government services. This significantly reduces latency."* (Ben Roberts). This demonstrates how LEO broadband providers leverage regional infrastructure to improve service quality.

The importance of IXPs was also emphasised: *"Building regional ground stations and connecting to internet exchange points allows LEO broadband services to remain competitive with terrestrial networks."* (Michuki Mwangi). LEO networks can ensure faster, more efficient data routing by interconnecting with major cloud and content providers such as Google and Microsoft.

However, a challenge remains in how IP addressing impacts user experience. One participant noted: *"The challenge with LEO broadband right now is that users often get an American IP address or the IP of the station where the service lands, which affects content localization."* (Tom Makau). This issue can lead to incorrect content region settings and potential access restrictions.

Over the years, the internet has evolved, with a shift in focus to placing content at near-consumption locations. The movement of services to cloud locations has also encouraged mega-scale companies to deploy their infrastructure globally. This study examines several enhancements that have improved the user experience, including reduced latency, increased throughput, and lower costs associated with accessing this content. This has been demonstrated by major CDNs that have deployed expensive infrastructure in various strategic areas, such as Google Global Cache, Amazon Web Services (AWS), and other social media platforms, such as Facebook and TikTok. The deployment of this decentralised infrastructure is primarily driven by enhancing user experience through reduced latencies and reduced long backhaul costs to far-end colocation sites.

Figure 10 Map of metros where at least one edge node (GGC) is present



Source: (Google Edge Network, 2025)

Figure 10 illustrates Google's efforts to decentralise content worldwide. This is primarily aimed at enhancing the user experience.

This content distribution has necessitated adjustments to the designs of broadband internet infrastructure, focusing more on access to content with low latency and lower costs. Terrestrial infrastructure is designed to connect to local terrestrial broadband provider core networks, and the provider can steer traffic to the nearest cache or cloud service site. This differs from LEO broadband design, which relies on ground gateways to offload content. These ground stations are strategically located to serve multiple regions. LEO broadband services face challenges in achieving latency and leveraging cost reductions from these locally hosted caches and cloud services.

Unlike the global peering and cache points, internet exchange points have been deployed globally. The main theme of the Internet exchange point is to keep local traffic local.

A world map illustrating the global distribution of 1000 sampling locations. The locations are marked by blue pins, which are most densely clustered in Europe, North America, and East Asia. The map also shows the distribution of 1000 sampling locations across the world, with a high density in Europe and North America, and more sparse distributions in South America, Africa, and Australia. The map includes labels for major continents and oceans.

The buildup of internet exchanges and peering caches worldwide undermines the principle of global connectivity provided by LEO broadband services. Suppose a significant portion of traffic is cached locally by Google, Amazon, and other CDNS. In that case, the demand for LEO's unique capabilities diminishes, particularly for cached content such as YouTube videos or Google Search Results. Recognising this weakness, Starlink has sought agreements with Google and Microsoft (SpaceX, 2021). However, this remains insufficient, as the deployment of ground stations can only be restricted to specific geographic locations. Ma et al (2023) measured performance in various locations and found that Starlink users experience much greater variability in throughput and latency compared to terrestrial network users, along with frequent outages. Beyond global access, data privacy laws enacted by most countries will likely increase the presence of CDNs in additional countries.

Integrating LEO broadband with existing internet routing protocols presents opportunities and challenges. Participants highlighted how smart routing strategies and regional interconnections can enhance network efficiency and pointed to technical inefficiencies in current implementations.

Despite this potential, current implementations face limitations. *"They haven't fully integrated smart routing*

yet. Right now, they are simply offloading traffic to the closest PoP." (Michuki Mwangi). This indicates that while LEO networks have the infrastructure for optimised routing, further refinements are needed to make routing decisions more efficient.

One potential solution is to integrate adaptive routing strategies that leverage local infrastructure. *"If they implement smart routing, where traffic bound for West Africa is routed via satellite instead of submarine cables, certain destinations could be reached much faster."* (Michuki Mwangi). This highlights how LEO networks could sometimes complement or even outperform traditional undersea cable infrastructure.

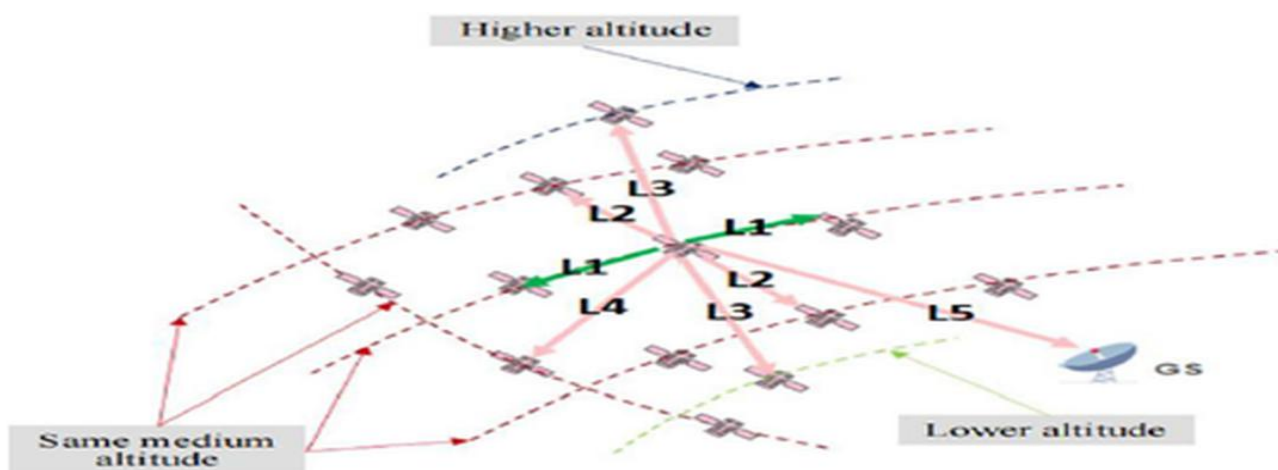
Additionally, future developments in LEO routing intelligence could further enhance efficiency. *"LEO satellites will likely develop advanced routing capabilities to determine the most efficient traffic paths—whether landing in Kenya or South Africa for better CDN access or routing directly to Europe."* (Michuki Mwangi). This suggests that continued innovation in routing algorithms will be key to maximizing the potential of LEO broadband.

While challenges remain, advancements in routing protocols, increased deployment of ground stations, and improved traffic optimisation strategies are expected to enhance the overall performance and efficiency of LEO broadband networks.

To deliver localised routing, LEO satellites may need to evolve to implement routing onboard. However, this comes with teething problems. To provide broadband services, traditional ISPs rely on both Interior routing protocols (IGP) and Exterior routing protocols, normally Border Gateway Protocol (BGP). Traditional networks are stable, and this protocol is designed to work in stable environments. Link flaps indicate the presence of connectivity issues. These protocols have mechanisms to identify the best link for routing traffic based on metrics or other human-configurable parameters.

LEO satellites typically have four links, L1-L4 and L5, which is the link to the ground station, as illustrated in figure 13.

Figure 12 LEO satellite with ISL Links



Source:(Han et al., 2022)

Table 8 ISL Link Types for LEO

Link Type	Description	Link Life	Link Metrics
L1	Between adjacent satellites in the same orbit	Steady	Steady
L2	Between two satellites in an adjacent orbit	Unsteady lasts tens of minutes	Keep changing
L3	Between two satellites on adjacent orbit shell	Unsteady lasts tens of minutes	Keep changing
L4	Between two satellites on the un-adjacent orbit	Unsteady lasts for a couple of minutes	Keep changing
L5	Between the Satellite and the ground station	Unsteady lasts for a couple of minutes	Keep changing

Source: (Han et al., 2022)

Table 8 and Figure 13 clearly show that the routing protocols, designed to work on stable traditional links, cannot be deployed on LEO without modification. LEO network links are dynamic. Using the current routing protocol without modification would lead to substantial flooding events due to link flapping, leading to less network usability (Han et al., 2022, p. 36). For delivering broadband consistently, routing tables must remain stable; however, with the LEO satellite constellation, there are challenges, as the LEO satellite would need a different routing table depending on its visibility to the earth station. To do this, source routing needs to be embraced. Westphal et al (2023) maps routing in LEO satellites into two categories: Dynamic routing and snapshot routing. With snapshot routing, the topology remains static for the duration the satellite is visible to the end user and is swapped out when the next satellite becomes visible. In this case, the snapshot table is computed just before the satellite becomes visible and is immediately loaded once a line of sight is established. The other alternative is dynamic routing, which means the topology is computed in real-time; this would result in a delay in forwarding packets as the network converges. The current routing protocols implemented on traditional networks would not converge in time, as the satellite is only visible for 7-10 minutes.

4.2.4 Spectrum allocation and interference

One of the significant technical challenges for LEO broadband is managing spectrum allocation and interference. Participants highlighted concerns about frequency congestion, cross-border interference, and conflicts with existing satellite and terrestrial networks.

A key issue is the availability of spectrum, which is crucial for the expansion of LEO broadband. *"One of the biggest challenges with wireless technology is the availability of spectrum. It's a finite resource, and both mobile terrestrial 5G operators and LEO providers want to use the same frequencies."* (Michuki Mwangi). The demand for limited spectrum has led to regulatory disputes and competition between different technology providers.

Interference was another primary concern. *"How do you use a beam that doesn't cross into another country's frequency? If you use Safaricom's frequency in Kenya, how do you make sure it doesn't interfere with MTN's in Uganda?"* (Ben Roberts). This issue underscores the complexities of managing frequency use across national borders.

LEO satellites also face interference from GEO satellites. *"They're using Ku and Ka bands allocated by the ITU, but even that has challenges because LEO satellites cross the GEO arc at some point."* (Ben Roberts). This highlights the need for careful spectrum coordination to prevent disruptions between different satellite systems.

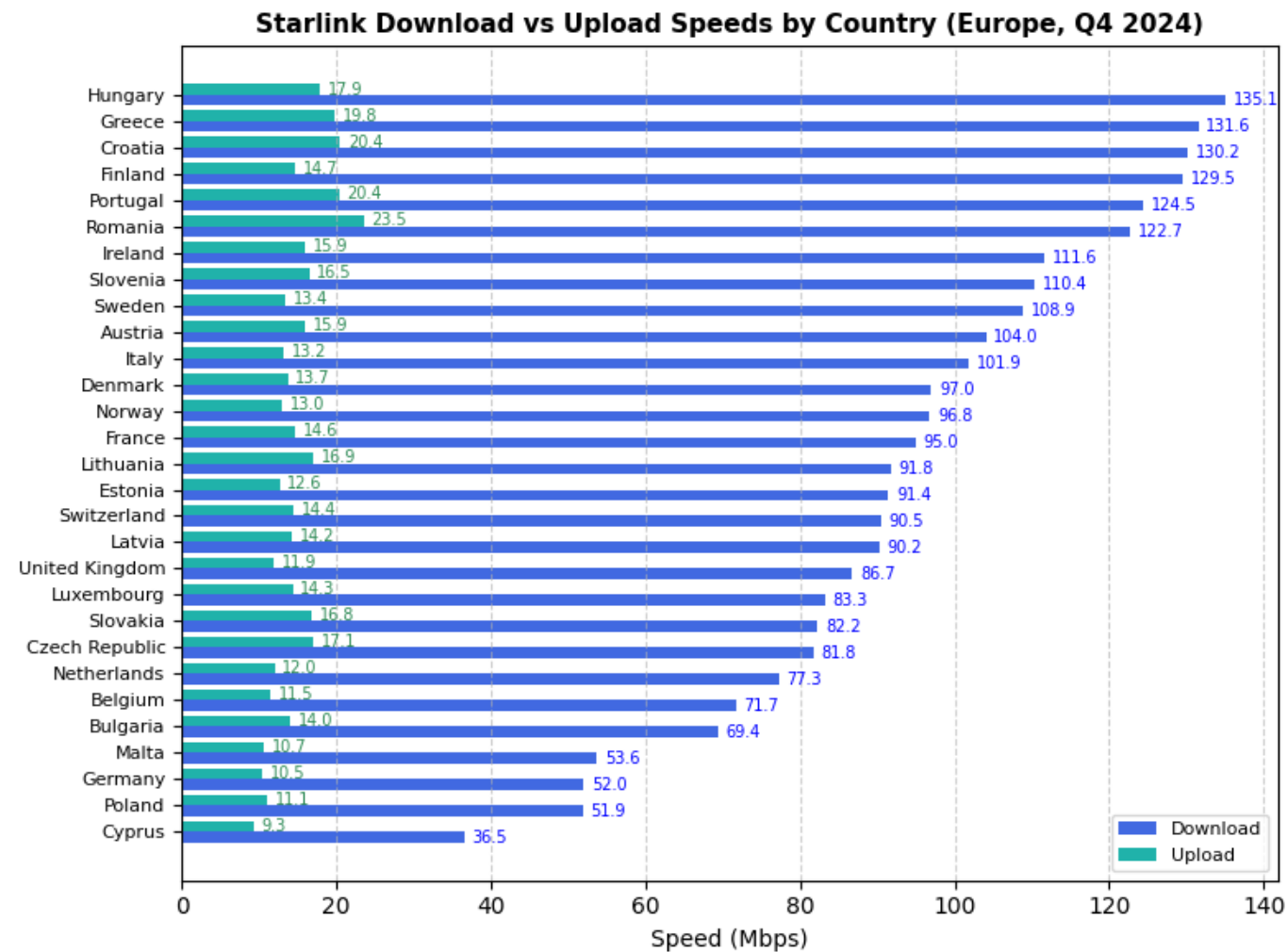
Additionally, evolving spectrum technologies, such as optical communication, can help alleviate some of these challenges. *"There's been a lot of discussion around using 3 GHz for LEO, but the challenge is that 5G terrestrial networks also want the same spectrum. New technologies like laser-based communication could help push more data while avoiding interference."* (Michuki Mwangi). These advancements may offer future solutions for optimising spectrum use.

Addressing spectrum allocation and interference will ensure seamless global connectivity as the adoption of LEO broadband grows. Collaboration between regulators, satellite operators, and telecom providers will be essential to mitigating these challenges and maximising the efficiency of satellite networks.

Ku and Ka bands face growing spectrum demand from various industries, including satellite operators, 5G providers, and broadcasters. Large amounts of spectrum are required to support high-data-throughput LEO broadband. The available spectrum is finite, leading to conflict and congestion; licensing disputes may slow deployment and create barriers to entry for new players. In May 2023, the FCC settled a dispute between Starlink and Dish Network in the US by voting to preserve parts of the 12 GHz spectrum band for advanced satellite services, denying terrestrial networks the use of this band for terrestrial 5G communication (Sbeglia Nin, 2023). The scarce frequency resources have led LEO broadband to attempt encroaching on frequencies assigned for other services. In March 2024, the FCC denied SpaceX's application to operate in the 1610-1617.775 MHz (Earth-to-space) and 2483.5-2500 MHz (space-to-Earth) bands, which are reserved for mobile satellite services (FCC, 2024 a). The FCC also permitted SpaceX to conduct communications in the 71.0-76.0 GHz (space-to-Earth) and 81.0-86.0 GHz (Earth-to-space) frequency bands (collectively, E-band) with the 7,500 Gen 2 Starlink satellites. With these frequencies, Starlink hopes to achieve more than 50 Gbps per satellite (FCC-E, 2024). LEO frequency capacity utilisation is now limited to Ku-band (12-18 GHz), Ka-band (26-40 GHz), and E-band (71-86 GHz). Despite all the frequency allocations, LEO still delivers a fraction of the capacity available on terrestrial links, with fiber capable of delivering 10 Gbps to 400 terabits per second per fiber (DWDM technology) without interference challenges. Some providers offer 10 Gbps broadband connectivity for homes (Spitfire, n.d.). This poses a challenge for future LEO scalability, especially if new applications start to require more than 500 Mbps of capacity. LEO user speeds depend on the number of active connections per satellite, and providing dedicated capacity can be very expensive for individual users. Figure 13 illustrates the uplink and downlink speeds measured over time for Starlink in various European cities. As observed, the increase in the number of subscribers

affects the downlink speeds available to the end user

Figure 13 Starlink medium Downlink and Uplink in Europe



Adopted from: (Ookla®, 2025)

Some services that require high throughput are Cloud and streaming services. It is worth noting that the uplink throughput is still below the recommended speed to access cloud services

Higher-frequency signals are easier to intercept or jam, posing security concerns for sensitive applications

4.2.5 Latency and Jitter

Latency is a critical factor in the performance of LEO broadband networks. Participants highlighted the advantages and challenges associated with latency in LEO satellite communications compared to traditional geostationary (GEO) satellites and terrestrial networks.

One of the key advantages of LEO broadband is its ability to offer lower latency compared to GEO satellites. *"Geostationary latencies are very high—300 to 500 milliseconds—making streaming and enterprise applications difficult. Enterprises were not happy with GEO because of the high latency."* (Bhavesh Mistry). In contrast, LEO satellites operate at significantly lower altitudes, thereby reducing signal travel time and enhancing real-time communication.

The deployment of local Points of Presence (PoPs) has further reduced latency for users in certain regions.

"One of the LEO providers has a PoP in Kenya, and latencies have dropped to about 30 milliseconds." (Michuki Mwangi). This demonstrates how localised infrastructure can significantly improve performance and user experience.

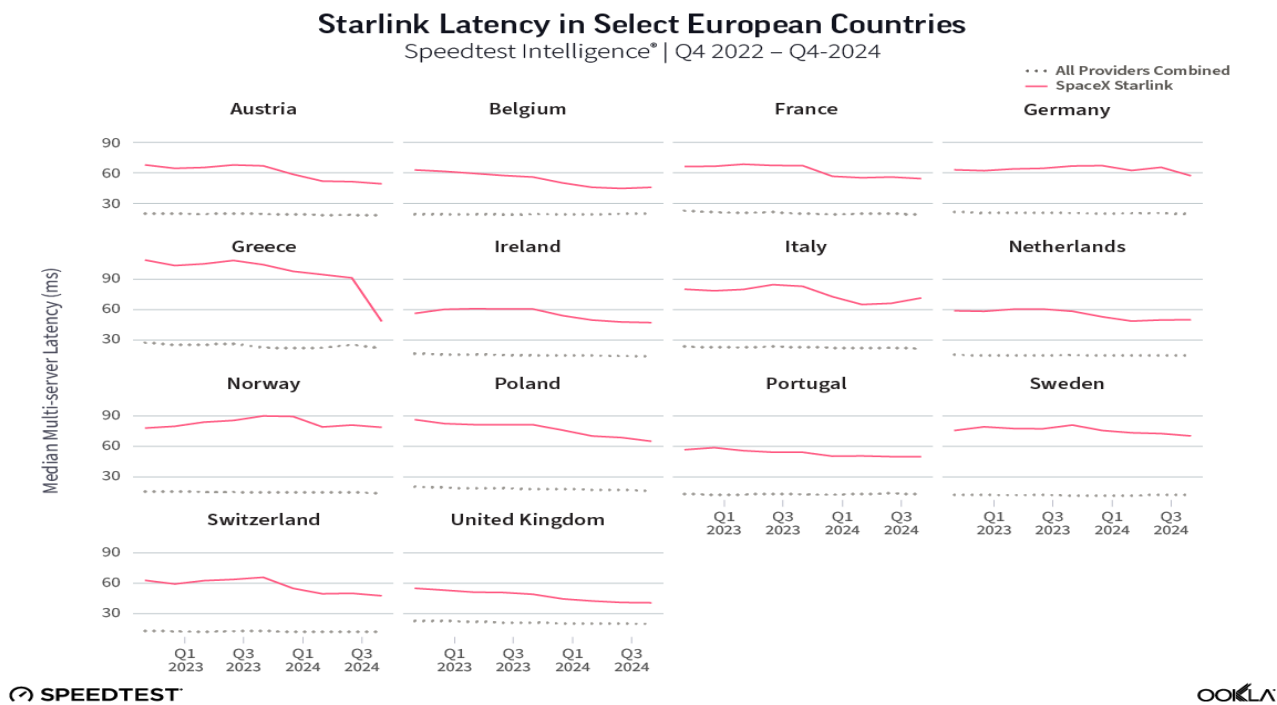
However, latency challenges still exist due to network congestion and routing inefficiencies. "Starlink dishes in Kenya now connect straight down to a Starlink PoP in Kenya, which helps reduce latency. But previously, when they routed traffic through other locations, latency was higher." (Ben Roberts). This highlights the importance of strategic network design in ensuring optimal performance.

Buffer bloat was also identified as a challenge for LEO networks. "Buffer bloat affected Starlink massively due to network distances and the architecture of user terminals." (Brian Longwe). This issue can lead to increased latency and performance degradation, particularly under high network load.

Despite these challenges, LEO broadband continues to improve its latency

Latency is the time that it takes for a packet to make it across the network from a sender to a receiver, and for the response to come back (Sundaresan et al., 2020). Global coverage over GEO and MEO satellites has had high latencies of between 250 and 350 ms over time. LEO satellite constellations at lower altitudes have reduced the round-trip time to 30 ms in ideal situations. While LEO constellations offer reduced latency compared to satellites, terrestrial infrastructure still provides better latency. The comparison below of latency between Starlink and traditional European providers illustrates the challenges of LEOs meeting customers' latency expectations.

Figure 14 Comparison between Starlink latency and other service providers



Source: (Ookla®, 2025)

Figure 14 illustrates that although LEO broadband provides low latencies, they are still higher than those

provided by traditional broadband services. LEO still needs to connect to terrestrial networks to access common content, thereby increasing the latency it offers its customers.

Content providers like Netflix and Google's YouTube have made efforts to minimize latencies, considering their proximity to consumers. This, coupled with caches of customised content in regions, makes latency an important factor when choosing broadband providers.

The term 'jitter' is a commonly used term to refer to variation in the latency of arriving packets over time (Sundaresan et al., 2020). Since LEO satellites move rapidly (around 7.5 km/s), user connections must frequently handover between satellites, causing latency fluctuations and jitter. Tiwari et al (2023) performed test to Google's 8.8.8.8 from Barcelona, Spain, and found that Starlink-like LEO networks could offer low (30-50 ms) latency at the median but are subject to high tail latencies. Such high latencies are a result of a complex mix of constellation geometry, orientation and location of the user terminal, deployment of gateways, satellite hand-offs, as well as other unforeseen variable factors in the terrestrial path segments (Tiwari et al., 2023). This can be attributed to frequent hand-offs, leading to a user terminal being mapped to a satellite, such that the latencies are persistently inflated for the entire duration that the terminal is connected to that satellite. These spikes in latency can cause jitter in sensitive applications such as voice and video streaming.

There is potential for improved latencies with LEO over long distances. Considering the speed of light travels faster in air than in fiber. It takes 3millisecond to transmit light from the Earth's surface to an LEO satellite orbiting at 1000 km (at the speed of light) above the Earth's surface. An equivalent distance of 1000 km between the LEO satellites is only 15% longer than their radial projection on Earth. However, the speed of light in fiber is 2/3rd of the speed in space. However, your signal will travel 1.47 times faster than in an optical fiber on the ground, resulting in a net gain (Westphal et al., 2023), LEO satellites can achieve a substantial latency reduction for long distances by allowing physically shorter paths, and operating at nearly the speed of light in a vacuum (Bhattacharjee et al., 2018). The net effect is that there will be lower latencies, especially with intercontinental connectivity over the oceans.

4.2.6 Network Standardisation

The need for standardised protocols and interoperability in LEO broadband networks is becoming increasingly critical as multiple satellite operators enter the market. Participants highlighted the challenges of proprietary communication systems and the importance of developing open standards to facilitate seamless integration across different networks.

One participant noted that proprietary systems currently dominate inter-satellite communications: *"Right now, everything's proprietary. The inter-satellite links used by Starlink satellites to communicate with each other are all proprietary, as is their communication with the ground. If we are going to have innovations like space-based data centers and CDNs, we will need open standards."* (Ben Roberts). This suggests that industry-wide collaboration will be necessary to ensure compatibility across different satellite systems.

Another challenge is the lack of interoperability between LEO constellations and traditional satellite

networks. *"Unlike the GEO world, where you could choose your encoding and interoperable standards between vendors, LEO broadband still lacks standardized protocols. This will inevitably need to evolve."* (Ben Roberts). This highlights the contrast between the well-established geostationary satellite industry and the emerging LEO sector.

Participants also emphasised the need for regulatory intervention to enforce standardisation. *"We can't have space ruled by one company with proprietary access to everything. Standards will evolve, and eventually, regulations will require the use of these standards, just like we've seen with phone charging plugs."* (Ben Roberts). This suggests that policymakers may play a crucial role in shaping the future of LEO network interoperability.

Additionally, as more competitors enter the LEO market, the push for standardization is expected to increase. *"Starlink is the pioneer, but with Amazon's Kuiper project and other future entrants, there will have to be some level of standardization, regulation, and coordination. Imagine hundreds of satellites from different operators—without interoperability, it could become chaotic."* (Bhavesh Mistry). This highlights the importance of industry collaboration in establishing best practices and regulatory frameworks for LEO broadband.

With different providers launching constellations and the high cost of deploying satellites, some level of interoperability where satellites from different vendors talk to each other is necessary in the future (Westphal et al., 2023, p. 16). 3GPP, a major SDO (Standard Development Organisation) that contributes to Satellite Networking from several angles, including Physical signalling, Radio spectrum, Radio Network, Use Case, Deployment, and System Architecture, has taken an interest in satellite standardisation. The challenges to standardise LEO constellations include the orbital parameters, which include altitude, inclination, and different orbital planes; the physical limitations such as satellite size, weight and power; different business models for different operators (Al Homssi et al., 2022, p. 21). Unlike the ITU (2025), which produces ITU-T Recommendations, which are standards defining how telecommunication networks operate and interwork, LEO broadband satellites lack unified protocols for satellite communication, making interoperability between different constellations difficult. The lack of standardization leaves the technology proprietary primarily to the vendors, slowing down growth and the entrance of new players.

The future potential for integrating LEO satellites with terrestrial technology has led standardisation bodies to begin work on standardising LEO satellites with terrestrial network standards. After 5G, satellites are being considered as both access networks and backhaul networks to existing terrestrial networks (Westphal et al., 2023, p. 11). Establishing a standard that works with all LEO providers will be necessary to operate this terrestrial network.

4.2.7 Scalability and Bandwidth Limitations

Scalability remains a significant challenge for LEO broadband providers, as increasing demand puts pressure on satellite networks. Participants highlighted several key issues, including bandwidth limitations, network congestion, and regional saturation.

One of the primary concerns is the rapid exhaustion of available capacity in high-demand areas. *"Starlink grossly underestimated the consumption, especially over Nairobi. Within three months after launching, they stopped offering additional setups."* (Tom Makau). This demonstrates the challenge of balancing supply and demand in urban markets.

Similarly, LEO broadband providers have had to pause new connections due to capacity constraints. *"In Kenya, Starlink had to stop new connections for two to three months because they had so much demand and not enough satellite capacity to cover the area."* (Bhavesh Mistry). This highlights the difficulties in scaling operations to accommodate a growing user base.

Network congestion further exacerbates performance issues. *"Initially, users could get speeds of 200 Mbps, but as congestion increased, speeds dropped to 40–60 Mbps—still better than some terrestrial services."* (Tom Makau). While LEO broadband offers competitive speeds, maintaining consistent performance as subscriber numbers grow remains challenging.

Participants also pointed to issues of geographic saturation. *"Places like Nairobi and other capital cities are completely sold out. Starlink has reported that they can no longer sell connectivity in some urban centers due to capacity limitations."* (Carol Kamaitha). This suggests that LEO broadband providers may need to expand satellite capacity or implement alternative strategies to manage demand.

Beyond consumer markets, scalability challenges also impact enterprise and government users. *"For large-scale deployments, LEO networks cannot yet compete with terrestrial networks in terms of bandwidth availability."* (Brian Longwe). This underscores the current limitations of satellite networks compared to fiber and other high-capacity terrestrial solutions.

Another technical constraint is spectrum availability. *"LEO satellites cannot concentrate too much spectrum in one location. A new European constellation has a total global capacity of just two terabits per second, which highlights the spectrum limitation challenge."* (Heinrich Nothnagel). This suggests that LEO broadband may struggle to scale effectively without innovative spectrum management strategies.

Despite these challenges, participants acknowledged ongoing efforts to enhance capacity. *"They've had to put in more satellites to keep up with demand."* (Bhavesh Mistry). However, LEO broadband providers must continuously invest in network expansion and optimisation to ensure long-term scalability.

As adoption continues to rise, addressing scalability and network limitations will be crucial to ensuring the sustainability and reliability of LEO broadband services.

4.2.8 Environmental concerns

One of the key environmental challenges associated with LEO broadband deployment is the increasing issue of space debris. Participants raised concerns about the sustainability of launching large satellite constellations and the potential risks posed by defunct satellites and falling debris.

A significant concern is the accumulation of space junk, which could pose operational and safety hazards. *"There's going to be many other technical considerations. So when we're going to get space junk?"* (Ben

Roberts). This highlights the growing challenge of managing decommissioned satellites and debris in Earth's orbit.

Participants also noted real-world incidents where falling debris posed a risk to people on the ground. *"We saw rings falling in Kenya that didn't kill anybody, but they could have. Space junk is going to be a problem, and you're going to see other issues like light pollution in the skies."* (Ben Roberts). This suggests that the uncontrolled re-entry of satellite components is an emerging concern that needs regulatory attention.

The rapid launch rate of LEO satellites further compounds the issue. *"These guys are launching satellites every day, and now we're having discussions around space debris. Aside from the first-generation Starlink satellites, I saw something last week about how a lot of them are now becoming a problem."* (Carol Kamaitha). This indicates that as older satellites become obsolete, the risks associated with orbital congestion and re-entry events will continue to rise.

Another concern is the lack of clarity regarding accountability for damage caused by space debris. *"There was the issue of space debris and what happens when they start falling—who compensates who? We had a recent incident in Machakos the other day."* (Michuki Mwangi). This highlights the legal and regulatory uncertainties surrounding liability for damages caused by falling satellite components.

As LEO broadband networks expand, addressing environmental concerns such as space debris management, satellite de-orbiting protocols, and regulatory frameworks for liability will be critical for the long-term sustainability of satellite-based connectivity.

4.3 Sustainability and Economic Considerations

4.3.1 Deployment costs

The cost of deploying LEO broadband networks is a significant barrier to large-scale adoption, particularly in rural and underserved areas. Participants highlighted the high expenses associated with infrastructure setup, satellite launches, and equipment procurement.

One of the significant cost components is the initial investment in satellite terminals. *"There have been two challenges I see with Starlink or with LEO today. One is the massive upfront cost. The antennas that sit on people's rooftops are not cheap, though they are subsidized. There is still a significant funding requirement."* (Bhavesh Mistry). Despite efforts to lower equipment costs through subsidies, this highlights the financial burden on providers and consumers.

The high cost of satellite launches also adds to deployment expenses. *"Each individual launch is so expensive. And then the satellite itself is extremely expensive."* (Brian Longwe). While satellite launch costs have declined over the past decade, they remain substantial in overall network economics.

Additionally, pricing transparency remains a challenge in the satellite industry. *"You cannot compare to the cost of satellite launch, and it always reflects in the cost per megabit. We've debated internally about Starlink's pricing, and sometimes the numbers don't make sense based on what we know about satellite costs."* (Carol Kamaitha). This suggests that the actual financial model behind LEO services may not always

be transparent, raising questions about the long-term sustainability of pricing.

The lifespan of LEO satellites also impacts deployment economics. *"Most of these satellites have a lifespan of 9 to 15 years. As they burn out, they need to be replaced by the next generation."* (Michuki Mwangi). Continuous investment in new satellites is required to maintain service availability, further driving long-term operational costs.

Another view is that deployment cost challenges affecting traditional broadband providers include the high cost per connection in rural areas, which makes it difficult for traditional ISPs to expand their coverage effectively. *"For traditional ISPs to cover rural areas, the cost per connection is very high because of the distance from the closest PoP or tower. If people in these areas cannot afford a \$10, \$15, or \$20 service, they still have to rely on mobile broadband."* (Bhavesh Mistry). This suggests that affordability remains a crucial factor in determining the viability of LEO services in low-income regions.

Despite these financial challenges, governments and organisations have implemented subsidy programs to support the adoption of LEO broadband. *"At the beginning, some countries in the Americas signed subsidy agreements in bulk, committing to long-term contracts to provide subsidized access using LEOs for various agencies."* (Michuki Mwangi). These agreements serve as anchor investments that help sustain deployment costs while ensuring connectivity for critical government services.

While LEO broadband presents a promising solution for expanding global internet access, its high deployment costs remain a key challenge. To achieve sustainability, operators must explore cost-reduction strategies, innovative financing models, and potential public-private partnerships to make satellite connectivity more economically viable.

LEO constellations have high deployment costs. The costs of launching LEO satellites include capex and ongoing operational costs.

$$Capex = C_{Launch} + C_{Station} + C_{Spectrum} + C_{Integration}$$

Where:

C_{Launch} – Satellite launching costs

The launch costs of satellites have drastically reduced.

$C_{Station}$ – Sum of all ground station building costs

$C_{Spectrum}$ – Spectrum Acquisition costs

$C_{Integration}$ – Costs of integrating to terrestrial infrastructure

For Operating costs, the formulae can be given by

$$Opex = O_{GS\ Energy} + O_{Acquisition} + O_{RD} + O_{Lab} + O_{maint}$$

Where:

$O_{GS\ Energy}$ – Sum of all the energy costs consumed by all ground stations

$O_{Acquisition}$ – Costs to onboard customers

O_{RD} – Costs for Research and development

O_{Lab} – Cost of labor

O_{Maint} – Maintenance costs

(Osoro & Oughton, 2021)

LEO broadband satellites range from small, cube-based devices (~100 kg) utilising consumer-grade hardware to larger, purpose-built satellites (100–1,000 kg) composed primarily of commercially off-the-shelf components. These larger satellites cost around \$500,000 and can be developed in about 18 months. Their rapid production, combined with lower launch costs than GEO satellites, has driven a surge in LEO deployments, with frequent launches occurring almost daily (TE Connectivity, n.d.). According to Astrology Jonathan McDowell's website, as of February 2025, Starlink had launched 7018 satellites in LEO (McDowell's, 2025). The rise in the number of satellites in Leo is attributed to a decrease in the costs of satellite manufacturing, specifically launch costs, as shown in Table 4. The commercially available components have made it easier and faster to mass-produce satellites. While launch costs have improved, especially with the Falcon 9, SpaceX reusable rocket allows SpaceX to reuse the most expensive parts of the rocket, which in turn drives down the cost of space access (SpaceX, n.d.). Despite the reduction in costs, launching satellites remains a costly endeavour.

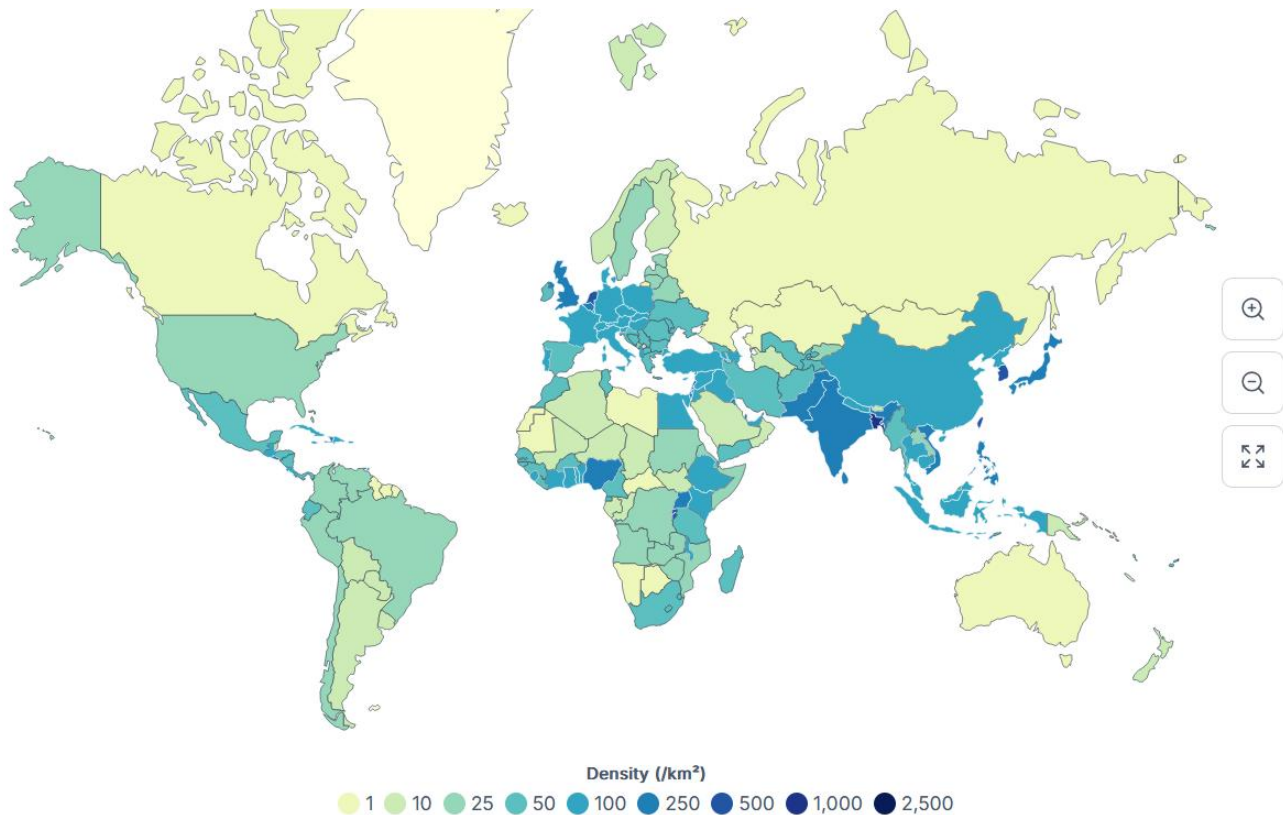
Table 9 Launch Vehicle Costs for LEO Deployment

Launch Vehicle	Provider	Estimated Cost per Launch (USD)	Payload Capacity (LEO)
Falcon 9	SpaceX	\$67 million	~22,800 kg
Falcon Heavy	SpaceX	\$97 million	~63,800 kg
Electron	Rocket Lab	\$7.5 million	~300 kg
Ariane 6 (A62)	Arianespace	\$75 million	~10,350 kg
Ariane 6 (A64)	Arianespace	\$115 million	~21,650 kg
New Glenn	Blue Origin	\$100+ million	~45,000 kg
Vulcan Centaur	ULA	\$100-200 million	~27,200 kg
Soyuz-2	Roscosmos	\$48.5 million	~8,200 kg
Long March 5	CNSA (China)	~\$90 million	~25,000 kg
H-IIA	JAXA (Japan)	~\$90 million	~15,000 kg

Source: (Choucair, 2023)

While the cost of deploying broadband networks can be costly, the costs of sustaining and keeping this network operational are also important. These costs apply to satellites that orbit the Earth, covering dense and sparsely populated areas. While the world is not uniform in terms of population density

Figure 15 Population density (people per square kilometre) by country in 2025



Source: (Countries by Population Density 2025.)

Traditional broadband services tend to be targeted at areas with both a need for service and the population uptake, which makes business sense for deployment. At the same time, predicting uptake in urban and suburban areas is possible. Both capex and opex will ultimately contribute to the per-MB pricing for the end consumer. For LEO broadband, there is no targeted deployment except the general satellite inclination to avoid areas with low-density populations.

4.3.2 Return on investment

The long-term return on investment (ROI) for LEO broadband remains uncertain, with participants highlighting both the potential and challenges of achieving financial sustainability. While LEO broadband has successfully entered various markets, including terrestrial, maritime, and aviation communications, concerns persist about whether its business model is viable without significant external funding.

One key concern is the heavy reliance on government subsidies, particularly from the U.S. Department of Defense. *"If you didn't have DoD funding, if you didn't have U.S. federal defense subsidies—really, defense contracts that act as subsidies—it's unclear if the business model would be sustainable."* (Ben Roberts). This suggests that military and government contracts are critical in sustaining LEO broadband providers, raising questions about their long-term independence from state funding.

Another participant echoed this sentiment, emphasizing the lack of transparency in funding sources. *"Funding that has been given to Starlink since inception—when you look at it, a huge amount of money came from the U.S. Department of Defense. What they're really paying for is getting soldiers connected to the internet."* (Brian Longwe). This reinforces the idea that defence-related contracts may be a primary revenue stream, rather than relying solely on consumer subscriptions.

Some participants questioned whether LEO broadband can generate sufficient revenue from direct consumer services. *"They probably need a few million subscribers, and even then, the ROI would take hundreds of years. My theory is that they are making money elsewhere for things we don't see."* (Brian Longwe). This implies that the commercial viability of LEO broadband remains uncertain, with potential hidden revenue streams playing a crucial role in its long-term survival.

Despite these concerns, LEO broadband has demonstrated strong market potential in aviation and maritime sectors. *"They've taken some market share in global maritime communications, which is worth about \$3 billion. The aviation market is another area where they could expand, but the quality of experience has been poor historically."* (Ben Roberts). Traditionally relying on satellite communications, these industries present significant growth opportunities for LEO providers.

LEO broadband complements, rather than replaces, fiber infrastructure, which could impact its long-term profitability. *"I personally don't think their ROI would be anywhere close to that of a dense fiber network from a fixed operator in a particular market."* (Bhavesh Mistry). This suggests that while LEO broadband fills connectivity gaps, it may struggle to compete with fiber in densely populated areas.

Some participants highlighted the role of large-scale customer agreements in sustaining LEO broadband. *"Depending on how they sign these agreements, they may have enough anchor customers to sustain operations at a global level."* (Michuki Mwangi). This suggests securing long-term contracts with enterprises and governments may be crucial to achieving financial stability.

Aviation connectivity was also identified as a high-growth revenue stream. *"For \$5 or \$10, passengers can have broadband connectivity on a full 10-hour flight, just like they would on the ground. If you count the number of annual passengers—more than 200 to 300 million—even if a third of them pay for in-flight internet, that's a huge revenue stream."* (Michuki Mwangi). This illustrates how widespread adoption in the aviation sector could provide a steady income source for LEO broadband providers.

LEO broadband presents significant market opportunities, its long-term return on investment remains uncertain. Heavy reliance on government contracts, lack of pricing transparency, and competition with terrestrial broadband raise concerns about its profitability. However, growth in the maritime, aviation, and enterprise sectors could help drive financial sustainability if LEO providers secure strategic long-term agreements.

Unlike their Geostationary counterparts and traditional broadband infrastructure, LEO satellites are exposed to harsher atmospheric conditions. Their lifespan is approximately 5-7 years after deployment. Starlink places the lifespan of LEO satellites at 5 years (Dobrijevic & updated, 2022). This reduces the payback period and

might affect the project's viability. Traditional broadband has a longer lifespan, which can extend the period before one can expect a return on investment.

Designing a LEO satellite that covers only a specific area, such as a continent, is not feasible. LEO satellites move around the globe in orbit. Since the Earth is not uniformly populated and most of its surface is covered by water, approximately 71% of the Earth's surface is covered by the ocean (Ocean, n.d.). The satellites will spend most of their time in orbit with low utilisation, leading to low returns. LEO broadband returns must be constrained when the satellite is in orbit, which determines the costs to the end consumers. These metrics contribute to the average rate per user required to ensure a return on investment.

While it is easy to analyse the subscription take-up rate based on the number of premises passed on traditional broadband, LEO broadband covers a large area in square miles. Hence, the subscription rate might vary depending on population density and internet availability. Despite traditional broadband, such as fixed broadband, targeting areas with potentially high subscription rates, the on-premises subscription rates are barely over 50%.

According to the National Telecommunications Agency, Brazil had 49.7 million accesses to fixed broadband in July 2024. Of these, 75.6% are made by fibre and 17.3% by coaxial cable. The satellite network accounts for only 1% of connections ("Checagem de Fatos," 2024). A similar trend in LEO subscription take-up could occur, and LEO broadband providers need to consider the costs of premises passed in their return on investment forecasting.

4.3.3 Ground Station/Gateway Deployment

Ground station deployment is crucial to LEO broadband infrastructure, enabling enhanced connectivity, reduced latency, and improved content delivery. However, participants highlighted the significant costs and logistical challenges of setting up these stations.

One of the primary concerns is the high cost of deploying ground stations in multiple countries. *"I don't think they can afford to do that in most countries. The business case won't be there to do it everywhere."* (Ben Roberts). This suggests that LEO providers must carefully select locations where the investment in ground stations is justified by market demand and economic feasibility.

Despite these challenges, ground stations are crucial for enhancing network performance by reducing latency and facilitating local content caching. *"To interconnect and serve content, they are being forced to put stations in-country or in-region because they know that latency and caching are critical."* (Bhavesh Mistry). By establishing ground stations closer to users, LEO providers can minimise the need for data to be retrieved from distant locations, enhancing the overall user experience.

Another major factor influencing ground station deployment is the dependence on local infrastructure and service providers. *"They have to incur massive costs for the ground stations, pay local fiber providers in different countries, and cover expenses for data and colocation centers."* (Brian Longwe). This highlights the financial burden of maintaining ground stations and relying on regional telecom operators for connectivity.

Participants also noted that regional ground stations are vital in improving connectivity within specific markets. *"Building regional ground stations has a huge benefit because it allows for direct connections to internet exchange points and content delivery networks (CDNs). For example, in East Africa, dropping traffic into Nairobi and connecting to local CDNs like Microsoft and Google significantly improves the end-user experience."* (Michuki Mwangi). This highlights the strategic importance of situating ground stations in key regions to improve service quality.

Routing inefficiencies remain a concern in areas without local ground stations. *"If the service isn't locally available, the traffic may need to take a complex route first to a European hub and then back. This adds latency and externalities that compromise the quality of service."* (Michuki Mwangi). This suggests that while regional ground stations can significantly improve performance, gaps in coverage may still result in inefficiencies in certain areas.

Ground station deployment is essential for optimising LEO broadband performance, but high costs, reliance on local infrastructure, and routing inefficiencies pose significant challenges. To enhance connectivity and service reliability, strategic investments in key regions, partnerships with local service providers, and continued expansion of ground station networks will be necessary.

Ground station costs contribute to the costs of LEO broadband. Ground station placement is determined by various factors, including costs (capital and operating expenditures), geographical considerations, technological limitations, climate and weather conditions, and resource availability, such as frequency allocation.

The initial capital costs include the expenses incurred to acquire or lease the location for the ground station. The gateways/ground stations need to be built in areas with reliable power, affordable internet (Fiber backhaul) connectivity, and proper connectivity to undersea cables to provide broadband internet services (Monzon Baeza et al., 2024). The LEO gateways must also be easily accessible for security and maintenance purposes. This will mostly place ground/gateways in urban areas with the infrastructure to support the ground stations for LEO broadband (Monzon Baeza et al., 2024). The costs for leasing and acquiring necessary space are higher in urban areas, eventually increasing the cost of LEO broadband to the end user.

Antennas used in these ground stations are also sophisticated, unlike the user terminal antennas, making them very costly. This would directly translate to the costs of operating LEO broadband.

Ground stations must also maintain connectivity to LEO satellites to facilitate constant broadband services. One of the challenges with satellite constellations is that LEO satellites are only visible to ground stations for a few minutes. Weather conditions in certain areas do not permit the placement of gateways. Using inter-satellite links can help reduce the number of ground stations and serve areas where weather conditions might not allow ground station placement.

4.3.4 Operational and support challenges

As LEO broadband scales, several operational and support challenges are emerging, particularly in customer service, satellite lifecycle management, and market structure. Participants highlighted concerns

about how these factors could impact long-term sustainability and user experience.

One key challenge is the lack of in-country customer service, which could create difficulties as LEO broadband expands. *"The model they're deploying is going to give them challenges—you know, not having in-country customer service. Some of these things will become issues when they scale."* (Ben Roberts). This suggests that while LEO broadband providers have successfully entered markets with minimal local support infrastructure, their ability to maintain customer satisfaction at scale remains uncertain.

Another primary concern is the short lifespan of LEO satellites, which require frequent replacements. *"The biggest technical challenge for LEO is that satellites have a limited life of five to six years. Every few years, satellites will fall out of the sky, and new ones will need to be launched. The complexity and cost of continuously refreshing satellites is massive."* (Bhavesh Mistry). This highlights a fundamental operational challenge—ensuring the continuous deployment of new satellites to sustain the network.

The simplified distribution model adopted by LEO providers has disrupted traditional Internet service provider (ISP) structures. *"Starlink has successfully removed many of the traditional layers between service providers and customers. There's no marketing team, no sales agents—you just buy it, install it, and it works. This allows them to cut overhead costs and offer better pricing."* (Heinrich). While this direct-to-consumer approach has streamlined operations and reduced costs, it also eliminates traditional support structures, which could pose challenges for technical assistance and troubleshooting.

Some participants viewed this disruption as a potential risk to traditional ISPs, as LEO broadband bypasses the typical service provider model. *"ISPs are at risk because these companies are essentially becoming global ISPs, cutting out many middlemen who previously played an important role."* (Heinrich). This suggests that traditional ISPs may need to adapt their business models to compete with the direct-to-consumer approach of LEO providers.

In summary, LEO broadband's operational model presents both advantages and challenges. While cutting out intermediaries has reduced costs and simplified access, the lack of local customer service and the high cost of satellite replacements remain significant concerns. Additionally, disrupting traditional ISP structures could reshape market dynamics, forcing existing providers to find new ways to compete in an evolving broadband landscape.

4.4 Regulatory Hurdles

LEO satellites are subject to both Outer Space law and domestic laws in states where they are launched and where they operate. Outer Space law provides frameworks for international space activities, while domestic laws ensure compliance (Akcali Gur & Kulesza, 2024). LEO broadband providers must adhere to the local laws in markets where they operate.

4.4.1 Resources

The allocation and cost of spectrum remain key challenges for LEO broadband providers, as they must navigate complex regulatory landscapes across multiple countries. Unlike terrestrial networks, which

typically operate within national jurisdictions, LEO satellites require spectrum rights across multiple regions, making regulatory approval more complex and costly.

One of the main challenges is the requirement for individual negotiations with regulators in each country. *"Because LEO broadband uses spectrum in every country, providers need to speak to each and every regulator in every market."* (Michuki Mwangi). This process varies significantly across regions, creating barriers to efficient global operations.

Another issue is that spectrum licensing is a significant revenue source for many governments, making some regulators reluctant to implement regional licensing frameworks. *"Spectrum carbonate costs have been a source of revenue for many countries."* (Michuki Mwangi). This can lead to higher spectrum costs, directly affecting service affordability and LEO broadband providers' business model.

Limited spectrum availability also poses challenges as governments strive to balance the expansion of LEO broadband with the needs of other industries, such as mobile networks. *"The cost of that spectrum will determine how they are able to package their services."* (Michuki Mwangi). This suggests that spectrum pricing plays a significant role in determining the viability of LEO services across various markets.

Achieving regional licensing agreements could help streamline spectrum allocation, reducing regulatory complexity and costs. *"Collaboration will be needed between member states to implement regional-wide licensing and spectrum allocation for a single service provider."* (Michuki Mwangi). However, such agreements remain limited, and most LEO providers must still obtain licenses on a country-by-country basis.

Additionally, competition for spectrum is intensifying, particularly with the expansion of 5G networks. *"In the U.S., the FCC authorized the use of the C-band for 5G, which created challenges due to interference concerns raised by the FAA."* (Tom Makau). This suggests that future spectrum conflicts between satellite and terrestrial operators could further complicate LEO broadband deployment.

Ultimately, spectrum allocation remains a critical issue for LEO broadband, affecting market entry and operational costs. LEO providers will face significant hurdles in securing affordable and consistent spectrum access across different regions without coordinated regulatory policies.

LEO satellites require frequency resources to operate in markets. Despite being positioned in high orbit, individual countries still require licenses to connect users to LEO broadband services. The International Telecommunication Union (ITU) and its Radiocommunication Sector (ITU-R), along with its executive arm, the Radiocommunication Bureau (BR), are the global agencies responsible for managing the radio-frequency spectrum and satellite orbital resources (ITU-R, n.d.).

However, the enforcement of the frequency use is left to member states. Obtaining these licenses can be difficult in some states, where the costs and conditions are very stringent. Offering LEO via Ku and Ka-band frequencies can also be challenging, as these frequencies are allocated for 5G and other terrestrial communication in various countries. n in various countries.

4.4.2 Regulatory and Licensing Barriers

The expansion of LEO broadband services has encountered significant regulatory and licensing barriers across different countries. These challenges range from strict ownership requirements, government mistrust, and regulatory fragmentation to unclear licensing frameworks that complicate market entry.

One of the major obstacles is that some countries require LEO providers to have local ownership. *"In South Africa, the debacle is that Starlink is asking for a waiver on the 30% local ownership requirement."* (Brian Longwe). This issue extends to other regions, where some governments demand local partnerships or shared ownership structures, which LEO operators are often unwilling to accept. *"They don't want to share with anybody. They want a license with 100% ownership."* (Brian Longwe).

Governments also exercise control over licensing to maintain regulatory oversight. *"Some countries will not grant licenses unless the provider has a physical presence or complies with specific conditions."* (Ben Roberts). In cases where governments demand in-country gateways for traffic control, providers face additional regulatory hurdles. *"Countries like Ethiopia, Uganda, and Tanzania require in-country gateways, and without them, they won't grant licenses."* (Ben Roberts).

Another regulatory issue is the lack of transparency in licensing policies. Some governments impose hidden or unwritten conditions, making it difficult for LEO providers to navigate regulatory frameworks. *"Special licensing conditions in some countries are not always written down, making compliance difficult."* (Ben Roberts). Additionally, high licensing costs can act as a deterrent. *"The cost of entry and operating in certain markets is high, with additional regulatory fees and compliance costs."* (Michuki Mwangi).

The fragmented regulatory landscape in Africa further complicates LEO broadband expansion. *"Africa's regulatory scene is highly fragmented, with each country having different requirements. This makes it difficult for LEO providers to scale efficiently."* (Tom Makau). Some countries resist granting licenses altogether, further limiting market penetration. *"South Sudan has not granted any licenses, and some governments are withdrawing support for LEO providers."* (Ben Roberts).

The disruptive business model of LEO operators has also led to government mistrust. *"Governments in Africa and Latin America are reluctant to work with Starlink because they bypass traditional regulatory frameworks."* (Carol Kamaitha). The ease of deploying and subscribing to LEO services without government intervention has made regulators uneasy. *"With Starlink, all I have to do is go online, register, and bring up my terminal. This rewrites the rules of regulation."* (Carol Kamaitha).

Some regulators have taken proactive steps to limit foreign LEO operators by enforcing local licensing requirements. *"We have seen markets where regulators require LEO operators to obtain local licenses or meet certain requirements that effectively lock them out of entering the market."* (Michuki Mwangi). This approach provides greater control over foreign satellite providers, ensuring they comply with national regulations.

Despite these barriers, some countries are adapting their regulations to accommodate LEO broadband. *"We've seen discussions around regional-wide licensing and spectrum allocation to single service*

providers." (Michuki Mwangi). However, uncertainty remains about compliance and long-term regulatory frameworks. *"There are ongoing discussions about Universal Service Fund contributions and other compliance measures for new operators like Starlink."* (Michuki Mwangi).

Overall, LEO broadband expansion faces complex regulatory and licensing challenges. Governments impose strict ownership requirements, high licensing costs, and fragmented policies, while some resist granting licenses altogether. However, emerging discussions on regional licensing and regulatory adaptation suggest a potential shift in policy approach as LEO broadband expands.

The provisioning of broadband internet via LEO is still relatively new, and most countries are still establishing guidelines and licensing regimes for this service. In Kenya, the regulator has proposed a review of the initial licensing fee from \$12500 to \$117000 and an annual operating license of \$31,250 or 0.4% of annual revenues (Citizen, 2025). This will likely be replicated across countries where LEO broadband access will be offered. The costs will be passed on to consumers in these countries, making the services unaffordable, especially those in remote locations. Apart from outright increases in the licensing fee, there are cases where licensing is denied, like Belize, where Starlink was denied a license to operate (lovefm, 2025). In Namibia, the Communications Regulatory Authority ordered Starlink to cease operations due to the lack of an appropriate license (Reuters, 2024).

4.4.3 Regulated internet

The arrival of LEO broadband services has disrupted traditional regulatory frameworks, raising concerns about government control over internet access, licensing requirements, and regulatory enforcement.

One of the primary motivations for requiring LEO operators to register locally is to ensure that governments can control or shut down the internet when necessary. *"Regulators are forcing Starlink to either register locally or find a partner in each country. The reason? To be able to shut off the internet when they want to."* (Bhavesh Mistry). This reflects how regulatory policies are shaped by political and security considerations rather than just market competition.

Some governments have historically exercised control over social media and digital platforms, further illustrating their interest in regulating LEO broadband. *"The government likes to have certain controls. They like to filter Facebook and other things—this is public knowledge. Facebook was blocked for a long time in Uganda."* (Ben Roberts). This suggests governments may impose similar restrictions on LEO broadband providers to maintain control over digital spaces.

A key regulatory challenge is that LEO operators are not physically domiciled in the countries they serve, making enforcement difficult. *"With LEO operators who are not domiciled in a country, regulation becomes tricky, especially if they come from powerful countries like the US."* (Tom Makau). This highlights the geopolitical complexity of regulating global satellite networks, particularly when major providers originate from politically influential nations.

The ability of LEO broadband to bypass national control mechanisms poses additional challenges for regulators. *"Regulators have no say if somebody sneaks a device into their country. Unless they run*

terrestrial jamming systems, they cannot interfere with Starlink's frequencies." (Brian Longwe). This suggests that traditional regulatory tools used for terrestrial ISPs are ineffective against satellite-based services, potentially undermining government oversight.

However, some regulators recognise that expanding connectivity through LEO satellites aligns with their long-term policy goals. *"Policymakers want everyone to have access to connectivity. Traditional ways don't always succeed, so regulators may have to open up further."* (Michuki Mwangi). This highlights a tension between enabling connectivity and maintaining regulatory control, especially in markets where traditional ISPs have struggled to expand coverage.

Despite regulatory concerns, governments prioritise cybersecurity, data protection, and equitable access. *"Data protection, cybersecurity, and ensuring affordable access remain key regulatory priorities."* (Michuki Mwangi). This suggests that while LEO broadband presents new regulatory challenges, existing frameworks will still apply to consumer protection and cybersecurity issues.

Overall, regulating LEO broadband remains a complex issue, balancing national control, international geopolitical influences, and the need for expanded connectivity. As satellite broadband adoption grows, regulators must adapt their policies to address security concerns and market opportunities, ensuring that LEO services operate within national frameworks without undermining government oversight.

The fear of governments losing control over the internet may lead to demands from countries to localise exit points for in-country traffic, which could force LEO providers to deploy gateways where governments control internet access. This can be driven by a fear of misuse of social media for activities that the establishment perceives as being against it. In South Sudan, the government banned social media for 3 months (CNN, 2025). LEO broadband could have gateways outside the government's jurisdiction, making it difficult to suspend internet services. Countries' demands to localise exit points for in-country traffic might force LEO providers to deploy gateways where governments control internet access.

4.4.4 Geopolitical restrictions

Geopolitical factors are increasingly influencing the deployment and adoption of LEO broadband services. Several countries are reconsidering their stance on satellite-based internet due to political affiliations, national security concerns, and regulatory interventions.

One of the key challenges is government pushback against LEO broadband providers, with some countries withdrawing from agreements or restricting market entry. *"You're seeing other countries withdrawing and saying, 'No, we're not going to do that anymore.' Mexico and South America are pulling out of some of these things."* (Ben Roberts). This suggests that regulatory uncertainty and shifting political priorities could impact the expansion of LEO services in certain regions.

The broader geopolitical influence of major tech companies also affects how LEO broadband is perceived. *"It's not only about Starlink. All major tech companies face geopolitical risks because they are either affiliated with the US or China."* (Bhavesh Mistry). This suggests that LEO broadband providers may face political scrutiny based on their country of origin, which could impact regulatory approvals and market access.

Concerns over espionage and surveillance further complicate the situation. *"Espionage, surveillance, and global supremacy in terms of tactical supremacy are key concerns."* (Brian Longwe). This highlights fears that LEO networks could be exploited for geopolitical advantages or intelligence gathering, leading some governments to impose restrictions on their use.

Spectrum allocation and cross-border interference also present geopolitical challenges. *"Remember, spectrum is within the geographic boundaries of a country. I cannot take that spectrum and use it in another market."* (Michuki Mwangi). Additionally, spectrum spillover at border points could lead to conflicts between neighboring countries. *"What happens if you start broadcasting your neighbor's country's spectrum into mine? These are the same issues we see with mobile operators at border points."* (Michuki Mwangi). These concerns suggest that international coordination will be necessary to avoid spectrum conflicts.

The largest LEO provider, Starlink, has faced political pushback in multiple regions, further illustrating the intersection of satellite broadband and global politics. *"There's also a lot of pushback in some countries—Ukraine and others—that have expressed dismay. Of course, the new American way of politics plays a role."* (Tom Makau). This suggests that LEO broadband providers could become entangled in broader diplomatic and trade tensions, potentially impacting their ability to operate in certain regions.

Geopolitical restrictions pose significant challenges for LEO broadband expansion. Issues related to regulatory pushback, national security concerns, spectrum conflicts, and political affiliations could limit market access for LEO providers in certain countries. As LEO broadband grows, international negotiations, regulatory harmonisation, and geopolitical considerations will play a crucial role in shaping its future deployment.

Regulatory hurdles arise from consumer destinations and the country in which the LEO provider is based. Companies in the United States are often restricted from doing business with specific countries or individuals. Due to geopolitical restrictions, Russia, Iran, and North Korea face limitations in accessing Starlink services.

4.4.5 Data privacy

Data privacy concerns are a significant challenge in adopting LEO broadband services. Participants highlighted issues related to a lack of user control over data routing, cross-border data flows, and data sovereignty.

One key issue is the customer's inability to control how their data is routed. *"Even though Kenya is going to have local gateways, you don't have control as a customer over how your data is routed."* (Ben Roberts). This suggests that despite the presence of ground stations, traffic routing decisions remain centralized and opaque to end users.

Another primary concern is the risk of data being routed through countries with weak or non-existent data privacy laws. *"That's going to be a problem. You don't know whether your traffic is landing in a country that doesn't have data privacy laws—it could be landing in Russia, for example."* (Ben Roberts). This highlights the uncertainty regarding data jurisdiction, which could pose risks for users in countries with strict data

protection regulations.

Additionally, even when ground stations are deployed locally, they do not store user data but serve as network infrastructure. *"Even if the ground station is in the country, it's not part of the operational or billing support systems. Those systems are sitting comfortably in the USA. Every single Starlink subscriber's data is stored in America."* (Brian Longwe). This implies that data sovereignty concerns persist, as user information is processed and stored outside the country of origin.

Some African governments have begun pushing back against LEO broadband providers due to concerns over data sovereignty. *"We've seen other countries in Africa push back, saying, 'We don't have data sovereignty if people are not going through a traditional local gateway.' Some data needs to be routed locally."* (Carol Kamaitha). This suggests that regulatory interventions may increase as governments seek to enforce local data processing requirements.

Data routing can create compliance challenges with strict data protection laws for countries without a ground station. *"If there's no ground station, data has to leave the country before coming back. That's a tricky issue for countries with strict cross-border data flow regulations."* (Michuki Mwangi). This underscores the potential conflict between LEO broadband operations and national data protection policies.

However, some participants noted that existing data privacy regulations have had little impact on major tech companies, suggesting that enforcement may remain weak. *"It will have no effect because the same laws have not affected Google, Microsoft, or other players regarding what they do with our data."* (Tom Makau). This raises questions about whether governments can effectively regulate LEO broadband providers in the same way they struggle to regulate global tech giants.

Data privacy and sovereignty remain key regulatory challenges for adopting LEO broadband. The lack of local data control, cross-border routing issues, and potential conflicts with national regulations may lead to greater scrutiny and regulatory interventions from governments seeking to enforce stricter data protection measures.

4.4.6 National security and critical infrastructure

The expansion of LEO broadband services has raised national security concerns, particularly regarding data sovereignty, surveillance risks, and the dependence of critical infrastructure on foreign satellite networks.

One of the primary concerns is the potential over-reliance on LEO broadband for essential services. Governments fear that if a foreign-owned LEO provider experiences outages or disruptions, it could cripple national infrastructure. *"If all your hospitals are using Starlink and then it goes down, people can't pay their bills, they can't get admitted, and they can't receive treatment."* (Ben Roberts). This dependency on external providers for critical services is seen as a national security risk.

Another primary concern is the lack of control over data and communications, particularly in relation to potential surveillance and espionage risks. *"Not only do they know exactly where I stay, but they can listen*

in on all my conversations and even turn that device into surveillance equipment." (Brian Longwe). This lack of transparency in how LEO providers handle data raises concerns about user privacy and state security.

Governments also struggle to enforce regulations on LEO providers that operate outside traditional licensing structures. Some companies bypass local regulations by selling terminals in neighbouring countries, making them difficult to track and control. *"There are thousands of Starlink kits in Sudan despite no formal approval from the government. People just buy them in Egypt or Dubai, bring them in, and set them up."* (Carol Kamaitha). This unregulated expansion further complicates national oversight.

Regulatory frameworks were initially designed to protect domestic telecom operators and national interests, but LEO broadband disrupts these models. *"The regulation of LEO players is now more about balancing national security interests with protectionist measures for licensed local operators."* (Tom Makau). Some governments impose licensing barriers or outright bans to mitigate these risks, while others attempt to integrate LEO services into their existing regulatory frameworks.

Despite these concerns, some governments are cautious, allowing LEO providers to operate while assessing long-term risks. However, the challenge of balancing national security, data sovereignty, and open-market competition remains unresolved.

4.5 Consumer Adoption and Accessibility Factors:

Edison Alliance, founded by the World Economic Forum (2024) focuses on three factors that affect Consumer internet adoption: affordability, usability, digital literacy, and accessibility. We take a three-dimensional approach to examining these factors. In addition to our in-depth interviews with experts, we investigate previous studies on broadband adoption.

4.5.1 Affordability

Affordability remains a key challenge for the widespread adoption of LEO broadband, particularly in rural and low-income areas. While LEO services offer competitive pricing in urban markets, the high cost of terminals and monthly subscriptions remains a barrier for many consumers.

In urban areas, LEO broadband is priced competitively compared to traditional ISPs. *"In Kenya, the pricing is very competitive to what ISPs would offer for a fixed fiber or fixed wireless service."* (Bhavesh Mistry). This affordability has increased competition among ISPs, particularly in middle- and upper-class urban neighborhoods.

However, in rural areas, affordability is a significant issue. *"For rural areas in low-income places where people struggle with a \$5 or \$8 monthly service, LEO might not be a solution because its pricing is still too high."* (Bhavesh Mistry). Similarly, another participant emphasized, *"It's not affordable for rural areas. It has to be super, super subsidized for this to be sold in a village in Kitale."* (Carol Kamaitha). These insights highlight that LEO broadband is currently inaccessible for many rural communities unless heavily subsidized.

The cost of terminal equipment is another major affordability barrier. *"Most people cannot afford \$500 for*

the terminal equipment and \$100 a month for access. That's out of the equation for the 2.6 billion people who are unconnected." (Michuki Mwangi). Even though LEO broadband promises fast speeds and low latency, the initial setup cost is a major deterrent for adoption among low-income users.

Traditional broadband services still offer more cost-effective options for many users. *"Now you can get unlimited connectivity for 8 Mbps for just \$20, whereas LEO broadband is still priced significantly higher."* (Tom Makau). This suggests that LEO services must further reduce costs to compete with existing broadband alternatives in lower-income markets.

Brian Longwe also highlighted the cost barriers for rural users, emphasising that the high cost of LEO terminals makes them unaffordable for most rural consumers. *"The terminal is far, far, far, far more costly than what your average rural user can afford."* (Brian Longwe). This suggests that even if LEO broadband were available, many rural users would still be unable to afford the required hardware.

Additionally, the cost of subscription remains a challenge. *"Yes, the subscription cost is quite reasonable in quotes, but it is probably still higher than what the average rural user can afford."* (Brian Longwe). This indicates that even if users can acquire the hardware, ongoing monthly costs make the service inaccessible to lower-income groups.

He also noted that rural internet adoption is closely tied to broader economic realities. *"Do I use the money that I've set aside for fertilizer for this year's crop, or do I use it for internet? In most of Africa, rural communities do not have the same disposable income as those in the US."* (Brian Longwe). This underscores the importance of contextualising LEO broadband pricing within local economic conditions, as many rural users prioritise essential needs over internet connectivity.

Furthermore, Brian noted that LEO broadband appeals more to high-income customers than the mass market. *"The high-end upper class and expatriates do end up using LEO services, but that's not our core market. We really didn't feel much of an impact in terms of customer loss when Starlink came. Only a few high-end customers switched."* (Brian Longwe). This reinforces the idea that LEO broadband primarily serves wealthier users, rather than bridging the connectivity gap for underserved populations.

While LEO broadband is competitively priced for urban middle- and high-income users, its affordability remains a significant barrier in rural and low-income regions. Without substantial price reductions, subsidies, or alternative financing models, LEO broadband may struggle to reach the most underserved populations who would benefit the most from improved connectivity.

The Broadband Commission's affordability target was initially set to "less than 5% of average monthly income," however in 2018, the Commission updated this target to specify that prices for entry-level broadband services should be below 2% of monthly GNI per capita in developing countries by 2025 (Broadband commission, 2025). Our findings indicate that affordability is the primary concern for most consumers. The current pricing of \$599 for the user terminal offered by Starlink is prohibitively expensive for users, especially in vulnerable areas. Starlink has offered solutions such as rent-to-own. LEO subscription costs now range from \$11 to \$120, above the normal spending power for vulnerable groups.

This affects both developing and developed countries, and it was found that 2.4% of the European Union's 450 million inhabitants could not afford an internet connection in 2022 (Euronews, 2023). To achieve sustainable development goals and objectives for broadband connectivity, governments and LEO providers must intervene to subsidise services so they can be affordable. In the United States, the Affordable Connectivity Program (ACP) has funding of \$14.2 billion and provides monthly subsidies of \$30 and a one-time device subsidy of \$100. Households in qualifying tribal lands receive a subsidy of up to \$75 per month (FCC, 2024). Most countries ensure broadband services are available to underserved populations by requiring operators to pay universal access or service funds annually (Garrity & Aminata, 2020).

LEO broadband's market entry and disruption force traditional service providers to lower costs and improve service availability. This is an attempt to maintain market share. Over the years, the pricing of the Internet to consumers has been stable. LEO is making broadband services affordable to a broader population by forcing prices down.

Given the diversity and areas LEO broadband covers, it is challenging to develop an average rate per user (ARPU) that fits all. The LEO providers must understand the areas in which they offer services. Countries with higher incomes might afford the services; however, areas with vulnerable groups may find it challenging to afford the services LEO offers.

4.5.2 Technology availability

While LEO broadband offers widespread coverage, its commercial availability remains limited in some regions due to regulatory, economic, and logistical barriers. *"In terms of coverage, it does cover the whole continent, but it is not available commercially for all countries."* (Brian Longwe). This indicates that while LEO satellites provide theoretical coverage, practical accessibility remains inconsistent due to market entry challenges in various nations. Some regions lack comparable alternatives to LEO broadband. *"They just didn't have that kind of service that compared to what LEOs were providing."* (Michuki Mwangi). This suggests that LEO broadband may be the only viable high-speed internet solution in certain areas, particularly in remote communities.

Additionally, community-based rural networks are exploring LEO broadband as a solution for underserved areas. *"I've seen this in Zimbabwe where we have a rural community network doing tests, and the same is happening in Kenya."* (Michuki Mwangi). These trials suggest growing interest in utilising LEO broadband to enhance rural connectivity, although its long-term feasibility remains to be proven.

4.5.3 Digital literacy

Even where LEO broadband is technically available, adoption is influenced by the level of digital literacy.

"A demand for digital capacity for connectivity will only be sustainable where there's a level of literacy, and people appreciate the need for internet and use it" (Carol Kamaitha). This highlights that internet adoption depends not solely on availability but also on users' ability and willingness to engage with digital services.

In some communities, internet access is not yet seen as essential. *"I can't go five hours without connectivity,*

but someone in a rural village doesn't mind. If it works, it works, if it doesn't, their way of life is not deeply impacted." (Carol Kamaitha). This suggests that demand for LEO broadband may remain limited in certain areas without strong digital education and awareness efforts.

Besides broadband services, Technology availability and digital literacy affect consumer adoption of broadband services. The populations of underserved areas are specifically affected by the lack of technology to access broadband internet, such as phones or computers, and the lack of digital know-how on internet usage. Apart from making the Internet accessible in these areas, there must be an effort to make these communities digitally literate. This not only affects developing countries, but in 2019, 30% of Californians did not have access to low-cost broadband, and more than 1 in 10 Californians don't have a computer (*California's Digital Divide*, n.d.) LEO broadband seems to address only the availability of broadband internet while ignoring the other factors contributing to the digital divide. However, experts agree that broadband availability is a first step in bridging the digital divide.

4.5.4 Easy to Deploy

The ease of deployment is a significant factor influencing consumer adoption of LEO broadband services. Participants emphasized how the straightforward setup process of LEO satellite terminals makes them an attractive alternative to traditional broadband solutions, particularly in areas with limited infrastructure.

One participant highlighted how LEO broadband simplifies installation compared to fixed-line services. *"In urban areas, it definitely increases competition for ISPs because you now have a service that someone can easily connect to—you just put an antenna on your roof."* (Bhavesh Mistry). This plug-and-play model allows consumers to access broadband without complex installations.

The user-friendliness of LEO broadband was another key advantage. *"I don't know if you've experienced this, but it is just so easy. They've made it so easy for anyone to install a small Starlink dish on the roof. They have an app, a tool, and the instructions are so clear."* (Bhavesh Mistry). This streamlined process lowers technical barriers to adoption, making it accessible even to non-technical users.

Another participant echoed this sentiment, emphasising the intuitive design of LEO broadband equipment. *"By design, they've made it very simple to deploy. The user experience and ease of installation are among the best we've ever seen."* (Michuki Mwangi). This suggests that LEO broadband providers have prioritised consumer-friendly setup processes to enhance market penetration.

LEO broadband services lower the entry barriers for new users by eliminating the need for professional installation and providing a seamless onboarding experience. This ease of deployment is expected to drive higher adoption rates, especially in underserved regions where traditional broadband infrastructure is lacking.

4.5.5 Enhanced services

Reliability and Speed Enhancements

Unlike traditional broadband providers, LEO broadband offers a stable, high-speed connection. *"It's a far*

more reliable and faster service. Faster meaning 100 Mbps, so you can stream HD content without worrying about slowdowns when your neighbor comes online, which is very common with Zuku." (Bhavesh Mistry). This suggests that LEO broadband provides a more consistent user experience than shared fiber or wireless connections.

Many traditional ISPs provide "up to" speeds, but performance can vary significantly. *"Safaricom sells you a 20 Mbps service, but it's shared 1:4. So you could get maybe 5 Mbps if everyone you're sharing with is online at the same time."* (Bhavesh Mistry). This highlights how LEO broadband's dedicated bandwidth model offers an advantage in reliability over some terrestrial ISPs.

The availability of LEO broadband is also pushing mobile operators to expand and improve their network offerings. *"We've also seen an effort to increase network reliability, as some users who switched to Starlink did so because of concerns over the quality of their previous network."* (Michuki Mwangi).

Increased Competition Leading to Service Expansion

LEO broadband has forced ISPs and mobile operators to expand coverage and improve services in areas they previously ignored. *"In other places where they had set them aside because they felt there was no market, they've now started reaching out and providing fixed broadband, both fiber and wireless, just to compete in those spaces."* (Michuki Mwangi). This suggests that LEO broadband is driving new investments in underserved areas.

The lack of uniform Quality of Service (quality of service) in terrestrial networks remains a challenge, giving LEO broadband an edge. *"If mobile operators do not scale up to provide uniform coverage, the challenge with current terrestrial networks is the lack of uniform quality of service across large regions."* (Tom Makau). This reinforces how LEO broadband could catalyse ISPs and mobile operators to standardize and enhance their service quality.

The entry of LEO broadband is reshaping the competitive landscape by pushing ISPs and mobile operators to improve network reliability, increase speed offerings, and expand service coverage. As competition intensifies, consumers are benefiting from more stable and faster internet services. However, traditional ISPs must address bandwidth-sharing issues and inconsistent quality of service to remain competitive against LEO providers.

4.5.6 Customer support challenges

The operational model of LEO broadband providers, particularly Starlink, presents significant challenges in customer support and service reliability, especially as they scale.

A significant challenge is the absence of local customer service infrastructure, which could impact long-term scalability. *"The model they're deploying is going to give them challenges—not having in-country customer service will be an issue as they scale."* (Ben Roberts). This suggests that the lack of localised support could lead to customer dissatisfaction as subscriber numbers increase.

Uncertainty Around Service-Level Agreements (SLAs)

Many enterprise customers hesitate to adopt LEO broadband due to the absence of formal SLAs. *"Starlink doesn't provide an SLA. They talked about introducing SLAs early this year, but I haven't checked. A lot of enterprise customers, including MNOs and AATLs, were reluctant because of this."* (Carol Kamaitha).

The lack of guaranteed performance commitments makes LEO broadband less attractive to businesses. *"If it works, it works—but even that is not committed. We ran a lot of tests where speeds and reliability varied."* (Carol Kamaitha). This uncertainty means that large organizations and telecom operators may be cautious in fully integrating LEO broadband into their networks.

The absence of localized customer support and clear SLAs could hinder the scalability of LEO broadband services. While the technology provides competitive speeds and coverage, enterprise adoption may remain limited until stronger service guarantees are in place. To improve market penetration, LEO providers may need to establish in-country support and formalize performance commitments for business customers.

4.5.7 Bandwidth Considerations

The availability of higher bandwidth at competitive pricing has been a key factor driving consumer interest in LEO broadband, forcing traditional ISPs to adjust their offerings.

Many consumers are attracted to LEO broadband due to its significantly higher speeds compared to traditional fiber or fixed wireless services. *"People are used to buying 20–30 Mbps from fiber operators, and suddenly you have this 100 Mbps package at a similar price. So people are willing to switch for blistering fast speeds."* (Bhavesh Mistry). This has pressured traditional ISPs to increase their speed offerings. *"Starlink is giving me 200 Mbps for \$50, while my old ISP was charging me the same price for only 20 or 40 Mbps."* (Brian Longwe). Some LEO broadband plans come with data caps, which limit high-speed access after a certain threshold is reached. *"The standard residential service has a cap. Once you hit that, you get degraded performance or you're done for the month."* (Brian Longwe).

Consumers prefer uncapped services and may choose providers based on flexibility. *"End users don't like caps. We at CTA don't put any usage caps. We say it's unlimited—use as much as you want as long as you pay for the pipe."* (Brian Longwe).

Affordability, availability, and higher bandwidth offerings are the primary factors influencing consumer choices. *"It is availability, costs, and the amount of bandwidth that matter the most."* (Michuki Mwangi).

LEO broadband has disrupted the market by offering higher bandwidth at competitive pricing, compelling traditional ISPs to enhance their service offerings. However, usage caps remain a challenge, and consumers may favour providers that offer unlimited data plans. Affordability and consistent high-speed performance will be key factors in long-term adoption and competition.

4.6 Future Trends & Industry Predictions

4.6.1 Geopolitical issues

Emerging Geopolitical Tensions

Growing hostilities between nations could lead to regulatory roadblocks for LEO broadband providers in certain regions. Brian noted, *"With the current trend in geopolitics and emerging hostilities, there is a high chance that LEO providers like Starlink may face restrictions in certain countries that don't align with them politically."* (Brian Longwe).

Potential Threats to LEO Constellations

There are potential threats that LEO broadband networks may face, such as geopolitical-driven sabotage or interference. *"If someone decided to sabotage a LEO constellation, these satellites are 1,500 kilometres in space, making them vulnerable."* (Brian Longwe).

As global tensions escalate, LEO broadband providers may encounter political and regulatory challenges, which could impact their expansion and operations in specific markets. Additionally, the risk of geopolitical interference, including targeted sabotage, remains a concern for the industry moving forward. The UN has raised an alarm over the increasing cases of jamming and spoofing of satellite navigation systems. (ITU, 2025).

4.6.2 New Market Entrants

New Market Entrants Likely to Cause Competition Amazon's Kuiper project may introduce new competition by bundling internet services with other products. Bhavesh noted, *"I have heard some rumors about how Amazon plans to bundle internet with other things through its Kuiper project. That will be interesting to see."*

There is also likely to be an evolution of satellite technology, which will increase competition and make broadband access more affordable and accessible. *"Technology evolution and new competition among market players are always good for increasing the availability of affordable and reliable access."* (Michuki Mwangi).

The expansion of LEO broadband services is forcing operators to adapt to underserved markets. This in turn, will make services available to the underserved arrears. *"Its reach creates an environment where operators who were previously reluctant to enter certain markets may now have to adapt."* (Michuki Mwangi).

Affordability remains a key goal, with benchmarks targeting costs at 2% of a country's Gross National Income (GNI) per capita. *"One of the goals anticipated is affordability reaching around 2% of monthly income or GNI per capita."* (Michuki Mwangi). Significant efforts must be made to reach most of the world and make broadband more affordable.

4.6.3 Increased Access and Digital Growth

Expanding LEO broadband and increased market competition will drive greater internet access and digital

adoption. *"We are going to see increased access because of more people getting online due to technology evolution and competition from new entrants in the market. More people will get connected."* (Michuki Mwangi).

The growing digital ecosystem will lead to increased adoption of online services. *"This will also lead to growth in the digital space, meaning more people will use digital services."* (Michuki Mwangi).

4.6.4 Satellite Replacement and Long-Term Sustainability

LEO satellites have a limited lifespan, necessitating frequent replacement, which in turn increases operational costs. *"The biggest technical challenge for LEO is that the satellite has a limited life of five to six years. Every few years, they fall out of orbit and need to be replaced."* (Bhavesh Mistry). The continuous need for satellite replenishment poses long-term financial challenges. *"You can imagine the complexity and cost of always refreshing satellites every five to six years—this is a massive cost that will hurt them in the long term."* (Bhavesh Mistry).

While companies like Starlink currently have sufficient resources, sustainability concerns remain. *"Right now, Starlink is okay. They have a lot of money and customers, but longer-term, how they operate and manage costs will be critical."* (Bhavesh Mistry).

Advancements in satellite technology may help extend operational lifespans and reduce costs. *"Of course, they are pushing technology—maybe future satellites will last longer."* (Bhavesh Mistry).

4.6.5 Evolution of Spectrum and Wireless Speeds

According to one participant, constraints in the electromagnetic spectrum will likely be overcome through innovation. *"The current constraints we see in the electromagnetic spectrum are going to be superseded—they will be innovated around, and we'll see more advancements."* (Brian Longwe).

Future wireless technologies may achieve speeds comparable to fibre-optic networks. *"Today, with 4G and 5G, we already see gigabit speeds. In the near future, we might be able to see 10 gigabit or even 100 gigabit speeds over wireless infrastructure."* (Brian Longwe).

Projects like Google's Taara demonstrate the potential for high-speed wireless connectivity. *"A good example would be Taara—Google's Taara."* (Brian Longwe).

5 Discussion

This research aims to find out how the emergence of low Earth orbit (LEO) satellite broadband internet influences the market dynamics, service quality, and customer preferences for broadband services traditionally provided by Mobile Network Operators (MNOs) and Internet Service Providers (ISPs).

It also focuses on how regulators are prepared to accommodate the new broadband services on Low Earth Orbit satellites.

5.1 Summary of Key Findings

This study identifies four key areas where LEO satellite broadband internet will face challenges and opportunities, influencing market dynamics, providing high-quality service, adapting to customer preferences for broadband services, and navigating regulatory challenges. The key areas include marketing opportunities, technical opportunities, sustainability and economic considerations, regulatory hurdles that LEO broadband faces in the market, as well as consumer adoption and accessibility factors.

Marketing Opportunities and Challenges

This study found that broadband is touted as a means to bridge the digital divide, despite the limitations of LEO. The current business model is primarily favourable to those who are already connected. In urban areas, it's not fulfilling the price gap to connect the population who cannot afford broadband services; however, in terms of providing broadband capabilities, LEO satellites' large footprint is reaching areas not covered by traditional broadband providers, hence extending broadband coverage. One more interesting note is that, although electricity availability is an issue, as LEO broadband dishes require power, the number of broadband users is not proportional to the population connected to the grid. Emphasis on expanding broadband coverage can be better explained by the ITU classifications of the regions as

- A) Market-driven areas (where private investment is viable),
- B) Policy-dependent areas (requiring regulatory adjustments),
- C) Subsidised zones (where incentives are needed), and
- D) Unfeasible regions (requiring ongoing support).

It provides a solid foundation for bridging the digital connection divide and expanding the reach of broadband services. To bridge the digital divide, LEO broadband must align its business model with the ITU classifications of broadband coverage.

Remote islands and landlocked countries were also found to benefit from the large LEO footprint as it introduces competition to existing monopolies. In these areas, lowering broadband prices is likely to increase broadband adoption.

Support for IoT, especially in logistics, was highlighted as one area where LEO broadband could offer new capabilities, especially with cross-border tracking. Automation, especially in agriculture, was also highlighted as an area where LEO can improve connectivity to IoT devices on farms.

LEO broadband has an advantage in providing in-motion Internet for the airline and shipping industries. The shipping industry has primarily relied on GEO satellites for broadband connectivity; however, it has been noted that the limited space on ships and the high broadband costs restrict the availability of capacity on these vessels. With LEO broadband, increased capacity is offered at a price that enables the crew to enjoy enhanced connectivity, and the small antenna facilitates more straightforward installation. It is worth noting that the phased array antenna simplifies connectivity to satellites, as manual tracking is no longer necessary. Offshore drilling activities will also benefit from better connectivity at a fraction of the cost provided by GEO satellites.

Partnerships between LEO providers and traditional broadband providers to provide backhaul to remote locations where it's expensive to deploy terrestrial infrastructure present a new opportunity for LEO providers. This enables traditional broadband providers to cover areas they could not reach with terrestrial backbones. Traditional broadband providers can utilise these backbones to develop business cases for future terrestrial infrastructure deployments.

Low latencies provided by LEO infrastructure were highlighted as a key advantage for providing backhaul to traditional ISPs. The backhaul provided by LEO can also be utilised for congestion avoidance, allowing ISPs to avoid installing expensive backhaul and instead rely on LEO to provide burst capacity when needed. Apart from the normal backhaul, there are cases where the terrestrial and undersea infrastructure fail. LEO providers can be beneficial as they are not susceptible to cable breakages, fires, or vandalism that affect terrestrial and undersea backbones.

LEO broadband providers face market viability and pricing challenges for their products. Most broadband providers have clustered their products into four categories

- a) Usage-based (prepaid)
- b) Usage-based (postpaid)
- c) Value-added services
- d) Limited revenue – free access

LEO broadband will need to adapt to this market in order to compete effectively. Most of these products are tailored to suit the demand of the subscribers. A plan that differentiates itself solely by bandwidth consumption will not align well with the needs of broadband consumers. There are concerns that most of the pricing offered by LEO could be subsidised and aggressive for the service. Some providers have been forced to adjust their plans to double capacity or add value to existing products to counter LEO services. In addition to the service plans, LEO broadband providers face challenges in repatriating billing revenue in some countries due to foreign exchange limitations.

Significant adjustments have been made in the competitive market since the entry of LEO. Competition has increased, with existing broadband providers adjusting their bandwidth by doubling the capacity offered or revising their fair use policies. Different models have emerged where providers allow users to share bandwidth over the same connection to the service provider. The direct-to-consumer business model employed by Starlink has also eliminated middlemen in the supply chain for broadband services. These factors have led regulators to require local licensing from LEO broadband providers to regulate market competition. The competition among LEO is also set to increase as more providers launch constellations to provide broadband services. Interestingly, providers are now investing in PONS that can deliver up to 10G to the end user, as evidenced by the increase in purchases of PONS in recent years. This will increase the availability of terrestrial broadband capacity to end users, creating competition for the current deliverable capacity via LEO broadband.

Despite competition, some providers have found areas where they can collaborate with LEO providers. LEO provides the necessary bandwidth for distribution networks in areas where traditional providers have less bandwidth. In other cases, some ISPs are acting as resellers, which allows Starlink not only to use their local distribution networks but can also be a way to get faster access to countries with stringent licensing rules. These partnerships can be established using an effective revenue share model.

Lastly, there is also an opportunity in unmanned vehicles, where LEO can provide bandwidth for control and monitoring. This can be particularly useful in remote regions without terrestrial coverage, enabling autonomous vehicles to operate seamlessly. LEO broadband's combination of low latency, global coverage and high-speed connectivity is expected to provide autonomous vehicles with the ability to operate in remote areas and enhance future automation.

In summary, LEO broadband presents new opportunities in broadband applications; however, LEO must overcome several challenges to capitalise on these advantages.

Performance Technical Opportunities and Challenges

While delivering broadband capacity LEO broadband has some opportunities such as the low latency as compared to GEO counterparts, there are also challenges that LEO broadband face such as weather conditions, the decentralised nature of internet content, the design of current routing protocols that are not designed to work with the mobile nature of LEO satellites, the allocation of spectrum within bands usable by other terrestrial links causing interference, Scalability and bandwidth limitations in densely populated areas, network standardisation and environmental concerns.

Our findings indicate that LEO broadband offers lower latencies of less than 50 milliseconds than GEO counterparts, which typically range from 300 to 500 milliseconds. Combined with POP deployments in strategic locations, this has dropped latency to below 30 milliseconds. Latencies are, however, sometimes affected by routing inefficiencies. With this improved latency, LEO broadband services still fall short of the service provided by traditional ISPs for shorter distances. Conde et al.'s (2024) study finds terrestrial networks offering lower latencies over shorter distances. However, the study mainly focuses on the propagation delay. LEO broadband is a means to access content. Hence, propagation delay is important, and the location of the content required by end users is also crucial. This also applies to emergency services numbers and services.

LEO broadband providers have attempted to deploy ground stations and partner with local internet exchange points and content distribution networks to improve user experience by reducing latencies. LEO's efforts are paying off as latency has improved with the deployment of local ground stations. Internet addresses provided by LEO broadband to end-users, mostly allocated by their home regional registries, pose routing challenges for geographic routing policies based on source routing. Before deployment, ground station deployments have several factors to consider, including a sufficient terrestrial backbone to offload traffic, weather conditions, distances between ground stations, and potential interference. This does not account for the costs of deploying ground stations, making it challenging to efficiently deploy ground stations.

that match the distributed content and Internet exchange points, and to provide consistent and comparable latency to traditional terrestrial broadband.

One way LEO can overcome the dynamic nature of content is by adopting a regional routing approach. Essentially, a router in the sky would use snapshot routing as LEO satellites come into position in various regions globally. This would enhance LEO broadband delivery through smart routing decisions onboard the satellites. Current routing protocols are designed with stable network links in mind. Westphal et al (2023) Li et al. suggest adopting current routing protocols to adapt to either dynamic or snapshot routing for LEO links. Li et al (2023) examine various types of LEO satellite routing algorithms and their associated challenges and opportunities.

The scalability of LEO broadband services is limited in densely populated areas. Due to network congestion, LEO broadband services temporarily pause future subscriptions in various cities. Spectrum, being a finite resource, is identified as one of the causes of scalability limitations. There are attempts to expand the LEO spectrum to other frequencies, such as E-band frequencies. There are limits to the extent to which LEO can deliver scalable bandwidth compared to terrestrial fibre infrastructure, especially fibre, which has virtually unlimited capabilities.

The technical capabilities to deliver broadband affordably using small antennas limit LEO satellites to high-band frequencies, which makes the service susceptible to weather conditions such as rain. A study on the impact of rain on KU-Band services in Cairo, Praia, and Port Louis found that, based on ITU-R P.837-7 rain intensity results, higher rain intensity was associated with increased service interruptions (Osoro, n.d.). Geomagnetic storms have also posed a challenge, particularly for LEO satellites that provide LEO broadband services. The Japanese ASKA satellite lost altitude and re-entered the Earth's atmosphere, while Starlink lost 39 satellites due to a magnetic storm. This can result in significant losses and unforeseen events that cannot be controlled or planned.

As the LEO industry grows, standardization efforts will be crucial for ensuring efficient and scalable network operations. Open standards will facilitate competition, enhance interoperability, and enable seamless global connectivity. Terrestrial and GEO broadband networks have benefited mainly from standardisation. Despite some differences in vendor implementation, the likelihood of interference and interoperability issues is manageable. LEO satellite communication is evolving from the traditional bent pipe deployment, where the satellite acts only as a repeater, to more complex topologies and models with intersatellite communication. The proprietary nature of LEO technology needs to be addressed for future growth. Currently, most providers maintain proprietary technology, which limits interoperability. The potential entry of LEO broadband using direct-to-cell technology has led standardisation bodies, such as 3GPP, to investigate interoperability with existing terrestrial networks. As more competitors enter the market, participants observed that some form of standardisation for the technology LEO uses to provide broadband services will be needed. Deploying ground stations or gateways is costly and time-consuming due to the varying jurisdictional requirements for LEO broadband. Standardisation of ground stations would enable rapid growth, as LEO providers would focus on launching satellites while leaving ground station deployment to

various entities. This can only be possible if ground stations are not proprietary technology, as is the case now. Different constellations are accustomed to working with only LEO satellites from a specific provider.

Space debris is likely to pose a significant challenge that needs to be addressed promptly. The number of satellites launched into space will require a proper satellite disposal mechanism. In cases where companies go bankrupt, there may be a need to restructure a large constellation. Iridium survived the deorbiting of its 66 satellites when the U.S Department of Defence awarded the company a tender to keep them in orbit (ABC News, 2000). This could be a concern in the future if effective space debris management policies are not implemented.

Sustainability and Economic Considerations.

Despite the drop in payload launch and satellite manufacturing costs. The cost to launch LEO constellations remains beyond the pricing capabilities of traditional broadband providers. This restricts participants to a few companies or state actors. The transparency in the costs is still not public knowledge, hence speculation that the end-user pricing is currently subsidised. The costs of deploying gateways and licensing fees make it more expensive to account for the additional costs incurred for operating the gateways. The lifespan of LEO satellites is only 5-7 years; this limits the time the satellite must generate revenues and break even before the end of its life. The user terminals are also not affordable to the average broadband user. Participants noted that even with subsidies, the upfront costs for consumers remain high, limiting accessibility in rural and underserved areas. This aligns with Westphal et al (2023) prior research suggests that cost-effective financing models and public-private partnerships are necessary for LEO's long-term viability. In cases where the LEO provider does not own terrestrial infrastructure, they are restricted to purchasing internet backbone connectivity from third-party providers, primarily incumbent broadband providers, making it complicated to obtain competitive pricing (ITU - Space Connect, 2025).

However, in unexploited areas such as the marine and aviation industries, LEO broadband provides better alternatives to the current broadband providers, making it an area with potentially high returns.

The long-term sustainability of the LEO business model is still in question, as interviewees pointed out the subsidy programs, especially from the US Department of Defence. There is also a lack of transparency regarding financial sources, and the subsidies put into question the LEO broadband business model. The large areas that LEO covers make it challenging to ascertain if the subscription rate is compared to the coverage area of the beams. Such information is not publicly available.

To compete with terrestrial infrastructure, LEO broadband providers must deploy numerous gateways to minimise subscribers' latency. Deploying this gateway, alongside payments to local fibre providers, increases operational costs and makes the return on investment a challenge for LEO providers.

Apart from operational and deployment challenges, LEOS faces support challenges for their many subscribers. This model may face challenges, as subscribers are often isolated in countries without local support. The simplified direct-to-consumer model has, however, given advantages to LEO by cutting out the middleman, hence cutting their cost of sales.

Regulatory Hurdles

LEO operators face many challenges when dealing with local regulators to activate services. This can be more evident with the new request by Starlink to the U.S. Trade Representative's Office "said that it has to pay foreign governments for access to spectrum, import duties on Starlink equipment and other regulatory fees, which "artificially" inflate operating costs abroad", According to Reuters article the request for intervention further states that "Foreign operators have used these anti-competitive policies to block or slow SpaceX from providing a better quality and lower cost service to customers in those countries"(Reuters, 2025).

LEO satellites are subject to Outer Space Law and domestic laws. This means that LEO broadband providers must comply with local regulations in each country where they intend to provide services, despite the physical infrastructure that is typically already in place. Most countries use licenses and spectrum resources to raise revenue, imposing significant costs on these resources. The competition for spectrum with other services from traditional broadband providers, such as 5G, is creating further challenges for LEO broadband operators in acquiring the necessary resources, country by country.

Apart from the spectrum, some countries require licensing to operate. Specific requirements in individual countries complicate the licensing requirements. Several countries, such as South Africa, have a 30% local ownership requirement for telecom licensing. In other jurisdictions, there are requirements that the company must have a local office and install local gateways. The slow adoption of the new LEO service business model could make some licensing regimes difficult to navigate. Without a distinction between a ground station used for LEO licensed individually and the LEO broadband user terminal, Individual licensing can make it impossible to deploy the number of terminals in the country, especially when factoring in the cost. There is some movement towards adjusting old licensing regimes to accommodate LEO broadband services, and in some jurisdictions, regional licensing is under consideration.

Concerns from governments that regulate internet access are shaping the conditions under which LEO broadband operates in some jurisdictions. Some governments have tight controls on content and sometimes access to the internet. The business model LEO is deploying makes it difficult, if not expensive, for the government to regulate the Internet. Some governments have made it mandatory to route all internet traffic through local gateways to comply with the terms of providing broadband services. The balancing act of expanding internet connectivity where traditional connectivity has failed to reach, versus losing control over internet access, is a delicate challenge that governments must navigate.

Geopolitical factors in home countries where LEO providers reside affect how LEO broadband providers can access global markets. Affiliation by country might determine where licensing is granted to operate. Concerns of espionage and surveillance also play a significant role in the acceptance of LEO services globally. Sanctions and trade restrictions can also impact shipping LEO broadband terminals to specific markets. Interestingly, satellite beams do not form along countries' borders. Hence, ensuring that LEO services are not utilised in these restricted or controlled areas is challenging.

The emergence of data privacy laws and enforcement in various jurisdictions is exerting some control over how end-user data can be used and where it can be stored. GDPR is one such law enacted to protect the personal data of EU citizens or residents. It applies to anyone who offers goods or services to EU citizens or residents, even if you are not in the EU (GPDR.eu, 2018). There is a fear that even with compliance in LEO home countries, traffic could be rerouted through countries with weak LEO laws. Enforcement of data privacy laws may also be challenging for small countries, as it often requires international cooperation and enforcement. Controlling which gateway traffic is routed might be difficult even in countries with ground stations and gateways.

The over-reliance on external infrastructure outside the jurisdiction might be a challenge, especially regarding National and critical infrastructure. Governments struggle to enforce regulations on telecommunication companies that reside outside their jurisdictions. There was fear that Starlink would cut off critical LEO access services to Ukraine (Shalal & Roulette, 2025). There are also cases where LEO broadband services are provided in the country without government approval, as regulators acknowledge it is challenging to regulate. It can only rely on data from the LEO operators (ITU Space Connect, 2025).

Consumer Adoption and Accessibility Factors

Factors affecting consumer adoption of LEO broadband services include affordability, the availability of the technology, digital literacy, ease of deployment, after-sales support, and bandwidth considerations.

For urban areas or those with high incomes, the pricing for LEO broadband services in the market is within competitive limits; however, they cannot expand beyond their existing customer base, which already uses traditional broadband services. Affordability is particularly relevant in rural areas with lower income brackets, where the majority cannot afford the subscription fees charged. The initial equipment costs for LEO terminals remain out of reach for most internet users. This has restricted access to Starlink services to residents with higher incomes. The suggestion that broadband pricing should be 2% of the Gross National Income by the Broadband Commission can only be achieved through interventions such as the Affordable Connectivity Program (ACP) in the United States or subsidies.

The availability of technology and the means of access are also important for consumer adoption. In some cases, there are just no other services to compare, which means no other option is available for the consumer. Regulatory, economic, logistical, and tariff barriers affect the end user's availability of technology and services. This impacts LEO broadband providers more as the market adopts and adjusts to accommodate the new business model.

Affordability and availability are factors, but digital literacy is crucial in broadband adoption. While LEO enables access to remote areas, parallel efforts are needed to improve digital literacy. A 2019 study in California revealed that 30% of Californians lacked computer access. So, despite efforts to make broadband available, digital literacy must be addressed.

Deploying LEO terminals has made it easier for consumers to access services without going through the traditional installation process of broadband providers. This, in turn, removes the installation fees charged

to consumers for broadband installations. This will make a difference, especially in underserved regions where transporting and installing technicians is difficult or costly.

The entry of LEO broadband has brought enhanced speeds, necessitating adjustments to speed and improvements in service levels by local broadband providers. Some have doubled capacity and improved and removed usage limits to increase competition. Unlike LEO broadband services, traditional broadband services have struggled to deliver consistent quality of service across large regions. Consumers are benefiting from the competition brought by LEO providers to local markets.

LEO providers' global nature and operational model present challenges in providing local customer support to consumers. As subscriber numbers increase, LEO providers will face challenges in providing basic support to subscribers worldwide without local support in different operating regions. This type of service may encounter difficulties in providing and fulfilling service-level agreements with consumers, which could result in substantial fines.

Future Trends and Industrial Predictions

LEO, being a new technology, presents many unknowns for the future; however, some are already emerging, such as Geopolitical issues, new market entrants causing competition, increased access and digital growth, satellite replacements, long-term sustainability concerns, and evolution in spectrum usage.

Emerging ties between LEO providers and government funding may limit access to LEO providers in specific markets. There are also observations that this issue may require attention due to potential geopolitical sabotage and interference. The UN has raised an alarm over the increasing cases of jamming and spoofing of satellite navigation systems, which could lead to disruptions in broadband internet.

Potential new market entrants in LEO broadband services will increase competition. Amazon's Kuiper project is just one of them, aiming to provide global high-speed internet service with over 3,000 satellites.

Increased competition among LEO providers may lead to increased access and digital growth, resulting in higher adoption rates of online services. This might enable broadband to reach more people at an affordable rate, reducing the broadband access gap.

Due to limited spectrum, there is a trend to move LEO services beyond the current Ku and Ka-band frequencies to E-band and V-band frequencies.

5.2 Implications for the Broadband Industry

The introduction of LEO broadband services has disrupted the global broadband industry. There is increased competition, a potential to expand the broadband industry to accommodate the underserved and unreached, opportunities for collaboration to enhance and supplement broadband industry connectivity, and spectrum management challenges.

Increase Competition and Market Disruption

LEO broadband providers can provide internet globally, covering large footprints quickly and making high-

bandwidth connections available to end consumers. This, combined with the direct-to-market approach, is changing how the Internet is provisioned in targeted markets. This has disrupted the normal flow and the intermediaries required to provide internet to end-users. This approach outpaces the business model used by traditional broadband providers, which must deploy their infrastructure. Unlike the GEO counterparts, the antennas used for providing LEO broadband are small and can easily be deployed by the end user. This again eliminates the intermediaries that provided installation services in the country for these services. As Bhavesh noted, “One of the advantages of LEO is its large footprint.” With more LEO broadband providers launching, the competition among LEO providers is likely to result in more affordable internet and increased capacity, thereby putting more pressure on traditional broadband providers to improve the reliability of their networks and innovate on more affordable solutions for their customers.

Expansion of broadband coverage globally

Broadband coverage has been driven by profit, and traditional broadband providers have only deployed networks where they could generate a return on investment, despite government policy support and, in some cases, subsidies. The LEO broadband model changes this as the large satellite footprint can reach areas with harsh terrain and low population density. In 2024, Starlink partnered with the government of Ontario, Canada, to connect fifteen thousand premises in Ontario that do not otherwise have a plan for fibre optics. (Hicks, 2024). Whole constellations will likely achieve global coverage, immediately changing broadband reach. Broadband coverage is expected to increase globally, particularly with the emergence of more entrants in Low-Earth Orbit (LEO) broadband.

Collaboration between broadband service providers

The broadband industry has struggled with infrastructure limitations in serving remote Points of Presence (POPs) that serve areas with harsh terrain, security concerns, and low return on investment. With LEO, traditional providers can utilise LEO broadband to provide connectivity to remote Points of Presence (POPs) and offer backup connectivity or additional capacity, making it economically viable to provide services to these zones. OneWeb and Bharti Airtel India have partnered to integrate LEO with fibre to provide resilient connectivity to cellular backhaul (Eutelsat ONEWEB, n.d.). Such collaboration might extend to reselling models where traditional broadband providers will act as channel partners. Starlink has made a reseller agreement with Paratus (an Internet provider in Africa) to resell Starlink services in Africa (Arevalo, 2024). As LEO broadband services evolve and expand, collaborations will likely emerge to integrate LEO into existing LEO infrastructure, enhancing service delivery to broadband users.

Challenges in Spectrum Management

As more LEO services are deployed, they require additional spectrum. This is evident with LEO providers exploring E-band and V-band frequencies to provide services. This is because the spectrum is scarce and may not be sufficient for all market players. For players in the market using fixed wireless broadband, interference from LEO networks can risk interference and degrade services. Regulators must consistently monitor and establish standards to prevent interference, as seen in the case between Dish Network and

Starlink, where the FCC voted to preserve parts of the 12 GHz spectrum band for advanced satellite services.

5.3 Implications for Policy and Regulation

Regulatory and existing policies cannot regulate the business models that LEO providers use to provide global broadband access. Most countries' regulations on satellite internet are based on GEO satellites. Countries must adjust their regulatory frameworks to accommodate the LEO broadband business model. Two levels of regulation are required: the first involves the space station and frequency used to provide services across different jurisdictions, and the second pertains to the user terminals used to access these services.

The space station will require a license from the ITU member state in which it is launched, but it will also need to be licensed to provide services in every jurisdiction, including the spectrum allocation.

The terminals and ground stations require licenses from the countries where they are based. Regulators must prepare and update regulations to license the new LEO constellations, enabling them to provide broadband services in their respective countries.

Regulators that still restrict internet access must strike a balance between regulating internet content and access and the benefits of broader broadband coverage. Some Regulators have shifted from per-terminal licensing to blanket licensing for LEO services' end-user terminals. There is also movement towards regional licensing for LEO services.

Policymakers also need to consider insurance for damages that may arise from debris falling from LEO satellite launches. In recent years, debris has fallen back to Earth, which could result in damage on the ground. This also applies to services that LEO providers do not deliver in jurisdictions where they do not have a physical presence, as specified in service-level agreements.

5.4 Limitations of the Study

While this study highlights LEO broadband services' opportunities and challenges, this section identifies some research challenges. These include sufficient data availability and rapid evolution in the LEO industry. Only a few providers are operational.

Data availability and new information

Due to the limited number of LEO providers, comparing data across providers was challenging; therefore, the findings may change as soon as new entrants publish their data. Data available is also volatile and changes daily, such as the number of satellites launched into orbit. The rapid change in information dynamics has necessitated adjustments in the study. Some information may become outdated within a short period.

Geopolitical uncertainty

This study tried to accommodate the evolving geopolitical uncertainty. However, information contained in the study might change in due course. Efforts were made to accommodate the changing restrictions due to data privacy laws; however, these are still being implemented in various countries. It is unclear how data privacy laws will eventually affect LEO broadband use. Trade disputes, international sanctions, or national security policies are also likely to change perception and future expert views on the adaptability of LEO broadband. The relevance of recommendations and conclusions may change due to shifting geopolitical dynamics.

Limited Scope of Experts

This study primarily interviewed experts in the African region, and the findings may not accurately reflect the challenges and opportunities faced in other regions, particularly those with developed economies and mature regulatory frameworks.

Limited access to proprietary information

Information on available LEO providers is generally proprietary and confidential, making access to this information challenging. This limited the extent to which the study could comprehensively analyse specific business models or long-term sustainability strategies. The research relied heavily on publicly available data and expert interpretations rather than internal corporate disclosures.

Interpretive Nature of Qualitative Analysis

The research methodology used is subject to the derivation and interpretation of themes. Despite the safeguards, personal biases and differences in interpretation may have affected some of the analytical decisions and findings. As the study is qualitative, findings may not be generalisable to the entire telecommunications industry. The reliance on predefined themes could have limited the emergence of novel insights, and lastly, the focus on industry experts may have underrepresented consumer perspectives. In the future, the mixed approach of qualitative and quantitative analysis can help to eliminate the bias.

5.5 Suggestions for Future Research

Some of our findings may require future research due to insufficient or unavailable information.

Sustainability of LEO broadband services.

Most companies that attempted to launch LEO services faced financial difficulties. As our interviewees pointed out, LEO providers are not very transparent. Future research can focus on the financial sustainability of LEO broadband. This should be possible as more commercial entities launch to LEO and publish financial data.

LEO Broadband and Data Privacy

LEO broadband is in its early stages of deployment. It is unclear how countries will enforce data privacy,

particularly in light of the enactment of data protection laws.

Service Scalability

This research could not comprehensively cover how LEO broadband providers will address the demand in urban areas, as the conclusion of these findings coincided with the pause of LEO broadband services in specific locations. Will this be a problem in the future, given the insufficient number of subscribers on LEO?

Regulatory harmonisation

During this study, some evolutions have still been made in how regulators accommodate LEO broadband. The future harmonisation of regulatory policy and requirements across regions can significantly affect the adoption of LEO broadband by lowering costs and increasing adoption.

Spectrum allocation

A fair share policy that will allow latecomers to benefit from LEO technology in the future by ensuring a fair share of resource allocations. There are indications and possibilities of hoarding resources for future speculation.

D2D via IMT

Introducing direct-to-device technology may enhance access to broadband for previously unreached populations, particularly in underserved areas. Future research can include direct-to-cell as part of the findings.

6 Conclusion

LEO broadband presents new market opportunities by covering underserved regions, including the marine and aviation industries. Its additional bandwidth also significantly supports IOT. Traditional providers can also benefit from LEO by incorporating it into their terrestrial infrastructure or collaborating with LEO providers. Consumers will likely benefit from the increased competition LEO broadband has brought to the market. The direct-to-market model adopted by LEO providers has altered the dynamics of LEO provisioning, making it more cost-effective for customers to install the services.

There are technical challenges that LEO providers have to overcome. Using Ku and Ka bands makes LEO broadband services susceptible to weather, which can affect the services, especially during rain. There have been challenges with latency, as LEO providers lack gateways to offload traffic locally, resulting in consumers experiencing deteriorated services. However, in some cases, LEO providers have installed gateways to improve these services. LEO providers have faced challenges in providing services in urban areas. They are sometimes forced to suspend subscriptions, which has been attributed to a lack of sufficient capacity on the currently deployed satellites. Due to the finite nature of the spectrum, LEO broadband providers have had to expand into other frequencies, such as E-band and V-band, as other services or providers already utilise these frequencies. The lack of standardisation is slowing LEO providers, as most providers use proprietary technology. Space debris poses a threat to LEO broadband service provisioning; however, there are ongoing efforts to implement global policies that would reduce space debris.

LEO broadband services have high deployment costs, which have restricted the number of entrants in the LEO broadband market. The return on investment for the LEO broadband service is currently unclear. One of the challenges is that LEO providers must deploy multiple ground stations or gateways to offload traffic in different regions. This increases deployment costs and exposes LEO providers to competition as they purchase internet backhaul from their competitors. LEO Global Services also exposes them to operational and support challenges, as they must keep the gateways running worldwide. This also includes monitoring centres to ensure proper service delivery worldwide.

Regulatory and licensing hurdles are among the challenges LEO broadband faces. LEO broadband providers must deal with 200 different regulators worldwide to provide global services. Licenses are sometimes prohibitively expensive and, at times, not granted to operate. Governments also use resources to raise revenue. Spectrum is scarce, and LEO providers must pay high fees to offer services. Some regulators have laws that need to be updated to accommodate the new LEO model, affecting the speed of LEO adoption in those countries. In some countries, the internet is regulated, and LEO providers are required to install local gateways to maintain internet control. There are also concerns about trusting LEO providers with national security and critical infrastructure in some countries. In some cases, geopolitical restrictions may prevent LEO providers from operating in certain countries. Data privacy laws are also likely to challenge LEO providers, as most countries enact and enforce them.

Leo's broadband availability is insufficient if consumers cannot utilise the service. This study finds several factors that hinder consumer adoption of LEO broadband services. One of the most prominent concerns was affordability; most consumers found the service prices to be higher than they could afford, especially the initial cost of the terminal. Digital literacy also emerged as a factor influencing consumer uptake of broadband services and, by extension, the LEO broadband services. Efforts must be made to improve digital literacy in conjunction with the increasing availability of technology. The issue of customer support is also crucial to service sustainability; most LEO providers lack a local physical presence, posing a challenge to after-sales support. Despite this, LEO's terminal is easy to deploy, allowing consumers to install the gadget themselves without requiring specialised technical skills. There were also service improvements from current service providers due to the competition that LEO had created.

7 Recommendations for the Broadband Industry

Based on the findings, the following recommendations are made to the broadband industry with emphasis on creating new markets, enhancing service adoption and promoting the sustainability of LEO broadband in the future

7.1 Collaboration

Better collaboration between LEOs and traditional broadband providers is necessary to leverage their complementary strengths and reach remote and underserved areas. LEO providers can enhance coverage and reach the underserved areas by leveraging existing broadband distribution networks and expertise. The direct-to-market approach limits the market penetration of LEO providers.

7.2 Address Regulatory Challenges

Harmonising regulatory frameworks is needed. This will encourage more providers and enhance services. Favourable regulatory frameworks globally will also encourage more entrants to the LEO broadband sector. The broadband industry must engage regulators and harmonise broadband regulations to level the playing field and discourage unfair competition. New licensing frameworks and spectrum coordination that enable regional licensing should be explored to reduce the number of regulators LEO providers need to engage with to provide global services.

7.3 Improve digital literacy

Digital literacy must be increased in underserved areas to expand reach and create a larger market for broadband services. LEO providers must be integrated into existing broader digital development strategies. This will expand the current broadband market and avoid the concentration of subscribers in limited locations.

7.4 Service Scalability and Affordability

The future sustainability of LEO will depend on innovations in managing network congestion, satellite lifespan, and spectrum efficiency. Future innovation in these areas will lead to the affordability of broadband services by increasing available bandwidth per user and increasing the amortisation period for new satellites. This will be achieved by allowing more players and lowering the initial costs of acquiring broadband services.

7.5 Address Environmental Factors

Space debris challenges and threatens the continued provision of LEO broadband services. Efforts to address space debris must continue to avoid future space collisions. Strategies to reduce orbital congestion, minimise launch emissions, and manage end-of-life satellite disposal must be implemented and enforced. The future of space usage sustainability will determine LEO broadband sustainability

7.6 Network Standardisation

The current deployments of LEO broadband are highly proprietary. In the future, equipment needs to be standardised to enable interoperability. Transparency and the ability to communicate across platforms also need to be achieved. Standardisation can bring more advantages, such as independent ground station/gateway construction. Ground stations are key to reducing latency and improving user experience. Allowing for multiple players to participate can increase LEO broadband adoption and reduce the initial costs for new entrants in the LEO market.

References

- ABC News, A. B. C. (2000). *Pentagon Moves to Keep Iridium Satellites Aloft*. ABC News. <https://abcnews.go.com/Technology/story?id=119255&page=1>
- Al, J.-A. K., Labari. (2024, October 29). Starlink's Lack of Ground Stations Are Limiting Access To Users in Ghana and Nigeria. *Tech Labari*. <https://techlabari.com/starlinks-lack-of-ground-stations-are-limiting-access-to-users-in-ghana-and-nigeria/>
- Akali Gur, B., & Kulesza, J. (2024). Equitable access to satellite broadband services: Challenges and opportunities for developing countries. *Telecommunications Policy*, 48(5), 102731. <https://doi.org/10.1016/j.telpol.2024.102731>
- Akhlaghpasand, H., & Shah-Mansouri, V. (2023). *Traffic Offloading Probability for Integrated LEO Satellite-Terrestrial Networks* (arXiv:2307.16406). arXiv. <https://doi.org/10.48550/arXiv.2307.16406>
- Al Homssi, B., Al-Hourani, A., Wang, K., Conder, P., Kandeepan, S., Choi, J., Allen, B., & Moores, B. (2022). Next Generation Mega Satellite Networks for Access Equality: Opportunities, Challenges, and Performance. *IEEE Communications Magazine*, 60(4), 18–24. <https://doi.org/10.1109/MCOM.001.2100802>
- Alam, J., Hossain, Md. S., Ansari, J., Imran, A. Z. Md., & Kamrul, I. (2019). Estimation of Rain Attenuation of Earth-to-Satellite Link over Nepal for K_u & K_a Bands. *2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 169–173. <https://doi.org/10.1109/ICREST.2019.8644484>
- Ana-Maria, N., Alexandru, M., & Cristian, P. E. (2021). Study of millimeter waves in 5G. *2021 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*, 1–4. <https://doi.org/10.1109/BlackSeaCom52164.2021.9527846>
- Andy. (2023, July 5). *How many satellites are orbiting the Earth in 2023?* | Pixalytics Ltd. <https://www.pixalytics.com/satellites-orbiting-earth-2023/>
- Anna. (2024, October 21). *Tracking Wildlife & Protecting Oceans with Argos at CLS Oceania*. Argos. <https://www.argos-system.org/tracking-wildlife-protecting-oceans-with-oliver-palin-at-cls-oceania/>
- Arevalo, E. (2024). *Paratus Group signs reseller agreement with SpaceX Starlink for Internet services across Africa*. TESMANIAN. <https://www.tesmanian.com/blogs/tesmanian-blog/africa>
- Aslam, S., Michaelides, M. P., & Herodotou, H. (2020). Internet of Ships: A Survey on Architectures, Emerging Applications, and Challenges. *IEEE Internet of Things Journal*, 7(10), 9714–9727. *IEEE Internet of Things Journal*. <https://doi.org/10.1109/JIOT.2020.2993411>
- Baptista, E. (2024, September 6). China's Geespace launches 10 low-orbit satellites, eyeing Starlink. *Reuters*. <https://www.reuters.com/technology/space/chinas-geespace-launches-10-low-orbit-satellites-eyeing-starlink-2024-09-06/>
- Baruah, Y., Roy, S., Sinha, S., Palmerio, E., Pal, S., Oliveira, D. M., & Nandy, D. (2024). The Loss of Starlink Satellites in February 2022: How Moderate Geomagnetic Storms Can Adversely Affect Assets in Low-Earth Orbit. *Space Weather*, 22(4), e2023SW003716. <https://doi.org/10.1029/2023SW003716>
- Beesemyer, J., Ross, A., & Rhodes, D. (2012, September 11). Case Studies of Historical Epoch Shifts: Impacts on Space Systems and their Responses. *AIAA SPACE 2012 Conference & Exposition*. AIAA SPACE 2012 Conference & Exposition, Pasadena, California. <https://doi.org/10.2514/6.2012-5221>
- Bhattacharjee, D., Aqeel, W., Bozkurt, I. N., Aguirre, A., Chandrasekaran, B., Godfrey, P. B., Laughlin, G., Maggs, B., & Singla, A. (2018). Gearing up for the 21st century space race. *Proceedings of the 17th ACM Workshop on Hot Topics in Networks*, 113–119. <https://doi.org/10.1145/3286062.3286079>
- Broadband commision. (2024). State of Broadband 2024: Leveraging AI for Universal Connectivity. *Broadband Commission*. <https://www.broadbandcommission.org/publication/state-of-broadband-2024/>
- Broadband commision. (2025). Broadband Advocacy Target 2. *Broadband Commission*. <https://www.broadbandcommission.org/advocacy-targets/2-affordability/>
- Butt, R. A., Mohammad, S. H., Idrus, S. M., & Rehman, S. (2015). *EVOLUTION OF ACCESS NETWORK*

FROM COPPER TO PON.

- Cable Communications Policy Act of 1984. (2024). In *Wikipedia*.
https://en.wikipedia.org/w/index.php?title=Cable_Communications_Policy_Act_of_1984&oldid=1232653304
- Cakaj, S. (2009). Rain Attenuation Impact on Performance of Satellite Ground Stations for Low Earth Orbiting (LEO) Satellites in Europe. *International Journal of Communications, Network and System Sciences*, 02(06), 480–485. <https://doi.org/10.4236/ijcns.2009.26052>
- California's Digital Divide*. (n.d.). Public Policy Institute of California. Retrieved February 3, 2025, from <https://www.ppic.org/publication/californias-digital-divide/>
- Cao, M., & Rafi, G. (2022). *New Analysis Shows Offline Households Are Willing to Pay \$10-a-Month on Average for Home Internet Service, Though Three in Four Say Any Cost is Too Much* | *National Telecommunications and Information Administration*. <https://www.ntia.gov/blog/2022/new-analysis-shows-offline-households-are-willing-pay-10-month-average-home-internet>
- Castro-Carrera, A. (2025). A systematic review of LEO satellite services for IoT. *Multidisciplinary Reviews*, 8(3), 2025083–2025083. <https://doi.org/10.31893/multirev.2025083>
- Checagem de fatos: Starlink não controla acesso à internet no Brasil. (2024, September 6). *Reuters*. <https://www.reuters.com/fact-check/portugues/GVXULGECVJIJPF7SIHKG73ERO4-2024-09-06/>
- Chen, P.-H., & Cho, R. L.-T. (2024). The Technological Trajectory of LEO Satellites: Perspectives From Main Path Analysis. *IEEE Transactions on Engineering Management*, 71, 3629–3640. *IEEE Transactions on Engineering Management*. <https://doi.org/10.1109/TEM.2023.3347921>
- Chen, Y., Ma, X., & Wu, C. (2024). The concept, technical architecture, applications and impacts of satellite internet: A systematic literature review. *Heliyon*, 10(13), e33793. <https://doi.org/10.1016/j.heliyon.2024.e33793>
- Chikwendiu, J. (2023, April 3). *Broadband Challenges and Opportunities in Affordable Rental Housing*. <https://pew.org/40TNHlh>
- Choucair, C. (2023, August 16). *How Much Does It Cost to Launch a Rocket? [By Type & Size]*. *Space Insider*. <https://spaceinsider.tech/2023/08/16/how-much-does-it-cost-to-launch-a-rocket/>
- Chronopoulos, A. (2024, May 22). *Space Venture 2023: Key Takeaways*. ESPI. <https://www.espi.or.at/news/space-venture-2023-key-takeaways/>
- Cisco. (2023). *Cisco Annual Internet Report—Cisco Annual Internet Report (2018–2023) White Paper*. Cisco. <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>
- Citizen. (2025, January 8). *Starlink now faces higher licence fees as Kenya targets satellite providers*. Citizen Digital. <https://www.citizen.digital/business/starlink-now-faces-higher-licence-fees-as-kenya-targets-satellite-providers-n355509>
- Clausen, H. D., Linder, H., & Collini-Nocker, B. (1999). Internet over direct broadcast satellites. *IEEE Communications Magazine*, 37(6), 146–151. <https://doi.org/10.1109/35.769289>
- CNN. (2025). *South Sudan suspends social media platforms over videos of Sudan killings* | CNN. <https://edition.cnn.com/2025/01/23/africa/south-sudan-suspends-social-media-platforms-intl/index.html>
- Conde, J., Martínez, G., Reviriego, P., & Hernández, J. A. (2024). Round Trip Times (RTTs): Comparing Terrestrial and LEO Satellite Networks. *2024 27th Conference on Innovation in Clouds, Internet and Networks (ICIN)*, 42–46. <https://doi.org/10.1109/ICIN60470.2024.10494421>
- Congressional Research Service. (2021). *Low Earth Orbit Satellites: Potential to Address the Broadband Digital Divide*. <https://crsreports.congress.gov/>
- Countries by Population Density 2025*. (n.d.). Worldpopulationreview.Com. Retrieved May 25, 2025, from <https://worldpopulationreview.com/country-rankings/countries-by-density>
- Danaher, C. (2025, January 31). *Norway seizes Russian-crewed ship on suspicion of causing 'serious damage' to undersea cable between Latvia and Sweden*. CNN.

<https://www.cnn.com/2025/01/31/europe/norway-seizes-russian-crewed-ship-on-suspicion-of-causing-serious-damage-to-undersea-cable-between-latvia-and-sweden/index.html>

Darwish, T., Kurt, G. K., Yanikomeroğlu, H., Bellemare, M., & Lamontagne, G. (2022). LEO Satellites in 5G and Beyond Networks: A Review From a Standardization Perspective. *IEEE Access*, 10, 35040–35060. IEEE Access. <https://doi.org/10.1109/ACCESS.2022.3162243>

digisat.org. (n.d.). *Intelsat IS-20 Satellite Footprint Maps- Ka-Band, C-Band, Ku-Band*. Retrieved April 24, 2025, from <https://www.digisat.org/intelsat-is-20>

Digital Regulation Platform. (2024). <https://digitalregulation.org>

Dobrijevic, T. P. from D., & updated, A. M. last. (2022, April 14). *Starlink satellites: Facts, tracking and impact on astronomy*. Space.Com. <https://www.space.com/spacex-starlink-satellites.html>

DOCSIS. (2024). In *Wikipedia*. <https://en.wikipedia.org/w/index.php?title=DOCSIS&oldid=1227782167>

Dorros, I. (1981). ISDN. *IEEE Communications Magazine*, 19(2), 16–19. <https://doi.org/10.1109/MCOM.1981.1090491>

Draim, J. E. (1991). Continuous global N-tuple coverage with (2N + 2) satellites. *Journal of Guidance, Control, and Dynamics*, 14(1), 17–23. <https://doi.org/10.2514/3.20599>

Dvornikov, S. V. (2023). Electromagnetic Accessibility of Starlink User Terminals. *Proceedings of Telecommunication Universities*, 9(6), 34–41. <https://doi.org/10.31854/1813-324X-2023-9-6-34-41>

Earley, L. D. (2021). Communication in Challenging Environments: Application of LEO/MEO Satellite Constellation to Emerging Aviation Networks. *2021 Integrated Communications Navigation and Surveillance Conference (ICNS)*, 1–8. <https://doi.org/10.1109/ICNS52807.2021.9441549>

Ericson. (2023). *Ericson 2023 report.pdf*. <https://www.ericsson.com/en/press-releases/2024/3/ericsson-annual-report-2023-published>

Euronews. (2023). Euronews. <https://www.euronews.com/my-europe/2023/08/01/millions-in-the-eu-still-unable-to-afford-internet>

European Commission. (2024). *Commission takes next step to deploy the IRIS² secure satellite system*. https://defence-industry-space.ec.europa.eu/commission-takes-next-step-deploy-iris2-secure-satellite-system-2024-12-16_en

EutelSAT ONEWEB. (n.d.). *Cellular Backhaul solution*. OneWeb. Retrieved April 4, 2025, from <https://oneweb.net/solutions/carrier-enterprise/cellular-backhaul-solution>

Expereo. (2024). *The Future of Low Earth Orbit*. Expereo. <https://www.expereo.com/resource/the-future-of-low-earth-orbit-expanding-possibilities-and-capabilities/>

Fahim, K., & Sinha, S. (2023). FTTX NETWORK DESIGN ARCHITECTURES. *International Research Journal of Modernization in Engineering Technology and Science*.

Fang, T.-W., Kubaryk, A., Goldstein, D., Li, Z., Fuller-Rowell, T., Millward, G., Singer, H. J., Steenburgh, R., Westerman, S., & Babcock, E. (2022). Space Weather Environment During the SpaceX Starlink Satellite Loss in February 2022. *Space Weather*, 20(11), e2022SW003193. <https://doi.org/10.1029/2022SW003193>

FCC. (2021). FCC Annual Broadband Report Shows Digital Divide Is Rapidly Closing. <https://docs.fcc.gov/public/attachments/FCC-21-18A1.pdf>

FCC. (2024). <https://www.fcc.gov/affordable-connectivity-program>

Fixed-data. (2024). *Fixed data subscriptions by internet speeds*. <https://datawrapper.dwcdn.net/Lw2KX/1/>

Freudenrich, C., & Pollette, C. (2022). *How Fiber Optics Work | HowStuffWorks*. <https://computer.howstuffworks.com/fiber-optic.htm#pt4>

Furht, B., & Ahson, S. A. (Eds.). (2010). *HSDPA/HSUPA Handbook*. CRC Press. <https://doi.org/10.1201/b10268>

Garrity, J., & Aminata, G. (2020). *The Last-mile Internet Connectivity Solutions Guide Sustainable connectivity options for unconnected sites*.

Gill, J. (2012, August 2). *Role of Digital Subscriber Line Access Multiplexer In Enabling the Communication over the Inter-Networks*. <https://www.semanticscholar.org/paper/Role-of-Digital-Subscriber-Line-Access-Multiplexer-Gill/0a0d6321a48e2e86c75e96834fd52ffa6d38eb76>

Google Edge Network. (2025). <https://peering.google.com/#/infrastructure>

GPDR.eu. (2018, November 7). *What is GDPR, the EU's new data protection law?* GDPR.Eu. <https://gdpr.eu/what-is-gdpr/>

Gruber, H. (2007). *European Sector Regulation and Investment Incentives for Broadband Communication Networks* (SSRN Scholarly Paper 976887). Social Science Research Network. <https://doi.org/10.2139/ssrn.976887>

Hadden, A. (2009). Mobile broadband—Where the next generation leads us [Industry Perspectives. *IEEE Wireless Communications*, 16(6), 6–9. <https://doi.org/10.1109/MWC.2009.5361172>

Haidine, A., & Aqqal, A. (2018). Introductory Chapter: Next Generation of Broadband Networks as Core for the Future Internet Societies. In A. Haidine & A. Aqqal (Eds.), *Broadband Communications Networks—Recent Advances and Lessons from Practice*. InTech. <https://doi.org/10.5772/intechopen.80508>

Han, L., Retana, A., Westphal, C., & Li, R. (2022). Large Scale LEO Satellite Networks for the Future Internet: Challenges and Solutions to Addressing and Routing. *Computer Networks and Communications*. <https://doi.org/10.37256/cnc.1120232105>

Hasan, S., Gorinsky, S., Dovrolis, C., & Sitaraman, R. K. (2014). Trade-offs in optimizing the cache deployments of CDNs. *IEEE INFOCOM 2014 - IEEE Conference on Computer Communications*, 460–468. <https://doi.org/10.1109/INFOCOM.2014.6847969>

Hawkins, A. J. (2025, January 5). *United Airlines now says first flights with Starlink Wi-Fi will take off spring 2025*. The Verge. <https://www.theverge.com/2025/1/5/24332356/united-starlink-wifi-flights-spring-2025>

Hicks, A. (2024, November 14). *Province partners with Starlink to bring internet to remote and rural communities*. CTVNews. <https://www.ctvnews.ca/northern-ontario/article/province-partners-with-starlink-to-bring-internet-to-remote-and-rural-communities/>

Hughes. (2022, May 19). *Celebrating 25 Years of Satellite Internet | Hughes*. <https://www.hughes.com/resources/insights/hughesnet/celebrating-25-years-satellite-internet>

Hybrid fiber-coaxial. (2024). In *Wikipedia*. https://en.wikipedia.org/w/index.php?title=Hybrid_fiber-coaxial&oldid=1248741342

Internet Exchange Map. (n.d.). <https://www.internetexchangemap.com/>. Retrieved January 21, 2025, from <https://www.internetexchangemap.com/>

Internet Society. (2022). *Perspectives on LEO Satellites—Using Low Earth Orbit Satellites for Internet Access*.

Internet Society-1. (2024). 2024 East Africa Submarine Cable Outage Report. *Internet Society*. <https://www.internetsociety.org/resources/doc/2024/2024-east-africa-submarine-cable-outage-report/>

ITU. (2023a). *Facts and Figures 2023—Internet use*. <https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-internet-use>

ITU. (2023b). *Measuring digital development*.

ITU. (2023c). *Press Release*. <https://www.itu.int/en/mediacentre/Pages/PR-2023-11-27-facts-and-figures-measuring-digital-development.aspx>

ITU. (2025, March). *Press Release*. ITU. <https://www.itu.int:443/en/mediacentre/Pages/PR-2025-03-25-radio-navigation-satellite-service-harmful-interference.aspx>

ITU - DatHub. (n.d.). *National broadband plans: National broadband plan exists—ITU DataHub*. Retrieved November 25, 2024, from <https://datahub.itu.int/data/?i=100113&e=SOM&d=Website>

ITU - Space Connect. (2025, March). *ITU Space Connect*. <https://www.itu.int/space-connect/march-2025/>

ITU ICT_Regulation_Toolkit. (2014).

ITU Space Connect. (2025, February). *Space Connect Episode 2 (February 2025) – ITU Space Connect*. <https://www.itu.int/space-connect/february-2025/>

ITU-R: Managing the radio-frequency spectrum for the world. (n.d.). ITU. Retrieved January 26, 2025, from <https://www.itu.int:443/en/mediacentre/backgrounders/Pages/itu-r-managing-the-radio-frequency-spectrum-for-the-world.aspx>

ITU-T. (2025). *ITU-T Recommendations*. ITU. <https://www.itu.int:443/en/ITU-T/publications/Pages/recs.aspx>

Kataoka, R., Shiota, D., Fujiwara, H., Jin, H., Tao, C., Shinagawa, H., & Miyoshi, Y. (2022). Unexpected space weather causing the reentry of 38 Starlink satellites in February 2022. *Journal of Space Weather and Space Climate*, 12, 41. <https://doi.org/10.1051/swsc/2022034>

Kaur Bakshi, I. (2023). THE EVOLUTION OF THE INTERNET: THE ARPANET TO THE WORLD WIDE WEB. *International Journal of Social Science & Economic Research*, 08(09), 2729–2739. <https://doi.org/10.46609/IJSSER.2023.v08i09.017>

Keller, J. (2024). The US Navy Is Going All In on Starlink. *Wired*. <https://www.wired.com/story/us-navy-starlink-sea2/>

Kernan, J. (2002). *Asymmetrical digital subscriber line (ADSL) an in-depth study*. [https://www.semanticscholar.org/paper/Asymmetrical-digital-subscriber-line-\(ADSL\)-an-Kernan/ff8177dd561331587812f95ded583abe60a35823](https://www.semanticscholar.org/paper/Asymmetrical-digital-subscriber-line-(ADSL)-an-Kernan/ff8177dd561331587812f95ded583abe60a35823)

Khodairy, S., Sharaf, M., Awad, M., Abdel Hamed, R., & Hussein, M. (2020). Impact of solar activity on Low Earth Orbiting satellites. *Journal of Physics: Conference Series*, 1523(1), 012010. <https://doi.org/10.1088/1742-6596/1523/1/012010>

Kim, A. (2020, September 30). *How an internet system from space is helping firefighters in Washington* | *CNN Business*. CNN. <https://www.cnn.com/2020/09/30/business/spacex-starlink-washington-scn-trnd/index.html>

Kohn, D. M. (1997). Providing global broadband Internet access using low-earth-orbit satellites. *Computer Networks and ISDN Systems*, 29(15), 1763–1768. [https://doi.org/10.1016/S0169-7552\(97\)00108-6](https://doi.org/10.1016/S0169-7552(97)00108-6)

Kos, T., Grgic, M., & Mandic, L. (2003). CATV broadband technologies. *Proceedings EC-VIP-MC 2003. 4th EURASIP Conference Focused on Video/Image Processing and Multimedia Communications (IEEE Cat. No.03EX667)*, 2, 829–834 vol.2. <https://doi.org/10.1109/VIPMC.2003.1220566>

Kramer, G., & Pesavento, G. (2002). Ethernet passive optical network (EPON): Building a next-generation optical access network. *IEEE Communications Magazine*, 40(2), 66–73. <https://doi.org/10.1109/35.983910>

lboulter@gsma.com. (2024, October 23). New GSMA report shows mobile internet connectivity continues to grow globally but barriers for 3.45 billion unconnected people remain. *Newsroom*. <https://www.gsma.com/newsroom/press-release/new-gsma-report-shows-mobile-internet-connectivity-continues-to-grow-globally-but-barriers-for-3-45-billion-unconnected-people-remain/>

Li, C., Zhang, Y., Cui, Z., Zhang, Y., Liu, J., Yu, Z., & Zhang, P. (2023). An Overview Of Low Earth Orbit Satellite Routing Algorithms. *2023 International Wireless Communications and Mobile Computing (IWCMC)*, 866–870. <https://doi.org/10.1109/IWCMC58020.2023.10182766>

Loutfi, S. I., Shayea, I., Tureli, U., El-Saleh, A. A., & Tashan, W. (2024). An overview of mobility awareness with mobile edge computing over 6G network: Challenges and future research directions. *Results in Engineering*, 23, 102601. <https://doi.org/10.1016/j.rineng.2024.102601>

lovefm. (2025). *PUC Denies Starlink's License Application but Allows Limited User Imports*. Love FM Belize News & Music Power. <https://lovefm.com/puc-denies-starlinks-license-application-but-allows-limited-user-imports/>

Ma, S., Chou, Y. C., Zhao, H., Chen, L., Ma, X., & Liu, J. (2023). Network Characteristics of LEO Satellite Constellations: A Starlink-Based Measurement from End Users. *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*, 1–10. <https://doi.org/10.1109/INFOCOM53939.2023.10228912>

Maxwell, K. (1996). Asymmetric digital subscriber line: Interim technology for the next forty years. *IEEE*

Communications Magazine, 34(10), 100–106. <https://doi.org/10.1109/35.544330>

Mayring, P. (2014). *Qualitative content analysis—Theoretical foundation, basic procedures and software solution*.

McDowell's, J. (2025). *Jonathan's Space Report | Space Statistics*.

<https://planet4589.org/space/con/star/stats.html>

Modem. (2024). In *Wikipedia*. <https://en.wikipedia.org/w/index.php?title=Modem&oldid=1249383413>

Monzon Baeza, V., Ortiz, F., Lagunas, E., Abdu, T. S., & Chatzinotas, S. (2024). Gateway Station Geographical Planning for Emerging Non-Geostationary Satellites Constellations. *IEEE Network*, 38(4), 158–165. <https://doi.org/10.1109/MNET.2023.3321531>

Mungadze, S. (2025, April 9). *VIDEO: MTN Group makes move on LEO satellite as race opens*. ITWeb. <https://www.itweb.co.za/article/video-mtn-group-makes-move-on-leo-satellite-as-race-opens/xA9POvNEQmpqo4J8>

Nahm, W. J., Boyd, C. J., & Montgomery, R. A. (2023). Satellite internet technology implementation for the practice of medicine and surgery. *The American Journal of Surgery*, 225(5), 941–942. <https://doi.org/10.1016/j.amjsurg.2023.01.018>

NCTA. (2024). Protecting Networks: Battling the Rise in Copper Theft and Vandalism. *NCTA*. <https://www.ncta.com/news/protecting-networks-battling-the-rise-in-copper-theft-and-vandalism/>

Ngetich, J. (n.d.). *IEBC risks failing to account for 1m votes as connectivity issue lingers*. The Standard. Retrieved December 13, 2024, from <http://www.standardmedia.co.ke/national/article/2001448045/iebc-risks-failing-to-account-for-1m-votes-as-connectivity-issue-lingers>

Nzioka, S. (n.d.). *Agency says object that fell from the skies is a separation ring from a rocket*. The Standard. Retrieved January 1, 2025, from <http://www.standardmedia.co.ke/eastern/article/2001508893/agency-says-object-that-fell-from-the-skies-is-a-separation-ring-from-a-rocket>

Ookla®. (2025). Ookla - Providing Network Intelligence to Enable Modern Connectivity. <https://www.ookla.com/articles/starlink-europe-q1-2024>

openfalklands. (2024, May 12). An update to the use of Starlink in the Falkland Islands; Part One, the Ascension Island story. *OpenFalklands*. <https://openfalklands.com/an-update-to-the-use-of-starlink-in-the-falkland-islands-part-one-the-ascension-island-story-part-1/>

Osoro, O. B. (n.d.). *An Evaluation of Link Performance Based on rain attenuation LEO - Africa.pdf*.

Osoro, O. B., & Oughton, E. J. (2021). A Techno-Economic Framework for Satellite Networks Applied to Low Earth Orbit Constellations: Assessing Starlink, OneWeb and Kuiper. *IEEE Access*, 9, 141611–141625. <https://doi.org/10.1109/ACCESS.2021.3119634>

Pereira, P., & Ribeiro, T. (2011). The impact on broadband access to the Internet of the dual ownership of telephone and cable networks. *International Journal of Industrial Organization*, 29(2), 283–293. <https://doi.org/10.1016/j.ijindorg.2010.07.004>

Qu, Z., Zhang, G., Cao, H., & Xie, J. (2017). LEO Satellite Constellation for Internet of Things. *IEEE Access*, 5, 18391–18401. <https://doi.org/10.1109/ACCESS.2017.2735988>

Rath, K. (2021). Broadband networks. *IEEE Multimedia*, 7(4), 10–13. <https://doi.org/10.1109/93.895149>

Reudink, D. O., Acampora, A. S., & Yu-Shuan Yeh. (1981). The transmission capacity of multibeam communication satellites. *Proceedings of the IEEE*, 69(2), 209–225. <https://doi.org/10.1109/PROC.1981.11953>

Reuters. (2024, November 28). Musk's Starlink ordered to cease operations in Namibia. *Reuters*. <https://www.reuters.com/technology/musks-starlink-ordered-cease-operations-namibia-2024-11-28/>

Reuters. (2025, March 14). Elon Musk's SpaceX asks US to address foreign trade barriers. *Reuters*. <https://www.reuters.com/technology/elon-musks-spacex-asks-us-address-foreign-trade-barriers-2025-03-14/>

Safaricom. (2025). <https://www.safaricom.co.ke/media-center-landing/frequently-asked->

questions/safaricom-home-fibres

Satellite Tracker. (2025). *Starlink satellite tracker*. Starlink Satellite Tracker. <https://satellitemap.space>

Savino, S. P. (2000a). Digital subscriber line: Leading technology revolutionizing access to the information highway. *Proceedings of the 2000 IEEE Engineering Management Society. EMS - 2000 (Cat. No.00CH37139)*, 453–457. <https://doi.org/10.1109/EMS.2000.872545>

Savino, S. P. (2000b). Digital subscriber line: Leading technology revolutionizing access to the information highway. *Proceedings of the 2000 IEEE Engineering Management Society. EMS - 2000 (Cat. No.00CH37139)*, 453–457. <https://doi.org/10.1109/EMS.2000.872545>

Shaban, A. R. A. (2017). *Ethiopia telecoms monopoly now Africa's largest mobile operator* | *Africanews*. <https://www.africanews.com/2017/11/16/ethiopia-telecoms-monopoly-now-africa-s-largest-mobile-operator/>

Shalal, A., & Roulette, J. (2025, February 22). Exclusive: US could cut Ukraine's access to Starlink internet services over minerals, say sources. *Reuters*. <https://www.reuters.com/business/us-could-cut-ukraines-access-starlink-internet-services-over-minerals-say-2025-02-22/>

Sharma, P. (n.d.). *International Journal of Computer Science and Mobile Computing Evolution of Mobile Wireless Communication Networks-1g to 5g as Well as Future Prospective of next Generation Communication Network*. Retrieved April 15, 2025, from <https://www.semanticscholar.org/paper/International-Journal-of-Computer-Science-and-of-to-Sharma/8e324078c7b0848c5e8c573861878cdbe417e89e>

Singh, E. I., Singh, E. J., & Singh, E. S. (2018). A Review on Rain Fade and Signal Attenuation by Rain in Ku and Ka Band Satellite Communication at MGC Campus Punjab India. *International Journal of Trend in Scientific Research and Development, Volume-2(Issue-6)*. <https://www.ijtsrd.com/engineering/computer-engineering/18727/a-review-on-rain-fade-and-signal-attenuation-by-rain-in-ku-and-ka-band-satellite-communication-at-mgc-campus-punjab-india/er-sukhjinder-singh>

Smith-Rose, R. L. (1950). The Speed of Radio Waves and Its Importance in Some Applications. *Proceedings of the IRE*, 38(1), 16–20. *Proceedings of the IRE*. <https://doi.org/10.1109/JRPROC.1950.232449>

Space Debris—NASA. (n.d.). Retrieved January 1, 2025, from <https://www.nasa.gov/headquarters/library/find/bibliographies/space-debris/>

Space Economics. (2023, August 24). The Role of Reusable Rockets in Reducing the Cost of Access to Space. *Space Economy Institute*. <https://spaceeconomyinstitute.com/2023/08/24/the-role-of-reusable-rockets-in-reducing-the-cost-of-access-to-space/>

Space launch market competition. (2025). In *Wikipedia*. https://en.wikipedia.org/w/index.php?title=Space_launch_market_competition&oldid=1268769928

SpaceX. (n.d.). *SpaceX*. SpaceX. Retrieved February 24, 2025, from <http://www.spacex.com>

SpaceX. (2022, February 9). *SpaceX loses 40 satellites to geomagnetic storm a day after launch*. <https://www.bbc.com/news/world-60317806>

Stallings, W. (2014). *Data and computer communications* (Tenth edition). Pearson.

Starlink. (2024a). *Starlink -F*. Starlink. <https://www.starlink.com>

Starlink. (2024b, June 25). *Understanding the Size of Starlink Satellites—Starlink Installation Pros*. <https://starlinkinstallationpros.com/understanding-the-size-of-starlink-satellites/>

Starlink | Technology. (n.d.). Starlink. Retrieved December 11, 2024, from <https://www.starlink.com/ie/technology>

Starlink Coverage Tracker. (2024). <https://starlink.sx/>

Starlink-P. (2025). Starlink. <https://www.starlink.com/ke/service-plans>

Statskenya. (n.d.). *Kenya: Population Density by County - Stats Kenya*. Kenya Data & Statistics. Retrieved April 11, 2025, from <https://statskenya.co.ke/at-stats-kenya/about/kenya-population-density-by-county-data-map-and-charts/27/>

- Su, Y., Liu, Y., Zhou, Y., Yuan, J., Cao, H., & Shi, J. (2019). Broadband LEO Satellite Communications: Architectures and Key Technologies. *IEEE Wireless Communications*, 26(2), 55–61. <https://doi.org/10.1109/MWC.2019.1800299>
- Sujay Vailshery, L. (2022). *Global IoT and non-IoT connections 2010-2025*. Statista. <https://www.statista.com/statistics/1101442/iot-number-of-connected-devices-worldwide/>
- Sundaresan, K., White, G., & Glennon, S. (2020). *Latency Measurement: What is latency and how do we measure it?*
- Švedek, V., Jurin, G., & Weber, M. (2011). Increasing availability of broadband access over copper network infrastructure. *2011 Proceedings of the 34th International Convention MIPRO*, 407–412. <https://ieeexplore.ieee.org/document/5967090>
- Taara. (n.d.). Taara. X, the Moonshot Factory. Retrieved February 21, 2025, from <https://x.company/projects/taara/>
- Taara in Africa. (n.d.). X. Retrieved February 21, 2025, from <https://x.company/case-study/taara-in-africa/>
- Tanveer, H. B., Puchol, M., Singh, R., Bianchi, A., & Nithyanand, R. (2023). Making Sense of Constellations: Methodologies for Understanding Starlink’s Scheduling Algorithms. *Companion of the 19th International Conference on Emerging Networking EXperiments and Technologies*, 37–43. <https://doi.org/10.1145/3624354.3630586>
- taoti_admin. (2017, April 12). Dig Once: A Solution for Rural Broadband. *USTelecom*. <https://www.ustelecom.org/dig-once-a-solution-for-rural-broadband/>
- TE Connectivity. (n.d.). *COTS Components in LEO Satellites*. TE Connectivity. Retrieved February 24, 2025, from <https://www.te.com/en/industries/aerospace/insights/cots-components-in-leo-satellites.html>
- The Sun. (2025, February 17). *Community Fibre went down leaving thousands of customers with no internet access*. The Sun. <https://www.thesun.co.uk/money/33417597/community-fibre-down/>
- Tiwari, S., Bhushan, S., Taneja, A., Kassem, M., Luo, C., Zhou, C., He, Z., Raman, A., Sastry, N., Qiu, L., & Bhattacharjee, D. (2023). *T3P: Demystifying Low-Earth Orbit Satellite Broadband* (arXiv:2310.11835). arXiv. <https://doi.org/10.48550/arXiv.2310.11835>
- Tomas, J. (2016, August 4). *Global internet of things connections to reach 27 billion in 2025*. <https://www.rcrwireless.com/20160804/internet-of-things/internet-of-things-connections-tag23>
- Tomás, J. P. (2024, February 6). *China Mobile claims it launched world’s first 6G test satellite*. <https://www.rcrwireless.com/20240206/featured/china-mobile-claims-launched-world-first-6g-test-satellite>
- Torres, A. B., José L. (2024). *Low-Earth Orbit Faces a Spiraling Debris Threat*. Scientific American. <https://www.scientificamerican.com/article/low-earth-orbit-faces-a-spiraling-debris-threat/>
- Wachtel, B. (n.d.). *T-1, T1, DS-0, DS-1, T-span, DSX, Channel Bank*. Retrieved January 9, 2025, from <https://www.dcbnet.com/notes/9611t1.html>
- Water Science School. (2019). *How Much Water is There on Earth? | U.S. Geological Survey*. <https://www.usgs.gov/special-topics/water-science-school/science/how-much-water-there-earth>
- Westphal, C., Han, L., & Li, R. (2023). *LEO Satellite Networking Relaunched: Survey and Current Research Challenges* (arXiv:2310.07646). arXiv. <https://doi.org/10.48550/arXiv.2310.07646>
- Wood, L. (2003). Satellite Constellation Networks. In Y. Zhang (Ed.), *Internetworking and Computing Over Satellite Networks* (pp. 13–34). Springer US. https://doi.org/10.1007/978-1-4615-0431-3_2
- World Broadband Association. (2024). *Digital Progress and Trends Report 2023*.
- World Economic Forum. (2024). *World Economic Forum’s EDISON Alliance Impacts Over 1 Billion Lives, Accelerating Global Digital Inclusion*. World Economic Forum. <https://www.weforum.org/press/2024/09/world-economic-forum-s-edison-alliance-impacts-over-1-billion-lives-accelerating-global-digital-inclusion/>
- WRC-2027. (2025). *WRC-27 | Federal Communications Commission*. <https://www.fcc.gov/wrc-27>
- Xiao, Z., Yang, J., Mao, T., Xu, C., Zhang, R., Han, Z., & Xia, X.-G. (2024). LEO Satellite Access Network

(LEO-SAN) Toward 6G: Challenges and Approaches. *IEEE Wireless Communications*, 31(2), 89–96.
<https://doi.org/10.1109/MWC.011.2200310>

Appendix A: Interview Questions

1) Competition & Market Dynamics

- What is the effect of LEO broadband to the existing broadband market? Are they creating expanding broadband?
- What challenges do LEO broadband providers face in the competition with traditional ISPs?

2) Performance & Technical Challenges

- From your professional perspective, how does LEO broadband compare to fiber, DSL, and mobile broadband regarding speed, latency, and reliability?
- Based on your experience, what are LEO broadband's most significant technical limitations, and how do they impact its performance and scalability?

3) Financial Sustainability & Economic Factors

- What factors influence LEO broadband services' affordability and long-term financial viability?
- How do LEO broadband costs compare to traditional broadband options in different regions?

4) User Adoption & Accessibility

- What factors drive or discourage customers from choosing LEO broadband over traditional internet services?
- Are there specific customer segments that benefit most from LEO broadband?

5) Regulation & Policy Challenges

- What are the most significant regulatory challenges that LEO broadband providers face in gaining market entry and maintaining operations?
- How do different countries or regions regulate LEO broadband differently?

6) Future Trends, & Industry Predictions

- Where do you see the LEO broadband industry heading in the next decade?
- What technological or market advancements could improve or challenge the growth of LEO broadband?

Appendix B: Interview Transcripts

Participant 1 MM

Name Of Participant: Michuki Mwangi

Profession: Regional Development Manager for Africa at the Internet Society (ISOC)

Date of interview: 10 March 2025

[Wycliffe Bahati] 10:02:06

That's great. So yeah, I sent you the just highlights Yeah, the question so maybe we can just go straight in Starting with the marketing dynamics and competitive landscape.

[Michuki Mwangi] 10:02:12

Questions. Yes.

[Wycliffe Bahati] 10:02:23

So how do you think the LEO broadband services are affecting the traditional pieces as we know it.

[Wycliffe Bahati] 10:02:33

And what do you think are the marketing opportunities and challenges?

[Michuki Mwangi] 10:02:39

Yes. So, well, the way I see Leo's is that it's an evolution of an existing technology We've had satellite services for many years and So Leo's is just an evolution of previous generational geostationary which were much higher and meals, which have also come into operation not too long ago

[Michuki Mwangi] 10:03:08

And so Leo's is sort of like just a an advancement of that technology.

[Michuki Mwangi] 10:03:14

And so the operators who are in the satellite space who are struggling to provide a competitive service given the evolution of technology. Because if you look at the technology used today, it's a lot more focused on, there's a lot more demand for real-time applications and high bandwidth applications.

[Michuki Mwangi] 10:03:41

And they're the geostationary and meals were not able to fully provide that functionality or support for those kind of application applications. And so Leo's sort of come in and bridge that gap that existed.

[Michuki Mwangi] 10:04:01

And the other thing is that it also allows access to locations where the availability of connectivity was not easy to get to.

[Michuki Mwangi] 10:04:14

I mean, by and large. Advancement, investments in the traditional connectivity where we are looking at cable and wireless, terrestrial wireless, if I could call it that.

[Michuki Mwangi] 10:04:29

There's been quite an expansion of coverage to many areas, but not all areas have made business case and business sense to expand the connectivity. So satellites were sort of left as a reserve to those areas including maritime environments and you know those kind of areas where

[Michuki Mwangi] 10:04:50

You can simply put a cable. There are many situations where you could think of people who are drilling offshore they depended on mostly on geostationary connectivity people on ships, people on airplanes. These are places where satellite was still serving

[Michuki Mwangi] 10:05:14

And had a role to play. Now, with the introduction of Leos, it now really brings that real-time application and use.

[Michuki Mwangi] 10:05:24

To this segments which were sort of being left behind.

[Michuki Mwangi] 10:05:31

And could not really use real-time applications. And also, when we look now at the larger percentage, the U the it's most recent data showed that there are about 2.4 billion people who are unconnected. Majority of them live in rural and underserved areas where

[Michuki Mwangi] 10:05:51

The business case hasn't really been made. And that's quite a huge population.

[Michuki Mwangi] 10:05:55

So how do you get them to get them get connected affordably And to also be able to experience a modern day internet, not a limited version of the internet. So Leo's provide sort of the technology itself makes this a possibility.

[Michuki Mwangi] 10:06:19

And so that is where we see the business case and the opportunities imagine for follow Leos, so to speak.

[Wycliffe Bahati] 10:06:31

Okay, that's great. One of the things is what Given that a lot of strays have been made to the traditional satellite, with traditional broadband services, things like 5G, 4G, being able to deliver high capacities This is... traditional providers.

[Wycliffe Bahati] 10:07:00

Would likely...

[Wycliffe Bahati] 10:07:04

Respond. To this kind of... new service as a challenge.

[Wycliffe Bahati] 10:07:12

So how do you see them responding to this Because I think they probably would perceive it as a competition.

[Michuki Mwangi] 10:07:22

Yes. So indeed they have, especially let's look at Kenya as an example.

[Michuki Mwangi] 10:07:28

We've seen a huge take uptick of of Leo's services, specifically Starlink, which is operational in Kenya in urban centers where there is quite competitive environment when it comes to both fixed and wireless and and uh fixed wireless and uh

[Michuki Mwangi] 10:07:56

Fixed connectivity broadband solutions like, you know, fiber, 5G and so on. There's been a huge uptake. So the reaction we've seen from the operators is that they've either gone on to provide more bandwidth at a lower cost so we've seen

[Michuki Mwangi] 10:08:14

People now you know 100 megabits to their home is the cost that used to be well above almost \$100 is coming down.

[Michuki Mwangi] 10:08:23

For both businesses and home subscribers. And it's also, you know, you can see even the data bundles that being provided by various Let me see that two or three mobile operators that have a 5G or 4G.

[Michuki Mwangi] 10:08:42

Has also come down. We've also seen an effort to increase the network reliability Because that in itself was one of the concerns that some of the users that went for styling card is that the network quality of the network

[Michuki Mwangi] 10:09:01

The reliability of the network was not up to the did not meet the expectations We've also seen them expanding service availability.

[Michuki Mwangi] 10:09:12

In other places where they sort of had set them aside because they felt there was really no market. But now they've actually started reaching out to those places and providing fixed broadband, both fiber and wireless sort of just to compete in those spaces because they know if they don't go in now, then they will have lost the market. And this is now outside, going outside the

[Michuki Mwangi] 10:09:35

Metro areas and the urban locations that have started going into a number of rural areas that have been quite surprised to find them operating in places where I thought generally you may not find some large FTTH providers. The other thing they've also done is they've sought

[Michuki Mwangi] 10:09:56

They've also started to leverage the leo's connectivity to actually expand their connectivity elsewhere so like if it's a small base station that they need to deploy, they won't be subscribing more than 200 or 400 megabits They're using that to sort of leverage the connectivity as backhole

[Michuki Mwangi] 10:10:15

Because they don't have the backhaul. Infrastructure. So they're able to deploy base stations in remote places So that's a conversation. I don't know of any particular site but i know of places where that actually been tested, tried and tested, it seems to be working

[Wycliffe Bahati] 10:10:18

Okay.

[Michuki Mwangi] 10:10:34

We've also seen in markets where they take a different approach where they go to the regulator and seek regulatory protection where they ask the regulator to either require the Leo's operator to have a local license.

[Michuki Mwangi] 10:10:53

Or to meet certain requirements that then locks out the Leos from entering the market.

[Michuki Mwangi] 10:11:00

Until the operators sort of make their own strategies and position themselves to compete with such an end.

[Michuki Mwangi] 10:11:09

A market entrant. In other markets, we've seen them partnering.

[Michuki Mwangi] 10:11:16

With operators. So for instance. We've seen in the US, they've gone into a partnership with Is it... forgetting the name.

[Wycliffe Bahati] 10:11:29

Boom. Yeah, the mobile provider.

[Michuki Mwangi] 10:11:32

Yes, the mobile operator. Yes. So that to deliver direct signal connection to the handset. And so those are all strategies because it's either you collaborate or you lose out. So it's understanding that I think it will all go down to understanding that this is a new technology. Just like if the first time

[Wycliffe Bahati] 10:11:39

Hmm.

[Michuki Mwangi] 10:11:55

We were spending a lot of connection connect a lot of connectivity was through copper and coaxial cables.

[Michuki Mwangi] 10:12:04

The first person to bring fiber, you don't fight them, you figure out how do you collaborate with fiber to to enhance your business. So the ones who actually realize that model are the ones who will thrive, the ones who will take

[Wycliffe Bahati] 10:12:11

Yes.

[Michuki Mwangi] 10:12:17

A different stance are the ones who are going to lose out.

[Wycliffe Bahati] 10:12:21

And, um. I don't know if you've seen the new development where they're trying to deploy direct to sell technology.

[Wycliffe Bahati] 10:12:33

I think there are two aspects to it. The first one is where they're trying to use the existing mobile cell, MSS, There are frequencies that were used for mobile satellite.

[Wycliffe Bahati] 10:12:49

But then there's also advancement they think about using the normal frequencies that we use for normal forms.

[Michuki Mwangi] 10:12:49

Yeah.

[Michuki Mwangi] 10:13:02

Yeah.

[Wycliffe Bahati] 10:13:04

I mean, the people were very skeptical that given the coverage of satellite inside buildings.

[Wycliffe Bahati] 10:13:10

It might be tricky to get to the same level for access to as a tradition, the mobile broadband What's your tick?

[Michuki Mwangi] 10:13:25

Yes. So I've seen those arguments and those proposals, and that's why there's the ongoing test in the US with Why is it's not T-Mobile? What's the name of the other

[Wycliffe Bahati] 10:13:38

I'm both. I've seen it first.

[Michuki Mwangi] 10:13:40

Anyway, yes. My take is that these tests By and large, it is not to provide coverage in buildings it's for buildings access to mobility services because it's a mobile they're providing mobile services to people in areas where the mobile coverage doesn't exist. And if you're in a large commercial building, there is micro cells in those almost each and every floor.

[Michuki Mwangi] 10:14:13

So you're not the target audience. The target audience for that or the target market is for a majority of the people who don't reside in urban centers.

[Michuki Mwangi] 10:14:23

They're in rural areas. And they are looking for mobile coverage. And right now the mobile coverage tends to be quite spotty. And the business case for deploying a base station to cover those kind of groups tends to be tricky. The other thing is also during

[Michuki Mwangi] 10:14:44

You know. Times when there are natural disasters. Communication in those kind of areas also tends to be spotty. So satellite seems to be the best way to work. So there are a number of scenarios where these services would be needed.

[Michuki Mwangi] 10:15:01

And so making sure that they're there's an alternative way of communicating. I'll give an example. In North America during the fires that were happening. They were actually burning down even terrestrial fiber cables. So there were a number of places that were offline because of fires.

[Michuki Mwangi] 10:15:22

Fast respawn as needed to stay in contact. Now, if the terrestrial wireless is not working, fixed is not working.

[Michuki Mwangi] 10:15:30

Satellite becomes the next thing. And so being able to sort of use that as a backup to provide and it's very easy for them to have a service to have a a contract that, oh, in case my backhaul service terrestrial backhaul service fields to this place

[Michuki Mwangi] 10:15:48

Then my customers are going to access the same signal over the wireless, then that becomes a business model that they are actually able to provide.

[Michuki Mwangi] 10:15:58

And I think it's quite interesting to see because it's interesting because as a mobile operator, there are some places that are here to get connected that you're not sure whether there is a business case.

[Michuki Mwangi] 10:16:09

So would you roll out the infrastructure first or would you want to test it out through a system that already exists and has determined is there a market before I actually go out in making that investment?

[Wycliffe Bahati] 10:16:19

Yes.

[Michuki Mwangi] 10:16:23

And once it does work, then you even make a determination to what extent do you actually start rolling out the terrestrial deployment And so again, there's a huge percentage of people that are yet to be connected And I think that's primarily the target.

[Michuki Mwangi] 10:16:41

For these kind of services.

[Wycliffe Bahati] 10:16:44

That's good. At least you've mentioned something interesting Fuck that you could use this to do your business case with minimum expenditure and determine whether it's viable to deploy your that's interesting.

[Michuki Mwangi] 10:17:01

Yeah.

[Wycliffe Bahati] 10:17:02

And I don't know if you've interacted with some of these users. What is the perception between the two services?

[Wycliffe Bahati] 10:17:09

Especially where like we've talked about upon areas where there's both So have you had any what from the user's perspective What's the perception?

[Michuki Mwangi] 10:17:22

Yes. So the first one I met and I met a few is a cost of of of uh bandwidth, the bandwidth pricing. I mean, this one provides 100 megabits low latency and the costing compared to what was prior previously available was quite this was seen as a very competitive product

[Michuki Mwangi] 10:17:44

So a number of people moved to it because of that The others felt that it was more reliable than what they had.

[Michuki Mwangi] 10:17:54

They felt that, you know, some of the, especially like here in Kenya, depending on where you are you may be solved primarily by one FTTH provider or one fixed wireless provider. So they felt that this performed better.

[Wycliffe Bahati] 10:18:11

Okay.

[Michuki Mwangi] 10:18:11

Then what they were experiencing. I've seen a few who've also tried to use it outside. Again, this is purely an issue of availability.

[Michuki Mwangi] 10:18:24

Availability of service. They just didn't have that kind of service that compared to what Leo's was providing. I've seen this in in Zimbabwe where we have uh rural community network that's also using doing tests here in Kenya as well, some community networks are also evaluating that as a service.

[Michuki Mwangi] 10:18:47

So overall. It is availability costs and amount of bandwidth that's available. Those are the key drivers.

[Michuki Mwangi] 10:18:55

For adoption by adoption the number of customers or users that I've spoken to

[Wycliffe Bahati] 10:19:03

Okay, I think that's fine then. Hmm.

[Michuki Mwangi] 10:19:05

And one more thing. These that June 25th issue that happened last year.

[Michuki Mwangi] 10:19:16

And that there's the These are a small, I don't know what percentage of users that feel that this is free from government controls.

[Wycliffe Bahati] 10:19:17

Mm-hmm.

[Wycliffe Bahati] 10:19:26

Actually, I was going to talk about it. Yeah, yeah, yeah, yeah.

[Michuki Mwangi] 10:19:28

Yes, yes. Yes.

[Wycliffe Bahati] 10:19:33

Yeah actually we come to that when you go to regulatory. It's a very interesting topic.

[Michuki Mwangi] 10:19:39

Yeah.

[Wycliffe Bahati] 10:19:40

So yeah, the next I would like us to discuss about the technological challenges and opportunities like any other technology.

[Wycliffe Bahati] 10:19:51

Usually there is limitations, these challenges that it faces So for your broadband travel.

[Wycliffe Bahati] 10:20:00

What do you see the... would be the technically i mean advantage And what are the challenges?

[Michuki Mwangi] 10:20:13

Yes. So I think by design, they've made it very simple to deploy. That's the one thing that, you know, it's like.

[Michuki Mwangi] 10:20:21

Take it home, buy it, take it home, put it somewhere, self-adjust.

[Michuki Mwangi] 10:20:25

And off you go like the user intuitiveness user experience for details of deployment I think is one of the best we've ever seen.

[Michuki Mwangi] 10:20:38

And so that makes it easier for users to just take, they don't need to be technologically savvy. They just can take it and actually deployed. So that's great.

[Michuki Mwangi] 10:20:49

It also comes with very low latencies. I mean, you know, geostationary upwards of 600 plus

[Michuki Mwangi] 10:21:01

This is... about 100 and something 110 milliseconds And that's to hopes outside to Europe and other places, which is very submarine cable latencies between Africa and Europe.

[Michuki Mwangi] 10:21:19

So that almost at par. Now, you may have seen some reports that Now, these are pop one of the Leos has a pop in Kenya. And so the latencies have dropped to about 30 milliseconds.

[Michuki Mwangi] 10:21:39

So that's still fantastic. I mean, that's very much within the limits of having a... fixed wireless technology service, right? Terrestrial terrestrial fixed wireless service overview have a point-to-point wireless link or a 5g 4G, you're sort of in the same

[Michuki Mwangi] 10:22:05

You know latency level Yes, fiber will have slightly lower you know latencies now the question is, is this a latency sensitive um audience, depending on the applications.

[Wycliffe Bahati] 10:22:06

Playgrounds.

[Michuki Mwangi] 10:22:21

So if someone is a gamer, they will probably prefer the fiber over the satellite.

[Michuki Mwangi] 10:22:29

But then if they are playing with people that are in Europe and maybe if they're hoping across the traffic to Europe through the uh the intersatellite connectivity that they have to Europe or to the US, then it's also an advantage. So it all depends on how the network is going to be designed. Like, will they have smart routing at the

[Michuki Mwangi] 10:22:53

At that level to know, oh, this traffic is for Kenya and they'll send it to the ground station and this is to the US and use the intersat light relay to set it to Europe and the US, then that will really give it an advantage in terms of the network design.

[Wycliffe Bahati] 10:23:02

Yeah.

[Michuki Mwangi] 10:23:08

But they haven't gotten there yet. They're sort of trying to offload to the closest pop.

[Michuki Mwangi] 10:23:16

So if they integrate that kind of smart routing where traffic to West Africa goes through the satellite.

[Wycliffe Bahati] 10:23:16

Yeah.

[Michuki Mwangi] 10:23:24

It will be in some situations much faster to get to certain destination via satellite and via submarine cable.

[Wycliffe Bahati] 10:23:31

So I'd like to just dig deeper a bit on that one because I know, again, like I said.

[Wycliffe Bahati] 10:23:40

You might say... we are very well advantaged to I couldn't understand this technology and the impacts.

[Wycliffe Bahati] 10:23:48

There's something interesting that like us to discuss.

[Wycliffe Bahati] 10:23:55

The sorry the The... We know now that we know we know now We have the Googles.

[Wycliffe Bahati] 10:24:07

Amazon's Basically, there's almost a community routine of...

[Wycliffe Bahati] 10:24:18

The way we used to do BGP sometime back It's not the same way. There's almost some determinant in terms of your geolocation and where you access the service from.

[Michuki Mwangi] 10:24:22

Mm-hmm.

[Wycliffe Bahati] 10:24:31

That's one. I just wanted to find out. What's your perception on... Because I'm sure you know the ground stations are very expensive they will not be able to deploy them in every country.

[Michuki Mwangi] 10:24:47

Yeah.

[Wycliffe Bahati] 10:24:47

So how do you think that will play out with the LEO broadband services?

[Wycliffe Bahati] 10:24:53

Again, the consumer now We've had a bit of... challenges sometimes where consumers become latency

sensitive So how do you see it playing out in the long run, given that things like ai

[Wycliffe Bahati] 10:25:15

The content. Terms of movies are more becoming localized.

[Wycliffe Bahati] 10:25:19

How would you see it playing out in the long run?

[Wycliffe Bahati] 10:25:23

With this technology.

[Michuki Mwangi] 10:25:25

So yes, so one of the things that they've done, which I think is great is that building regional building regional regional ground station is is going to have a good benefit on the service in that It means, let me give an example of a region. So East Africa.

[Michuki Mwangi] 10:25:47

It means that if they're dropping everything into Nairobi. And then connecting to the internet exchange point.

[Michuki Mwangi] 10:25:56

And also having either bilateral interconnects with the various large CDNs that have already built POPs here.

[Michuki Mwangi] 10:26:05

The large city and so your microsoft, your Google, your Kamai and so on, it means that the end user experience remains very competitive to a terrestrial connect.

[Michuki Mwangi] 10:26:20

Connectivity. So that's very comparable. And that's actually the way to design the network where you want to offload the users to the closest points where they can actually have access.

[Michuki Mwangi] 10:26:34

To the content that they are trying to reach to. The other thing is also being able to have the intelligence within the network to determine, well.

[Michuki Mwangi] 10:26:46

This service is not locally available in this region. So the best thing for me is to route it to the place that doesn't, because if you drop them you've already take a 30 millisecond then you drop them on the ground

and then at the terrestrial path and the submarine path for them to get to Europe.

[Michuki Mwangi] 10:27:04

And then for it to get back to the same ground station and then back up.

[Michuki Mwangi] 10:27:10

There's a lot of externalities involved that could compromise the quality of service.

[Michuki Mwangi] 10:27:19

What we'll then need to do is probably in future the satellites or the LEO satellites will probably have some intelligent way of routing and making those kind of determinations.

[Michuki Mwangi] 10:27:31

And routing the most efficient way to deliver traffic where they know, oh yeah, there are lots of CDNs in Kenya. There are lots of CDNs in South Africa.

[Michuki Mwangi] 10:27:40

So anything in sub-Sahara, I'm either landing in Kenya or South Africa to give them the best performance in terms of access to content and reducing latency. Anything that's not served in those markets, then I'm going to drop it off directly to Europe.

[Michuki Mwangi] 10:27:56

Which again will probably be faster, maybe 60 milliseconds, but without the issues of going through networks that are terrestrial so the options will be there. They will have to do that deterministic sort of analysis to determine how best to route the traffic. But yes, leveraging internet exchange points. So they are connected to exchange points in

[Michuki Mwangi] 10:28:23

In Kenya. They are also deployed in exchange points in West Africa.

[Michuki Mwangi] 10:28:30

Or a region. I know in other parts of Europe, they've connected to the exchange points. Even in the US, they have connected to internet exchange point. In South Africa, I know they had reached out to connect the internet exchange points. They've just not been able to provide services locally because of the regulatory issue.

[Michuki Mwangi] 10:28:49

In South Africa. But for the neighboring countries. They already connected to the exchange point. So that

means they may not be providing service to users, subscribers in South Africa.

[Michuki Mwangi] 10:29:02

But they're able to offload traffic into the South African interconnection ecosystem. And so that's how they're able to provide some level of service.

[Wycliffe Bahati] 10:29:13

Great. And so what about the challenges?

[Michuki Mwangi] 10:29:20

Well, so the challenges here we start with First and foremost, it's weather related, right?

[Michuki Mwangi] 10:29:31

On cloudy days, the service will not perform optimally.

[Michuki Mwangi] 10:29:39

Second. By design and some of the Leos, at least the current Leo provider that's in the market.

[Michuki Mwangi] 10:29:47

Is more designed for a single home use or single use. So either a single enterprise or single deployment that's where the business model is.

[Michuki Mwangi] 10:30:00

And efforts to try and use it differently. Have faced a number of limitations. So for instance.

[Michuki Mwangi] 10:30:09

If you use the router that comes with the deployment the home deployment the moment you reach I forget whether it was like 100 devices it sort of falls over So it has a small memory. And so because it's doing that and translation and all that.

[Wycliffe Bahati] 10:30:29

Okay.

[Michuki Mwangi] 10:30:30

I forget the actual number, but we had someone test it and say, yeah, once you reach a certain threshold.

[Michuki Mwangi] 10:30:37

It will not be able to support more subscribers than that. So they also the access point is it's basically

designed for you put it at home and you're okay. If you decide to use it for a party Then you start to quickly uh

[Michuki Mwangi] 10:30:52

Running into limitation. So if you want to then deploy for 100 megabits is still a substantial amount of capacity.

[Michuki Mwangi] 10:31:02

And so if you want to serve that in the corporate enterprise, for instance.

[Michuki Mwangi] 10:31:10

Then you will need to add on additional devices that can help with that so you'll need to have a separate router to be able to so that you're not depending on the the CPE that comes with a dish. So you're using a more powerful cp

[Michuki Mwangi] 10:31:28

To do the nutting and then everything else goes through a different device. So that's the only way you need to scale. So we've faced a couple of those. If you need to do more than 100 megabits.

[Michuki Mwangi] 10:31:41

I mean, theoretically, they say 200 megabits but In reality, some of the tests that have been done show about 100, 120 mbps so if you need 200, Then you have to buy another one and then you do load balancing.

[Michuki Mwangi] 10:32:00

Across multiple of those devices to be able to get more bandwidth.

[Michuki Mwangi] 10:32:04

So by design, essentially. It has its limitations and scaling up, then the cost goes up.

[Wycliffe Bahati] 10:32:15

Yeah, and we've seen cases where they stopped selling the service um because of capacity.

[Wycliffe Bahati] 10:32:24

Sometimes like i think in jeky and sometimes

[Michuki Mwangi] 10:32:25

Yes. Yes, in Nairobi as well.

[Michuki Mwangi] 10:32:29

I'm told even... Since we... they stopped selling the service. I don't think they've resumed. I was speaking to someone last week And they say that it's still difficult to once once it's sold out it's had they haven't been able to upgrade the capacity so new subscribers are still struggling

[Michuki Mwangi] 10:32:51

Nairobi and some other parts. I think Kajiato was one of them and i forget the other parts in

[Wycliffe Bahati] 10:32:54

Okay.

[Wycliffe Bahati] 10:32:59

Oh, okay. And... You know, if you look at the kind of look at the the capacity available on fiber It's almost... Not meaningless, but... scraped in terabits, right?

[Wycliffe Bahati] 10:33:19

Do you see the neoc broadband reaching those levels just to be able to be apple to apple with fiber.

[Wycliffe Bahati] 10:33:29

Because again, I also just wanted to highlight the thing I saw some report where over the years in some countries

[Wycliffe Bahati] 10:33:42

The last mild equipment. Has been moving away, moving from moving megabits to gigabits and now people are deploying 10G devices Next to the user terminal means that In a few years, users will start having 10G connections in their homes.

[Wycliffe Bahati] 10:34:05

I mean, do you see this technology ever scaling up to

[Michuki Mwangi] 10:34:13

So for it to scale to that level, it's it's a number of things. One of the biggest challenge that we often face with wireless technology is availability of spectrum.

[Michuki Mwangi] 10:34:28

And that's number one like you need to have spectrum. The other thing that will also need to is evolution in the the way wireless equipment handles spectrum to push more.

[Wycliffe Bahati] 10:34:38

Okay.

[Michuki Mwangi] 10:34:46

Bandwidth. So those are the evolutions that are needed i mean it's wireless so it's only it's going to have to work within the constraints of the existing spectrum Yes. So it's if the engineering evolves in that you can push a lot more bandwidth within that spectrum, then yes.

[Wycliffe Bahati] 10:34:57

The next space.

[Michuki Mwangi] 10:35:09

Have we been successful with terrestrial In some instances, yes.

[Michuki Mwangi] 10:35:17

You will probably see like that tara laser technology that's able to push up to 10 gigabit over 10 kilometer to even 20 gigabits over 10 kilometer But it becomes harder for mobility, but for fixed, there are some ways. So I think evolution of the technology is inevitable

[Wycliffe Bahati] 10:35:21

Okay.

[Michuki Mwangi] 10:35:45

To what extent or how soon we'll be able to get there Then there's the economics of it. So there's two phases of that. There's the economics of it.

[Michuki Mwangi] 10:35:57

Fast right now what they are deploying means that the technology in use is able to essentially push on ground 100 megabits, 200 megabits tops So it's dependent on both the satellites in the sky and on the ground.

[Michuki Mwangi] 10:36:15

So to move to the next technology, it means the chips on the satellite will have to change.

[Wycliffe Bahati] 10:36:23

I didn't go back.

[Michuki Mwangi] 10:36:23

And also means that the dishes, the CPEs we have on the ground.

[Michuki Mwangi] 10:36:28

Will also have to change. To be able to support.

[Michuki Mwangi] 10:36:33

I would say it's more generational. So meaning that if we were to call this depending on if you're tracking like the actual what's going on.

[Michuki Mwangi] 10:36:46

They are now rolling out their second or third generation satellites.

[Michuki Mwangi] 10:36:55

Which are better, more efficient, et cetera, than the first generation.

[Michuki Mwangi] 10:36:59

Most of the ones they are rolling out now are anywhere between nine years and 15 years of operation.

[Michuki Mwangi] 10:37:08

That's the lifespan of most of the satellites. So as they burn out, as they fail, so nine ten nine to 15 years, they will get replaced by the next generation.

[Wycliffe Bahati] 10:37:13

There.

[Michuki Mwangi] 10:37:20

And so it is only when the generational phases get there and the technology has evolved to be able to push more bandwidth through the limited spectrum that's available.

[Michuki Mwangi] 10:37:34

Or we start releasing spectrum additional spectrum so that they can push more bandwidth is only when that change will happen so you know there's been conversations around opening up more of the three gigahertz space opening up more of the six gigahertz space uh

[Michuki Mwangi] 10:37:56

To be used also by Leo's like there's been a lot of conversation around which spectrum there's the three gigahertz is where I know there's been a lot of conversations around using that but then using that but the fight comes in, mobile terrestrial 5G operators were on the same spectrum.

[Michuki Mwangi] 10:38:18

Satellites, LEO's operators also want to use the same spectra. It's only a finite resource. So question is.

[Michuki Mwangi] 10:38:27

Then how much can they fit in into the spectrum that they've been allocated.

[Michuki Mwangi] 10:38:32

It's an ongoing conversation and evolution in that technology that will then see more bandwidth being pushed through.

[Wycliffe Bahati] 10:38:42

So now I think since you've gone to the economic aspects, I think we can just move to that point Now... I mean, there's been debate about The costs of deploying the LEO broadband services.

[Wycliffe Bahati] 10:39:02

Vis-a-vis... the terrestrial infrastructure.

[Wycliffe Bahati] 10:39:07

Apart from... the... the underserved areas.

[Wycliffe Bahati] 10:39:14

Which Leo are good at.

[Wycliffe Bahati] 10:39:20

Do you think that...

[Wycliffe Bahati] 10:39:24

This LEO broadband services will eventually make a business case Right now, we know that this It's not very clear. I mean, it's not the company does not actually release a lot of information.

[Wycliffe Bahati] 10:39:41

But there's speculation that... The costs of launching these constellations and running them may be very expensive And maybe the return of investments might not be as attractive as it looks.

[Wycliffe Bahati] 10:39:57

I don't know what your take is on that.

[Michuki Mwangi] 10:40:02

E. Like with all technologies at the very beginning they are very expensive to deploy. In fact, the business case struggles for various reasons.

[Wycliffe Bahati] 10:40:08

Okay.

[Michuki Mwangi] 10:40:18

My Thoughts are that the goal that has been set to with more i'll say with more LEO operators coming into the market

[Michuki Mwangi] 10:40:34

The potential scale of the constellations.

[Michuki Mwangi] 10:40:38

For them to have global coverage because that's that's one of the things that has been proposed.

[Michuki Mwangi] 10:40:47

There is either two things that are likely to happen. The... there will be a... a sustainability issue.

[Michuki Mwangi] 10:40:59

In terms of the cost of deployment versus sustainability is not going to be met.

[Michuki Mwangi] 10:41:07

And so they have to revisit that market. We've seen some countries where some government subsidies.

[Michuki Mwangi] 10:41:17

I remember the very beginning there were some countries in the countries in the Americas that had signed And they had signed subsidy programs and in bulk so they were sort of serving as anchor customers where they were procuring or paying for like a 10-year agreement to provide subsidized access

[Michuki Mwangi] 10:41:43

Using Leo's to various agencies and so on. Because for instance, governments um when they want to provide full coverage for their agencies, depending on the locations, etc.

[Michuki Mwangi] 10:41:55

They want something that they can rely on. This did offer that kind of solution. So depending on how what the terms and and and um the business model that they signed with these operators at the beginning.

[Wycliffe Bahati] 10:42:11

Mm-hmm. Just a moment.

[Michuki Mwangi] 10:42:15

Okay.

[Michuki Mwangi] 10:42:20

So depending on how they sign these agreements. They may have enough customer anchor customers to sort of sustain this at a global level.

[Michuki Mwangi] 10:42:32

But I foresee that. By and large the business deals that we are not visible to the end user subscription.

[Michuki Mwangi] 10:42:47

Is... is sort of a value add.

[Michuki Mwangi] 10:42:55

Because what we may not know is how has this actually disrupted the existing business on geosatellite.

[Michuki Mwangi] 10:43:10

And meals. Because the geosatellite business was big.

[Michuki Mwangi] 10:43:18

And it remained big. I forget the actual numbers but the annual in terms of revenues when i look at organizations like Intel Sat and the rest.

[Wycliffe Bahati] 10:43:21

Yes.

[Michuki Mwangi] 10:43:28

They were making a lot of money.

[Wycliffe Bahati] 10:43:30

Yes, they're launching things and they're pretty expensive, but they keep launching them.

[Michuki Mwangi] 10:43:33

And they are very exactly So this is... It is not an exact... it doesn't replicate it doesn't take away from the

geosatellites because they said an advantage is that geosatellites have that Leos don't have.

[Michuki Mwangi] 10:43:54

But the complementary nature of it means they've they are targeting the same subscriber base.

[Michuki Mwangi] 10:44:01

By and large. So militaries around the world will be picking on those are big contracts. Those are contracts that keep them going for some time.

[Michuki Mwangi] 10:44:12

The shipping, maritime. Industry is really, really big.

[Michuki Mwangi] 10:44:21

The... the airline industry.

[Michuki Mwangi] 10:44:23

Now, you know, where it's become You know, people want to be on planes and don't want to be disconnected. So you've seen the number of airlines that are now providing high-speed broadband connectivity for \$5 for an entire flight, which was previously not possible.

[Wycliffe Bahati] 10:44:39

Yeah.

[Wycliffe Bahati] 10:44:44

Thank you.

[Michuki Mwangi] 10:44:45

And \$5 for or \$10 for a full 10 hour flight and you have broadband connectivity like you are still on the ground.

[Michuki Mwangi] 10:44:54

That's transformational. Now, if you count the number of users, the number of passengers in a year.

[Michuki Mwangi] 10:45:03

And multiply by, you know, I think the last number I saw was like.

[Michuki Mwangi] 10:45:09

More than 200 or 300 million passengers fly in a year even if you take a third of them spending five dollars

in a year and that's shared So the industries that were traditionally dependent on satellite just got something better.

[Michuki Mwangi] 10:45:29

And so that's enough to keep the businesses going. The end user for me is the other additional value. Like if you think of any business where you have multiple income streams.

[Wycliffe Bahati] 10:45:32

Okay.

[Michuki Mwangi] 10:45:41

I think they've they've... package this in a way that it will be sustainable.

[Wycliffe Bahati] 10:45:41

Do you?

[Michuki Mwangi] 10:45:50

And if we have more end user subscriptions. This is more of like the cherry on top of the cream. So it sort of like makes the deal even better.

[Wycliffe Bahati] 10:46:00

Is that... And they have did this to talk about in terms of economics is Of course, 10 world countries this is if you look at the costs of constellations.

[Michuki Mwangi] 10:46:01

Yes.

[Wycliffe Bahati] 10:46:19

So it's a challenge. So you have the, I mean, countries either who are already in space or We have economic muscle to do this, we find these people basically launching these constellations.

[Wycliffe Bahati] 10:46:37

From these countries. So we have the traditional broadband provider.

[Wycliffe Bahati] 10:46:46

Locally built locally built mostly maybe employing local resources And maybe the money remains in the local banks I mean, with this type of new, I mean, it's not really new because like you said geo has done this for some time

[Wycliffe Bahati] 10:47:09

The target market with folio vis-a-vis the effect they have on traditional providers.

[Wycliffe Bahati] 10:47:16

It won't impact did it have on... Especially, let's say third world countries If you look at the terms of jobs.

[Wycliffe Bahati] 10:47:33

The ability of deploying additional infrastructure for local providers.

[Wycliffe Bahati] 10:47:41

I mean, in terms of money circulation Is there any impact that will connect broadband can change.

[Michuki Mwangi] 10:47:54

So yes, in that, again, this is, I would say a temporary situation.

[Michuki Mwangi] 10:48:01

Temporary in that if you go back to the late 90s and early 2000s when you know we didn't have a lot of operators. We didn't own any submarine cables. We didn't own the satellites.

[Michuki Mwangi] 10:48:21

The internet access here was, there's a local service provider who provides the service, but the bulk of the money went to the international operator.

[Wycliffe Bahati] 10:48:29

Okay.

[Michuki Mwangi] 10:48:30

And then as time went by. Local operators got more organized. They learned how to keep traffic local they That's brought some cost savings by building IXPs. Then they realize, oh, we can also build and own our own submarine cables so that the issue of Teams, CICOM,

[Michuki Mwangi] 10:48:51

And a few other submarine cables sort of went on to impact the amount of money that stays within the market.

[Michuki Mwangi] 10:49:00

What I am seeing now, my sort of initial take is that Leo's is coming to bridge a gap.

[Michuki Mwangi] 10:49:11

The reason it makes a lot of sense in underserved and unconnected areas is there's not been a technology that's easy to deploy and taking to those far-flung areas where digital adoption is low.

[Michuki Mwangi] 10:49:28

But the moment digital adoption goes up.

[Michuki Mwangi] 10:49:34

It makes a business case for a terrestrial operator to come in and extend the connectivity to that place and make a business case.

[Michuki Mwangi] 10:49:43

So in such a case, then there will be competition and where there's competition, it means to a large extent you will see a balance, a balance of either a certain percentage of people using the terrestrial operator for other reasons.

[Michuki Mwangi] 10:50:00

And a few or a certain percentage will be remaining using the satellites.

[Michuki Mwangi] 10:50:06

The technology evolution, the use case, the applications and all those things, the marketing and all those things will play a longer term role in how users maintain continue to adopt the service or which way the end users sort of lean towards in terms of

[Michuki Mwangi] 10:50:24

Do they end up taking the terrestrial service providers or do they use the LEOs?

[Michuki Mwangi] 10:50:30

So that's more long-term strategic right now quite frankly, if you look at the leos that have taken root in the country. They have done very little marketing in the country, if you look at Kenya or across the board, it's almost the same way we used to say, oh, look at

[Michuki Mwangi] 10:50:50

Some of the international operators, service providers like the cdns They did not invest a lot of marketing in the country, but by word of mouth.

[Michuki Mwangi] 10:50:59

People got to know the service exists. And then over time as the business case has become more viable.

[Michuki Mwangi] 10:51:07

The local operators will now stand finding ways of packaging the local solutions to target that audience that they want to to sort of take up the services.

[Michuki Mwangi] 10:51:21

I think in them short and medium term.

[Michuki Mwangi] 10:51:25

Yes, there will be outflows, capital outflows to the service providers.

[Michuki Mwangi] 10:51:35

We've had that for a long time with geos and meals.

[Michuki Mwangi] 10:51:40

Um with uh same thing you could say about the same applied to DSTV or when, you know, like for satellite television so that has been there.

[Michuki Mwangi] 10:51:55

But to places where majority of the subscribers are likely to be, which is remote and less armed areas.

[Michuki Mwangi] 10:52:02

I think it's very transitional. Meaning that it's short term to mid term but in the long term that will change as more of those places get more connectivity.

[Michuki Mwangi] 10:52:13

The other thing that is also worth noting is there's an emergence of small operators who are actually doing using the leo's so they do a mesh network they do fiber network but because there's no problem fiber back hole. They're using LEOs.

[Michuki Mwangi] 10:52:36

And that's why I feel that's transitional. So the moment there's a strong market and people want one gig capacity in that area.

[Michuki Mwangi] 10:52:45

Then someone will then they will switch to the fiber. Yes.

[Wycliffe Bahati] 10:52:50

Okay. All right so and um yes I'll just go back to the consumer adoption issue i know we've discussed it a bit So what do you see?

[Wycliffe Bahati] 10:53:11

That's the... The key factor that would make a decision a consumer maybe to a to say, okay, I have a fiber Leo, I prefer to use Leo.

[Wycliffe Bahati] 10:53:26

I mean, I know you've talked about the reliability experience But I mean, apart from that, is there any other factors that you see like the consumers would consider

[Michuki Mwangi] 10:53:44

Um... Well, so the key factors here is The pricing and bandwidth available that's going to be key.

[Michuki Mwangi] 10:53:55

We've also seen reliability, but that varies I would imagine that if you're Let's give an example of the Kenyan highlands around Limuru area and thereabouts where it's rainy during like the season we are going into versus someone who is in northeastern Kenya where, you know, it's sunny throughout the year

[Michuki Mwangi] 10:54:23

The experience is going to be very different. And so reliability for someone in northeastern will find yes That's a very reliable service.

[Michuki Mwangi] 10:54:32

In the reef valley area of Kenya will be like, yeah, it works in certain times of the year And, you know, so they will want to have a backup what would sort of determine this, I think, is the application use.

[Michuki Mwangi] 10:54:47

A user who's a heavy user really needs backup because they are in the financial sector, for instance.

[Michuki Mwangi] 10:54:58

Health sector And a few other, I'll say critical users who need who needs consistent connect like reliable access then this serves like a really good I'll say alternative or backup solutions so that you're getting quality of service, it's reliable. So if one fails

[Michuki Mwangi] 10:55:25

Part of the other challenge we face is that or fiber in Kenya is really unreliable. By global standards, our

fiber connectivity is really, really unreliable.

[Michuki Mwangi] 10:55:37

There was a complaint that has been pushed by by a number of fiber operators in that we have just far too many cats.

[Michuki Mwangi] 10:55:50

Like we are cut because of road construction. They are cut because of you know, people just doing their own things like someone is doing the fiber has passed through someone's farm they were clearing a hedge and accidentally cut some fields like sabotage like it was in the news recently, you know, like it's

[Michuki Mwangi] 10:56:15

There are just too many cats. In fact, the reliability on I forget who had published, but they were all struggling to maintain a 90% uptime.

[Michuki Mwangi] 10:56:28

On connectivity. Power is a second issue.

[Michuki Mwangi] 10:56:33

That our power is really unreliable. So they have to make a huge investment on the power.

[Michuki Mwangi] 10:56:38

So those two combined cuts and power bring down the reliability of fiber considerably.

[Michuki Mwangi] 10:56:50

A critical user, someone who really depends on broad connectivity for their business uh health education.

[Michuki Mwangi] 10:57:01

Finance and so on. Will suddenly start having a dependence on satellite or on leo Because it's a really good fallback.

[Michuki Mwangi] 10:57:15

In case you have outages. On fiber so so what you'll find in at least in most of the emerging markets Until we get to a point where we are able to stabilize.

[Wycliffe Bahati] 10:57:19

It would get weird.

[Michuki Mwangi] 10:57:27

The quality of our fiber infrastructure. In terms of reliability.

[Michuki Mwangi] 10:57:35

It becomes a really good alternative. And I've spoken to people in the north northeastern side of the country where yeah the long distance fibers tend to have a lot of cuts as well. So for users who are in remote areas and depend on that.

[Michuki Mwangi] 10:57:52

Before it's identified and fixed. The satellite black hole the Leos will really be a lifesaver for them.

[Wycliffe Bahati] 10:58:05

So the... The other... I think I've seen praising of praising the dish.

[Wycliffe Bahati] 10:58:17

And I don't know if it's... It's really helpful.

[Wycliffe Bahati] 10:58:23

Affordable to affordable to the common because there's a market that can afford that and you find mostly maybe their corporates Or some... somebody was not doing so badly.

[Wycliffe Bahati] 10:58:41

But for the mass market Because in Kenya i saw they're selling it at, is it 49 000 from the terminal.

[Michuki Mwangi] 10:58:54

Yeah.

[Wycliffe Bahati] 10:58:55

Not sure if uh because i I'm wondering, I thought initially they would... I think they had a proposal to try and make it spread the cost of the subscription.

[Wycliffe Bahati] 10:59:08

But I think they are coming back to just sell it.

[Wycliffe Bahati] 10:59:12

I was one one time. Do you see it affecting the end quote?

[Michuki Mwangi] 10:59:17

Yes. So... Yes. So when it comes to barriers to entry that prevent broader adoption.

[Michuki Mwangi] 10:59:26

In the emerging markets. The model that the model some of the Leos had.

[Michuki Mwangi] 10:59:35

Was for single use, single home entry. Meaning that the way it makes business case for them is that Every homestead should have its own leos.

[Michuki Mwangi] 10:59:47

Now, when we look at the affordability index. You will see that most people cannot afford \$500 for the terminal equipment \$100 a month for access. That's out of the equation.

[Michuki Mwangi] 11:00:04

Not for the 2.6 billion, 2.4 billion that is unconnected. That's out of the equation.

[Michuki Mwangi] 11:00:11

However, is it possible that You can use the same technology so to land it as a hub Meaning learn their 100 or 200 megabit.

[Michuki Mwangi] 11:00:24

So spend the 500 or \$1,000 for the terminal equipment.

[Michuki Mwangi] 11:00:30

And then \$100 or \$200 for monthly subscription and then distribute that to more users.

[Michuki Mwangi] 11:00:36

Who have a lower entry point of \$20. For terminal equipment.

[Michuki Mwangi] 11:00:42

And maybe three dollars a month, then that makes sense. And so for the work we've been doing with what we call community networks, where it's basically think of the way in rural areas you have a borehole, the community comes together, drinks a borehole.

[Michuki Mwangi] 11:00:58

And then pipes water to every homestead and the people contribute monthly to pay for the maintenance.

[Michuki Mwangi] 11:01:03

Taking a similar approach where the community comes together, sets up a community-based organization that's uh sets up the network. And so you can do wireless fixed wireless or hotspots and things like that to every home state. A man used now the leo's as a way of getting back hold to the internet.

[Michuki Mwangi] 11:01:24

One leo or you know two bundled or three bundle deals to sort of give you that 200 300 megabits access that you can redistribute to the entire community. So that business case makes sense. But they didn't have this business case.

[Michuki Mwangi] 11:01:41

Going into market. In fact, as we spoke to them about two, three years ago before they launched.

[Michuki Mwangi] 11:01:46

We asked, would you have a business case that works this way?

[Michuki Mwangi] 11:01:49

And one of them say, no, we don't have that because our business model is based on We can only make this affordable if every user buys a terminal.

[Michuki Mwangi] 11:02:00

And so looking at the monthly subscription of every user, that about \$100.

[Michuki Mwangi] 11:02:05

One of them said, actually, that made a really interesting case scenario for them too.

[Michuki Mwangi] 11:02:11

Consider. And so we are hoping that as we they deploy and come online they will have a sort of like a retailer or reseller.

[Michuki Mwangi] 11:02:23

Kind of model where you can have a hub it allows more connections.

[Michuki Mwangi] 11:02:30

And you can then redistribute it. And make a business case for getting more of the unconnected. So that's where we see that playing a really important role. The single terminal deployment by Homestead does not scale for emerging markets.

[Michuki Mwangi] 11:02:46

And where the bulk of the 2.4 billion people are

[Wycliffe Bahati] 11:02:52

Okay. And... So I think we can just move on to regulatory hurdles

[Wycliffe Bahati] 11:03:05

One thing about this service, inasmuch as it's licensed at the point of launch in the countries where they're launched.

[Wycliffe Bahati] 11:03:15

They literally have to deal with every jurisdiction in terms of spectrum dungeon.

[Wycliffe Bahati] 11:03:22

I mean... What have you seen to be one of the challenges that this LEO providers are facing in countries where they have to deploy these services.

[Michuki Mwangi] 11:03:35

Yes. So you're absolutely right. Because of Spectrum, I mean, it uses spectrum in every country.

[Michuki Mwangi] 11:03:43

So that means for that particular footprint of that country. They need to speak to each and every regulator in every market.

[Michuki Mwangi] 11:03:54

Spectrum carbonate costs. It has been seen or at least has been a source of revenue for many countries.

[Michuki Mwangi] 11:04:07

So it is rig spectra is Yes, it's a critical resource for each and every country. And yes, I met there is a limit.

[Michuki Mwangi] 11:04:21

Limited spectrum available. Now, there are some challenges that they will face in some countries.

[Michuki Mwangi] 11:04:30

First start with is the spectrum available?

[Michuki Mwangi] 11:04:36

Does the country have a good spectrum management system? At the regulator level.

[Michuki Mwangi] 11:04:44

Has it been previously allocated to someone else or not? So those are sort of some of the initial things they have to go through. If it's been allocated to someone else.

[Michuki Mwangi] 11:04:53

Then um it may not be available. If it's available then there's a question of how much spectrum they need and how much is it going to cost.

[Michuki Mwangi] 11:05:03

The cost of that spectrum then will determine how they are able to package.

[Michuki Mwangi] 11:05:11

Yes. The other thing is, what is the cost of entry and operating in that market?

[Michuki Mwangi] 11:05:17

Because as you'll find in some markets they will have a regulatory conversation around local ownership like we've seen in South Africa.

[Michuki Mwangi] 11:05:27

They need to have up to 30% local ownership. And 70% international. We had something similar here in Kenya, but it's sort of been reviews and it's not enforced, so to speak.

[Michuki Mwangi] 11:05:42

Then there's also the question of compliance once they are licensed.

[Michuki Mwangi] 11:05:51

To make sure even before I go to compliance, there is a question of selling directly to end users versus going through

[Michuki Mwangi] 11:06:04

A local operator to operate or a local reseller to resell the service. That then sort of impacts it will be a regulatory issue.

[Michuki Mwangi] 11:06:16

Which then impacts compliance. Then what happens in non-compliance? So there's a lot of conversations around those issues of compliance what does it mean?

[Michuki Mwangi] 11:06:29

What are the issues they need to look at from And then once they look at all those issues.

[Michuki Mwangi] 11:06:36

Is it a major cost to doing business in that country?

[Michuki Mwangi] 11:06:40

Or not. I recall some conversations. I think it was in 2022 Was it 2022, 2022, 2023 where one of the regions, Sadakh, Southern African Development.

[Michuki Mwangi] 11:07:01

Commission was looking at trying to see Kansasada come together and jointly instead of having them to do each country like do something at a sub-regional level.

[Michuki Mwangi] 11:07:15

And issue a license at that level. So there were conversations around that.

[Michuki Mwangi] 11:07:22

In practice, it would be difficult because each country has its own spectrum. And so there's a lot of

[Michuki Mwangi] 11:07:32

Collaboration that will be needed between the member states. Of a region to reach a point where they are doing regional wide licensing and spectrum allocation to a single service provider.

[Michuki Mwangi] 11:07:47

And so on and the revenue share that will be then given based on spectrum allocation and and so on. There are even issues like contribution to a universal service fund.

[Michuki Mwangi] 11:07:59

All operators contribute to universal service via license. So what is the terms?

[Michuki Mwangi] 11:08:04

So you probably saw the discussions uh with I think it was telling in Kenya around their contribution to what will be their contribution to the Universal Service Fund.

[Michuki Mwangi] 11:08:16

That being moved upwards so that they can contribute more. So there are a lot of regulatory issues that will keep popping up when... this is uh because again it's a new operator. It's a new um area that is doing a lot of interest. So I expect that this to happen.

[Michuki Mwangi] 11:08:40

The good thing is the spectrum issues were already addressed at the global level.

[Michuki Mwangi] 11:08:46

At the um While Radio Congress in 20 was it 2020 2019 or 2020 2020, yes.

[Wycliffe Bahati] 11:08:48

Thank you.

[Wycliffe Bahati] 11:08:56

2020, I think.

[Michuki Mwangi] 11:09:01

Yeah oh it was just 2021, I think, after COVID. Because 2020, nobody went anywhere.

[Wycliffe Bahati] 11:09:09

Oh.

[Michuki Mwangi] 11:09:11

Yeah, I think so. So once the regulatory, the spectrum, because that's where the biggest fight was on spectrum.

[Michuki Mwangi] 11:09:23

So now the spectrum issues are very localized.

[Wycliffe Bahati] 11:09:29

Then.

[Michuki Mwangi] 11:09:29

Yeah, they are very, very localized. So it all depends on what the countries looking for from a spectrum perspective.

[Wycliffe Bahati] 11:09:40

And so... Given... This... I mean, do you see an opportunity for countries for example if they can make... the regulatory terms attractive.

[Wycliffe Bahati] 11:09:58

That some of these major constellations would consider using their country's full.

[Wycliffe Bahati] 11:10:07

Launching this launching this these services i think uh i don't know if rwanda has done something like that for a constellation so for regulators in africa i'm not sure how vast or how fast they're moving to to adapt to

[Wycliffe Bahati] 11:10:29

To this new technology and try to take advantage of it instead of Gotcha. Mm-hmm.

[Michuki Mwangi] 11:10:38

Could you say frame that question again? I probably didn't.

[Wycliffe Bahati] 11:10:41

So, so... Basically, a country that follows the spectra with ITU, for example, under

[Wycliffe Bahati] 11:10:53

I'm a venture capitalist or somebody who wants to do the constellation There's an option of I can do it in.

[Wycliffe Bahati] 11:11:05

In the US and Europe. So do you see regulators looking at it from that perspective in terms of making more attractive laws that are would allow some of this constellations to use their countries.

[Wycliffe Bahati] 11:11:23

As in terms of ownership of these space stations.

[Michuki Mwangi] 11:11:30

Um... I don't know.

[Michuki Mwangi] 11:11:35

Whether that is a viable Because it still boils down to one thing.

[Michuki Mwangi] 11:11:42

That... there's a finite spectrum available.

[Michuki Mwangi] 11:11:47

It is being competed upon by both mobile and satellite. And the question is, and I'll say this candidly.

[Michuki Mwangi] 11:11:58

You can choose to sort of mask it because I don't want to quote anyone.

[Michuki Mwangi] 11:12:04

It's who pays the top dollar for the spectrum.

[Michuki Mwangi] 11:12:10

If you pay the top dollar for the spectrum, the regulator will let you do whatever.

[Michuki Mwangi] 11:12:19

And I'm saying that because I've spoken to a regulator who was candid enough to tell me exactly that.

[Wycliffe Bahati] 11:12:19

Yeah. Yep.

[Michuki Mwangi] 11:12:26

If you give me a billion dollars for the spectrum. I will give you to do anything you want with it.

[Wycliffe Bahati] 11:12:26

Mm-hmm.

[Michuki Mwangi] 11:12:33

But remember, that spectrum is within the geographic boundaries of the country. I cannot take that spectrum and use it in another market.

[Michuki Mwangi] 11:12:43

It is because the way ITU regions located is that the spectrum And it has a start and end based on the region and then down to the country.

[Wycliffe Bahati] 11:12:50

Okay.

[Michuki Mwangi] 11:12:54

So when it comes to the national level, if you pay the money I'll give you the spectrum.

[Michuki Mwangi] 11:13:01

And that's about what's about a wide fixed wireless mobile wireless connectivity remains expensive because there is a lot of money that rides in the spectra.

[Michuki Mwangi] 11:13:20

And in actual sense, if you ever look at any network.

[Michuki Mwangi] 11:13:27

In their books, one of the biggest assets they have in their books is spectrum.

[Michuki Mwangi] 11:13:35

So if they are to wind up the company. They will sell the spectrum.

[Wycliffe Bahati] 11:13:36

Maybe so.

[Wycliffe Bahati] 11:13:40

Mm-hmm.

[Michuki Mwangi] 11:13:42

The companies value the spectrum is the value it's like a building And... Funny enough, spectrum is like when I say like a building is like the moment you get it, it has a certain value. The moment you start using it, the value grows.

[Michuki Mwangi] 11:14:00

How widely used the spectrum is if you were to sell that to a new business, like under new acquisition the spectrum number is not what you initially paid for as the license fee.

[Michuki Mwangi] 11:14:14

When you are querying it. So, um.

[Michuki Mwangi] 11:14:18

So it is a very... I don't know how to frame it, but spectrum is one of those things that is gold to the business And these goals to the regulator. So it's a very sensitive uh part of the service and of the technology.

[Wycliffe Bahati] 11:14:31

Okay.

[Wycliffe Bahati] 11:14:42

Okay. And what do you think the regulators need to do?

[Wycliffe Bahati] 11:14:49

Change into the way that because previously they've been dealing with geo services what do they need to adjust, change?

[Wycliffe Bahati] 11:14:58

With a weekly or services.

[Michuki Mwangi] 11:15:04

So there are actually a number of issues. That we have identified relating to geos.

[Michuki Mwangi] 11:15:12

But those issues then sort of escalate to or rather they cascade upwards because they are not unique to a country.

[Michuki Mwangi] 11:15:22

And I can share with you this paper which we wrote in in 20 um Let me see we can put it in a chat room here.

[Wycliffe Bahati] 11:15:27

Be nice.

[Michuki Mwangi] 11:15:34

So yeah, we did that in at the end of 2022, which was a perspective

[Michuki Mwangi] 11:15:44

This is... Yes.

[Wycliffe Bahati] 11:15:44

Okay. Yeah, I've actually seen this. Very interesting.

[Michuki Mwangi] 11:15:50

Yeah. Yes, so there are a couple of issues here to do with um the impact I don't know whether you've seen recently there was a conversation i think that was in 2023, if I'm not mistaken 2022 or 2023 where they said if the full deployment of the constellations are complete

[Michuki Mwangi] 11:16:15

How does that impact the

[Michuki Mwangi] 11:16:20

The geos Because, you know, it almost covers the world. So it affects the um the signal quality for geos and NEOs who are above.

[Michuki Mwangi] 11:16:33

So that was one. And, you know, because it does cost like an interference like a block right So that was one. There was the issue of space debris And what happens when they start falling who compensates who And we had a recent incident, I think, in Machakos the other day

[Michuki Mwangi] 11:17:02

Then there was a question around interference So same typical issues we've had with interference on terrestrial Now, what happens when you have multiple competing geo providers.

[Michuki Mwangi] 11:17:18

I mean, LEO providers and spectrum issues Because now they'll be operating in similar spectrum, right? Their ranges will be not too far apart.

[Michuki Mwangi] 11:17:27

So how do you manage that? Now it's all fine with because there's only one.

[Michuki Mwangi] 11:17:33

But there are two others that are meant to come online.

[Michuki Mwangi] 11:17:36

So all these are issues that are bound to happen, come up once you have more competitors in that space.

[Michuki Mwangi] 11:17:44

And so the national level issues those will be alongside they'll be alongside competition once they have multiple operators. So interference and things like that.

[Michuki Mwangi] 11:18:01

But then also some of the other issues will be What happens if you start broadcasting your neighbor's country's spectrum into my Especially at border points. So the same problems we have with the spectrum at border points between mobile operators that you're getting

[Michuki Mwangi] 11:18:20

This you remember the issues like if you go too close to a border you start going into roaming and things like that.

[Wycliffe Bahati] 11:18:26

I mean.

[Michuki Mwangi] 11:18:26

Yes. So border points will become a really interesting conversation for for geos because which spectrum do you use? So all those are again not there will be to me that's why I keep saying it's just a new technology similar issues now just facing

[Wycliffe Bahati] 11:18:34

Yeah.

[Michuki Mwangi] 11:18:47

Satellite space industry, which we didn't really essentially have in the past

[Wycliffe Bahati] 11:18:53

And on... What about the new data privacy and same security laws.

[Wycliffe Bahati] 11:19:03

Do you think that there'll have to be some kind of change in the way we look at that are privacy with the legal procurement.

[Michuki Mwangi] 11:19:14

So... Yes. So this is certainly one of the topical issues because of cross-border data flows specifically.

[Michuki Mwangi] 11:19:30

In those scenarios, then... a country that doesn't have a ground station.

[Michuki Mwangi] 11:19:36

Then what happens? Where there's an issue around cross border data for those. And unfortunately, like, you know, two government agencies that are you trying to use the same service in the same country, there is no ground station means it has to leave the country go

[Michuki Mwangi] 11:19:54

Into another place before it comes back. So that will be a very tricky issues for countries that have data privacy laws that strict on... cross-border data flows.

[Wycliffe Bahati] 11:20:14

So like also what about the issue of tight controls like for example still have countries like Ethiopia that still

have very tight controls.

[Wycliffe Bahati] 11:20:27

On the internet. I mean, I think the regulators there have not even started to move towards opening the space.

[Wycliffe Bahati] 11:20:37

Do... think they'll find... a way to work with Leo Bergman.

[Michuki Mwangi] 11:20:46

I think in the long term. Because it's an issue of regulation and it's an issue of regulation when you speak to a lot of the policymakers and regulators, they all want everybody to have access to.

[Michuki Mwangi] 11:21:00

Connectivity. The reality that they also do see is that traditional ways of getting connectivity don't always succeed in bringing connectivity to everyone.

[Michuki Mwangi] 11:21:17

And so for them to succeed in that goal, because if you look at all the regulators, they all talk about they have a vision or an objective of creating an enabling environment so that people everywhere have access to telecommunications technologies.

[Michuki Mwangi] 11:21:36

And so... as they open up, Ethiopia being a good example or as they open up and other markets they will hit a point where they realize Unless they open up further to open up further going back to my earlier comment.

[Michuki Mwangi] 11:21:57

Leo's is just a technology. Regulations are meant to be technology agnostic.

[Michuki Mwangi] 11:22:03

Or technology neutral. So if they are living by that mantra of they're not there to regulate the technology. They are there to regulate how the technology is applied from or the adoption of actually they look at they're regulating.

[Michuki Mwangi] 11:22:30

The application of technology in a market in a way that without being, you know, like specific to the technology, but just saying this is how the technology gets applied.

[Michuki Mwangi] 11:22:41

Without being specific about the technology. Then what they will be looking at is looking at how do people access connectivity?

[Michuki Mwangi] 11:22:51

Regardless of what technology lies behind it. What are the issues of concern?

[Michuki Mwangi] 11:22:57

Data protection, cybersecurity, and all those adhering to making sure that everybody has access to equitable access, affordable access.

[Michuki Mwangi] 11:23:09

Inclusivity and so on and so forth. All those issues is what the regulation is there to protect.

[Michuki Mwangi] 11:23:14

Protect the end user from high to use? What technologies use is a different conversation.

[Michuki Mwangi] 11:23:20

And so if they do so, look at it from that perspective.

[Michuki Mwangi] 11:23:24

And it doesn't matter whether it's Viba, whether it's Leo's, Geo's, Muse.

[Wycliffe Bahati] 11:23:29

Yes.

[Michuki Mwangi] 11:23:29

It's the same. It's being used applied for.

[Wycliffe Bahati] 11:23:37

Yeah, all right great so Yes, as we close, I would just like you to give a kind of what you see at the will be the future outlook in this market.

[Wycliffe Bahati] 11:23:51

With this new entrance.

[Wycliffe Bahati] 11:23:56

What do you see changing in the next maybe three, five years.

[Michuki Mwangi] 11:24:03

Well. Technology evolution new competition in terms of market players are always good for increased availability of affordable and reliable access.

[Michuki Mwangi] 11:24:21

And so When I look at Kenya and I look at the emerging markets.

[Michuki Mwangi] 11:24:29

More the evolution of this technology. Its availability you know like it's it's rich really creates an environment where for operators who operators well you know for instance reluctant to go to certain markets or places because they felt they have to adapt.

[Michuki Mwangi] 11:24:52

And so we are going to see increased increased access because of more people getting online because of the technology evolution the competition that results because of new entrants coming into the market. So more people will get connected.

[Michuki Mwangi] 11:25:11

As a result. And then also a growth in the digital space. So meaning the digital ecosystem more people using digital services.

[Michuki Mwangi] 11:25:23

As a result of more people coming online. So more innovation more digitization of services really growth of the digital ecosystem as a whole.

[Michuki Mwangi] 11:25:36

In many places. I think it fast tracks in some way some of the goals that were sort of said to be seen by let's say 2035.

[Michuki Mwangi] 11:25:50

We are likely to start seeing them by 2030. Because of this.

[Michuki Mwangi] 11:25:56

Not all, but some, meaning that there are some milestones we are likely to hit earlier than projected before.

[Michuki Mwangi] 11:26:05

Courtesy of, again, new entrance evolution in the technology So those two things are really As always, more competition and evolution in technology really does move things faster.

[Michuki Mwangi] 11:26:18

And so we'll hit a few milestones. I can't predict which one specifically But some of the ones that I anticipated will start seeing is one of the goals was for affordability to be affordability at about 2%.

[Michuki Mwangi] 11:26:38

Of monthly income of GNL gross national income per capita.

[Michuki Mwangi] 11:26:45

There will be the cost of access for everyone I expect the target was 2030 I expect that we will see that for being realized earlier for a number of countries as a result.

[Michuki Mwangi] 11:27:01

Not all for various other reasons like licensing issues, delays and things like that.

[Michuki Mwangi] 11:27:09

We could also growth in the digital ecosystem in that it contributes more to GDP and adoption we'll see more of that coming on.

[Michuki Mwangi] 11:27:24

Earlier in some markets than anticipated because some countries I'd like, let's give an example. The AU had a agenda 20 2060, I think is the call is the name of the agenda So we will see some of the goals in that.

[Michuki Mwangi] 11:27:45

2060 agenda probably around 2035. We'll start seeing milestones being achieved.

[Michuki Mwangi] 11:27:52

Yes, there are milestones they have for 2035. But I think that like the 2040 milestones will probably start seeing them earlier So, you know, when you start talking about things like smart cities and things like that.

[Michuki Mwangi] 11:28:10

Those would then become much easier to deploy. Because these ubiquitous access if you have a solution, it's easy to deploy it country-wide.

[Michuki Mwangi] 11:28:21

Or regional wide because there's coverage everywhere, right? So those kind of things will become of

possible monitoring of weather you know all those challenges that we faced. There's a number of milestones that we are likely to so but you have to break them down and see

[Michuki Mwangi] 11:28:42

Which one specifically. I haven't had a chance to look at all of them in detail.

[Wycliffe Bahati] 11:28:46

Yeah, okay. Around digital inclusivity. And affordability, I think.

Participant 2 BR

Name Of Participant: Ben Roberts

Profession: Former Chief Technology Officer Liquid Technologies (Expert in broadband services)

Date of interview: 03 March 2025

[Wycliffe Bahati] 15:05:04

I'm doing this thesis and they said the sub The topic I'm investigating is the effect of law of orbit satellite internet.

[Wycliffe Bahati] 15:05:15

To traditional broadband service providers.

[Ben Roberts] 15:05:22

Yeah.

[Wycliffe Bahati] 15:05:23

Yeah, so I just have like almost like about four sections which you can go through I mean, briefly. Yeah. So the first one is I just wanted to know from your perspective um What do you think are the market opportunities and challenges that LEO providers will have to compete with

[Ben Roberts] 15:05:31

Okay.

[Wycliffe Bahati] 15:05:45

Traditional broadband service providers.

[Ben Roberts] 15:05:49

Well, I think, okay, so... I think that... The challenge that they'll have is if they try and compete.

[Ben Roberts] 15:06:01

With traditional service providers. And they have starlink and entering the market.

[Wycliffe Bahati] 15:06:01

Mm-hmm.

[Ben Roberts] 15:06:05

Took a decision to do that. And they've seen the challenges, right?

[Ben Roberts] 15:06:12

You know, this technology is transformative in that it can provide fast broadband.

[Ben Roberts] 15:06:23

Anywhere in Kenya. At a price that many people can afford a higher income households can afford, right?

[Wycliffe Bahati] 15:06:25

Mm-hmm.

[Ben Roberts] 15:06:33

And that's the benefits of it. However.

[Ben Roberts] 15:06:40

You know, they've made a deliberate play entering the market.

[Wycliffe Bahati] 15:06:45

Mm-hmm.

[Ben Roberts] 15:06:47

To compete with what we call traditional service providers, but particularly in cities right So... Yeah, but they've done strange things, right?

[Wycliffe Bahati] 15:06:54

Yes.

[Ben Roberts] 15:07:00

First of all, let's take cities. Nairobi uh his sumo mombasa, Nakuru These are the cities in Kenya.

[Wycliffe Bahati] 15:07:06

Mm-hmm.

[Wycliffe Bahati] 15:07:11

Mm-hmm.

[Ben Roberts] 15:07:11

And clearly, there are many broadband users in their own There's probably maybe a million with a fixed broadband connection of some kind, whether it be Safaricom Fiber, JTL, Zuku power or your small friendly neighborhood reseller There's a lot of people with a fixed broadband connection.

[Wycliffe Bahati] 15:07:16

True.

[Wycliffe Bahati] 15:07:36

Mm-hmm.

[Ben Roberts] 15:07:36

Starting from... Prices around the 1500 shilling mark.

[Ben Roberts] 15:07:44

Up to, you know, higher for the fastball band packages from SafariCom.

[Ben Roberts] 15:07:50

Now... the bandwidth consumed By those guys... You know, people using those services is in the terabits, right? I mean, we're looking at We're looking at... You know, people are having backbones, Liquid having, you know, one terabit backbone safaricom maybe having two terabit backbone. Airtel having...

[Ben Roberts] 15:08:15

You know one terabit, you know, we're looking at you're adding up all of the... bandwidth coming into Nairobi.

[Wycliffe Bahati] 15:08:23

Mm-hmm.

[Ben Roberts] 15:08:23

Which is serving businesses and homes We're looking at 5, 10 terabits per second, right?

[Ben Roberts] 15:08:31

Now, Starlink, for all its strengths. Can't provide that much bandwidth over Nairobi. The bandwidth There's

a limitation in the bandwidth.

[Wycliffe Bahati] 15:08:33

Yeah.

[Wycliffe Bahati] 15:08:39

Proof. Mm-hmm.

[Ben Roberts] 15:08:40

Of those leo beams. And so there's never, there was never enough capacity to serve, you know, not even 100,000 people in No Obi.

[Ben Roberts] 15:08:52

Certainly not a million people in Nairobi. So the fact that they came they came to Nairobiome Well, they came to Kenya and they've marketed the thing In Nairobi, they put billboards in Ngong Road.

[Ben Roberts] 15:09:07

They've put it into NYVA supermarkets in Karen Hub. They've sold it on Jumia.

[Wycliffe Bahati] 15:09:11

Mm-hmm.

[Ben Roberts] 15:09:12

They've flooded us with social media advertising on TikTok and WhatsApp and Facebook.

[Ben Roberts] 15:09:18

So they've deliberately marketed their program, their product In cities, in Nairobi particularly right Now, that's why they've sold out. That's why they've now... Yeah, people started complaining that the speeds were going down and now they announce they're sold out.

[Ben Roberts] 15:09:32

And they can't sell anymore. And then that isn't just all of Nairobi they can't sell anymore.

[Ben Roberts] 15:09:39

You know, they can't sell anymore for the The whole area, Kiamu, you know, a whole area around Nairobi, even Caggiado, you know, Caggiado, they can't really sell it, you know.

[Ben Roberts] 15:09:50

So they've basically sold out because they decided... took a conscious decision to do that.

[Wycliffe Bahati] 15:09:58

Mm-hmm.

[Ben Roberts] 15:09:59

When it first launched, I don't know if you ever saw my tweet. Maybe we even talked about it. I wrote this tweet.

[Ben Roberts] 15:10:04

And I said, if you've got Safaricom, Zuku, JTL, Liquid Telecom, Fiber at your house, and you're lucky enough.

[Ben Roberts] 15:10:09

To have a fiber connection. Don't even think about throwing away that connection to get Starlink.

[Ben Roberts] 15:10:15

And if you do, be prepared to shed premium tears. That's what I wrote. And that's because I knew And I wrote that for a reason. And that's because there's not enough bandwidth on the Starlink.

[Ben Roberts] 15:10:26

So, you know, if Starlink had gone into, you know, decided to pitch their product in rural areas selling it through agrovets, through agricultural shows or whatever it is you know whatever you need to do to create a distribution chain in rural areas.

[Wycliffe Bahati] 15:10:35

Mm-hmm.

[Wycliffe Bahati] 15:10:45

Mm-hmm.

[Ben Roberts] 15:10:46

Then they would have done far better. You know, they might have had to put more money into into marketing they might have to You know, get local partners who would share some revenue But they would have done far better for themselves than what they've done.

[Wycliffe Bahati] 15:10:50

Boom.

[Ben Roberts] 15:11:04

But they wanted to go direct. They wanted to get easy customers, get people in Naobi, social media advertising So that's the challenge they've faced, right? Is that they've basically gone up against.

[Ben Roberts] 15:11:17

Product that we've got almost infinite bandwidth With a limited bandwidth and they don't have enough bandwidth to compete with traditional service providers.

[Wycliffe Bahati] 15:11:25

Okay. Okay.

[Ben Roberts] 15:11:25

In the cities. That's simple as that yeah

[Wycliffe Bahati] 15:11:28

Yeah, actually. You're right.

[Wycliffe Bahati] 15:11:32

Because the idea was, if you look at the initial idea of standing, they talked about being able to reach the areas which traditional providers have not been able to reach but looks like that changed very quickly.

[Wycliffe Bahati] 15:11:47

When they realized that they could sell this in cities. Okay, so I'm just wondering, how do you think this traditional providers liquid How are they adapting to this situation now that we have this new guy trying to sell capacity in the cities?

[Ben Roberts] 15:11:51

Yeah.

[Ben Roberts] 15:12:07

Well, how they've responded, which is a good response and it should have been done earlier.

[Ben Roberts] 15:12:13

Was that they've added more bandwidth to the packages. You know, we've seen significant reductions in internet transit bandwidth over the years, but we haven't really been seeing Much disruption in the bandwidth that people are giving them.

[Wycliffe Bahati] 15:12:13

Mm-hmm.

[Ben Roberts] 15:12:27

There isn't a problem. We have a problem in Kenya of the resellers, as you know right where Sound fun.

[Ben Roberts] 15:12:41

The first package into a building Then somebody buys that and they share it with all their neighbours. We have that problem, right?

[Ben Roberts] 15:12:47

Now, because of that problem. This is one of the reasons that... Service Pride has been very conservative in their top-level packages.

[Wycliffe Bahati] 15:12:47

Yeah.

[Ben Roberts] 15:12:55

And until, you know, when Starlink came out The fastest package you could buy in Kenya is 140 megabits per second was the fastest from JTL 140. And then the fastest home broadband you could get from Safaricom was 100.

[Wycliffe Bahati] 15:12:59

Mm-hmm.

[Ben Roberts] 15:13:10

And 100 from liquid and others, but 140 was the very fastest you could buy in the country. I spoke to a colleague in South Africa about nearly six months, eight months ago.

[Ben Roberts] 15:13:20

And I said, how many people are people doing... In a one gigabit broadband. When you have a fiber t-on link.

[Ben Roberts] 15:13:27

Technically, you can deliver one gigabit, right? Now, in UK, you can't buy anything less. In UK... You'll pay about 50, £60 a month For full fast fiber, one gigabit per second. And the government is subsidising that. So The government is paying service providers

[Wycliffe Bahati] 15:13:31

True.

[Ben Roberts] 15:13:46

Every new house you connect with one gigabit fast. Fiber.

[Ben Roberts] 15:13:53

Subsidizing the installations And anyway, there were at least five ISPs in South Africa offering one gigabit per second.

[Ben Roberts] 15:14:02

So now it's the vicom off. Hello? Yeah, so Safari can't increase their packaging.

[Ben Roberts] 15:14:09

They're offering 500 and 1 gigabit in some areas. And Liquid are actually offering maximum 300, which is... more conservative, but they've got a much bigger FUP than Safaricom.

[Ben Roberts] 15:14:20

Safaricom have got a fair usage policy of of one terabyte. So if you use One terabyte, they limit your bandwidth.

[Ben Roberts] 15:14:26

Whereas liquid is eight terabytes. So it's much, much, much, much bigger fup But a slightly lower top line rate.

[Ben Roberts] 15:14:32

So that's the response that we've had.

[Wycliffe Bahati] 15:14:38

Okay. Saldin... There's also a study that was done

[Ben Roberts] 15:14:41

Rather than rationing the bandwidth they're using the What should I say?

[Wycliffe Bahati] 15:14:47

It's fine. Just proceed.

[Ben Roberts] 15:14:51

Yeah, so I'm just moving on to Wi-Fi between 4G and then it's sort of breaking up so So what I'm saying is, you know.

[Wycliffe Bahati] 15:14:55

Okay.

[Ben Roberts] 15:14:58

Fiber providers were, although they've got technology that can deliver one gig.

[Ben Roberts] 15:15:02

They were limiting it to 100 meg because they were being conservative.

[Ben Roberts] 15:15:06

And when Stalin comes along. Sounds more attractive to consumers You know, they've responded by offering higher bandwidth.

[Ben Roberts] 15:15:13

That are closer to the technological limits that the technology provides. Because they have the technology advantage. So now they're actually giving that to customers, the technology advantage that they have over Starlink.

[Ben Roberts] 15:15:25

Because it seemed to the consumer It seemed to the consumer like Starlink had a technology advantage, but they didn't. Actually, fiber has the technology advantage.

[Wycliffe Bahati] 15:15:26

Actually... Hmm.

[Ben Roberts] 15:15:34

But the package The package was less attractive.

[Wycliffe Bahati] 15:15:39

Actually, you're right because there's a study that shows that there's a trend that the higher capacity equipment, fiber equipment is being sold to the edge to the age nearing the homes so we have almost some countries that are almost talking about 10 g level

[Ben Roberts] 15:16:01

Yeah.

[Wycliffe Bahati] 15:16:04

When they are passive optic. Networks so it's true that it's true We do.

[Ben Roberts] 15:16:10

Yeah. Now, if you give a home in UK, if a home in UK has one gigabit.

[Ben Roberts] 15:16:15

They won't flatline one gigabit all day long, right? They only use it when You know, they come home and they watch Netflix or something, right?

[Wycliffe Bahati] 15:16:16

Mm-hmm.

[Ben Roberts] 15:16:22

You get natural sharing. The average people use is only probably 10. You give them one gigabit, but they only use 10 gigabits on average.

[Wycliffe Bahati] 15:16:26

Mm-hmm.

[Ben Roberts] 15:16:32

Once you average it out. But it's all about where the bottlenecks are in the network.

[Wycliffe Bahati] 15:16:35

Okay. And... Okay. And... Do you see... maybe in the future providers being used Right now, they're focused on selling maybe to end users. Do you see them being used as backhaul to traditional providers to reach those remote areas where they can't build trust?

[Wycliffe Bahati] 15:16:59

To restore infrastructure. H.

[Ben Roberts] 15:17:03

Yeah, I mean, this has happened. They've announced a strategic partnership with Africa Mobile Networks, who are in west africa So if you look for that amn african mobile networks So AMN are using them to connect to towers for telco services. They had been using traditional VSATs.

[Wycliffe Bahati] 15:17:16

Mm-hmm.

[Ben Roberts] 15:17:23

But now they've converted a few hundred of their sites to Starlink to provide extra bandwidth for 4G here at those sites.

[Wycliffe Bahati] 15:17:32

Yeah. And another question i have is Okay, apart from Just... the unavailability of terrestrial infrastructure in some areas.

[Wycliffe Bahati] 15:17:47

What they call now the digital divide where some areas are not accessible by traditional providers because of maybe they're not profitable Now, my question is, such areas, okay, there are those that probably is just because of terrain But I see like the other factors that make those areas not

[Wycliffe Bahati] 15:18:10

Constructive to traditional providers. For example, a lack of basic things like infrastructure power I mean, if Starlink or the LEO providers are mainly building to target these areas.

[Wycliffe Bahati] 15:18:29

Do you see them actually improving any I mean, improving the access gap between people who have internet and those who truly don't have any.

[Ben Roberts] 15:18:42

Yeah, so it's definitely going to improve the access gap. But at the moment.

[Ben Roberts] 15:18:50

We are... we're imposing on ourselves policies that are giving Starlink an advantage over our homegrown ISPs.

[Ben Roberts] 15:19:00

So take something like whey leaves, right? So if you want to get internet to Wagia.

[Wycliffe Bahati] 15:19:00

Mm-hmm. Mm-hmm.

[Ben Roberts] 15:19:09

You know, you have to pay to get a fibre there, you have to pay a very large amount of money in whey

leaves.

[Ben Roberts] 15:19:16

To kenhurst, where Gia County, Tagarista County, everywhere you go along the way you have to pay all these whaling fees As well as, as you've mentioned already, high cost of power. You've got to have repeaters You've got high costs of power and all these things.

[Wycliffe Bahati] 15:19:22

Yeah.

[Ben Roberts] 15:19:34

To get internet to Wajir, you really are adding a layer of complication.

[Ben Roberts] 15:19:40

And cost to the journey you know build redundant networks and everything else. It doubles up and you're really adding a lot of cost all these things, particularly the whale leaves, it's just a tax Whereas Starlink don't encounter that. All Starlink need is just the user, the customer to have a power supply for his

[Wycliffe Bahati] 15:19:42

Including security.

[Ben Roberts] 15:19:59

However, they are very power hungry, right? So the power used by a Starlink compared to that used by a power link or mawingu link is much more But You know, because they're coming direct from space they're not having all these extra charges that we're imposing upon ourselves

[Wycliffe Bahati] 15:19:59

Mm-hmm.

[Wycliffe Bahati] 15:20:09

Mm-hmm.

[Ben Roberts] 15:20:18

That is making it less, making it harder to invest. So they're getting an advantage.

[Ben Roberts] 15:20:24

Over us. And it's a technology advantage, but also a policy advantage because we're taxing taxing the fuck out of whale leaves.

[Ben Roberts] 15:20:33

And then making this space-based service more attractive.

[Wycliffe Bahati] 15:20:40

And how do you explain, I saw there's an article that came out the other day that Sterling has become the second largest supplier of broadband in Nigeria with about 65,000, I think, terminals or something and then Compare that with other

[Wycliffe Bahati] 15:21:03

African countries, okay, let's just say Kenya because other African countries are quite behind

[Wycliffe Bahati] 15:21:11

What would make it attractive in Nigeria? Very mean attractive.

[Ben Roberts] 15:21:16

So Nigeria is much more underserved than us, right? I don't believe it's second because MTN and Airtel are providing far more broadband.

[Ben Roberts] 15:21:24

At least and there are others, there's Globicom and others as well, but there's no way, you know, MTN and Airtel have got far more It's mobile broadband or whatever. They've got a farm on 65,000 customers.

[Ben Roberts] 15:21:40

But Nigeria has a huge bigger gap, right? There was a piece of work done about three or four years ago by the broadband commission that was estimating the amount of capital that had to be spent by country to give universal broadband in Africa by 2030.

[Ben Roberts] 15:21:56

And 25% of the capital needed to spend in Africa was Nigeria.

[Wycliffe Bahati] 15:22:01

Mm-hmm.

[Ben Roberts] 15:22:02

Because... they haven't got They haven't got like, you know, we had Zuku, how long ago? When did Zuku come to Kenya and was like rolling out Zuku?

[Ben Roberts] 15:22:12

Suki was there, JTL was there Yeah, it's a long time ago, right? So, you know, large scale rollouts of fiber So they haven't had that. It's now starting to starting to catch up.

[Wycliffe Bahati] 15:22:14

Early 2000.

[Ben Roberts] 15:22:27

But look at South Africa now. People are fiberizing every estate up and every, you know, huge investment going into the fiber in South Africa.

[Ben Roberts] 15:22:35

Kenya has had spurts. It's not so big at the moment.

[Ben Roberts] 15:22:37

It had big spurts from JCL, Safaricom, and then it was only Zuku, JTL, Safaricom.

[Ben Roberts] 15:22:45

It's kind of slowed down a bit in terms of the amount of new fibre being built, whereas South Africa is ramping up and ramping up.

[Ben Roberts] 15:22:52

But Nigeria they haven't had the same investment. Therefore, you've got you know, probably the same problem. Lagos is a much bigger city, but you've probably got you know, nice areas where people can afford broadband.

[Ben Roberts] 15:23:08

And, um. Starlink is coming in as the only option or the better option.

[Ben Roberts] 15:23:13

So there was more of a gap, more of an access gap in Nigeria, we've got 97% 4G coverage in Kenya of population Nigeria hasn't got rap.

[Wycliffe Bahati] 15:23:23

Hmm. Okay. And I just want to talk about, I know you've done some work on Coyote, do you think?

[Wycliffe Bahati] 15:23:34

Big blue broadband will help in that space.

[Ben Roberts] 15:23:39

Am I certain? There were some... There were some innovative There still are some innovative startup companies around the space of IoT.

[Ben Roberts] 15:23:48

One of them was called Swarm and was bought by Starling. It was bought by SpaceX.

[Ben Roberts] 15:23:55

But there are others. So there are companies who are launching You know, LEO satellites for the purpose of IoT devices.

[Ben Roberts] 15:24:05

Um so You know, it will only really work outdoors But it can give global coverage around IoT devices. It means you can track a bag.

[Ben Roberts] 15:24:17

Or track a container You know, a container leaves the port in london and gets onto the high seas and then you know arrives in Mombasa means you can track it all the way on a single platform, right?

[Ben Roberts] 15:24:31

You know, they drive to Nairobi at the ICD. It's very powerful as to what you can do with IoT applications from satellite because you've got that global coverage with leos

[Wycliffe Bahati] 15:24:44

And is it likely to replace the and all this workaround it's called LoRa1 I think I've forgotten it.

[Wycliffe Bahati] 15:25:00

This communication protocol used by IoT devices to communicate over very long distances?

[Ben Roberts] 15:25:06

There's a few. I mean, Laura WAN, Sigfox um Others, you know, NB-IOT I don't think it's likely to replace because... Because satellite stuff isn't going to work indoors.

[Wycliffe Bahati] 15:25:07

It's a...

[Wycliffe Bahati] 15:25:13

Mm-hmm.

[Ben Roberts] 15:25:20

So some of these terrestrial networks using low frequencies are going to... are going to... work very well outdoors. It might be great in agriculture or something else, but they're not going to work within a building, within a hospital or something else like this. So you'll have different applications, right?

[Ben Roberts] 15:25:41

But The real benefit is an IoT coverage in in areas where

[Ben Roberts] 15:25:53

You have low density low coverage, high seas, whatever. I mentioned deserts. Mount Kenya will be, you know, you can track You can track things. Just remember that these bands that these IoT devices using, they were using the same sort of bands, but these

[Wycliffe Bahati] 15:26:00

Okay.

[Wycliffe Bahati] 15:26:07

And...

[Ben Roberts] 15:26:10

Ism band 800 megahertz somewhere somewhere 868 megahertz It's only a 2 megahertz band So again.

[Wycliffe Bahati] 15:26:16

Mm-hmm.

[Ben Roberts] 15:26:19

That'll work very nicely if you're tracking a container across the Pacific Ocean.

[Ben Roberts] 15:26:24

But if you've got... 50 million devices in Beijing It's not going to work, right? Because you're going to get from a satellite You'll have too much noise. You'll have too much of an accessible area. There'll be too much platform noise. You won't be able to discern it.

[Wycliffe Bahati] 15:26:36

Grand. Mm-hmm.

[Ben Roberts] 15:26:41

So all of these things, nothing is going to replace anything you know, all we're doing is supplementing coverage for terrestrial networks So I see it as a compliment. And by the way, perhaps to your first question, the market opportunity for ball ban in Kenya

[Wycliffe Bahati] 15:26:49

Okay.

[Ben Roberts] 15:26:56

Let's say there's a million homes that have got a fixed broadband connection.

[Wycliffe Bahati] 15:26:57

Mm-hmm.

[Ben Roberts] 15:27:00

There's at least 8 million homes. Kplc have got 8 million customers So there's 8 million homes. They've got a KPLC line in them.

[Wycliffe Bahati] 15:27:09

Mm-hmm.

[Ben Roberts] 15:27:09

You know, so and power. So there's 7 million homes that don't have a broadband connection that have power. There's other homes that don't have power.

[Ben Roberts] 15:27:17

Maybe another six or eight million homes that don't have power. But you see a huge market opportunity right It shouldn't be around fighting for what's already there.

[Ben Roberts] 15:27:27

It should be around connecting the unconnected, addressing the access gap rather than rather than trying to

[Wycliffe Bahati] 15:27:35

Compete for the same market.

[Ben Roberts] 15:27:37

Compete for the same market, yeah.

[Wycliffe Bahati] 15:27:40

Okay. Yeah, so I think I'm just off to the next section um there's the I don't know if since, what are they facing these leo providers okay in this case, starting in terms of technical challenges.

[Wycliffe Bahati] 15:27:59

I mean, to deliver because i know you've talked about finite space in terms of bandwidth.

[Wycliffe Bahati] 15:28:07

I don't know if you've seen any other technical challenges they're facing.

[Ben Roberts] 15:28:12

I think... the model they're deploying is going to give them challenges you know not having in-country customer service some of these things will get them will get them challenges when they get to scale.

[Ben Roberts] 15:28:26

But they're exploring Africa as a new territory. So it's okay.

[Ben Roberts] 15:28:32

You know, rain is still there, so KU Bands and KA band. Ka band uplink still affected by rain. Ku band downlink is still affected by rain.

[Wycliffe Bahati] 15:28:38

Mm-hmm. New Year.

[Ben Roberts] 15:28:39

So that has not changed. Um and so um It's working very well, but I will say it's extreme the system itself is extremely complicated.

[Ben Roberts] 15:28:50

So these 6,000 satellites whizzing around the sky, all talking to each other.

[Wycliffe Bahati] 15:28:52

Mm-hmm.

[Ben Roberts] 15:28:55

Is a very complex system. You know, this is why that it places such a barrier to entry that you don't you'll get a few number of customers companies doing that platform but they're going to have competition.

Amazon is going to come with competition

[Ben Roberts] 15:29:15

You know, so... Let's see what happens.

[Wycliffe Bahati] 15:29:19

Okay, and... Over the years, you know, if you noticed Initially, in Africa, for example.

[Wycliffe Bahati] 15:29:31

Most of our content is to come from Europe, so almost 80 to 90 percent But over the last few years, that has really changed.

[Wycliffe Bahati] 15:29:41

Almost the inverse because of maybe the CDNs being deployed of the internet exchange points.

[Wycliffe Bahati] 15:29:49

Does that affect... the latency that Starlink is able to provide to the end users

[Ben Roberts] 15:29:58

It has been at the moment because Starlink had been downlinking in nigeria.

[Ben Roberts] 15:30:04

So if you've been using a style in Kenya, it's been downlinking in Nigeria.

[Ben Roberts] 15:30:07

However, they're in the progress of bringing up pops, they put an IP pop into into a... Kenya and then the process of commissioning uplink pops as well. So that's to say if you're in If you're in Kenya.

[Ben Roberts] 15:30:24

Your styling dish will connect to a satellite and then we'll connect straight down to a Starlink pop in kenya And then they most likely will take advantage of the same fact that content is here in Kenya, they will connect them directly to Netflix and whatever and Google

[Ben Roberts] 15:30:38

And others. Once they get their... They're full gateways running up in country they're going to take advantage of the same the same phenomenon you're talking about.

[Ben Roberts] 15:30:54

They're already getting it from Nigeria. So when you downlink in Nigeria, they'll only be connecting to some of this content in Nigeria, but Kenya's probably got more content than Nigeria. You know, so by being here, they'll get more content. They'll also get the local content. So such as your

[Ben Roberts] 15:31:13

You know local e-government, whatever, you know, reduce the latency right. Because they'll appear locally at the KIXP. And then if you're going to connect with your Starlink you want to connect and e-citizen, it will then connect directly through that local peering to the local concept like that e-government services

[Ben Roberts] 15:31:30

Local banks whatnot

[Wycliffe Bahati] 15:31:35

That's, for example, when we say, okay, I've seen they're deploying the they've deployed the ground stations here. So what would happen to a country like Uganda, Toronto?

[Wycliffe Bahati] 15:31:48

Tanzania where they don't have the ground stations.

[Ben Roberts] 15:31:51

They haven't granted licenses. Tanzania, Uganda, Ethiopia.

[Wycliffe Bahati] 15:31:54

Yeah.

[Ben Roberts] 15:31:57

South Sudan have not granted licenses.

[Wycliffe Bahati] 15:32:00

Okay, so you see a situation where they'll be deploying this. I know they've done quite a bit of them in the US.

[Wycliffe Bahati] 15:32:06

Is likely as demand grows, they'll be able to deploy most of these tram stations in most countries.

[Wycliffe Bahati] 15:32:12

Subject to licensing.

[Ben Roberts] 15:32:12

No, I don't think they can afford to do that in most countries. I think it's a high cost.

[Ben Roberts] 15:32:18

And I think... the business case won't be there to do it in every country.

[Ben Roberts] 15:32:23

However, some countries will not allow them. To get a license if they don't have one.

[Wycliffe Bahati] 15:32:23

Okay.

[Ben Roberts] 15:32:30

You know knowing having some understanding of the cost of this That will mean that some countries will be left behind.

[Wycliffe Bahati] 15:32:40

Okay, and the standardization i mean uh address to infrastructure has really benefited both terrestrial and geo infrastructure has really benefited from standardization.

[Wycliffe Bahati] 15:32:52

New seat coming to the LEO providers, being able to work together to standardize so that they can Crewstock and serve each other.

[Ben Roberts] 15:33:05

It's a great question. I do feel there's a need. For that, I mean, I see... people are starting to talk of in space in space innovations. I saw a post yesterday about somebody who wanted to build a a data center in outer space.

[Ben Roberts] 15:33:20

I've seen another company talking about doing CDNs in outer space.

[Ben Roberts] 15:33:26

So, you know, if you're going to have that kind of thing, data centers CDNs.

[Ben Roberts] 15:33:33

You know, you're going to need open standards. For communication within space.

[Ben Roberts] 15:33:38

Right now, everything's proprietary. So those inter-satellite links, the Starlink satellites used to talk to other Starlink satellites It's all proprietary communications And likewise to the ground as well, all proprietary.

[Wycliffe Bahati] 15:33:48

Mm-hmm.

[Ben Roberts] 15:33:50

Very unlike the geo world where we could choose your communication on the ground You could choose your encoding your DVB or whatever you're using on the signal, you can choose your interoperable standards, work between vendors, work between systems, work between satellites.

[Ben Roberts] 15:34:08

So I think it's inevitable that that is going to evolve standards for internet satellite links.

[Ben Roberts] 15:34:14

Standards for things, architectures that are based in space We can't be having space ruled by one company having proprietary access to everything that isn't going to work.

[Ben Roberts] 15:34:25

And, you know, things will evolve from that point of view but you know it'll start with standards and then there'll be regulations imposing that you must use these standards just like we've seen with you know uh charging plugs where you've seen a free-for-all and everybody is using their own connector on their phone chargers

[Ben Roberts] 15:34:42

And you're seeing the European, you know, you're seeing standards come USB-C is a standard for a type of charging plug But you've seen European Union decide we're going to regulate USB-C is the only allowed standard for charging plugs for

[Ben Roberts] 15:34:59

For mobile phones. You're going to see that sort of shit happening.

[Wycliffe Bahati] 15:35:04

Yes. And... Okay, so on the economic considerations I know you've done quite a bit of this network business cases and stuff like that.

[Wycliffe Bahati] 15:35:16

Do you see, if you take away the subsidies because i'm This is not the first time a constellation has been tried there was the test Testra or something like that.

[Wycliffe Bahati] 15:35:29

Before we failed, went bankrupt. So without the subsidies, because I know Stanley able to access quite a bit of this Because they get it out of the Congress fund.

[Wycliffe Bahati] 15:35:40

For connecting and developed and connected Without that, do you see a return on investment in this type of constellation?

[Ben Roberts] 15:35:51

I don't have visibility of their numbers. And, you know, nobody does.

[Ben Roberts] 15:35:56

Even SpaceX, you know, Starlink is within SpaceX. So Starlink um Starlink is a... You know, it's a product within SpaceX, which is a rocket company. And even that's a privately held company.

[Ben Roberts] 15:36:11

So it's not public like some other companies are owned by Elon Musk. We don't have the same visibility.

[Ben Roberts] 15:36:20

However, you know, I'm skeptical. As to the viability of of this thing. They've got about three or five million customers globally, it seems, you know, and that's people taking their home broadband packages they've probably taken some market share of the um

[Ben Roberts] 15:36:37

Of the global maritime market. And the Global Maritime communications market was worth about \$3 billion. They've taken some market share of this.

[Ben Roberts] 15:36:46

That potential to take it all. There's an aviation market Which is, you know, communications for aviation is less Because the quality of experience has been bad.

[Ben Roberts] 15:36:58

Aviation communications. Probably been under realizing their revenue.

[Ben Roberts] 15:37:03

So, um. You've got really three markets. You've got terrestrial broadband You've got maritime broadband, maritime communications, you've got aviation communications.

[Ben Roberts] 15:37:16

For the civilian space. But Starlink have been have been doing and taking advantage of other things for the defense space And, you know, whether they are really paid commercial rates or whether these are just US government subsidies, it's hard to know.

[Ben Roberts] 15:37:35

These satellites are actually replaced every five years. So it's hard to know without having a deep understanding.

[Wycliffe Bahati] 15:37:35

Mm-hmm. Yeah.

[Ben Roberts] 15:37:40

As to whether the business case for this thing has um If the numbers add up, if you didn't have DoD, if you didn't have US federal defence subsidies, right? They call it defence contracts, but really it's defense subsidy. It's US government subsidies.

[Ben Roberts] 15:37:59

Through defense funding. So it's a grey area and you're seeing other countries as well have been talking about doing that. You see now with Trump.

[Ben Roberts] 15:38:06

You're seeing other countries are withdrawing and saying, no, we're not going to do that anymore.

[Ben Roberts] 15:38:09

You're seeing Mexico and South America pulling out of some of these things.

[Wycliffe Bahati] 15:38:11

Yet.

[Ben Roberts] 15:38:15

You know, so it's hard to say, and I don't know. I am skeptical, but I don't have visibility of the numbers.

[Wycliffe Bahati] 15:38:22

I think it will come open when the European ice squared or something Once that is launched, maybe the

cost will become transparent.

[Ben Roberts] 15:38:30

Yeah.

[Wycliffe Bahati] 15:38:34

So we'll know whether it's sustainable or not.

[Ben Roberts] 15:38:35

Yeah, I mean, exactly. Once there's competition, you'll see movement in prices and you'll understand market forces you know.

[Ben Roberts] 15:38:44

Amazon will launch, but Amazon is a listed company so maybe they'll have to report a bit more than Starlink do so Yeah.

[Wycliffe Bahati] 15:38:53

Yeah, so I think you've covered about ground station in terms of costs at least it's not very, it's a bit expensive to deploy the antennas are very expensive. And like you said, it's not easy for them to deploy everywhere.

[Ben Roberts] 15:39:09

Mm-hmm.

[Wycliffe Bahati] 15:39:09

So that comes to now we come to the regulatory hurdles these guys face.

[Wycliffe Bahati] 15:39:14

Because the other day. I saw someone in the regulator from Rwanda.

[Wycliffe Bahati] 15:39:25

Seeing this they have to literally apply for a license to operate in every country.

[Wycliffe Bahati] 15:39:31

And they have to pay for the spectrum And, you know, the other licensing fee. So do you think it will be a big problem Because a place like Ethiopia It may not be very easy.

[Ben Roberts] 15:39:53

Okay, so there are some special licensing considerations of some countries.

[Ben Roberts] 15:39:59

Around having local gateways. Uganda, Ethiopia.

[Ben Roberts] 15:40:04

Are countries that have some of these special licensing conditions are not always written down.

[Ben Roberts] 15:40:09

Sometimes they are written down, but you have governments who like to have internet gateways within the countries. Ethiopia is one, Uganda is another, Tanzania is another.

[Ben Roberts] 15:40:20

So I don't think those countries are going to... give licenses while Starlink does not meet their requirements.

[Ben Roberts] 15:40:28

So, you know, yes, they do allow VSAC services in Uganda.

[Ben Roberts] 15:40:33

However, it's not a mainstream service. It's a niche service. But if you're coming in and saying you're going to have mass broadband.

[Ben Roberts] 15:40:40

And the government likes to have certain controls they like to filter Facebook and other things you know this is not it's not It's public knowledge, really. Facebook was blocked for a long time in Uganda.

[Ben Roberts] 15:40:54

You know, things like this. So the government likes to have these controls.

[Ben Roberts] 15:40:57

So if the government doesn't have these controls. Then the government, I don't think will grant licenses. So that's regulatory challenge number one.

[Ben Roberts] 15:41:05

It's countries that have... strict control over internet gateways, it's going to cause a problem for Starlink.

[Ben Roberts] 15:41:14

Now we are going to face As well, we're having this direct device So there's going to be other regulatory hurdles around allocations, because those will standard IMT bands, right? Gsm IMT bands So you're going

to have areas of crosstalk over boundaries

[Wycliffe Bahati] 15:41:32

Okay.

[Ben Roberts] 15:41:35

How do you use a beam that doesn't cross into, you know, you use Safari Comm's frequency in Kenya How do you make sure that doesn't cross into Uganda and interfere with MTNs?

[Ben Roberts] 15:41:45

In allocated frequency in Uganda. These types of things are going to be a challenge.

[Ben Roberts] 15:41:48

So that's a big challenge. Then you are going to have... challenges around sharing of other states in the sharing of um or having ground multiple ground stations in the country by multiple providers.

[Ben Roberts] 15:42:05

So Amazon Kuiper have just been granted permission to have earth stations in Kenya, sorry, in UK.

[Ben Roberts] 15:42:11

And there's a long, long, long report by Ofcom And it's worth having a read and you can Google the Ofcom report on Amazon Kuiper's application.

[Wycliffe Bahati] 15:42:11

Been.

[Ben Roberts] 15:42:21

It did require some coordination and agreement. You know, between Starlink and Amazon Kaipo and others But right now you're seeing in Africa, you're seeing a land grab really where Starlink are placing their stations and others are going to come along and they're going to say, hey.

[Ben Roberts] 15:42:37

I want to be in that area. I'd like to be this place in the same. Well, you can't because of interference.

[Ben Roberts] 15:42:42

So there's that consideration of interference of ground stations And there's going to be many others. There's going to be many other technical considerations. So when we're going to get space junk.

[Ben Roberts] 15:42:52

Space junk, you know, satellites They burn up and they should land without without without causing damage.

[Ben Roberts] 15:43:03

But at some point, maybe we saw rings falling in Kenya that didn't kill anybody, but they could have done you. So space junkies are going to be a problem. You're going to see then You know, pollution light pollution in the skies. There's a whole bunch of things

[Wycliffe Bahati] 15:43:08

Yeah. Yeah.

[Ben Roberts] 15:43:16

Environmental frequency coordination business model, you know, in consideration of national hurdles Some of these things can be coordinated at a global level by the ITU. But at the moment.

[Ben Roberts] 15:43:31

They're launching into space based upon say-so from the FCC. Fcc is an American regulator.

[Ben Roberts] 15:43:38

You know, they're not they're not they're not getting Permission from every regulator in the world before launching a fleet of satellites.

[Wycliffe Bahati] 15:43:39

Yes.

[Ben Roberts] 15:43:46

So we have... They're using KU bands and KA bands which have been allocated by the ITU But even that is having challenges with the geos Because those Leo satellites crossed the geo arc at some point.

[Wycliffe Bahati] 15:43:56

Mm-hmm.

[Ben Roberts] 15:44:03

And then they have to turn down the power they transmitter.

[Ben Roberts] 15:44:07

The power, you know, they have to not interfere with the geos. So the more and more satellites you get The

more complicated it's going to become.

[Wycliffe Bahati] 15:44:08

Mm-hmm.

[Ben Roberts] 15:44:17

And... I'm not good enough at physics.

[Ben Roberts] 15:44:20

To understand what's going to happen, but something is going to happen. I'm telling you that.

[Wycliffe Bahati] 15:44:22

And... okay yeah and of course there's now talk of lasers as an alternative way to move away from radio frequencies But you see how that goes. What about...

[Ben Roberts] 15:44:38

Yeah, I'm skeptical about that one. Yeah, that one I'm skeptical on because they can't see through clouds, Wiki, right?

[Ben Roberts] 15:44:44

I mean, you know, if you're clouds you've got to If you're looking at laser light and you're shining through a cumulus cloud.

[Ben Roberts] 15:44:50

You're looking at 200 decibels of attenuation, right? I mean, you know, it's just basically light doesn't go through clouds.

[Wycliffe Bahati] 15:44:55

It's challenging. Yeah.

[Ben Roberts] 15:44:56

Even the sun, even the sun won't shine through through a cloud, right? You know, so...

[Wycliffe Bahati] 15:45:02

Yeah. What about... Data privacy do you see it an issue

[Ben Roberts] 15:45:09

Yeah, I do. I do. I do, because you're looking at data privacy laws around some certain types of services.

[Wycliffe Bahati] 15:45:11

The data privacy laws

[Ben Roberts] 15:45:20

That are... Saying data can't leave the country for certain types of services.

[Ben Roberts] 15:45:27

Now, if you use a geo service. Yes, you can still use satellites for services.

[Ben Roberts] 15:45:34

Geo services. If you have a local gateway and you have a local terminal You know that that terminal is going to go up to the satellite and then down again to your gateway.

[Wycliffe Bahati] 15:45:42

Yeah.

[Ben Roberts] 15:45:43

Now, even though Kenya is going to have local gateways. You don't have control as a customer as to how how your data is routed.

[Ben Roberts] 15:45:55

If they turn that local gateway off, they'll route it to Nigeria or Europe or anywhere they fancy.

[Ben Roberts] 15:45:59

So some of the other constellations, I think Telesat is launching a Leo constellation.

[Ben Roberts] 15:46:03

You know talking about for enterprise customers, giving much more control as to how your traffic is routed.

[Ben Roberts] 15:46:09

Because that's going to be a problem. You don't know whether your traffic is routed, whether it's landing in a country that doesn't have data privacy laws it could be landing in you know russia for you know right

[Wycliffe Bahati] 15:46:21

I hope chain later. Okay. So, I mean, I don't know the you've talked about the affordability

[Wycliffe Bahati] 15:46:33

If you compare the local prices versus what Leo Stalin is giving.

[Wycliffe Bahati] 15:46:42

Do you think it's because I mean... You could be having, you could be offering higher capacity But then the cost is just not what the consumer can afford.

[Wycliffe Bahati] 15:46:56

Do you... The consumers have seen it as a concern for consumers with pricing.

[Wycliffe Bahati] 15:47:02

Especially the cost of equipment.

[Ben Roberts] 15:47:05

Yeah, so I'd even look at the price of the bandwidth i mean A farmer in Texas pays \$100 a month for a Starlink link, unlimited.

[Ben Roberts] 15:47:15

But in Kenya they came and launched at 65,000 and they brought in lower packages And then they launched a 50 GB package. So they are um You know, they're lowering prices in Africa to capture market share.

[Ben Roberts] 15:47:29

Probably selling below cost. But as I said, as I said, you've got a whole business plan that's that's being you know, they're adding up all the revenues of the 2 million farmers in texas You know, all their customers in the USA, let's say

[Ben Roberts] 15:47:46

2 million, 3 million customers in the USA. They've got the maritime market, they've got the aviation market, they've got the DoD sponsors.

[Ben Roberts] 15:47:51

Subsidies. So they're looking at that and then they're discounting African ones to get market share.

[Ben Roberts] 15:48:01

Other people in Kenya are probably doing some cost up based pricing Yes, we are trying to be competitive, but also looking at what is the cost and doing cost up based pricing.

[Ben Roberts] 15:48:11

Whereas Starlink may not be doing cost-up based pricing. They may just be doing market share based

pricing.

[Ben Roberts] 15:48:15

So what is the market share to do this? They're also susceptible to to currency fluctuations, and you saw them double prices in Nigeria. They launched at very low prices.

[Ben Roberts] 15:48:25

And then the NYRA crash, and then they doubled the prices yeah

[Wycliffe Bahati] 15:48:26

And then...

[Wycliffe Bahati] 15:48:31

So I think... Generally, there's somebody who said I mean, all the technology sometimes you just have to wait it out and see how it goes So I'm just wondering if you have any view, what do you think will happen next five to ten years

[Wycliffe Bahati] 15:48:51

We'll learn. The leo broadband services become mainstream providers Or, I mean, they just maintain a small share of the... the markings.

[Ben Roberts] 15:49:08

I mean, for Africa, the Africa, they have a good chance of of taking a big piece of the rural market, right? I mean, growing a rural market that isn't there Because you've got a rural area where people are not having fast broadband they're just having

[Ben Roberts] 15:49:22

4g and 5g but You know, I think everyone will find their own way, but we are going to have As the market dynamics settle in, we're going to have regulatory challenge. We're going to have regulatory interventions.

[Ben Roberts] 15:49:36

Around the frequency coordination around the around the other things, the data protection and the you know the the local gateways, you're going to have Probably regulatory interventions around pricing.

[Ben Roberts] 15:49:49

Most likely. So a lot's going to pan out. It's very early days of tech.

[Ben Roberts] 15:49:53

So for me, it's too early to call. To answer your question. I think it will become a mainstream technology But, um.

[Ben Roberts] 15:50:02

I'm skeptical of the approach. That is being taken right now where you know one company dominates it all without really giving much value to local partners or anything else I think something's going to change in that regard.

[Wycliffe Bahati] 15:50:18

And... Do you see... future collaboration between the traditional providers and and leo providers in terms of substitutes. I mean, working hand in hand to to, for example.

[Wycliffe Bahati] 15:50:41

Using Leo for black hole, for example, instead of going to Moiale by terrestrial.

[Wycliffe Bahati] 15:50:47

Using Leo as a backhaul for the local providers.

[Ben Roberts] 15:50:52

I think that's one collaboration. I think we'll see a number. We're already seeing collaborations on the D2D, so the director device, you know, they're working with T-Mobile in the US here.

[Ben Roberts] 15:51:02

So, you know, we will see collaborations You see some of the companies, broadband pump, some company like Paratus in Africa is one of the first.

[Ben Roberts] 15:51:12

To start selling Starlink. Services to their customers, enterprise customers So we'll see more of that. And I think really If Starlink is going to become a mainstream. If Starling is going to get to millions of subscribers in kenya It probably is going to have to take his approach it can't just be

[Ben Roberts] 15:51:34

You know, everything's a board. You pay by credit card. It'll have to make more investments in markets.

[Ben Roberts] 15:51:41

You know, to achieve those things. The model they're coming in with probably isn't going to be, you know, gain, as you say, become mainstream. It's not going to get them to millions of subs.

[Ben Roberts] 15:51:54

You know, because that again countries will start seeing that as a national security risk as well, right?

[Ben Roberts] 15:52:02

Imagine you had... Imagine you had you're a Kenyan. You're the Kenyan. You're the president of Kenya and let's say there's you know, 100,000 people using Starlink to watch cat videos. You probably don't care.

[Ben Roberts] 15:52:18

But if you've got then all your hospitals are using Starlink and then you know If Stalin goes down, people can't pay you there and they can't pay the hospital bills. They can't be admitted, they can't be whatever, they can't get treatment.

[Ben Roberts] 15:52:31

Imagine that. You're going to be concerned about that because you've got no control over it.

[Ben Roberts] 15:52:39

So it will become

[Ben Roberts] 15:52:44

Great technology, but the models are going to change around it.

[Wycliffe Bahati] 15:52:48

Okay.

[Ben Roberts] 15:52:48

Intel Stat when it first came out. Interstat launched, it was owned by governments. So it was privatized eventually.

[Ben Roberts] 15:52:55

But Incel Sat was a company operating satellites in Kenya. Kenya was a shareholder in Intelsat.

[Wycliffe Bahati] 15:52:56

Yes.

[Wycliffe Bahati] 15:53:04

And I think that model was working very well because there wasn't a fear that there wasn't fear Something... Like I'm sure right now what's happening in Ukraine I'm not sure if tomorrow they'll still be having Starlink.

[Ben Roberts] 15:53:17

Law. Yeah.

[Wycliffe Bahati] 15:53:21

What is...

[Ben Roberts] 15:53:23

Yep.

[Wycliffe Bahati] 15:53:26

Yeah, so I think I've covered most of the... the topics that I wanted to cover.

[Wycliffe Bahati] 15:53:38

I guess unless you have something you can to add in terms of What are the biggest uncertainties you think of.

[Wycliffe Bahati] 15:53:51

Facing this LEO providers for long-term success.

[Ben Roberts] 15:53:55

Yet to be proven if the business model is fireball yeah and it's yet to be proven. And much as Much as Starlink and Elon Musk and all these companies you know i i sawing high at the moment.

[Ben Roberts] 15:54:10

You know, there might be throwing too high, right? So if you think about the share price of Tesla.

[Ben Roberts] 15:54:16

The share price of Tesla was way, way, way higher than Toyota.

[Ben Roberts] 15:54:20

Toyota's revenue is far greater. They make far more cars. But Tesla's market cap is like 20 times Toyota.

[Ben Roberts] 15:54:29

So you've got something that's overvalued. Probably, right?

[Wycliffe Bahati] 15:54:33

Most likely, yeah.

[Ben Roberts] 15:54:34

Yeah. Now, now, you know, now you've got a company that makes cars and is overvalued because of the cult of Elon Musk I'm saying that SpaceX is possibly in the same and Twitter and other things. You've got things that are

[Ben Roberts] 15:54:49

Basically being overvalued. So if Elon Musk dies tomorrow What happens to that business empire?

[Wycliffe Bahati] 15:54:56

Oh yeah it would be a challenge. I mean, the other thing I'm wondering is, I mean, if you look at the sustainable development goals There's a lot of hope.

[Wycliffe Bahati] 15:55:15

I mean, that the the small percentage that totally do not have internet. But my worry is that These people, some of them even just the digital literacy you know needs to be addressed because you could have the satellites flying around and be able to provide the internet but

[Wycliffe Bahati] 15:55:39

If you don't have the literacy to be able to use the service could also be a big factor.

[Ben Roberts] 15:55:46

Yeah, I mean, you're saying that already. As I say, we've got 97 4G coverage in Kenya and not everybody's using the internet right you know the the When the 2019 census asked people, how many internet users in your house

[Wycliffe Bahati] 15:55:54

Okay.

[Ben Roberts] 15:56:00

And it was less than 40% of people, you know, households said they were internet users.

[Wycliffe Bahati] 15:56:08

Yeah, so it's bigger challenge there. Oh, okay. So I think that's it.

[Ben Roberts] 15:56:10

Yeah.

[Wycliffe Bahati] 15:56:16

Once I've done everything, I'll be able to share with you my findings.

[Ben Roberts] 15:56:22

Okay. No worries. Good luck with your thesis then, Wycliffe.

[Wycliffe Bahati] 15:56:22

Oh. But to have...

[Ben Roberts] 15:56:29

Yeah, just let me know if you've got any other questions.

[Wycliffe Bahati] 15:56:31

Thank you.

[Ben Roberts] 15:56:33

All right. Take care.

Participant 3 BM

Name Of Participant: Bhavesh Mistry

Profession: Former Expert in broadband services

Date of interview: 03 March 2025

[Wycliffe Bahati] 14:05:13

Okay, so I think we're gonna start. I will just try to be as brief as possible because i think we have some quite a bit. So the first one is um One of the themes I identified is the on market

[Wycliffe Bahati] 14:05:28

Dynamics and competition landscape So I don't know how you feel or think the leo broadband will affect the market and in the broadband sector. In fact, in terms of competition.

[Bhavesh Mistry] 14:05:44

Sure. Yeah, so I think I'll give you my view. And I've already worked with a number of ISPs that are using the Leo solutions like Starlink and others. So I think The LEO satellites bring a good connectivity option at a very reasonable price

[Bhavesh Mistry] 14:06:03

For many parts of the market yeah so about urban and rural. I think in urban areas it definitely increases competition for the ISPs because you now have a service that someone can easily connect to because you just put an antenna on your roof you don't need to

[Bhavesh Mistry] 14:06:20

Wait for ISP to connect you. And you can get a pretty decent low latency, good bandwidth service.

[Bhavesh Mistry] 14:06:27

At a reasonable price. So for example, in Kenya, the pricing is very competitive to what ISPs would offer for a fixed or a or affixed to wireless or a fixed fiber kind of service.

[Bhavesh Mistry] 14:06:39

So in the rural scenario where people can forward connectivity. When I say afford i'm saying anything like \$15, \$20 a month for a service it definitely increases competition.

[Bhavesh Mistry] 14:06:52

For rural areas where in low income places where people cannot afford a \$20 service and would be struggling with a five dollar or an \$8 monthly service then Leo might not be a solution for them still because the pricing of the Leo solution is still a bit higher.

[Wycliffe Bahati] 14:06:52

Okay.

[Bhavesh Mistry] 14:07:08

So it wouldn't really increase competition for the really low end of the market in urban areas.

[Bhavesh Mistry] 14:07:14

And then on the rural areas, I think one of the advantages of LEO, of course, is that you have a huge coverage footprint.

[Bhavesh Mistry] 14:07:20

So you can connect a very large region very quickly And that is one of the differences between your traditional ISP so i think in the rural areas if someone can afford the service then there is definitely they will definitely be taking a lot away from your traditional isps because

[Bhavesh Mistry] 14:07:39

For traditional ISPs to cover rural areas, the cost per connection is very high.

[Bhavesh Mistry] 14:07:43

Because of the distance from the closest pop or tower But again, if in rural areas you're looking at people who do not or cannot afford a 10, \$15, \$20 service, then you still have to depend on either mobile broadband

[Bhavesh Mistry] 14:07:58

Or some traditional ISP in that rural scenario.

[Wycliffe Bahati] 14:08:02

Okay. And... How do you think the traditional ISPs are adapting to this new entry.

[Bhavesh Mistry] 14:08:12

Yeah, it's interesting. So I've seen two different, I've just seen two different ways of how things are working out so one Of course, there are those ISPs that are seeing LEO solutions. Let me use Starlink as an example as a competition.

[Bhavesh Mistry] 14:08:25

And what they are trying to do then is to try and either lower pricing or offer other incentives to the customer that styling cannot offer.

[Bhavesh Mistry] 14:08:34

So for example, zero installation cost because they're selling has an antenna that you have to install and that costs money but some of these ISPs are saying, okay, I'll install the service for you for free so you don't have any upfront cost.

[Bhavesh Mistry] 14:08:46

And then my monthly service is going to be similar to or maybe slightly lower than what Starlink is offering.

[Bhavesh Mistry] 14:08:53

So I've seen that happening. It is possible for ISPs that are a little bit more well-funded because you know to pay to give someone a free installation means you have to absorb that cost.

[Bhavesh Mistry] 14:09:05

And you need to have the funding to do that you know installation can easily cost you know can easily cost 80 60 to \$80 per home, depending on where the location is.

[Bhavesh Mistry] 14:09:15

And then I've seen the other ISPs who are actually working with Starlink. So one of the things that I've seen with ISPs doing is, and I have actually almost three of them now.

[Bhavesh Mistry] 14:09:26

They're actually using the styling connection, the Leo connection as an upstream.

[Bhavesh Mistry] 14:09:30

To serve their customers. So what they will do is they will say okay my customers cannot afford to individually get a Starlink connection because the cost of the installation is too high or the monthly cost is too high but i can share

[Bhavesh Mistry] 14:09:44

A Starlink link to my customer. So they will take a Starlink connection put it in their pop and they will do some kind of Wi-Fi or sort of low cost internet connection and start offering services that are actually using the styling uplink.

[Bhavesh Mistry] 14:09:58

The styling antenna. So I have, for example, I'll give you a specific example in kakuma in where the refugee camp is i have an ISP that's doing that is doing In Kakuma it is very difficult to get any kind of internet. I mean, the Safariacom service is there, but it's not that great

[Bhavesh Mistry] 14:10:15

So Starlink is the option and this is what the ISPs are doing.

[Wycliffe Bahati] 14:10:21

Okay, so the uplink rooster link and Okay.

[Bhavesh Mistry] 14:10:25

Yes. They then provide internet. It's like a sambasa. You know, the sambasas here in Kenya where they buy a safaricom link and then they provide this as internet. So that's what some of these folks are doing. They're taking a Starlink link and then providing the service to individual homes

[Wycliffe Bahati] 14:10:34

Oh.

[Bhavesh Mistry] 14:10:41

As a local sambasa, yeah.

[Wycliffe Bahati] 14:10:41

Okay. Okay, get it. Okay. And have you, I mean... What's the perception of... customers in the difference between the two services.

[Wycliffe Bahati] 14:10:58

Leo broadband and traditional. I don't know if you've interacted with any customers

[Bhavesh Mistry] 14:11:02

Yeah, I have. I have people I know who are using Leo broadband services And they're very, very happy with it because they are very happy with they have traditionally have trouble with Safari Chrome or Jami Telekom or someone else who's providing a terrestrial connection.

[Bhavesh Mistry] 14:11:17

And they are people who need internet all the time. So, you know, nowadays a lot of people work from home.

[Bhavesh Mistry] 14:11:22

There's a lot of people doing gig work, online work, and you need a constant internet service that works.

[Bhavesh Mistry] 14:11:29

The problem still with Safari Chrome today, despite everything they've done is you still have areas even in Nairobi where connectivity is a problem. Outages are too many you know all those kinds of things. Same with Jamieta, same with Sukuk.

[Bhavesh Mistry] 14:11:41

So there are a lot of customers I know in urban areas are very happy with the Starlink connection.

[Bhavesh Mistry] 14:11:47

Because they understand that maybe during heavy rain or there might be some interruptions but generally It's a far more reliable and faster service. Faster meaning 100 Mbps so you can stream HD content, you know, you don't have to worry about

[Bhavesh Mistry] 14:12:03

When your neighbor comes on board online, then your service becomes slow, which is very common with Zuku.

[Bhavesh Mistry] 14:12:08

If you're in a neighborhood, once your neighbors are back home, everyone's service goes down in speed. So none of those issues so i have come across people who are very happy.

[Bhavesh Mistry] 14:12:17

The LEO services that they've experienced so far.

[Wycliffe Bahati] 14:12:22

Okay, so the You just mentioned this part where when people come back home, the service goes down

[Wycliffe Bahati] 14:12:33

The traditional ISPs feel pressure now to deliver on a contracted bandwidth services?

[Bhavesh Mistry] 14:12:39

Of course, they feel too precious and not only traditional ISPC, even likes of Safaricom and Jami. So because what has happened is what the fixed providers have been able to provide is what people have sort of accepting because you don't get anything better. So for example.

[Bhavesh Mistry] 14:12:53

Safaricom sells you a 20 Mbps service But it's shared one to four. So you know that although it is at 20 mbps you could get maybe 5 mbps in case everyone that you are sharing with us on at the same time.

[Bhavesh Mistry] 14:13:07

That is being challenged. So when Stalin comes and says, hey, I'll give you 100 mbps service to your house.

[Bhavesh Mistry] 14:13:13

And you know you get 100 mbps the pressure on the ISPs and the service providers is you either have to up your game, meaning you have to deliver more bandwidth.

[Bhavesh Mistry] 14:13:22

To these homes and make sure that you are meeting the the offering you know just uh saying oh i do 20, but you're actually giving five So there's a lot of pressure there and this is why you see For example, Safaricom has doubled their bandwidth for most of their customers at the same cost.

[Bhavesh Mistry] 14:13:38

So for example, if you are giving 20, they're now saying we'll pay, we'll give you 40 Mbps by the same price And they're being challenged to keep that bandwidth at 40.

[Bhavesh Mistry] 14:13:47

Because they know people have an option with Starlink.

[Wycliffe Bahati] 14:13:53

And when is... When is it that a customer decides Okay, I will... it's not viable for Starlink. I have to keep my fiber connection Or how... When fiber?

[Wycliffe Bahati] 14:14:10

Having an upper hand. In this type of in this competition.

[Bhavesh Mistry] 14:14:14

Look, I think it's an interesting question. Let me talk about two types of customers so If you're an enterprise where you need an enterprise high bandwidth, high capacity, good quality of service, etc.

[Bhavesh Mistry] 14:14:29

Then you can still go to the traditional Safaricom and get an enterprise package. And you know, the enterprise package is way more expensive but it comes with a better quality it comes with a better customer service etc so if you're an enterprise

[Bhavesh Mistry] 14:14:44

For which internet is mission critical or business critical, then I think they are still remaining.

[Bhavesh Mistry] 14:14:49

With the traditional fiber operators, they can pay for fiber to their premise and all of that stuff.

[Bhavesh Mistry] 14:14:54

Those are not the people who are going to Starlink. The people who are going to Starlink are either individual homes, high-end individual homes or small businesses you know one or 2% businesses that Safaricom is not able to offer anything competitive. They are the ones who are going to Starlink.

[Bhavesh Mistry] 14:15:11

And they have been, I mean, the decision really comes down to price yeah because you are in a competitive market. So someone will say, look.

[Wycliffe Bahati] 14:15:12

Okay.

[Bhavesh Mistry] 14:15:18

I can get a 4,000 shilling service from Safaricom in Kenya shillings I can get a 3000 shilling service from Starlink, or I can get a 4,000 shilling service from Starlink that is way more higher capacity you'll go for it. So a lot of the decisions around speed and capacity

[Bhavesh Mistry] 14:15:36

And of course, a lot of people have bad experience with Safari Chrome or with Jeremy, which is also a factor that plays in.

[Bhavesh Mistry] 14:15:42

So I have a neighbor who just stays down the road and they say safari come here, but he's had a lot of problems with Safari come in terms of service uptime that he just decided, you know what, I'll just put starling

[Bhavesh Mistry] 14:15:52

And I don't need a Safari Chrome service anymore. Enterprise. So it all depends. I mean, but price is a factor and quality of service is a factor

[Wycliffe Bahati] 14:16:01

So what I'm getting here is if you can afford it, then Starlink is much better than traditional so yeah so it's

[Bhavesh Mistry] 14:16:09

Yes. Yes. Correct. And you have to understand Starling's journey also. So when they launched about a year and a half ago.

[Bhavesh Mistry] 14:16:16

The pricing was very high it was almost very high. Twice of what safaricum offers they have successfully managed to lower the price to attract customers.

[Bhavesh Mistry] 14:16:25

Both on the service, on the monthly service and even the equipment. So now the equipment is much lower. So that is what changed the dynamic. So when they lowered the price.

[Bhavesh Mistry] 14:16:33

To be closer to what the traditional ISPs were offering, then they saw this huge uptick in services because people were willing to now make that shift People don't have more money to pay, but for the same money, if you give a better service and better speed

[Bhavesh Mistry] 14:16:49

They'll move.

[Wycliffe Bahati] 14:16:51

Okay. All right. Then I think we can... move to the next topic, which is the service quality and the technological challenges.

[Wycliffe Bahati] 14:17:02

So on this part In terms of performance In this case, I think we can... talk about styling because it's mostly now the one with the majority So, um.

[Bhavesh Mistry] 14:17:16

Yes.

[Wycliffe Bahati] 14:17:19

Compare speed, latency, and reliability. Okay, we've talked about reliability. Maybe the speed and latency.

[Bhavesh Mistry] 14:17:27

Yes, yes, sure. Go back to the traditional, yes.

[Wycliffe Bahati] 14:17:28

Compared to one the conditional SPZ and 2, the original geostationary broadband providers.

[Bhavesh Mistry] 14:17:38

Yes, sure. Yeah, so I think, look, so latency we are looking of So if you are sitting at home with a fiber connection from a traditional ISP, your latency to the internet. I'm not talking locally, not to your local exchange point, of course, to your local exchange point will be a very low latency because you're going to a

[Bhavesh Mistry] 14:17:56

A point of presence in Nairobi or within the country but your your latency to say a point of presence in London where you're connecting to the internet is almost the same.

[Bhavesh Mistry] 14:18:08

Or maybe even slightly higher than what Starlink can offer you.

[Bhavesh Mistry] 14:18:12

So from a latency point of view, it is actually very competitive because the quality of service you'll get and everything is very competitive to what you would get from a fiber-based service provider.

[Bhavesh Mistry] 14:18:24

When you talk about... When you talk about when you talk about Yeah.

[Wycliffe Bahati] 14:18:30

The geo provide the ones who have been traditionally providing broadband wedge Okay.

[Bhavesh Mistry] 14:18:34

I'll come to geostationary. I'll come to geostationary. But when you talk about speed between the two.

[Bhavesh Mistry] 14:18:39

Then you know Starlink is able to offer very high speed packages, 20, 30, 50, 100 Mbps 200 mbps so that is something that is unheard of in the residential market in Kenya, for example.

[Bhavesh Mistry] 14:18:52

What I'm seeing is people are used to buying 20, 30 MBPS from the fiber operators and suddenly you have this 100 mbps package which is similarly priced. So you find people are willing to say, oh, by the way, I can have blistering fast speeds. Maybe I can watch HD content. So that has shifted the game a little bit. So forcing the

[Bhavesh Mistry] 14:19:09

Traditional ISPs to offer more. But I think that that advantage of the speed has been one of the key advantages for the Leo guys.

[Bhavesh Mistry] 14:19:16

And how comparing to fiber. Now going to geostationary, I mean, look, geostationary latencies are very, very high almost to the point where you really can't do any streaming you only use geostationary if you really have no other option.

[Wycliffe Bahati] 14:19:17

Okay.

[Bhavesh Mistry] 14:19:29

You know 300 400, 500 milliseconds to the internet enterprises were not really happy with geostationary because the latency is too high for enterprise applications.

[Bhavesh Mistry] 14:19:38

So for those people who move to Starling because of latency, there is a massive difference, like a fifth you know a fifth of a fifth of what they were experiencing before It's so much, so much faster.

[Bhavesh Mistry] 14:19:50

And same with the speed. I mean, geostationary could give you massive amounts of bandwidth, of course.

[Bhavesh Mistry] 14:19:56

Gigs maybe sometimes, but it was mostly being used where you don't have any option. I think styling definitely the leo definitely takes away most of what geostationary was offering from that point of view.

[Wycliffe Bahati] 14:20:11

So I think I'll combine the next three. One is... What technical challenges do you see standing half to deliver the quality services that we're just mentioning.

[Wycliffe Bahati] 14:20:29

And... what would make it a mainstream?

[Wycliffe Bahati] 14:20:35

Broadband service provider.

[Bhavesh Mistry] 14:20:38

Yes, I think there are multiple challenges. So of course, before I come technical you know there's all this regulatory problems, you know, that for every country that Starlink wants to offer the local regulator wants them to register.

[Bhavesh Mistry] 14:20:49

Directly or through a partner in the country because you're beaming internet into the country. You have end users on your service so that is there and I think they're slowly you've seen some countries in Africa have approved, some are still pushing back

[Bhavesh Mistry] 14:21:01

It's going to take a while, but I know they will get there. They have enough muscle and enough influence to push their way through across all the regulators in Africa and get it done. So I think that is a challenge, but I think they've been pretty successful with it, especially in Africa.

[Bhavesh Mistry] 14:21:16

When you look about challenges to do with capacity and reliability and quality and all of those things, technical ones.

[Wycliffe Bahati] 14:21:17

Mm-hmm.

[Bhavesh Mistry] 14:21:24

There have been two challenges I see with Starlink or with leo today. So one is of course the initial upfront cost. So it's a massive investment. Of course, Starlink, we're able to put that money in it's a massive investment.

[Bhavesh Mistry] 14:21:39

The antennas that sit on people's rooftops are not cheap, but of course they're subsidizing it. So there is a bit of sort of funding requirement that they need.

[Bhavesh Mistry] 14:21:49

If they want to be successful as a leo provider. And then the other one which was experienced recently by Starlink in Kenya was they had to pause new connections for a period of, I think, two, three months because they had so much demand.

[Bhavesh Mistry] 14:22:03

And they did not have enough capacity on their satellites to cover this area.

[Bhavesh Mistry] 14:22:08

So that means, of course, they have to keep on adding more satellites to cover a particular geographical area because they have so much demand.

[Bhavesh Mistry] 14:22:15

And this is the nature of how Leo works because you know the leo footprint is much smaller than a geo footprint.

[Bhavesh Mistry] 14:22:20

Coverage wise. So we see if geo was covering one particular region, you will need at least 10 Leo satellites to cover the same region.

[Bhavesh Mistry] 14:22:27

With equivalent capacity you know so so there is a huge capacity demand in some areas. They've had to put in more satellites to do that.

[Bhavesh Mistry] 14:22:35

And that has been a technical challenge for them, which means they had to pause new connections for a while, which is not a very good experience.

[Bhavesh Mistry] 14:22:42

Because customers instantly start saying, oh, by the way. This is not what we thought it was. Why are we being asked to pause you know things like that there to explain it.

[Bhavesh Mistry] 14:22:51

But the biggest technical challenge i believe for Leo is going to be that the satellite has a limited life of five years.

[Bhavesh Mistry] 14:22:58

Or soak. What that means is every five years or six years or the satellite will fall off the sky And a new one has to be put in.

[Wycliffe Bahati] 14:22:59

Mm-hmm.

[Bhavesh Mistry] 14:23:07

So you can imagine the complexity and the cost of always refreshing your satellites every five to six years is a massive cost.

[Bhavesh Mistry] 14:23:17

And that is definitely going to hurt them in the long term. That's why right now Starlink is okay. They have a lot of money. They've got a lot of customers, but I think longer term how they operate and how they manage the costs.

[Wycliffe Bahati] 14:23:18

Yeah.

[Bhavesh Mistry] 14:23:30

Of course, they are pushing technology. Maybe the satellites will last longer and all of that stuff.

[Bhavesh Mistry] 14:23:35

But that is going to be a big technical challenge. Now, on the positive side, one of the key technical obstacles that they managed to succeed in very well is how to install a dish.

[Bhavesh Mistry] 14:23:47

I don't know if you've experienced this, but it is just so easy. They've made it so easy for any person to go and install a small styling dish on the roof they have an application they have an app on the phone, they have a tool. The instructions are so clear.

[Bhavesh Mistry] 14:24:03

That they have enabled people to automatic. You don't need a technician. You don't need IT knowledge. You don't need anything. You can follow the instructions.

[Wycliffe Bahati] 14:24:04

Mm-hmm.

[Bhavesh Mistry] 14:24:11

And you can get a service working. And that, I think, has been one of the key barriers that they have really addressed on the technical side.

[Bhavesh Mistry] 14:24:19

To really get customers on board and i give them The hats off to them for doing that.

[Wycliffe Bahati] 14:24:25

Okay. And do you think that um as a... space is now structured. It looks like this leo providers are reoperating almost independently Most technologies have benefited a lot from standardization.

[Wycliffe Bahati] 14:24:49

You know the organization like the 3GPP coming up with the standardization so you interpretability comes in handy And then for, for example, you just mentioned about the five-year lifespan of satellite.

[Bhavesh Mistry] 14:24:49

You know. Yes.

[Wycliffe Bahati] 14:25:10

So do you think that maybe in the future the standardizing some things would make it easier for them to operate was feeling like they intercept like Lee.

[Bhavesh Mistry] 14:25:27

Yes, I think it is definitely going to be required. It will come. So if you have remember that styling styling sorry sorry you got cut off, yeah.

[Wycliffe Bahati] 14:25:28

Things and things like that.

[Wycliffe Bahati] 14:25:34

Oh, you had me. Okay, okay, okay, I can hear you.

[Bhavesh Mistry] 14:25:35

Yeah, I do, yeah. You can hear me? Yeah, I was saying standardization is definitely going to be required and it will come. You have to remember that Starlink is a pioneer. So they are the first ones to really put

[Bhavesh Mistry] 14:25:47

A constellation of LEO satellites out there They are doing stuff that no one else has done. But I think if you think about now Amazon Squeeper project is supposed to go live and others in future who might want to do this.

[Bhavesh Mistry] 14:25:59

Ast and others, they will definitely have to be some standardization, some regulation, some coordination because you can imagine You have hundreds of satellites from Starlink and then you have hundreds of satellites from Kuiper.

[Bhavesh Mistry] 14:26:12

And all of these things are moving around on the low Earth orbit and imagine the chances of collisions and debris falling and all those things. So I think it's definitely going to be required.

[Bhavesh Mistry] 14:26:23

And it will most probably have to be done by agencies that control space together with regulators on land because there's also the interference element that you mentioned you know So I think it's going to come. It just hasn't yet because everyone is like

[Bhavesh Mistry] 14:26:39

New to how this is working. Today we only have styling out there, so it's not really a big issue But it will have to come. That's a given. I don't think we're going to be And whether it's going to be space agencies or

[Bhavesh Mistry] 14:26:53

Bodies like itu or 3GPP? I don't know to be honest.

[Bhavesh Mistry] 14:26:58

Because, you know, it's not terrestrial telecom so it's not the domain of IT and others. But who knows, maybe they'll work together to create some policies and regulations for LEO systems.

[Wycliffe Bahati] 14:27:03

Yeah.

[Wycliffe Bahati] 14:27:12

Yeah, and... I'd like you to talk about what you see in terms of return of investments. I know you've mentioned the challenge of the five-year But I know you have experience a lot in network design and we kind of business cases

[Wycliffe Bahati] 14:27:31

Do you think that... they can sustain it.

[Bhavesh Mistry] 14:27:37

I think it's, look, it's difficult to say. I think that's still a an open question because it's not been around long enough and because it's cutting edge technology no one has deployed anything like this before. So it's very difficult to compare

[Bhavesh Mistry] 14:27:51

And say the ROI is going to be this ROI is going to be that. I'm sure they have their own internal numbers.

[Bhavesh Mistry] 14:27:56

That they have worked out. I don't know. Longer term, I still believe that fiber optics is still your cheapest option to deliver any gigabyte of data to a consumer or an enterprise however That only makes sense dense urban areas where you have a lot of it and

[Wycliffe Bahati] 14:28:10

Mm-hmm.

[Bhavesh Mistry] 14:28:19

There are also challenges in that case. So I think light, using light is really the cheapest medium.

[Bhavesh Mistry] 14:28:26

To deliver data, radio frequency or RF comes second in that case. And of course, Talink also uses some type of RF frequencies to reach the antennas on ground, et cetera.

[Bhavesh Mistry] 14:28:36

So I don't think Leo or satellite systems will get to that price point.

[Bhavesh Mistry] 14:28:41

And therefore will not get to the ROI. That you have with fiber today.

[Bhavesh Mistry] 14:28:46

But they are solving for an important element of places where you cannot get fiber, places where are remote and rural places in disasters. So I think it's a complementary to it should be seen long term as complementary to fiber optics and others.

[Bhavesh Mistry] 14:29:01

But I don't think, I personally don't think that their roi would be closer to a dense fiber network from fixed

operator in a particular market.

[Wycliffe Bahati] 14:29:12

Yeah, and this is a bit deeper in terms of technical You know that you know that Over the years we've seen... with CDNs or content distributors get strategically relocate their services to where users are. For example, now the Google cache in

[Bhavesh Mistry] 14:29:33

Yes. Yes. In the data center, yes, correct.

[Wycliffe Bahati] 14:29:37

That's Amazon. Dns. So you have all these services located everywhere okay So one, I would like you to address um this If you think about this situation and the capacities that are required to access some of these services.

[Bhavesh Mistry] 14:29:38

Yes.

[Wycliffe Bahati] 14:29:59

Vis-a-vis latency and then Well, on that you can look at how internet services are now designed in a way that routine is basically almost geolocated. So if you find you're coming from an IP address in africa you find you are likely to be rooted to

[Wycliffe Bahati] 14:30:24

Maybe South Africa, right? How, how... Yes.

[Bhavesh Mistry] 14:30:28

Yeah, so I'll tell you something that's very interesting. I don't know if many people know this yet, but Starlink themselves have already started putting earth stations or ground stations in specific countries in africa and are starting to do cross-connects to the content providers, so the CDNs.

[Bhavesh Mistry] 14:30:47

So for example, in Kenya, Starlink is already in the process of setting up a Earth station.

[Bhavesh Mistry] 14:30:52

In Nairobi and that will connect it will be connected directly to a the CDNs in the data centers.

[Bhavesh Mistry] 14:30:59

And so you can see that the direction they're going is basically what everyone does when it comes to routing traffic and serving content, which is You still have to be as close to the CDN providers as you can.

[Wycliffe Bahati] 14:31:00

Okay.

[Bhavesh Mistry] 14:31:11

And the only way to do that is to have a ground station or earth station in that country.

[Bhavesh Mistry] 14:31:15

So eventually think of this, you'll have this constellation around the earth but at so many different points, you'll have ground stations where the traffic is going.

[Bhavesh Mistry] 14:31:23

And connecting to the CDN locally. To interconnect to serve content and caching data. So it won't all be like you're going to a place in Europe to get the data. There are being forced to put those stations in in-country or in region because they know that the latency and the caching

[Bhavesh Mistry] 14:31:40

Is going to be important. So from an internet routing, I'll be very honest, I don't see this any different from your traditional networks where you have fiber networks, you have your access and your aggregation and your core and then you have these interconnects with other providers and interconnects with CDNs.

[Bhavesh Mistry] 14:31:57

It's the same. It's just that there's a satellite layer, there's a LEO satellite layer, but that's still coming to ground stations to connect to the same CTNs.

[Wycliffe Bahati] 14:32:06

Okay, so actually you brought up the ground station um Now, like you said, the regulators Now they want to license basically these services in each and every country, right?

[Bhavesh Mistry] 14:32:24

Yes. Yes.

[Wycliffe Bahati] 14:32:26

And there's the issue of controlling the internet. I think now a lot of governments are becoming a bit jittery, but especially social media And so there's that aspect of trying to control the internet.

[Bhavesh Mistry] 14:32:38

Yes.

[Wycliffe Bahati] 14:32:44

And from wondering. I'm not sure, but I think the ground stations are not very cheap.

[Wycliffe Bahati] 14:32:49

To deploy so In terms of regulatory hurdles, I mean, how will Starlink adjust to this because Uganda will not be very happy that they are only having the ground station in kenya And they cannot control the internet the way they want.

[Bhavesh Mistry] 14:32:50

Correct. Correct. That's easy.

[Wycliffe Bahati] 14:33:11

So will it be like they have to put in train stations everywhere where they, I don't know if that would be economical.

[Bhavesh Mistry] 14:33:20

No, I think the model will be very similar to what Facebook and Google have done. So basically, you don't have a cache everywhere, but you have a cache close to a submarine cable.

[Bhavesh Mistry] 14:33:29

That is in close proximity to that country. So today, for example, Google doesn't have a cache in Uganda, but has it in Kenya.

[Bhavesh Mistry] 14:33:35

So for those people who are serving traffic in Uganda get their cached content from Kenya. So I think the model is very similar where Starlink will decide which are the main or most important points where they will need to put a ground station to connect to the content delivery networks.

[Bhavesh Mistry] 14:33:50

But this is why I said earlier that The regulators are forcing Starlink to either register locally in each country or to find a partner in each country through which they can provide service. And the reason they are doing this is primarily

[Bhavesh Mistry] 14:34:04

To be able to shut off the internet when they want to.

[Bhavesh Mistry] 14:34:07

So today also, this is why Starlink, for example, has to registered in Kenya and South Africa and all these countries because part of that regulation is that if the regulator says turn off the internet, they're able to do that because they have a license. Without the license, they can't operate.

[Wycliffe Bahati] 14:34:08

Mm-hmm.

[Bhavesh Mistry] 14:34:23

The regulators will anyway get their way because they will force all the Leos to register in the country through directly or indirectly which is how they will enforce any shutting down.

[Bhavesh Mistry] 14:34:35

And I don't think it is dependent on where the caches or where the ground station is. Now, of course, a person in Uganda might say, oh, I'm not happy that this is sitting in Kenya. But the question is if they experience a service that is good enough

[Bhavesh Mistry] 14:34:49

Do they really do they really care that this is in Kenya? Not really.

[Bhavesh Mistry] 14:34:55

The case is whether they can enforce shutdown. I think they can do that through the license they've given styling to operate in that country.

[Wycliffe Bahati] 14:35:03

So actually, I've seen a lawyer argue that um countries should not be very concerned of where they are The ground station they should be concerned about if they can control it

[Bhavesh Mistry] 14:35:19

Well, look, I don't know. Look, as I said, for the regulator, I know regulators typically want to be able to shut down internet in a country And the way they do that is by licensing operators in that country because they can tell the operator

[Bhavesh Mistry] 14:35:32

I'm giving you a notice to shut down otherwise you'll be penalized or whatever And that can still be done with the Leos because the Leos are forced to register in the country.

[Wycliffe Bahati] 14:35:43

Okay.

[Bhavesh Mistry] 14:35:43

Because you can imagine, let's say there's a situation in Uganda where the regulator asks Starlink to shut down the service and they don't.

[Bhavesh Mistry] 14:35:51

Once everything is settled down, say the situation or the event has passed, they will be able to withdraw the license for Starlink and make it an illegal operator.

[Bhavesh Mistry] 14:36:01

You see, so I think there is still enforcement through licensing which they're able to do.

[Wycliffe Bahati] 14:36:07

Okay yeah so uh okay fine so just as we conclude um Where do you see maybe next five years?

[Wycliffe Bahati] 14:36:18

How will this play out?

[Bhavesh Mistry] 14:36:20

Yeah, so look, I'm waiting for Kuiper to launch. I actually, it's very interesting. I have a former colleague who I was working with who actually moved to the Amazon Kipo project in the US.

[Bhavesh Mistry] 14:36:30

And I know that we know that he moved there because they were sort of gearing for launch and all that stuff. So I am waiting for it because I think Coper's launch will bring some good competition.

[Bhavesh Mistry] 14:36:40

To Starlink, but also I think it will help stabilize the whole ecosystem of Leo's like what will happen to regulation how pricing will be thought about because it's always good to have competition. It's very good for the consumer.

[Bhavesh Mistry] 14:36:53

And I think I'm waiting for Quipa, I think, to launch and be able to offer some competitive services.

[Bhavesh Mistry] 14:36:59

What is also interesting is going to be I have heard some rumors, which I don't know if they are true

is how Amazon is going to bundle internet with other things because quipa is amazon project and that will be very interesting to see

[Bhavesh Mistry] 14:37:14

That you now start seeing internet being just something that's bundled with some other services that you're being given.

[Bhavesh Mistry] 14:37:21

And not just internet for the sake of it being internet.

[Bhavesh Mistry] 14:37:23

You know, like that. You know, Amazon has a prime video service. Amazon has shopping services so That combination, I think, will bring some good innovation in that sector maybe good value to customers. And then it'd be interesting to see. I'm excited to see

[Wycliffe Bahati] 14:37:25

Okay. Okay.

[Bhavesh Mistry] 14:37:39

How this plays out over the next five years.

[Wycliffe Bahati] 14:37:43

Okay. And do you see, because one of the

[Wycliffe Bahati] 14:37:50

The points which Leo has been, in fact, if you look at uh starlink and most of them base their main business strategy is to main target areas that have not been reached by internet.

[Wycliffe Bahati] 14:38:10

Do you see... it's contributing to the digital bridging the digital gap to those areas.

[Bhavesh Mistry] 14:38:18

Yes. So I think the answer is yes, but there's still a way to go.

[Bhavesh Mistry] 14:38:21

I'll say yes, because I think Starlink is now able to deliver in the digitally in the unconnected and underconnected areas with a good enough service that is very similar to internet service provider or mobile network. I think the traditional geo couldn't do that.

[Bhavesht Mistry] 14:38:39

Styling can you know it's low latency high speed etc So that one part they'll definitely solve. Secondly, as I said, styling is very easy to install.

[Bhavesht Mistry] 14:38:47

So again, for places where have traditionally been starved of the internet, this is a very good option.

[Bhavesht Mistry] 14:38:52

So those two fronts they have managed i think The thing that will still remain a challenge for them is the affordability element.

[Bhavesht Mistry] 14:39:01

Like, are they able to make the price of the service low enough for someone in the rural areas of those people who are under-connected, be able to afford that service. And I think that is still a challenge today.

[Bhavesht Mistry] 14:39:14

Even what they have today is not low enough. And this is just the nature of affordability in our markets yeah So I think that is long term to be seen. And, you know, it's one thing to lower pricing and it's the second thing to keep prices low.

[Bhavesht Mistry] 14:39:28

In the long term. You can lower prices because you want market share, but can you sustain that going back to the ROI discussion?

[Bhavesht Mistry] 14:39:35

Can you sustain that? I think that is where the real test comes. So I think, yes, they have the ability to bridge the digital divide and bring something that people who are not able to get before. But I think it is yet to be seen whether it's going to be

[Bhavesht Mistry] 14:39:49

At the price point that people can afford in those areas. And second, longer term they can sustain that.

[Wycliffe Bahati] 14:39:56

And would you see any major collaborations either between traditional providers and leo providers happening in the future.

[Bhavesht Mistry] 14:40:07

Yeah, so we've seen a bit of it or at least I've seen a bit of it. As I said, the smaller ISPs have I've embraced Starlink and I've tried to use it and tried to use it as a backhaul option instead of just seeing it as a competition.

[Bhavesh Mistry] 14:40:22

But I don't see the big boys. I talk the big talcos. I don't see them really working with Starling because they see it purely as competition today.

[Bhavesh Mistry] 14:40:30

Unless it's something, unless it's a place where they are not able to provide they are not able to serve with geo today and they can use telling. So I'll give you an example. Parts of DRC today There's still remote parts of DRC that mobile operators towers are being backhauled using geo.

[Bhavesh Mistry] 14:40:46

Solutions back to their main centers yeah like northern DRC and others. So I think those are places where Starling can definitely bring a big change because lower latency and higher speed would be great for those towers to be connected back.

[Bhavesh Mistry] 14:40:58

So that area I see collaboration but then of course you see competition on the more urban areas where they're seeing Starlink taking away some of their customers on fixed services.

[Wycliffe Bahati] 14:41:11

Okay so And finally, I think the everything is frame it.

[Bhavesh Mistry] 14:41:21

Yes.

[Wycliffe Bahati] 14:41:22

Okay, there's a debate on how much of this settlers can fit up there.

[Wycliffe Bahati] 14:41:28

So, I mean, and we know that, like you said, FIPI is still there In terms of delivering capacity Because, I mean, in a few years ago you would not imagine leaving Taiwan MB connection at home But now people are delivering up to one gig. In some countries, people are talking about 10 gig.

[Bhavesh Mistry] 14:41:37

Yes.

[Bhavesh Mistry] 14:41:52

Yes, yes, correct.

[Wycliffe Bahati] 14:41:53

So... I mean, and technology changes very fast because now I've seen the fiber manufacturers are now... the glowing 10G equipment almost to next to a home I mean, and local.

[Bhavesh Mistry] 14:42:09

Yes, yes.

[Wycliffe Bahati] 14:42:14

The consumer demands also changes. So especially with AI, you know, there's a lot of You could go bandwidth consumption with these AI applications.

[Wycliffe Bahati] 14:42:25

So in the near future. How do you look at the kind of availability of the space and how much surplus can fit in there plus the capacity that will be demanded by the consumer.

[Bhavesh Mistry] 14:42:43

Yeah, so I think, look, it's a very valid statement. There's limited amount of space. You cannot keep on pumping satellites up there because they will eventually collide or have problems. But you also have to remember On the other side, technology keeps on improving which means

[Bhavesh Mistry] 14:42:57

I would expect the Starlink satellites will keep on getting smaller, they'll keep on living longer They will also be integrating new technologies to do more capacity. So I think it's really, it's going to go in both directions. Yes, there's limited space, but

[Bhavesh Mistry] 14:43:12

As long as the LEO companies are innovating on the satellite footprints and size and life and everything, I think it will continue serving the needs of the user.

[Bhavesh Mistry] 14:43:22

Now, to the valid question about the needs of the user, you're right. I mean, people started at one Mbps, 10, 20, now 100, 1 gig, 10 gig.

[Bhavesh Mistry] 14:43:31

And that's going to continue. Ai is actually accelerating that whole thing because the amount of bandwidth required to serve AI-based content on your phone is massive already and you can see phones already are starting integrating some of the ai tools so

[Bhavesh Mistry] 14:43:45

That demand will continue growing. The way there's Moore's law around silicon there's also there's also There's also a law around bandwidth growth that talks about how bandwidth continues growing year on year.

[Bhavesh Mistry] 14:44:00

And you see this is almost doubling currently. So bandwidth consumption keeps on doubling every year.

[Bhavesh Mistry] 14:44:06

And I think this is why This is why I still say that fiber, because the thing about fiber optic networks is that the light that is in the cable is infinite. You can literally push more and more, right? You've not anywhere near its boundaries

[Wycliffe Bahati] 14:44:20

So, uh.

[Bhavesh Mistry] 14:44:23

But with other technologies, you are getting to the boundaries. With radio frequency you're getting to the the boundaries of what you can do today in terms of capacity. So I think this is where anything that is fiber or light will still continue being

[Bhavesh Mistry] 14:44:35

A preferred option because imagine you move from one gig to 10 gig for a home. All you have to do is change the equipment. The fiber cable remains the same because the light that goes through can carry terabits, not even gigabits you know if you want or better of zettabits or whatever the next version is so

[Bhavesh Mistry] 14:44:51

I think, yeah, there's always going to be like, okay, here there'll be fiber here, there'll be wireless, there'll be a bridge somewhere.

[Bhavesh Mistry] 14:44:58

There'd be some leo somewhere but Yeah, the rest to bandwidth increment is going to continue and i think At the same time, the satellite companies will continue innovating to try and push more out of the satellites also.

[Wycliffe Bahati] 14:45:13

Yeah, I think I've basically covered what I need. No, actually, it's the last one The issue of... Do they call it geopolitics or this came to my mind just When I saw how... Zelensky was treated.

[Bhavesh Mistry] 14:45:27

Mm-hmm.

[Wycliffe Bahati] 14:45:35

And then I thought, well, okay, maybe Elon Musk might kind of just decide to turn off the stunning satellite in Ukraine.

[Bhavesh Mistry] 14:45:45

In Ukraine. Yeah.

[Wycliffe Bahati] 14:45:48

So do you think it's a concern

[Bhavesh Mistry] 14:45:52

Yeah, look, I'm actually the last person to give a comment on geopolitics because I stay very far away from anything in that space in that space Look, I think it is a concern. I mean, if you ask my personal opinion, yes, because

[Wycliffe Bahati] 14:46:03

Yeah.

[Bhavesh Mistry] 14:46:05

You suddenly have these people like musk who are who are owning styling but owning so many other businesses that are so influential but now are in politics in some way because he's sort of partly quasi-government like he's doing all this stuff for Trump so yes

[Bhavesh Mistry] 14:46:21

There is a real risk about how politically affiliated some of the business owners are and these businesses are and what they can do.

[Bhavesh Mistry] 14:46:31

And that is going to be something that remains on the mind of all the countries that will be using a service like Starlink you know that you have these people who could turn off could do certain things with that service

that will affect you as a country.

[Bhavesh Mistry] 14:46:44

This is why I think going back to the earlier statement you made about why every country wants to have a ground station themselves. They all want to protect themselves, but you can't do it because a lot of this innovation is coming from

[Bhavesh Mistry] 14:46:56

The developed world or the Western world today, not from the developing world. So you sort of have to remain a little bit dependent on them and have to manage the geopolitics by whatever means you want to do, by affiliations, I guess, with whoever you want to be.

[Wycliffe Bahati] 14:46:58

Yes.

[Bhavesh Mistry] 14:47:12

But it is a factor. And it's not only for styling. I'll be very honest with you i think all major tech companies, if you look at the big tech companies, there is always a geopolitical risk.

[Bhavesh Mistry] 14:47:23

Because they're either assimilated to the US or affiliated to China.

[Bhavesh Mistry] 14:47:26

And there is geopolitics there. And that is going to be the case.

[Wycliffe Bahati] 14:47:31

Yeah, so I think this is just here to stay. You just have to navigate it.

[Bhavesh Mistry] 14:47:34

It's here to stay. It's here to stay and it's sort of the future that we have to manage yeah

[Wycliffe Bahati] 14:47:41

Yeah, okay, then I think I've covered most of the the topics that I wanted to cover.

[Bhavesh Mistry] 14:47:49

Sure. Yeah, no, I just had one or two questions for you. So I don't know, currently, what are you doing? Where are you working? Where are you based? I didn't ask you that initially when you started off. So I just wanted to know how things are on your end.

[Wycliffe Bahati] 14:47:50

Unless there's something that you had

[Wycliffe Bahati] 14:48:02

Oh, I know I work for... I don't know if you know bccw global

[Bhavesh Mistry] 14:48:07

Yes, NOPCW. Yes, that's the last I knew. I remember everyone saved your number for that. So you're still with PCCW?

[Wycliffe Bahati] 14:48:13

Obviously this is w um Now you take care of East Africa, the islands North Africa and West Africa.

[Bhavesh Mistry] 14:48:15

Okay.

[Bhavesh Mistry] 14:48:19

Okay, okay, okay. Okay, okay. You're still in Nairobi or are you somewhere else?

[Wycliffe Bahati] 14:48:24

Yeah, yes. Based in Nairobi.

[Bhavesh Mistry] 14:48:28

Okay, okay, cool. All right. Good. Maybe you should meet our coffee sometime.

[Wycliffe Bahati] 14:48:30

Yeah, yeah. Yeah, yeah, yeah. Yeah, I will do that. I'll make sure. I'm going to undistribute.

[Bhavesh Mistry] 14:48:37

Yeah, I think let's meet up. It'll be good to talk. I mean, I... Of course, you know, as much as I'm working on data, I'm always trying to stay in touch with what's happening in the submarine space and the data center space, because these are all areas of interest for me also.

[Bhavesh Mistry] 14:48:51

So it's always good to know, I was speaking to some Google folks the other day internally around their data center strategy and all this stuff.

[Bhavesh Mistry] 14:48:58

So it's always good to know what's happening in the industry.

[Wycliffe Bahati] 14:48:59

Mm-hmm. No problem.

[Bhavesh Mistry] 14:49:01

So yeah, let's plan to meet up sometime. Then the second question is regarding this study that you're doing.

[Wycliffe Bahati] 14:49:04

Yeah, yeah.

[Bhavesh Mistry] 14:49:08

For your own course or are you doing this How are you doing this?

[Wycliffe Bahati] 14:49:13

So this is for my thesis. Yeah.

[Bhavesh Mistry] 14:49:16

Okay, so what are you studying right now? What is this season about?

[Wycliffe Bahati] 14:49:21

I'm doing... I've been doing MSC computer science.

[Bhavesh Mistry] 14:49:24

Of the science okay and from which from which university

[Wycliffe Bahati] 14:49:30

The IU International University of Applied Sciences. It's based in berlin.

[Bhavesh Mistry] 14:49:35

Okay, okay, okay, okay, interesting. So will you publish this information or is this going to be submitted as part of your thesis?

[Wycliffe Bahati] 14:49:37

Yeah.

[Wycliffe Bahati] 14:49:43

So it will be submitted as part of the thesis. I will apply for it to be redacted and published at some point.

[Bhavesh Mistry] 14:49:47

Okay.

[Bhavesh Mistry] 14:49:52

Okay, when you do that, please share it with me. I'm always happy to read like what what's the output of the whole thesis and the whole to understand the whole thing and what your conclusion was yeah because i did something similar. There was a person who was doing a thesis two years ago and

[Wycliffe Bahati] 14:49:53

Yeah.

[Wycliffe Bahati] 14:50:00

I will.

[Bhavesh Mistry] 14:50:07

Similarly, they interviewed me in my personal capacity. And then when they published it, they shared it with me because it's always good to read.

[Bhavesh Mistry] 14:50:13

The whole thesis at the end of it. So please do share with me when it's when it's eventually uh public, yeah.

[Wycliffe Bahati] 14:50:14

Yeah, yeah.

[Wycliffe Bahati] 14:50:21

Definitely, I will. I'll look for you. I'm going to add this tomorrow, but next week we'll chat. Maybe you can have coffee and then we can discuss it.

[Bhavesh Mistry] 14:50:22

Sure, sure. Okay.

[Bhavesh Mistry] 14:50:28

Yeah, I'm around. So yeah, happy to do that. Yeah. Okay. All right. Thanks, Cliff.

[Wycliffe Bahati] 14:50:32

All right. Thank you very much.

[Bhavesh Mistry] 14:50:35

All right. Bye then. Yeah.

Participant 4 BL

Name Of Participant: Brian Longwe

Profession: Expert with over 25 years in broadband services

Date of interview: 10th March 2025

[Brian Longwe] 11:38:09

54 different markets, each with their own dynamics regulations different competitive structures and so on.

[Brian Longwe] 11:38:22

I would say that i would say the LEO networks are installing to be specific.

[Brian Longwe] 11:38:28

Because at the moment it's really the only one that is of substance. There's amazon Quipa which is on the way.

[Brian Longwe] 11:38:36

It's on the way. It's not yet operational. There is one way which is i think also still not really 100%.

[Brian Longwe] 11:38:47

So yeah, I would say that the impact of networks like Starlink being felt differently.

[Brian Longwe] 11:38:57

In different markets and are being responded to differently in different markets. However, there is one common trend that at least we can pick up in some of the more let's call it um actually call it The common thread that you can see across most of the areas where Starlink is available

[Brian Longwe] 11:39:17

Because it is because it is In terms of coverage, it does cover the whole continent, but it is not available commercially for all countries.

[Brian Longwe] 11:39:25

Due to various uh you know various results. Some of them are regulatory challenges how does they've just decided we don't want to go for that market right now maybe because it's unstable all and over.

[Brian Longwe] 11:39:41

So one common theme that at least I've been able to observe across the markets where Starlink is present.

[Brian Longwe] 11:39:52

Is that it is being used predominantly in the capital cities by the urban elite.

[Brian Longwe] 11:40:01

So as much as this was positioned as a network. That is supposed to meant to be unconnected. As much as this was positioned as a network that is going to alleviate rural connectivity issues and challenges In Africa, that is not the case.

[Brian Longwe] 11:40:22

On one hand, there is the issue of cost. The terminal is far, far, far, far more costly.

[Brian Longwe] 11:40:31

Than your rural user can afford.

[Wycliffe Bahati] 11:40:35

Yeah, yeah.

[Brian Longwe] 11:40:35

That's first and foremost. Secondly, there is also the cost of the subscription the cost of the subscription yes as much as it quite you know reasonable in quotes, I'll put this in quotes it is probably still higher than the average rural

[Wycliffe Bahati] 11:40:39

Hmm.

[Brian Longwe] 11:41:01

Home or user can be able to afford you know back there now they are choosing between Do I use the money that I'm going to I've set aside to buy seeds for this season's planting Or do I buy, do I pay for a standing subscription

[Brian Longwe] 11:41:20

Do I use the money that I've set aside for the fertilizer for this year's crop or do i so you know in the rural communities in most of africa it's not the same as the rural community in the USA. The rural community in the US, those ones are commercial farms. Those are ones that are called rural. Some of you is sitting on

[Wycliffe Bahati] 11:41:20

Hidden.

[Wycliffe Bahati] 11:41:42

Yes.

[Brian Longwe] 11:41:43

Thousand acres of wheat. You know um but because it's out there far away and because of course now he's in an area where there are many farms 2,000 acres is his plot and it's sitting somewhere in the middle. Then there's another one up next to him of maybe 800 acres

[Brian Longwe] 11:42:00

And there's one homestead there you know those are the ones now who have problems getting connectivity because it doesn't make sense to roll any kind of fiber infrastructure.

[Brian Longwe] 11:42:11

Or even 3G, 4G infrastructure out to a place which is so big and it has five subscribers.

[Brian Longwe] 11:42:18

You know? So rural in a developed country and rural in a developing country are two totally different things let's get that first of all clear um And, you know, I think that stalling was built with the western rural mind not the african rural

[Wycliffe Bahati] 11:42:44

Which almost translates to African amna. Oh.

[Brian Longwe] 11:42:45

So... Yeah, that's right.

[Brian Longwe] 11:42:49

If we talk about now market and competition how is this affecting the competitive landscape?

[Brian Longwe] 11:42:58

I think I would say that at the beginning. Most of the broadband operators or providers, the internet service providers felt threatened.

[Brian Longwe] 11:43:09

By solid entry. Many of them lost customers, that's a fact.

[Brian Longwe] 11:43:14

Because there was a lot of buzz, there was a lot of hype.

[Brian Longwe] 11:43:17

And a lot of people said, ha. You know, Starlink is coming in they can give me 200 mbps download only fifty dollars While here, this guy who I've been using for the last so many years is giving me 20 mbps download for the same price. So it's giving me 40 mbps download for the same price

[Brian Longwe] 11:43:40

So I'm getting 10 times more bandwidth from stomach Let me switch.

[Brian Longwe] 11:43:45

So there was a mad rush, you could call it a gold rush.

[Brian Longwe] 11:43:49

Was coming in a manner of speaking And we've experienced the same here in Malaw.

[Brian Longwe] 11:43:58

Right. Now, for my company in Malawi, we are an internet service provider that targets the middle income and lower income bracket.

[Brian Longwe] 11:44:08

Working class, middle income and working class We don't go for the clean.

[Brian Longwe] 11:44:15

The high-end upper class and expatriates and so on. Some of them do end up using our services, but that's not our core. So our pricing and our packaging of services is really looking at the middle class and the working class. Now, we really didn't feel

[Brian Longwe] 11:44:32

Much of an impact in terms of customer loss when Starlink came. We did have a few of the higher end customers who switched.

[Brian Longwe] 11:44:40

Which was to be expected. Funny thing is that they've all now started coming back Because for Malawi especially, two things have happened.

[Brian Longwe] 11:44:48

One, just as with every other African country, there was saturation The capacity of the network coverage over

the cities was quickly saturated.

[Brian Longwe] 11:45:03

And Staling said, okay, we are not taking any more residential subscriptions. Let's stop it.

[Brian Longwe] 11:45:09

The second thing that happened for Malawi, which is different to other countries.

[Brian Longwe] 11:45:14

Is that because of the economic situation of the country. Which has resulted in a major foreign exchange crisis.

[Brian Longwe] 11:45:27

Static is enabled to repatriate the funds that they collect in terms of subscriptions.

[Brian Longwe] 11:45:33

So as a result, they've also decided to further restrict new subscriptions because of that.

[Brian Longwe] 11:45:44

Because they can't get money out they are being paid in local currency by subscribers into a local bank but they can't get that money out so that has become a complication.

[Brian Longwe] 11:45:56

So that has further sort of you could call it narrowed the scope of services that are being provided. They actually said we are not going to offer any subscriptions, any residential subscriptions. We will only offer business subscriptions of a higher value.

[Brian Longwe] 11:46:11

And that's what the situation is in Malawi. Having said that, I will say that we've noticed an interesting trend in that some of the customers who initially ran away and went to Starlink, our residential subscribers.

[Brian Longwe] 11:46:28

Have started coming back one by one in the recent months.

[Brian Longwe] 11:46:31

Why have they started coming back one by one? Because Starlink's service is with higher download speeds inputs but it's not unlimited.

[Brian Longwe] 11:46:46

The standard residential service has a cap. Once you hit that cut.

[Brian Longwe] 11:46:51

Then you have a degraded performance or you actually now are done for the month on the standard service.

[Brian Longwe] 11:46:59

It has a usage cap and many of them are now hitting the usage cap.

[Brian Longwe] 11:47:04

And then ending up scratching their heads Because now they have to buy a mobile data bundle.

[Wycliffe Bahati] 11:47:04

Very quickly.

[Brian Longwe] 11:47:10

For the remainder of the month or the remainder of the month whatever so they're like hey okay this thing is not working for me So this is what we've been able to observe in the marketplace

[Wycliffe Bahati] 11:47:24

So the word sterling claims because I realized that claims When they came in the market, they were quite clear on the data usage cap.

[Wycliffe Bahati] 11:47:36

But I think maybe they realize it's sensitive So I've seen like in other countries, they removed that they're saying they're saying unlimited So it means it's actually unlimited, but in quotes there's just Just like any other fixed or fixed or mobile broadband there's a cap

[Wycliffe Bahati] 11:47:56

That you need to do.

[Brian Longwe] 11:47:58

Yeah, so there are many networks that actually use those caps And the fact is consumers don't like caps.

[Brian Longwe] 11:48:07

End users don't like caps. We as CTA, we don't provide it. We don't put any usage caps. We say it's unlimited.

[Wycliffe Bahati] 11:48:08

To...

[Brian Longwe] 11:48:18

Uncapped. Use as much as you want as long as you pay. The only thing that you'll be buying paying us for is the pipe, the size of the pipe.

[Wycliffe Bahati] 11:48:20

Mm-hmm.

[Brian Longwe] 11:48:26

You buy a 10 meg pipe you'll be able to use up to 10 meg worth of whatever throughout the month You buy a 20 megaby pipe, that's what you'll get.

[Brian Longwe] 11:48:35

So we sell on the basis of bandwidth, not on the basis of usage caps. And for us, that has been quite effective. I know other markets have similar, but you still have cases where in many different markets.

[Brian Longwe] 11:48:52

Providers still use those FUP, fair user policy caps And they limit or restrict once the user hits that cap, then they get a degraded service so you know maybe it's 28 bps until you hit your cup then it goes down to 2 mbps after which now it becomes impossible to use

[Brian Longwe] 11:49:14

So, so... you know one of the things that i'll say was an immediate impact regardless of how consumers behaved I think one immediate impact in the marketplace is that everybody increased the bandwidth that they offer.

[Brian Longwe] 11:49:33

Either they introduced a new package which has bandwidth that is higher and closer to looking like Starlink in order to try to keep or to capture back some of those customers who were going or they just increased bandwidth across the board on all their packages. So I think that's a common trend that has been also observed in many countries.

[Brian Longwe] 11:49:58

Kenya, you saw Safari come introducing a 100 mbps product Whoever heard of suffering from providing that you know the same thing happened to other providers.

[Wycliffe Bahati] 11:50:04

Yes.

[Brian Longwe] 11:50:10

And we've seen similar across other markets where mobile operators, other ISPs, fiber operators all bumped up the capacity of their packages, the bandwidth that's available on their packages, because in the face of Starlinks.

[Brian Longwe] 11:50:33

200 mbps download speed you know the previous offerings looked very weak, very feeble.

[Brian Longwe] 11:50:42

And questions were being asked. You know if starlink can give us this much at this price That means you've been stealing from us all this time.

[Brian Longwe] 11:50:54

You know so there was this kind of general vibe, you know, you know, the buzz in social media and whatever was that So all this time we've been taken for a ride by these guys.

[Brian Longwe] 11:51:10

They've been cashing in. To a certain extent is true. They were good at their game.

[Wycliffe Bahati] 11:51:16

Yeah, of course. It's a business everybody's in business to make money Yeah. Mm-hmm.

[Brian Longwe] 11:51:18

Yeah. Yeah.

[Brian Longwe] 11:51:23

That's right, yeah. So, yeah.

[Wycliffe Bahati] 11:51:26

So yeah so Yeah, so I think you covered quite much on the market challenges and opportunities there What if... So generally speaking, and I mean, that's been one of my observations is that?

[Wycliffe Bahati] 11:51:51

Inasmuch as the Leo broadband.

[Wycliffe Bahati] 11:51:57

Service has been... touted as going to break the digital divide.

[Wycliffe Bahati] 11:52:04

It's actually almost coming for the same customers mostly who existing providers are serving.

[Wycliffe Bahati] 11:52:14

So what do you think they need to do to actually find this still, I'm assuming they're still able to reach some of those customers, but because of the other factors like cost So, um.

[Wycliffe Bahati] 11:52:28

What do you think they need to do to be able to a kind of... achieve what they are mostly advertised that they're going to provide services that are not built in areas which have not been reached.

[Wycliffe Bahati] 11:52:49

By normal providers. What do they need to do?

[Brian Longwe] 11:52:55

Now, the LEO providers themselves

[Wycliffe Bahati] 11:52:59

Yeah. Because remember that they are also getting advantages. For example, there's a Congress fund which funds These provide us for reaching out to the digital divide. I think styling is one of the beneficiaries.

[Wycliffe Bahati] 11:53:16

Of that money and it's a lot of money they're given So if they're getting this money this for example u.s government that they are going to break the digital divide But then only to come back and squeeze out the existing providers from the existing market.

[Wycliffe Bahati] 11:53:40

I think the business model doesn't seem to fit But they intended to achieve.

[Brian Longwe] 11:53:51

So I think... I would say that there are parts of Stalag business model that we see.

[Brian Longwe] 11:54:01

And there are parts that we don't see. Okay. On the face of it.

[Brian Longwe] 11:54:09

We see the consumer offerings that they are giving And they keep on talking about wanting to get into the rural or being able to build a network that provides global coverage and here I'm using global with a lot of emphasis.

[Brian Longwe] 11:54:26

Because that's essentially what they've sold. Global coverage. Almost every inch of every country is under their network. That's the way it's been designed.

[Brian Longwe] 11:54:38

So that's one part of the on-premise and one part of their business case, the one that is visible.

[Brian Longwe] 11:54:46

However, there is the other part that is invisible. And having been in this business for a long time.

[Brian Longwe] 11:54:52

I do know that i do know that there are other arrangements that a provider will get into with the government or with governments to be able to advance those government agendas.

[Brian Longwe] 11:55:07

And if you look back carefully and you track.

[Brian Longwe] 11:55:12

Funding that has been given to Starlink Since inception, you will see that a huge amount of money was given out by the Department of Defense.

[Brian Longwe] 11:55:23

The US Department of Defense. So what they're paying to get soldiers connected on internet?

[Brian Longwe] 11:55:30

Or what were they paying for? So my theory, and this is a conspiracy.

[Wycliffe Bahati] 11:55:33

Yes, so... Mm-hmm.

[Brian Longwe] 11:55:36

Or whatever you know maybe it's not backed by But anecdotal is that Starlink is also providing a huge amount of surveillance.

[Wycliffe Bahati] 11:55:37

Native obsidian, actually.

[Brian Longwe] 11:55:47

To the US government. And that may also be the reason why they're getting a huge amount of funding.

[Brian Longwe] 11:55:54

Almost unrestricted funding year after year.

[Wycliffe Bahati] 11:55:57

Yeah.

[Brian Longwe] 11:55:59

Right. To support and to enable If you look at the cost of deployment, any provider no matter whether they are a Wi-Fi provider no matter whether they're a 4D provider, a fiber provider.

[Brian Longwe] 11:56:16

Or maybe your provider has to put capex down they have to put down income capital expenditure to build the infrastructure, to build the network.

[Brian Longwe] 11:56:29

That will give them their customers. Then they have to take that capex and put it into their model, their business case model and then see, okay, these are the number of subscribers that i need to be able to get a return on investment for my investors.

[Brian Longwe] 11:56:47

And typically, for most cases in the internet space the return on investment on network infrastructure is between 5 to 10 years.

[Brian Longwe] 11:56:57

And here I'm talking about your conventional wireless fixed wireless access fiber infrastructure.

[Brian Longwe] 11:57:07

Copa infrastructure. With satellite If you look at the price of a single satellite.

[Brian Longwe] 11:57:14

One of those that has been put up, not just the cost of the satellite itself but even just the launch the deployment cost.

[Wycliffe Bahati] 11:57:22

The dinner.

[Brian Longwe] 11:57:23

You know with fiber, we have to look at the civil rights. How much does it cost to dig and lay that fiber?

[Brian Longwe] 11:57:29

That's the cost of deploying. With fixed wireless access infrastructure, we have to include the cost of the engineers who are going to climb that tower, the riggers who are going to you know make this infrastructure active It's launch costs.

[Brian Longwe] 11:57:46

Each individual launch is so expensive. And then the satellite itself is extremely expensive. And they say they have how many?

[Wycliffe Bahati] 11:57:47

Yeah. Yeah.

[Brian Longwe] 11:57:55

A couple several hundred or is it a few thousand that already deployed.

[Wycliffe Bahati] 11:57:58

Then none of them seven i think of As of last month, there are around 7,000.

[Brian Longwe] 11:58:04

Yeah, 7,000 satellites. Now, if you count the individual cost one of them costs what a few hundred thousand dollars almost or is it a couple of million dollars?

[Wycliffe Bahati] 11:58:12

Yeah. One lunch.

[Brian Longwe] 11:58:14

A single one.

[Wycliffe Bahati] 11:58:18

You said it's about \$5 million or something.

[Brian Longwe] 11:58:21

The launch yeah but each satellite, the cost of production of a single satellite is also, I think, close to about a million Now, if you tested 7,000 times a million times that's 7 billion. Write it down just for the satellites. And then each launch

[Wycliffe Bahati] 11:58:36

Yeah.

[Brian Longwe] 11:58:37

Puts up about 60 to 200 satellites or spins out you know and they've had a few failed launches in the last like the past few whatever.

[Brian Longwe] 11:58:46

So when you look at the amount of cost that is being incurred to put these things up there.

[Wycliffe Bahati] 11:58:47

Yeah.

[Brian Longwe] 11:58:53

And you look at the amount of subscribers and subscriptions that they are their business case probably gives them an ROI of about 300 to 400 years.

[Brian Longwe] 11:59:03

Strictly speaking, to be completely honest with you, if they were going to make a business case out of internet subscriptions.

[Brian Longwe] 11:59:11

They would probably have to have some few million subscribers and the business the return on investment will be about some hundreds of years out So my thesis or my let me not use the word thesis, my theory is that they are making money or they're actually getting paid from elsewhere for other things that we don't see.

[Brian Longwe] 11:59:34

And the first and foremost in the past, one of the main purposes and uses for geo saturated infrastructure has been surveillance.

[Wycliffe Bahati] 11:59:48

Boom.

[Brian Longwe] 11:59:49

Now, who in their right mind would not use a satellite that is many, many, many, many, many thousands of kilometers closer because a geo satellite is operating at 30,000 kilometers now here you have a satellite that's at 1 500

[Brian Longwe] 12:00:04

Kilometers which means you can see more You can see clear. You can see better.

[Wycliffe Bahati] 12:00:05

Yeah. Yeah.

[Brian Longwe] 12:00:10

You know um you have much shorter round trip time on your surveillance.

[Brian Longwe] 12:00:16

So you're more accurate, you're more precise. Who did use that for surveillance? So, you know.

[Wycliffe Bahati] 12:00:19

Yeah.

[Brian Longwe] 12:00:24

Um For me, I would say the internet side of it is really more one facade. That is the nice sweet and friendly outward face that Starlink has been given to justify its existence and they have to of course you know um continue to keep that up.

[Wycliffe Bahati] 12:00:43

Yeah.

[Brian Longwe] 12:00:51

But they're losing money, probably. And, you know, we talked about this stellar constellation. We have not even talked about the art stations, the ground stations.

[Wycliffe Bahati] 12:00:54

Yeah, it's true because it's true

[Brian Longwe] 12:01:03

Because they have to incur massive costs. For the ground stations i'll have to pay local fiber providers in different countries for those ground stations, they have to pay data and collocation centers for hosting those

ground stations those are also huge expenses.

[Wycliffe Bahati] 12:01:03

Or leak. Yeah, the muscles. Good.

[Wycliffe Bahati] 12:01:18

That's right.

[Brian Longwe] 12:01:19

For this global network so um you know i've been building infrastructure around africa for 28 years.

[Brian Longwe] 12:01:26

It's not easy and it's not cheap. So a network like Starlinks, believe me you um There's more to this than meets the eye.

[Wycliffe Bahati] 12:01:30

Mm-hmm.

[Brian Longwe] 12:01:43

Yes.

[Wycliffe Bahati] 12:01:43

Right, right. Because before that before sterling remember we had daily desic and all those guys were trying to actually do this commercially Honestly, commercially and it was horrible.

[Brian Longwe] 12:01:48

Right. Yes.

[Wycliffe Bahati] 12:01:52

So this one looks a bit, it's a debate it's a debate that is going on actually that There's usually that is what is the actual cost In fact, people have tried to decipher from the the failings of Starling at the FCC,

[Brian Longwe] 12:01:53

Didn't work. Didn't work. Yeah.

[Wycliffe Bahati] 12:02:13

That how much it costs, but it's still not very clear how much it costs to actually do these things but of course people are doing reverse engineering And coming up with these costs.

[Wycliffe Bahati] 12:02:26

Yeah and um

[Brian Longwe] 12:02:26

Yeah, and maybe just related to the same because we are talking about LEOs in general I would say that if you look at that Countdown.

[Brian Longwe] 12:02:41

Strategies. That other countries have not backed up.

[Brian Longwe] 12:02:47

If you look at the amount of money that China is putting into LEO deployments to my recollection, I think there are two companies in China.

[Brian Longwe] 12:03:00

Have promised to deploy LEO. The most recent, I think, was announced about two months ago or three months ago And each of them are promising to be able to deploy networks that are two to three times more dense than Starlinks.

[Wycliffe Bahati] 12:03:00

Okay.

[Brian Longwe] 12:03:22

And they've all had launches. Of pilot load today. We've all had pilot launches and they are well on their way to being able to deploy.

[Wycliffe Bahati] 12:03:24

Mm-hmm.

[Brian Longwe] 12:03:32

I can guarantee you, China is not interested in connecting rural Africa or rural America or rural Europe.

[Brian Longwe] 12:03:40

This is A. Battle for global supremacy.

[Wycliffe Bahati] 12:03:40

No, no.

[Brian Longwe] 12:03:45

Espionage, surveillance. And global supremacy in terms of tactile supremacy.

[Brian Longwe] 12:03:54

So, you know, I know this is not supposed to be a conversation about geopolitics But these are some of the things that are driving these deployments. We can't just assume that these are good people who want to connect all of

[Wycliffe Bahati] 12:04:09

Like the undivided. Yes. And then connections.

[Brian Longwe] 12:04:09

Africa. Yeah, which is why they will make some uneconomic decisions sometimes and which is why sometimes you find them you know okay so the regulator in this country doesn't want to give us a license it's okay let's continue with the others.

[Wycliffe Bahati] 12:04:15

Yeah.

[Brian Longwe] 12:04:28

You know?

[Wycliffe Bahati] 12:04:29

But again, I think what will actually put the lid off this whole... thing is, I think when i think I squared, the one for Europe.

[Wycliffe Bahati] 12:04:46

Will come online if at all, they manage to pull through I think then the real costs will in the open people will know because i think the the europeans will not hate the Chinese will not tell you the Americans are doing their thing. But I think the Europeans, because it's a Europe thing funded by European Union.

[Wycliffe Bahati] 12:05:09

The grill costs will come out and maybe we'll now know exactly if this is making sense or not.

[Wycliffe Bahati] 12:05:15

Yeah, so... Yeah, the other thing Audit does talk about is the technicals.

[Brian Longwe] 12:05:16

Yep.

[Wycliffe Bahati] 12:05:24

Challenges and opportunities that are coming with these LEO deployments In terms of speed, latency i mean There's been a lot of observation that They're coming to improve the service.

[Wycliffe Bahati] 12:05:42

From what we have. I don't know what you feel on the ground in terms of the technical capability if they have an advantage over what they they the technologies we are using for traditional broadband.

[Brian Longwe] 12:05:59

Well, first and foremost, I'll say that Starlink have recruited some of the best in class right they have some of the best network engineers and programmers in the world working on improving and refining their network, especially the areas to do with latency routing protocols and the

[Brian Longwe] 12:06:25

Handoff between the different satellites as they fly overhead. I was one of the early adopters of Starlink when it first came into the continent. And of course, some of the first things I checked was latency and other network parameters network metrics such as buffer bloat etc

[Brian Longwe] 12:06:46

And it really had the pathetic score. It really had very poor performance.

[Brian Longwe] 12:06:51

But if you look at that, if you look at the network today against and this is going back to about 2022 when I first, when Starlink was first announced in Malawi and i bought i was one of the first people to buy a terminal

[Brian Longwe] 12:07:05

And get it deployed and check it. If you look at the network performance and operation You know, two years ago and you look at it today.

[Brian Longwe] 12:07:16

It's totally different it's night and day. They've been able to tackle and overcome some of the major buffer bloat challenges Actually, I have a whole book. I received a shipment of um for this book here above a bloat and Beyond. It's a very, very good um

[Brian Longwe] 12:07:35

Book that's written by Tokhoylan Jorgensen, which actually unpacks from the University of Kaustat in europe um which unpacks the issues of buffer bloat. It's actually considered nowadays as a as a a key resource on understanding buffer bloat. So buffer bloat is really it affects all networks

[Brian Longwe] 12:08:00

And it affected Starlink massively, massively. In fact, it was amplified by the distance distances of the network from the network the cp the terminals the user terminals But if you look at the amount of work they've been able to do on that and how they've been able to

[Wycliffe Bahati] 12:08:07

Hmm. Mm-hmm.

[Brian Longwe] 12:08:23

Address backup load challenges. It's incredible. So they've been able to optimize. They've been able to refine and they've been able to improve quality of service issues to the extent that even the latency itself they've been able to improve in terms of reducing latency substantially.

[Brian Longwe] 12:08:45

To below i would say 50 milliseconds if i'm not wrong uh in some cases in many cases Which is massive, which is huge because that now begins to compare with a lot of terrestrial networks.

[Brian Longwe] 12:09:03

So in that respect, I would say that you know, there's been a lot going on if you think about this from the perspective of you know um you know let me use a buzzword iot If you think of this from the perspective of being able to control

[Wycliffe Bahati] 12:09:27

Thank you.

[Brian Longwe] 12:09:28

Autonomous vehicles. If you think about this from the perspective of being able to control drones and other types of unmanned vehicles, whether they are unmanned aerial vehicles unmanned submarine vehicles unmanned terrestrial vehicles you can see.

[Brian Longwe] 12:09:53

Here now that there begins to be a big case for having a network like this that can actually provide coverage and enablement for a let's use the example of a UAV, an unmanned aerial vehicle.

[Brian Longwe] 12:10:13

That can transit multiple borders and countries thousands of kilometers to carry out the mission.

[Brian Longwe] 12:10:21

Whether it's a strike. Whether it's surveillance, whether it's delivery of something or collection of something. So there's a huge tactical advantage that comes with being able to use such a network.

[Brian Longwe] 12:10:38

With such coverage that can allow you to deploy in a moment's notice any kind of Otonomas vehicle.

[Brian Longwe] 12:10:51

Literally anywhere in the world. Right. So yeah.

[Brian Longwe] 12:10:58

I think we have yet to see in my view.

[Brian Longwe] 12:11:07

We are yet to see. What capabilities will come. I think there's going to be continuous innovation i think besides the teams that are working internally within you know companies like Starlink or this kind of optimizations and innovations you have a huge

[Brian Longwe] 12:11:29

Army of innovators working externally and working on developing ways in which these technologies can be used to support different forms of different types of activity.

[Brian Longwe] 12:11:46

So besides the conventional internet access. I would say that i would say that the capabilities of such networks are going to be probably predominantly used. Maybe there'll be a higher percentage of the use case for such networks that goes within the IoT space in terms of being able to

[Brian Longwe] 12:12:07

Enable automation and operation of machines.

[Brian Longwe] 12:12:12

Whether they are small machines like agricultural sensors.

[Brian Longwe] 12:12:17

Or whether they are big machines like a drone that is carrying strike missiles.

[Brian Longwe] 12:12:24

Yeah.

[Wycliffe Bahati] 12:12:28

Now, the other thing I want to find out is we've seen... The user evolve over time because from when the user used to use megabits connection We've seen... network trying to deploy now gigabits.

[Wycliffe Bahati] 12:12:51

To the near the end user So out of six sub countries which actually have providers who advertise 10G connections.

[Wycliffe Bahati] 12:13:05

To the home. And if you're aware about the kind of now the ai consumption on devices and So... With this in mind and think about the spectrum available for this LEO broadband providers.

[Wycliffe Bahati] 12:13:28

Do you see a belief? Them being able to catch up or compete in the future in the same market as us.

[Wycliffe Bahati] 12:13:35

The trust providers.

[Brian Longwe] 12:13:40

Well, no. I would say, again. It comes back it comes back to purpose it comes back really to the objective of the network and what it's supposed to achieve, what it's supposed to do.

[Brian Longwe] 12:14:01

With the current capacities that are available on the network.

[Brian Longwe] 12:14:08

We've been able to see that streaming video is working perfectly.

[Brian Longwe] 12:14:12

Without buffering. We've been able to see even in some cases live broadcasts being achieved using is technology. We've been able to see machines being controlled without too much of a challenge with increasing levels of precision and accuracy. I would say again

[Wycliffe Bahati] 12:14:21

Moving forward.

[Brian Longwe] 12:14:38

If that is maybe what you would call the heavier purpose Then... the current capacities are sufficient. However, having said that.

[Brian Longwe] 12:14:53

Innovation doesn't wait for anybody. And innovators are working. If we look at the

[Brian Longwe] 12:15:04

The rate of innovation that currently we are seeing you know during our time We learned about Moe's law.

[Brian Longwe] 12:15:18

Which basically told us that every 18 months the capacity and the performance of processors will double.

[Brian Longwe] 12:15:28

That has changed completely. I don't know if they've come up now with another law.

[Brian Longwe] 12:15:33

Um because Right now, what we are seeing is that every six months it's not just doubling. It's almost quadrupling it's even going up by several other factors.

[Brian Longwe] 12:15:45

A good example is the recent announcement by Nvidia's uh you know ceo during, I forget the name of the event, towards the end of last year where he revealed this new low cost ai chip Last month, we had from Microsoft

[Brian Longwe] 12:16:08

That they've now had a breakthrough in quantum physics, in quantum innovation And we now have a new chip they say which breaks the laws of nature.

[Brian Longwe] 12:16:22

I found that announcement. You know, they said that You know, according to the laws of nature we understand you know uh physical substances as both existing in three different forms liquid, gas.

[Brian Longwe] 12:16:46

And solid. They say that they've created a new a new form.

[Wycliffe Bahati] 12:16:53

In bold.

[Brian Longwe] 12:16:54

A new phone. And that's what they're using in this chip.

[Brian Longwe] 12:16:57

And that now this cheap is going to it it has potential to do calculations that i don't know how many thousands times faster that the current world's fastest supercomputer So I would say that You know?

[Brian Longwe] 12:17:18

The current constraints that we see.

[Brian Longwe] 12:17:23

On frequencies. The current constraints that we see in the electromagnetic spectrum are going to be it superseded they're going to be superseded you know innovated around And we'll see more and we'll see faster Today, with 4G, 5G technology, actually with 5G technology, we're able to see gigabits

[Brian Longwe] 12:17:50

Being transmitted. Of a wireless infrastructure.

[Brian Longwe] 12:17:56

Over the same frequency that 20 years ago could not even transmit one megabit per second.

[Brian Longwe] 12:18:06

Simply because of the level of technology that was available back then.

[Brian Longwe] 12:18:09

Since then, innovation has increased and multiplied. So I would say that innovation is not waiting for anybody. There's going to continue to be more and more innovation and These things are going to continue to change.

[Brian Longwe] 12:18:28

So you never know. While today you know you can only see the likes of 10 gigabit 100 gigabit speeds being realized of fiber It could be that it could be that in... the very near future, we might be able to see those kind of speeds being realized

[Brian Longwe] 12:18:52

Of our wireless infrastructure. A good example would be a good example would be Tara, Google's Tara.

[Brian Longwe] 12:19:01

You know the free space optics they're already doing 10 gigabits per second 20 gigabits per second over three space optics using lasers right The only challenge that they are not able that they are struggling with is struggling distance and particles.

[Brian Longwe] 12:19:22

Because if you have too much dust, it affects If you have rain, you lose the link completely.

[Brian Longwe] 12:19:31

So, so, so, so, so the point I'm making here is you know these are technologies that are being applied in LEO technology.

[Brian Longwe] 12:19:41

Starlink is using something very similar to the Tara chip. The inter-satellite communication is using lasers. That's how they do the handoff.

[Brian Longwe] 12:19:49

And out there in space you don't have these particles you don't have dust you don't have rain it's almost a near vacuum. So it's a perfect atmosphere or perfect environment for very high speed transmission.

[Brian Longwe] 12:20:03

And this is probably only going to, it's probably only going to increase So, you know.

[Brian Longwe] 12:20:11

It may have been in my lifetime, it may not, but at least I know for sure that in the next 10, 20 years we're going to see a complete shift And what currently we think of as fast is going to look very, very, very slow.

[Brian Longwe] 12:20:26

I remember how excited I was when I set up the first internet connection you know at uh from net africa in nairobi a 64k link all the way to Pipex UK.

[Brian Longwe] 12:20:42

And we send a pin over this link. It was a 64k pipe but we served 3 000 users email accounts.

[Brian Longwe] 12:20:51

With that 60 occupied. And people were amazed. Oh, the email has reached immediately Look at where we are today.

[Brian Longwe] 12:21:02

64k is not even enough for anything and we were running an entire isp or 64K, you know?

[Wycliffe Bahati] 12:21:04

You can eat it.

[Wycliffe Bahati] 12:21:15

Yeah, it's interesting actually Thanks. Mm-hmm.

[Brian Longwe] 12:21:16

This is just 20 years ago, 25 years ago you know loyal maybe it was 30 years ago because it was 1997.

[Brian Longwe] 12:21:24

You know, just the other day So decades have a big you know big difference. I think time moves faster in the world of innovation.

[Brian Longwe] 12:21:36

You know so yeah um I would say right now LL networks cannot compete in terms of access with the current bandwidth that you have available on terrestrial networks. For sure.

[Brian Longwe] 12:21:50

However, technology is changing and innovation is moving very fast. You know, there might even already be applications that are working using allios at faster but they are still classified because as mentioned previously These guys have been receiving massive funding from military

[Brian Longwe] 12:22:13

So there's a high chance that a lot of the use cases and applications that are currently in play are very classified and we don't know anything about them. What we see is just what we are shown.

[Wycliffe Bahati] 12:22:16

Yeah.

[Wycliffe Bahati] 12:22:25

Thank you.

[Wycliffe Bahati] 12:22:30

So, and... I know you've mentioned it a bit.

[Wycliffe Bahati] 12:22:35

So say... what would be the impact of this service?

[Wycliffe Bahati] 12:22:42

On the economy, for example, like you've mentioned mugawi you know at least malawi they are unable to retract the funds.

[Wycliffe Bahati] 12:22:51

But... If, for example, they manage to break the affordability and are able to deploy What do you see it happening to the local service provider i mean uh If, say, 50% of your customers It would go to Starlink.

[Wycliffe Bahati] 12:23:12

And... Of course, these payments Apart from what the regulator collects.

[Wycliffe Bahati] 12:23:18

Most of it, I mean, is repatriated. I know it's still the same with our current setup where we still have to pay.

[Wycliffe Bahati] 12:23:26

Full capacity, mostly millennials. How would this affect local service provider in terms of the jobs local economy In terms of the jobs, technological transfers.

[Wycliffe Bahati] 12:23:43

Skills if this was to be a model that works.

[Brian Longwe] 12:23:50

Well, I would say that you know uh The only thing that's constant in this world is change, right? This is something that I've been able to to lead through and see in my own technology career which now is spending about 32 years.

[Brian Longwe] 12:24:13

Um... Same thing when it comes to issues around aerals.

[Brian Longwe] 12:24:21

One thing that has, especially the mobile network operators shaking and losing sleep is what is being called the direct to sale.

[Brian Longwe] 12:24:32

Capabilities of these earlier networks right the fact that an LL network can be able to interact directly with a mobile phone a mobile phone cell phone in other words you don't need to buy an expensive advanced cp to use the services.

[Wycliffe Bahati] 12:24:42

So...

[Brian Longwe] 12:24:53

To make a call or to get data That is sending shakes and quivers through every single mobile network operator in the world and many of them I've already filed.

[Brian Longwe] 12:25:09

You could call it very strong objections with their country's regulators with their governments and are trying to demonize this capability and say this is a threat to national security Many of them in many countries I think we saw it most evidently or most

[Brian Longwe] 12:25:34

The place where I think I can recall in recent past where it was really you know publicized was sephorical.

[Brian Longwe] 12:25:43

The way that Safari com responded And the way that they communicated around this issue

[Brian Longwe] 12:25:52

You know to the regulator, to communication authority in Kenya. So this is something that is something that it's disruption disruptive Having said that.

[Brian Longwe] 12:26:06

If you look even with the current stunning data service.

[Brian Longwe] 12:26:11

You will find that in almost every market where Starlink is available.

[Brian Longwe] 12:26:16

A new group of workers has been created.

[Brian Longwe] 12:26:23

Now you have styling installers. A new business case has come into the market Starlink technicians companies that provide support For Starlink, companies that deploy this Starlink whatever's there's a whole

value chain of distributors and resellers that is beginning to emerge.

[Brian Longwe] 12:26:43

And so on and so forth. There's even now technical certification programs that technicians are going to and getting certified so that they can now be you know recognized as a static installer So that is probably going to increase.

[Brian Longwe] 12:27:04

And as we get more networks available, as soon as AWS, Amazon Kuipa comes online, as soon as the others become online.

[Brian Longwe] 12:27:12

You will have now a whole ecosystem of businesses and individuals that are working and are getting employment and income.

[Brian Longwe] 12:27:24

To support and sustain these technologies. So from an economic standpoint, I would say that you have to have winners and you have to have losers whenever technology changes right If you go back 20 years or not even that far back. Let's just go back 10 years.

[Brian Longwe] 12:27:45

10 years ago, if you looked at the number of employees in Africa that multi-choice DSTV we had and the number of subscribers they had.

[Brian Longwe] 12:27:57

It was massive. Today.

[Brian Longwe] 12:28:02

How many do they have? A fraction of what they used to have. They are no longer the key player. They are no longer the main network that is providing Africans with their entertainment. Now you have Netflix and all of these other online streaming whatever's.

[Brian Longwe] 12:28:17

So that's disruption. That is how the old has to go out to make way for the new.

[Brian Longwe] 12:28:26

And you've had a shift. So now instead of the entertainment needing a satellite dish and an installer to install that satellite dish All that's needed is an internet connection.

[Brian Longwe] 12:28:41

So that revenue shifted from multi-choice to the ISPs. Tomorrow it might shift from the isp to the LEOs. And then you never know what will come next. It will shift from the LEOs to whatever.

[Brian Longwe] 12:28:55

I cannot predict the future. I'm not a prophet. I wish I could. I would be the richest man in the world.

[Brian Longwe] 12:29:03

I'm just talking from what I've seen happening over the period of my technical career, my technology career, and especially here in Africa.

[Brian Longwe] 12:29:13

That change is the only thing that is constant. And, you know, it's circular.

[Brian Longwe] 12:29:21

When something new is coming in It threatens the hold.

[Brian Longwe] 12:29:26

And the old either has to change or die As the old dies, the new is able to come in and take its place.

[Wycliffe Bahati] 12:29:32

Well, mm-hmm.

[Brian Longwe] 12:29:36

So you know And, you know, jobs will be lost when the old dies, but there are new jobs created by the new with different skill sets for different people and that's probably where you now find the friction.

[Brian Longwe] 12:29:54

I don't want to lose my relevance. But maybe a part of the old.

[Brian Longwe] 12:29:58

I should just retire and forget about And let these young people now be the ones to come in and take over and run with all this ai stuff And LEO and other types of you know quantum you know quantum is now the new buzzword

[Brian Longwe] 12:30:13

All these quantum you know And it's just It's a fact of life. It's the way of the world.

[Wycliffe Bahati] 12:30:14

And...

[Wycliffe Bahati] 12:30:22

Okay. And there's something where just something else I wanted to find out what's your opinion?

[Brian Longwe] 12:30:24

Yeah.

[Wycliffe Bahati] 12:30:33

You know, we... I realized that... the way the internet has operated over the last 20 years has shifted We've seen... content being almost distributed everywhere closest to the user You can see the CDNs.

[Wycliffe Bahati] 12:30:59

Amazon facebook trying to serve this content next to this next the use of Now, apart from that.

[Wycliffe Bahati] 12:31:12

We almost have the internet now, especially the Taiwan providers doing what's called geopolito geolocation routing.

[Brian Longwe] 12:31:23

Geofencing and all that.

[Wycliffe Bahati] 12:31:23

So basically. You find using native communities or something we find that they try to automatically restrict not restrict as such, but provide content from certain regions to certain regions.

[Wycliffe Bahati] 12:31:41

So i try to look at the LEO provider as it is now. It's almost basically layer two. I don't think that might root us in the skies yet.

[Wycliffe Bahati] 12:32:07

Pairing this with the service you're getting from your you know from your provider from the local fixed or mobile broadband provider What would be the impact of the user at the at the user end is when you start using leo given that the ground's tissue is

[Wycliffe Bahati] 12:32:22

I'm not likely to be in every location where we have this XPs or content.

[Brian Longwe] 12:32:31

No, this is a very good question. Weekly and um it's an area or a topic that I am very passionate about and I've been looking at closely in recent years.

[Brian Longwe] 12:32:52

If we go back and look at the history of the internet.

[Brian Longwe] 12:32:57

From the genesis and work our way up to the current to the present.

[Brian Longwe] 12:33:03

You will see that you will see that back in the day, the internet was pipes.

[Brian Longwe] 12:33:10

The internet was connectivity. There was a network that had networks connected to it. And in order to be part of that you needed to have a pipe that connected to it.

[Brian Longwe] 12:33:23

Right. Isps essentially what the internet isps controlled 80-90% of what was known back then as the internet. You couldn't get connected to the internet unless you had an ISP.

[Brian Longwe] 12:33:39

If you need an email address, you had to go to the ISP. If you needed a website, you have to go to the ISP if you need it.

[Brian Longwe] 12:33:45

You know any kind of access you had to go to the ISP.

[Brian Longwe] 12:33:51

That has changed completely. The ecosystem has become much more complex And it has um got totally new players. And in recent years, what has become very clear is pipes have become a commodity.

[Brian Longwe] 12:34:06

There are very many different ways of getting access. And what has now been the main driver has been contacted.

[Brian Longwe] 12:34:14

And content has now you know constituted probably what we can call 80 to 90 percent of what we call the internet.

[Brian Longwe] 12:34:23

So you have players like cloudflare Akamai Facebook's this with their CDN.

[Brian Longwe] 12:34:30

Netflix. These are the people these are the organizations In essence, but are driving the internet now That's because that's because they hold or control the content that I pose.

[Brian Longwe] 12:34:47

Want to get to so they can call the shots And these are the ones who are doing all of this geofencing And your routine.

[Brian Longwe] 12:34:58

Of their staff you know when you open netflix in malawi and you look at what is available to you, it's totally different from when you open netflix in kenya or when you open Netflix in London or when you open Netflix in the USA.

[Brian Longwe] 12:35:15

In fact, there are certain online streaming services that are not even available Amazon Prime.

[Brian Longwe] 12:35:23

Hulu and others disney you know um and so on and so forth. They're only available in the US, in Europe, in some markets in South Africa.

[Brian Longwe] 12:35:33

They've restricted the access based on the geo. Now, what does this mean?

[Brian Longwe] 12:35:39

It means that the content determines the content determines where the content owner determines where this content is going to be available and how it's going to be delivered Over there. And in my view, I would say that africa is way behind. We have not been able to

[Wycliffe Bahati] 12:35:51

16.

[Brian Longwe] 12:36:03

Ourselves to start creating and building our own content delivery networks.

[Brian Longwe] 12:36:08

What is African content? If you look at these platforms, what do they have on them?

[Brian Longwe] 12:36:15

They have western music Western movies, Western news.

[Brian Longwe] 12:36:21

Western media. Where is our african where is our music, our African movies, our African news, our African media.

[Brian Longwe] 12:36:32

It's sitting on those same platforms and piggybacking.

[Brian Longwe] 12:36:37

On the rules that have been created that they've created for their customers.

[Brian Longwe] 12:36:43

It's time or it's about time that Africa started to create Africans started to create platforms CBNs that can be able to serve african content on african terms and be available on african you know basis if you look at the consumption of nigerian nollywood movies for example

[Brian Longwe] 12:37:06

It's across the board. You will go to Zambia, you'll find them watching Nollywood and enjoying You go to Kenya, you find them enjoying Noli. You go, you know, wherever you are. You look at South African produce content, it's big.

[Wycliffe Bahati] 12:37:15

Yeah. Uganda, the same.

[Brian Longwe] 12:37:23

In most countries right so africans are enjoying consuming African content and it has continental appeal you will not find it being enjoyed in the West.

[Wycliffe Bahati] 12:37:35

Yeah, completely.

[Brian Longwe] 12:37:35

Because it doesn't appeal to them. I want to watch content with people that look like me, talk like me, have cultures that are similar to me.

[Brian Longwe] 12:37:45

And so on and so forth. Anyway, let me stop Let me stop ranting about the whole content theme But just maybe close it by making this point.

[Wycliffe Bahati] 12:37:45

Yeah, we go to McDonald's like me.

[Brian Longwe] 12:37:55

That the network infrastructure is nowadays being driven by the content demand and the content owners And this is also what has resulted in the huge explosion of data centers.

[Brian Longwe] 12:38:10

However, now we're beginning to see things changing in data centers you know.

[Brian Longwe] 12:38:17

Many of the CDNs are beginning to take more of an edge you know trying to work directly with ISPs trying to get you know a little bit more precise in their targeting.

[Brian Longwe] 12:38:33

And so on and so forth. So it's an interesting space and one that I think we will continue to see a lot of change happening in.

[Brian Longwe] 12:38:45

Yes, there's a point I wanted to make. Which was based on content.

[Brian Longwe] 12:38:53

Over the years of the internet's growth. And evolution.

[Brian Longwe] 12:39:00

One country that stands out in terms of a country that has an almost completely insular ecosystem is Korea.

[Brian Longwe] 12:39:11

Right Koreans the internet in Korea is totally different from everything else Koreans have content that is in the

Korean language And you will find that you know there's a totally different ecosystem and infrastructure that is all within the country.

[Brian Longwe] 12:39:33

Koreans have come up with their own CDNs. They've come up with their own exchange points. They've come up with their own access networks that govern and drive all of what they consume. You will not find Korean content being consumed, for example, heavily in Kenya

[Brian Longwe] 12:39:51

Unless it's by the Korean community in Kenya. The reason I'm saying this is if you now go and look at how much is streaming being used in Korea.

[Brian Longwe] 12:40:02

It's probably not being used at all. They don't need it. Yeah.

[Wycliffe Bahati] 12:40:06

Mm-hmm.

[Brian Longwe] 12:40:08

To them, everything they need for their internet is here in the country.

[Brian Longwe] 12:40:12

So the reason why people are using internet in Africa is because the internet is still out there And they need networks that get them there.

[Brian Longwe] 12:40:24

The submarine cables Starlink, another type networks. If we bring the internet here Just like Korea has done.

[Brian Longwe] 12:40:34

We are with all the content we need is here. All the media we need is here.

[Brian Longwe] 12:40:39

Are we going to need these networks? So that's a key dynamic.

[Brian Longwe] 12:40:43

I think that needs to also be factored into anybody thinking or talking about internet.

[Brian Longwe] 12:40:50

And Korea is a perfect example They don't depend on submarine cables.

[Wycliffe Bahati] 12:40:52

Thank you.

[Brian Longwe] 12:40:55

Because the content is here. I don't need to go outside of the country.

[Brian Longwe] 12:41:00

To read my newspaper. I don't need to go outside of the country to reach my movies my movie you know all of my movie subscriptions. I don't need to go to the country to listen to radio.

[Wycliffe Bahati] 12:41:05

Yeah.

[Brian Longwe] 12:41:12

It's all here on any device it's all here. So that's a perfect example.

[Wycliffe Bahati] 12:41:15

Yep. Yeah.

[Brian Longwe] 12:41:19

Yeah.

[Wycliffe Bahati] 12:41:19

Yeah, so in a way, I think given given what's happening is in as much as we still have the western influence a lot of content I was just analyzing and i saw Initially.

[Wycliffe Bahati] 12:41:41

My traditional observation used to be that we used to fetch about 80% of content from europe Now, there's a local service provider called Silkenet and I was looking at their traffic profile.

[Wycliffe Bahati] 12:41:54

It's actually the reverse. 80% of the traffic is coming from It's not local but it's local Only 15, 20%.

[Brian Longwe] 12:41:55

Yeah, yeah, yeah.

[Brian Longwe] 12:42:00

From the local internet. Yeah. Yes.

[Brian Longwe] 12:42:06

But those CBNs, but those CBNs are the foreign CBNs. So that's a phenomenon that has come about through CBN's global CBNs coming and putting their content delivery networks.

[Wycliffe Bahati] 12:42:10

Yes, exactly.

[Wycliffe Bahati] 12:42:20

Yeah, okay.

[Brian Longwe] 12:42:20

Into local exchange points and local data centers. So in a sense.

[Brian Longwe] 12:42:26

Those CDNs still have to be connected to the mothership. So yes, in terms of traffic flow in countries like Kenya where exchange points have really been able to optimize And I remember there was a big ISOC put out a very good report

[Brian Longwe] 12:42:46

About two years ago about the shift the 80-20 shift that for Kenya and specifically Nigeria.

[Brian Longwe] 12:42:54

The shift is now it's 80% local, 20% international as opposed to before where it was 20% local and 80% international but the actual content itself is still

[Brian Longwe] 12:43:13

Foreign right it's still Netflix. It's still music it's still BBC, it's still Al Jazeera, it's still CNN, it's still you know super sports it's still uh you know fox media it's still universal movies.

[Wycliffe Bahati] 12:43:13

It's similar. Yeah.

[Wycliffe Bahati] 12:43:19

Google.

[Wycliffe Bahati] 12:43:25

Okay.

[Brian Longwe] 12:43:29

And so on and so forth. So there is a nuance there's a nuance around the content And so there's traffic and then there's content There's the nationality of the traffic and there's the nationality of the content of the content

[Wycliffe Bahati] 12:43:31

Mm-hmm.

[Brian Longwe] 12:43:49

And if you look deeper, you will find that that is now where it begins to become a distinction.

[Brian Longwe] 12:43:55

And I think that's the problem that needs to next be that's the next hurdle that Africa needs to overcome.

[Brian Longwe] 12:44:03

That's the next hurdle. That Africa needs to sort of mitigate against. And that's the reason why When the subm

[Wycliffe Bahati] 12:44:03

Okay.

[Brian Longwe] 12:44:12

We saw it perfectly last year. West Africa was offline.

[Wycliffe Bahati] 12:44:14

And...

[Brian Longwe] 12:44:18

For a week you know how many eight countries in west africa because three cables got cut at the same time No, it was about a month.

[Wycliffe Bahati] 12:44:20

Yeah.

[Wycliffe Bahati] 12:44:29

Yeah, and it was bad because i think We also had the cuts

[Brian Longwe] 12:44:29

Three cables At the same time and they were offline.

[Brian Longwe] 12:44:34

Because internet is out there, it's not in here. If you go and you cut off the submarine cables serving korea nobody will notice yeah they will not notice because they will still be able to access their local content. They'll still be able to see their local news. They'll still be able to communicate with each other. They'll still be able to use the email.

[Wycliffe Bahati] 12:44:43

You might not notice it.

[Brian Longwe] 12:44:55

They don't use gmail like we do we get it out there you know so so i think that's another angle but that's another discussion for another day.

[Wycliffe Bahati] 12:45:00

Yeah, absolutely. It's true.

[Brian Longwe] 12:45:07

Yeah.

[Wycliffe Bahati] 12:45:08

So and then also just to get from how do you see how AI are regulators adapting to this uh you know this sterling or leo broadband service You know, it's... tricky because, you know, unlikely the deal, like... You could easily spot the dish. How do you license this service how are they adapting to

[Wycliffe Bahati] 12:45:41

Have you seen any changes how they try to license these LEO providers?

[Wycliffe Bahati] 12:45:44

Vis-a-vis how they used to license the old service providers.

[Brian Longwe] 12:45:50

Yeah, so I think especially for Africa, there have been different ways in which regulators have been responding.

[Brian Longwe] 12:45:59

A good contrast would be the way that the South African regulator responded to Starlink against the way that the Malawian regulator, for example, responded to Starlink.

[Brian Longwe] 12:46:13

So to date Starlink has not been able to get clearance to operate in South Africa.

[Brian Longwe] 12:46:21

And they've actually gone in their own way in their own way retaliate by on their network imposing geo blocks that prevent a styling device from another country from being taken into South Africa to be used.

[Brian Longwe] 12:46:40

Because that's what was happening. A lot of that was happening.

[Brian Longwe] 12:46:43

Because of the dispute between them and the regulator. Whereas with malawi in one meeting overseas Starlink was given had to blanché and though i told you know send us a request.

[Brian Longwe] 12:47:01

Within a week we'll give you authorization And that's literally what happened.

[Brian Longwe] 12:47:06

The regulator said, you know. We are not satisfied with the level of reach and coverage and access that our existing licensees are providing.

[Brian Longwe] 12:47:19

Here is Talik, they are saying that if you know they already have coverage over every inch of the country and if we allow them tomorrow that anybody anywhere in the country will have internet okay let's do it. So they gave them an authorization to begin operating. They gave them a very fast strat license with special provisions.

[Brian Longwe] 12:47:41

Now, of course, there's been a bit of friction between them recently.

[Brian Longwe] 12:47:49

Especially on issues of tariffs, because Starlink has been saying we started with some heavily subsidized tariffs But we now want to change them because the foreign exchange devaluations in malawi have literally eroded the dollar value to almost uh you know too much

[Brian Longwe] 12:48:06

Because they're being paid in local currency and the quatcha has devalued so much against the dollar that what we're getting is no longer what we thought it was at the beginning. So there've been a bit of back and forth and friction. Although recently there was actually

[Brian Longwe] 12:48:23

Agreements but the key point here being that That was a two different regulators who have responded in two different ways right Now, I think you have the in-between, which is where on one hand, a regulator is very resistant and imposes, like in South Africa, the reason for the

[Brian Longwe] 12:48:43

The debacle is that Starlink are saying we want a waiver on the 30% ownership requirement.

[Brian Longwe] 12:48:54

Local ownership requirement. And the south african regulator is saying who are you Who do you think you are to come and tell us to wave the 30% local ownership requirement no you either comply or you stay away.

[Brian Longwe] 12:49:12

And Stalin, I said, okay, we'll stay away. And we'll even restrict and block to make sure nobody can use our network in your country.

[Brian Longwe] 12:49:21

On the other hand. Malawi has given 100, has given them permission and they've said we will waive that requirement for you.

[Wycliffe Bahati] 12:49:22

Yeah.

[Brian Longwe] 12:49:30

And it's been the same in any every other country where Starlink is operating. They don't have a local Because it's their policy. We will not share with anybody. Nobody else will have a share in our thing, in our whatever. Give us a license to give it to us with a 100%.

[Brian Longwe] 12:49:47

Ownership of the local entity by starting uh global styling holding or whatever it is so you know That's a contrast.

[Brian Longwe] 12:50:00

You know and that's i think uh pretty much how the regulators are responding. Some of them are taking an

extreme view. Some of them are saying this is a good thing for us. Let's waive this.

[Brian Longwe] 12:50:11

And let them come in.

[Wycliffe Bahati] 12:50:13

Okay. I also did find out because you know they What is interesting about the satellite BIM is... it probably does not follow the country's boundaries How are realtors making sure that apart from stopping this thing at customs from coming in.

[Wycliffe Bahati] 12:50:39

How are they making sure that nobody can use it severely on the boundaries of the or the countries where they are

[Brian Longwe] 12:50:48

There's nothing you can do about it. Which is one of the reasons why It's the perfect surveillance tool.

[Brian Longwe] 12:50:58

Nobody can risk these things are 1,500 kilometers in the sky short of you figuring out a way to create some kind of barrier in disguise over your airspace. There's nothing you can do.

[Wycliffe Bahati] 12:51:11

Yeah. And then...

[Brian Longwe] 12:51:14

So regulators have no say and no control if somebody manages to sneak that device in to their country and that device is allowed to be operated.

[Brian Longwe] 12:51:27

By starlink. That's it. It will work.

[Brian Longwe] 12:51:31

You know um Yeah, so, so, so, yeah.

[Wycliffe Bahati] 12:51:33

Yeah.

[Brian Longwe] 12:51:37

Regulators have very little. Actually, I would not say very little they have no zero control unless they run some kind of terrestrial jamming jammers that interfere with the frequencies that starlink is using for the ground for the ground terminals.

[Wycliffe Bahati] 12:51:52

It's really expensive.

[Brian Longwe] 12:51:57

And that would be extremely expensive.

[Wycliffe Bahati] 12:52:00

Yeah. What about data privacy? And... Yeah, so cybersecurity and other privacy I mean... these laws the data privacy laws are now being enforced by most countries Some of them are quite stringent, requiring that data be stored locally now with the

[Wycliffe Bahati] 12:52:26

Starlink. Or low broadband is most likely that maybe the ground stations will not be in those countries.

[Wycliffe Bahati] 12:52:35

I mean, how do regulators adopt to this?

[Brian Longwe] 12:52:40

Well, I would like to say that even if the ground station is in the country, that ground station is not doing any storage. The ground station is only part of the network infrastructure.

[Brian Longwe] 12:52:50

It's not part of the bss or oss it's not part of the operational support systems or the billing support systems those billing and operational support systems are sitting comfortably.

[Brian Longwe] 12:53:02

In the USA. So every single Starlink subscribers data is sitting in America.

[Brian Longwe] 12:53:11

Like it or not. So there's no way that there's no way

[Brian Longwe] 12:53:21

Any local Kenyan or malawian or Nigerian data protection law can be imposed on starlink with any significance.

[Brian Longwe] 12:53:35

Because... You know, they're not even physically present in your country.

[Wycliffe Bahati] 12:53:35

Yo.

[Brian Longwe] 12:53:39

If you want to go and arrest them. Or violation of you know using brian's subscription information the wrong way You can't find them. Elon Musk is sitting comfortably in the US actually nowadays. He spends most of his time in Washington.

[Brian Longwe] 12:53:58

You know um so so that's a reality yeah Yeah.

[Wycliffe Bahati] 12:53:59

Yeah, so actually And I think, like I mentioned before, imagine having your name And now with this terminal, they can actually know exactly where you stay.

[Brian Longwe] 12:54:13

Not just know exactly where I stay, but they can listen in on all my conversations they can sloop into everything. They can even turn that device into a you know uh a surveillance equipment.

[Wycliffe Bahati] 12:54:19

Earlier emails.

[Brian Longwe] 12:54:31

You know, definitely. There is a risk and there is no way you will know whether they are doing it or not because the network is not somewhere where you can touch it.

[Brian Longwe] 12:54:42

All right. It's not somewhere you can touch it. And even if you go and you close or shut down the power to a ground station.

[Brian Longwe] 12:54:53

That doesn't mean the network stops. The satellites just simply three root and use another ground station in another country so these are very real the risks, which I don't think our people have really thought so

[Wycliffe Bahati] 12:55:12

Yeah, that's why that's why I'm wondering, because this is not I know these regulators have been used to licensing geosol client services, but this is different to me.

[Wycliffe Bahati] 12:55:25

This is a bit... Yeah.

[Brian Longwe] 12:55:25

Yeah, totally. With the geosatellite service, the signal is originating from the ground.

[Brian Longwe] 12:55:33

So you have that ground station. You can now impose whatever controls or measures we want on that ground spatial.

[Brian Longwe] 12:55:45

The satellite in the air is simply acting as a relay.

[Wycliffe Bahati] 12:55:49

Really, yeah. Mm-hmm.

[Brian Longwe] 12:55:50

To another gravitation or to another terminal. So, you know, in a sense you have control in quotes.

[Brian Longwe] 12:56:00

Now, it's the other way around. The network is in the sky.

[Brian Longwe] 12:56:05

You don't have any control.

[Wycliffe Bahati] 12:56:07

Yeah, it's interesting yeah so just finalizing I mean, what did you sum up like what do you see the future of this uh leopard print.

[Wycliffe Bahati] 12:56:21

I mean, it's hard to tell, but... What's your feeling given your experience?

[Wycliffe Bahati] 12:56:26

The next five years, where will you be? Will you have everybody using it? Is it just a pump that will... Good

comment.

[Wycliffe Bahati] 12:56:36

Go down or will innovation make it relevant.

[Brian Longwe] 12:56:44

I would say that In the next five to 10 years, I would say LEO is going to just become one of the other ways that people can connect to the internet.

[Brian Longwe] 12:56:57

It will be one of the options or the alternatives that's available And one will have to think carefully about Well, not carefully, but one will have to weigh different considerations as they decide which connection do I want?

[Brian Longwe] 12:57:12

What does this give me that this one doesn't give me?

[Brian Longwe] 12:57:17

And, you know. The LEO constellations we will have different pros and cons.

[Brian Longwe] 12:57:26

And consumers will have to decide You know, do the pros and cons of this particular LEO constellation work in my favor and give me what I'm looking for Or should I stick with a terrestrial provider that gives me what I'm looking for.

[Brian Longwe] 12:57:47

Yeah, so I think the marketplace is just going to have more options going forward.

[Brian Longwe] 12:57:55

And that's how it's going to be. With the current trend in geopolitics and the hostilities that I'm seeing beginning to emerge now.

[Brian Longwe] 12:58:07

Between different countries and different regions. There is a high chance that you may even now begin to find LEOs like stalactic.

[Brian Longwe] 12:58:19

Beginning to find certain roadblocks being put in their way.

[Brian Longwe] 12:58:24

By different countries that don't see eye to eye with them.

[Brian Longwe] 12:58:29

And it might get a bit nasty, especially up there in space But let's wait and see.

[Brian Longwe] 12:58:36

Let's wait and see. If you really think about it If somebody decided to launch encounter you know well not let's let's not call it counter but if somebody decided to sabotage For example, the LEO constellation of static Yeah, these things are 1,500 kilometers out there in the sky

[Brian Longwe] 12:59:00

It's not like you have anti-missile

[Brian Longwe] 12:59:09

Countermeasures similar to what you have on the ground currently protecting most countries.

[Brian Longwe] 12:59:14

You know so by the time you realize what's going on you may find that you may find that some of these a good number of these satellites may have been you know messed up.

[Brian Longwe] 12:59:26

And I wouldn't put it past some of these enemies of so-called the West.

[Brian Longwe] 12:59:33

To have such plans in the making. And that becomes a risk Because the landmark is brought down Yeah.

[Wycliffe Bahati] 12:59:38

Noxian is already opened. So that happened, I think Russia and India have tested and desert flight missiles.

[Wycliffe Bahati] 12:59:50

I think Russia used to bring down one of an old satellite and it has cost i think 70 of the From what around 70 or 80% of the debris in space in the leo space were caused by that particular

[Wycliffe Bahati] 13:00:05

Explosion the russians did. Okay, because they're taking longer than usual, I think, to come back to the, I suppose, atmosphere and banner.

[Wycliffe Bahati] 13:00:14

So what you're talking about is real. And this is very tricky given the very tricky the politics.

[Brian Longwe] 13:00:17

Yeah, no, no.

[Brian Longwe] 13:00:27

Yeah, and no one, no one think what will happen in the future. So we just have to wait and see.

[Wycliffe Bahati] 13:00:27

Okay.

[Wycliffe Bahati] 13:00:34

Yeah, okay, so I think I'm good. I think you've covered most of what I... I had my questions.

[Brian Longwe] 13:00:34

Yeah.

Participant 5 CK

Name Of Participant: Carol Kamaitha (CK)

Profession: Expert with over 25 years in Satellite services

Date of interview: 10th March 2025

[Wycliffe Bahati] 10:43:49

Number one. Competition and market dynamics from your experience.

[CK] 10:43:51

Mm-hmm.

[Wycliffe Bahati] 10:43:56

How has the production of leo breadcrumb affected business strategies for ISPs and landlords?

[CK] 10:43:57

Yeah.

[CK] 10:44:03

Okay, so I... I can say maybe a couple of things there. One being... From a strategy point of view. So we've seen a lot of isps being forced to re-look at their own internal strategies in terms of where they add value

[CK] 10:44:24

So let's maybe take a step back just to be clear in terms of my understanding and you're understanding the

same when you're talking about Leo.

[CK] 10:44:32

So the only active commercially viable LEA operator today, Starlink. The rest of the guys are filed so you know Kuiper with Amazon they filed their done a few prototypes. I think they must have launched at least two satellites just because you know you filed by with ITU and then you have to

[CK] 10:44:52

Do some things to show that you're building a constellation. And they build the hiring people, put their systems in place, etc so for all indications there.

[CK] 10:45:02

Constellation will come good. And so I feel like when I talk about Leo I'm limited to Starlink.

[CK] 10:45:12

In terms of that they're the only active operator. Is that what you're looking at as well or do you want to have a more general discussion okay

[Wycliffe Bahati] 10:45:17

Correct. For now, it's the link. I think one way also has some model number about 600 circulates so but mostly the one that is commercially doing anything in the broadband world properly styling.

[CK] 10:45:24

Okay.

[CK] 10:45:29

Yes, yes.

[CK] 10:45:36

Yes, darling, yes. Because OneWeb also have the... the constellation which has been marked by a couple of delays I think they're trying to replace their generation one generation one. We haven't seen them as vibrant and they don't have as much scope.

[Wycliffe Bahati] 10:45:37

Yeah.

[CK] 10:46:00

But then if you look at both of them, they'll go to market. So we can start before we talk about the strategy on

the downstream with ISPs and mnos we can look at what strategies is Leo are the Leo operators adopting.

[CK] 10:46:14

So the starting strategy has been very, very disruptive for ISPs.

[CK] 10:46:18

Because Starlink is not going through any local ISPs. As a business strategy.

[CK] 10:46:27

Their desire and i know because I've had conversations a couple their desire is to be the largest independent ISP globally.

[CK] 10:46:36

So theirs is to enter a country and go direct. And Nana has come up with many fights with the MNOs like we've seen in Kenya with safaricom.

[Wycliffe Bahati] 10:46:41

Yeah.

[CK] 10:46:46

I think we've seen other countries in Africa that have pushed back And said, look, we don't have data sovereignty if you're just coming and people are not going through the traditional model and the traditional model being is a local and country gateway. There's data sovereignty. There's some data that needs to be routed locally.

[CK] 10:47:05

But with Starlink, the government doesn't even know what is being communicated.

[CK] 10:47:10

So that is Basil is a threat and that's why a lot of African nations were pushing back the ones that had the muscle. But when we had these major fiber crisis last year, we had to think about six countries in the west coast of Africa that went offline because of the cable cuts.

[CK] 10:47:28

And then we had some issues in Yemen. That affected part of the east africa traffic Then you had some issues that affected the Southern Africa traffic and i'm talking about fiber only We saw that governments changed and we saw governments that are taking a hard stand, the likes of Ghana

[CK] 10:47:46

Coming overnight and saying, okay, we can compromise and adopt starling and so on and so forth. But then that has also backed something else where now the fiber operators are getting together to figure out How do they create more redundancy?

[CK] 10:48:02

To avoid this kind of situation so there feels like there's a wall there of who gets what.

[CK] 10:48:08

But then I'm going ahead of myself. I think I'm answering multiple questions but Coming back now to the market dynamics on competition. So that's how I'd look at it. So what's the strategy of the leo's?

[Wycliffe Bahati] 10:48:13

Yeah.

[CK] 10:48:19

And others ask Leo strategy intersect with the ISB strategy. So we're telling From a personal view, and I tell you the number of ISPs that are completely and totally pissed off.

[CK] 10:48:31

Like they're like, just like we don't have a choice because it's Starlink and they're as big as they are.

[CK] 10:48:38

But they're being cut out of the value chain. We've seen what has happened. And now when you look at now more globally What are the areas, one of the verticals at Starlink or Leo went in in a big way so

[CK] 10:48:52

We take a step back. We look at the satellite industry there is your ISP broadband type of customers.

[CK] 10:49:00

Then another have one line of ISP, you have MNOs and telcos.

[CK] 10:49:04

So MNOs will be the MNOs, telcos being the Incubin, telcos are out there.

[CK] 10:49:08

So that's one bucket because they go direct to the end users and they have They also go to enterprises, SME, medium enterprises etc which is government and government is huge, especially the likes of the US

government.

[CK] 10:49:26

That governments in Europe, whether you're looking at Italy, France, all those big european entities are there. So selling to government is now the big contender in satellite. Any satellite operator that's going to be successful has to be selling to the US government and the European government.

[CK] 10:49:45

African governments and Latin American governments, the biggest challenge they have is that they don't have the money.

[CK] 10:49:52

The need is there, but there's an issue of money Okay, and then you have another bucket where you talk about aero arrow or mobility.

[Wycliffe Bahati] 10:49:53

Mm-hmm. Okay.

[CK] 10:50:03

So in-flight connectivity And anywhere, anything that is moving that requires satellite connectivity. So whether it's aero, your aircraft Whether it's... ships, whether it's a cruise liners or connectivity at sea, you know, the super yaches and all these other big anything that is on water or on the air.

[CK] 10:50:28

And that's also a very big business. It's a highly profitable business with the insert light.

[CK] 10:50:33

Because you won't get fiber in the middle of the ocean.

[CK] 10:50:36

You won't get fiber in the sky. So whenever you think about the satellite industry.

[CK] 10:50:42

The broadband, which is ISP, telco mno is a very tough space to compete because there are so many alternatives.

[CK] 10:50:50

Fiber on the backbone whether it's delivered by a any of the wireless solutions that you find, right?

[CK] 10:51:00

But government, because of the special comms and sometimes places where you have provide connectivity for government, areas where there's no connectivity that's still a prime area.

[CK] 10:51:12

Market area for satellite and the mobility part is also very big. So when Starlink, for example, came in, some of the markets they forecasted very, very strongly was within the mobility space.

[CK] 10:51:26

The mobility space is organized in a very different way. So you have the integrators.

[CK] 10:51:30

Because there are a lot of other things like health and safety and limited real estate on the ships and the boats and all the other dynamics that come in.

[CK] 10:51:39

So you have specialized installers or integrators that come into that space. And then now you have the space segment that comes in.

[CK] 10:51:48

But the strategy that starlink used On one end, they try to go directly And then they realized that they did not have the full expertise to deliver the managed services that are very important to to the likes of the cruise liners and all those guys and even aircraft aircraft you can't go and just tear apart an aircraft.

[Wycliffe Bahati] 10:52:08

Okay.

[CK] 10:52:13

You have your Panasonic, you have those that are like about three key players that do the integration But Stalin disrupted everything. If anything, when you think about strategy and competition The biggest disruptor has been Starlink. Onweb has towed the line, and I've seen the strategies of going direct to ISPs, they've gone to resellers.

[CK] 10:52:35

And there's been a big debate around what is the right approach? There's some people who hate the Starlink approach Because they don't care about the middleman, in this case, ISPMO.

[CK] 10:52:49

They don't even care about the government to some extent because if you go like to Sudan today because of all the civil unrest that has been there.

[CK] 10:52:57

There are thousands of styling kids. The government hasn't given formal approval, but people are just buying them. You buy them in Egypt, you buy them in Dubai, bring and set up.

[CK] 10:53:06

So this is that thing of they're not even listening to what the government is saying.

[Wycliffe Bahati] 10:53:07

Yeah.

[CK] 10:53:10

And so there's just a general sentiment around Starlink that It's just that they have rich And because we have the challenge of poor coverage.

[CK] 10:53:20

And there is demand that governments are struggling to even regulate competition in this case or put into place what we knew like in highway Where you had to get an ISP, get approval, get all those things.

[CK] 10:53:35

That has sort of gone out of the door. So... there's a lot more I can say, but let me measure myself.

[Wycliffe Bahati] 10:53:36

Yeah.

[Wycliffe Bahati] 10:53:44

Okay, so I just... So just order it right now.

[CK] 10:53:44

Right. Stick to the questions.

[Wycliffe Bahati] 10:53:49

Because one of the the major I mean, marketing points or okay Major... things that Starlink or the leo providers are trying to say is that we are going to bring on board new markets and underserved markets.

[CK] 10:53:49

Oh.

[CK] 10:54:10

Yep.

[Wycliffe Bahati] 10:54:10

And if you've seen SpaceX, I mean, Starlink they get special funding from like congress guest will be able to break this digital divide Do you think that... that the styling approach is actually bridging the digital divide.

[CK] 10:54:21

Yeah.

[CK] 10:54:26

Mm-hmm.

[CK] 10:54:33

So it definitely is not. Yeah, so it's not. And this is the, I think this is a very interesting area of research and I've seen enough There's this belief that a satellite is for rural areas.

[Wycliffe Bahati] 10:54:33

I mean, connecting the unconnected.

[CK] 10:54:47

And if I look at what we did like with O3B, the meal constellation Most people are like, okay, we'll go to the rural areas.

[CK] 10:54:56

But like in Africa. I could say like 98% of our business was in capital cities.

[CK] 10:55:04

Because there's still a demand. If you look at now, like Starlink, where I live, I live in Upper Hill.

[Wycliffe Bahati] 10:55:05

Okay.

[CK] 10:55:10

And I was trying to get a styling kit early last year.

[CK] 10:55:13

But the area is sold off. It's completely sold out. So you cannot get connectivity in urban centers.

[CK] 10:55:20

And that if you look at how Starlink, what Starlink has been reporting is there are places they cannot sell into capital cities anymore.

[CK] 10:55:27

Because they're completely sold out. So then it goes to the rationale of the rationale why people buy connectivity. So even when you have like a leo solution so the leo solution has multiple benefits. One being the low latency And it's fairly affordable.

[CK] 10:55:42

But it's not affordable for rural areas. Like it has to be super, super subsidized for this to be sold in a village in Kitale, for example.

[CK] 10:55:52

But in Nairobi, in the urban areas in Lovington, every house can easily get a styling kit so that when it rains.

[CK] 10:56:00

And we're having issues with the whatever. I mean, maybe not rain, but when there are issues.

[CK] 10:56:06

With your local connectivity, you're still able to connect And then that speaks to digital literacy as well, because with all the research that has been done.

[Wycliffe Bahati] 10:56:14

Okay.

[CK] 10:56:15

That a demand for digital capacity for connectivity will only be sustainable where there's a level of literacy and people appreciate the need and they use. Now, I can't go five hours without connectivity But this guy in a rural village in Italy.

[CK] 10:56:30

Doesn't mind. If it works, it works, it doesn't. His way of life is not deeply impacted.

[CK] 10:56:35

Now, mine is because if I need to pay, if I need to order gas.

[CK] 10:56:39

I need connectivity. I need to go online. I need to do whatever I need to do most of the services even now government offices government office services whether it's an easy citizen or whatever, I need to be online for a lot of the things that I do.

[CK] 10:56:52

So if I look at how what Starlink has been reporting is, first of all, they're pretty much sold out. I know they're sold out like in London.

[CK] 10:57:00

That was one of the things I reported a couple of months back. Then places like Nairogis sold out most of the capital cities. And there's a question that people now raised I remember having that online debate asking the same question you're asking.

[CK] 10:57:14

Shouldn't they be focusing on rural areas? Why are they eating into the market share?

[CK] 10:57:20

Of your likes of Safari com. Exactly. But I think it begs a deeper question, which is why do people feel that they need to go to Starlink?

[Wycliffe Bahati] 10:57:20

Existing Yeah.

[CK] 10:57:28

To buy this service? Are they so dissatisfied? With what is being provided by the current you know the current vehicle.

[Wycliffe Bahati] 10:57:31

Oh. Yeah, I think... Yeah, we'll come to that maybe later and see to internalize that.

[CK] 10:57:40

Yeah.

[Wycliffe Bahati] 10:57:41

So actually, what you put in the note is very interesting because yesterday I talked to, you know, there's a guy called Brian Longo. I don't know if you know him, Malawi.

[Wycliffe Bahati] 10:57:50

And he said he runs it. He runs an ISP called Convert. And he said when Starlink came the government said they will have less coverage. The Starlinks are saying that they can reach everyone immediately. But the moment he they came in.

[CK] 10:57:51

You sound familiar, yeah.

[CK] 10:58:09

Mm-hmm.

[Wycliffe Bahati] 10:58:12

The first people they went to who took off his upper class customers.

[CK] 10:58:17

Exactly.

[Wycliffe Bahati] 10:58:20

Soon. Anyway, so he says they're coming back now. So what you say the observation is very valid and um So the other thing I wanted to ask you, I mean, this is very inspiring. What do you think about the IoT market.

[CK] 10:58:25

Yeah. Yeah, yeah.

[Wycliffe Bahati] 10:58:39

How will stalling impact?

[CK] 10:58:41

Ah. Okay, and this is for Africa globally?

[CK] 10:58:46

So. Generally. So... I think generally, I anticipate it'll be the first market is supposed to take this up would be like in the US, for example.

[Wycliffe Bahati] 10:58:47

Generally, I mean... Yeah.

[CK] 10:59:00

Or Europe and in the way I understand it from the way I understand from it's the applications that come with it.

[CK] 10:59:08

So we've had this whole long discussion around self-driven cars and how they need to work.

[Wycliffe Bahati] 10:59:13

Correct.

[CK] 10:59:14

And for you to even have a self-driven car, you have to have IoT because then you need all these sensors that tell you you're going right or left or there are people, ETC, ETC.

[CK] 10:59:22

So I think I sort of feel like IoT will be application driven And connectivity also will influence.

[CK] 10:59:29

If we had enough connectivity in Africa, then we'd get to that point of we think, okay, how do we get more application?

[CK] 10:59:35

Although having said that, I know I see small applications of IoT, but not in a huge scale you know like if you go to an office there are people who say like, you know, as long as you have your health and safety. There's a level of applying IoT like in manufacturing

[CK] 10:59:52

Where you maybe don't need people there all the time, but you have devices and gadgets that are able to send and send back feedback etc But it's the deployment of IoT, which still feels very complex for Africa. First, you need the devices.

[CK] 11:00:09

You need the application that runs on this. People need to be educated. I know in the mining sector in Africa and coming from like a lot of the mines we used to service with SES, OTB, they used to have some level of IoT where you have just sensors

[CK] 11:00:27

But then you need an integrator who has the capacity to do this You need to specially build some of these, customize them to the local environment.

[CK] 11:00:36

So I almost feel in Africa we are such a long way from IoT. You'll see it growth in very specific sectors, manufacturing, mining.

[CK] 11:00:47

Leading is special because they have the money. And they are able to adopt technology quickly. And when you look at a lot of mines in Africa, not even a lot, majority, I'd say 98%, are not minds are minds that are owned

[CK] 11:01:03

By entity sitting in Canada. Or introducing in the US or in Europe somewhere. So instead thinking about Anglo and Ashanti So thinking of... copper, a lot of the copper mines They are listed in the Canadian Stock Exchange because they are parent companies are there. So technology naturally then just flows. Whatever they're doing in Europe, then they can actually do it locally.

[CK] 11:01:27

When you look at solid manufacturing, you go to industrial just here in Nairobi.

[CK] 11:01:32

You won't find a lot of these automation. We're still labor intense.

[CK] 11:01:36

And that's the thing. As long as we have people there's almost a feel of why do we need IoT in manufacturing when you have Kamau and Jorge.

[CK] 11:01:44

Who are going to carry everything and put their lives at risk because that's where we are. And then there's a digital literacy as well.

[CK] 11:01:52

So I haven't read up too much on IoT, but everything I read tells me generally that um application is still formed.

[Wycliffe Bahati] 11:02:00

So I was looking... I was looking at IoT in terms of you know we use ILT for remote monitoring. For example,

just an example in the polar regions where this very cold or extreme conditions, we tend to have sensors there

[CK] 11:02:17

Mm-hmm.

[CK] 11:02:26

Yeah. Yeah. Mm-hmm.

[Wycliffe Bahati] 11:02:30

That relay information like whether So I was looking at that market because generally it's really been served mostly with the geocatellites most of the time.

[CK] 11:02:33

Yeah, yeah.

[CK] 11:02:44

Mm-hmm.

[Wycliffe Bahati] 11:02:45

So I was trying to see if actually this leo circulates would enhance capability because With Jio, you had limited circulate capacity and expensive so now we Neil you have more capacity.

[CK] 11:02:53

Yeah. Yeah.

[CK] 11:02:59

Yes, yes, yes, yes.

[Wycliffe Bahati] 11:03:04

Being delivered so So are we able now to get more images, yeah.

[CK] 11:03:05

I agree, yeah. Yeah.

[CK] 11:03:10

More so So I agree with you. I think from a physics point of view, just as you've described it, from a science point of view.

[CK] 11:03:18

We should be able to get more. But then it comes to the commercial. So if Starling has the choice to sell that capacity to the North Pole.

[CK] 11:03:27

Vases are densely populated city. Miami, for example.

[CK] 11:03:33

Where there's a lot of demand, I think it'll come down to the commercials.

[CK] 11:03:39

It's now a commercial strategy, but from a sciences point of view.

[CK] 11:03:43

It makes absolute what you're saying. It makes complete sense. But the commercials.

[Wycliffe Bahati] 11:03:48

Okay.

[CK] 11:03:50

Are the ones that will be because even with leo The thing with satellite capacity, it's still a finite capacity. That's why Starlink is saying, for example, they can't sell in Nairobi anymore, the council in London. So if they got a few more megabits, would they put it here or put it in the North Pole?

[CK] 11:04:08

I think he'll go to the question of who pays the most.

[CK] 11:04:12

Or what fits their commercial strategy. The same with OneWeb. So I've seen the OneWeb like coverage map across Africa.

[CK] 11:04:19

Especially around the equatorial area, DRC part of kenya. They have very limited capacity just because of the science. We think the especially over the Equatorial region.

[CK] 11:04:33

You tend to have less capacity comes back to how you, if I can remember correctly.

[CK] 11:04:40

There's always the separation between whether it's Leo, Mio, and with the geosatellite so you avoid interference. So there's a bit of offsetting they do around the equatorial region.

[CK] 11:04:52

But they're very, very limited. Again, so one web needs to make that decision. For example, are they going to put all that capacity in DRC, whatever they have.

[CK] 11:05:01

Or do they give you to a place like the Northwood Auto applications that may not have the same commercial attractiveness. But science correct commercial And that's the thing with the satellite. I think sometimes we underestimate how... how much commercials play a role on the strategy

[CK] 11:05:24

Offset letter. So that's GeoMe or leo because of the finite nature.

[CK] 11:05:30

Of the satellite capacity unlike fiber which is easy you can go fire up, maybe deploy fire is the fire up payers or whatever It almost feels like geo capacity is limitless. Of course, it's not limitless.

[CK] 11:05:44

But it's a lot more than satellite capacity. Um yeah

[Wycliffe Bahati] 11:05:52

And now still yes that marketing dynamics and competition Do you see any chances or opportunities for collaboration?

[CK] 11:05:59

Yep.

[Wycliffe Bahati] 11:06:03

Between the traditional manure's ISP and the LEO providers. In terms of 4K.

[Wycliffe Bahati] 11:06:12

Can we complement each other?

[CK] 11:06:16

So I wouldn't see that if you look at this purely from a logical maybe business strategy points of view ISPs are

there because the big operators.

[CK] 11:06:31

Are not able to offer some level of service. So if you took him an SES or you take Starlink one web utilise whatever the big players The main reason they go through ISPs is for the local licenses because that means the local government has institutionalized that for you to do this, you have to pay a certain amount of license fees.

[CK] 11:06:52

So for a lot of these big operators, the Leos. They'd much rather go through a partner who understands the local terrain, has people, ETC.

[CK] 11:07:02

There's always issues around equipment, getting equipment in country, clearing, deploying. So for the big operators, it's it's a positive thing.

[CK] 11:07:12

That they don't have to get involved with. Shipping and logistics and ETC.

[CK] 11:07:17

Although I say all that. Acknowledging that Starlink has come and done the complete opposite.

[CK] 11:07:24

And somehow they've managed. And maybe I'll talk about how they've disrupted that model.

[CK] 11:07:28

So ISPs as well, apart from the regulatory, satisfying regulatory conditions or requirements, providing feet on the ground for integration, export, import, deployment, implementation, and all those things.

[CK] 11:07:44

They carry the local financial risk, billing, collection, ETC, and they pay these guys.

[CK] 11:07:49

So for a lee operator or a satellite operator. That's the value.

[CK] 11:07:56

And that's what has always kept and people thought when like Starlink came, they did the same thing.

[CK] 11:08:01

Pasadlink has come in and said, we don't care. We'll ship you equipment.

[CK] 11:08:06

We'll send you a kit that is so easy You as Carol, can install it in your own house.

[CK] 11:08:12

So you don't need an engineer. Unlike before where we had teams of VSAT engineers and all these types of other engineers who come and do installation troubleshooting ETC.

[CK] 11:08:23

And everything is on a digital platform. And the expectation is that Kaipah will do this because they have the background of Amazon.

[CK] 11:08:32

And Amazon being great at shipping stuff, you know, they just leverage on amazon they can put a Kuiper kit in every house all over the world. They are good at logistics.

[CK] 11:08:44

In times of regulatory because Starlink has come and used their financial muscle to say, we don't care, you're the Kenyan government.

[CK] 11:08:54

People will go buy those kits and they'll figure it out as they go. I mean, I know people in Ethiopia of all places They just want a way of getting it across the board and not Ethiopia, for example.

[CK] 11:09:05

Heavily, heavily regulated when it comes to this kind of communication.

[Wycliffe Bahati] 11:09:08

Greatly. Yeah.

[CK] 11:09:10

So I don't believe that there's a country in the world now that doesn't have a single kid, legal or illegal.

[CK] 11:09:16

And the kit is small enough that it can be smuggled into the country.

[CK] 11:09:20

Or it can access the Buddhas easily. So on one hand, there are all those values that the ISPs have always hold held dear And they've known that the big operators will come calling on them.

[CK] 11:09:36

But I don't feel and understanding the styling strategy and anticipating what the one-web strategy is going to be It doesn't seem evident that doesn't evident there is there are compelling reasons to do this now.

[CK] 11:09:51

There's some exceptions that I've seen in Africa. Is Malawi for Starlink. So I know that in malawi At least what I understood. I don't know what I had from from people in the industry that because of the challenge of

[CK] 11:10:08

Access to dollars and affordability and payment systems. Malawi is not as developed as Kenyan. Some of the countries, maybe even Nigeria, where you're able to do a lot of online payments. So the financial services sector are still behind.

[Wycliffe Bahati] 11:10:21

Yeah.

[CK] 11:10:23

So there's this guy and it's not the guy you talked about i think it's a different company, but I remember. And I can send this to you later.

[CK] 11:10:30

And what he did is he did You know, he got the silent guys. They had a conversation. He said, look, I will de-risk you financially.

[CK] 11:10:38

Send me a couple of hundreds of kits. And those kids, I will sell them downstream And I will accept payment. I'll pay you, but I'll accept payment.

[CK] 11:10:48

So this guy is just like an equipment bender of sorts.

[CK] 11:10:51

But the service subscription fees are not necessarily paid through him. I think some of them are, some of them are not.

[CK] 11:10:58

What he did is he went to Starlink and said, I'll fix for you your access into the Malay market and find your way to do financial payments.

[CK] 11:11:06

In good time, if those financial payment issues are sorted out.

[CK] 11:11:10

Starlink had not known for loyalty in the market. In fact, a lot of the ISPs that I know are pissed off.

[CK] 11:11:16

They said, okay, let's do this thing together. They talk about a bunch of things and they disappear And then guys are fired or here there's a new guy there and everything changes you know changes So it's a very brash

[Wycliffe Bahati] 11:11:32

Yes.

[CK] 11:11:32

American way of doing business. But I would say that that that's still to be seen.

[CK] 11:11:39

But right now, with all the indicators It's not clear that it is clear Starlink or Kuiper will see value in the ISPs.

[CK] 11:11:51

Or the MNOs. And the MNOs, now it's a whole different ballgame because of the direct-to-device conversation.

[CK] 11:11:56

That now all these discussions. Of course, we know it's not going to happen overnight But there are all these discussions about the direct to device. What does that mean for all these base stations? What does this mean for areas that have low coverage? And we've seen the steps like the Kenyan government has taken saying, look, Starlink, if you guys come here.

[CK] 11:12:13

You have to pay a very heavy license fee and stuff like that. And those are the options. The regulators, governments can put into place very punitive

[CK] 11:12:27

License fee regimes and all those things But for me, I think it's a losing battle. And the only way is if we can fix the internet penetration or the availability of connectivity, if that can be fixed.

[CK] 11:12:40

And the local operators are doing a good job. Then it doesn't have to be worried about Starlink. But as long as the gap is there and demand exists.

[CK] 11:12:49

People break the rules and the government doesn't have their capacity to police everybody everywhere.

[CK] 11:12:55

You know so

[Wycliffe Bahati] 11:12:56

Actually, there's a guy in ruin it's called George works with Rwanda on these regulatory authority and you know saying that um They actually went ahead and did a blanket license but then To be honest, it says they say They don't know if every styling that is out there is...

[Wycliffe Bahati] 11:13:19

It was Jimmy is actually illegal. You've got it.

[CK] 11:13:22

No, they don't. But then also you have to understand Rwanda because Rwanda has a differentiated relationship And again, these are things we pick from the grapevine. So Rwanda is a is a special styling market.

[CK] 11:13:36

They will do business there and they will not be hindered.

[CK] 11:13:41

And some of the things I had and i saw firsthand What are the strategies that Starlink has, which is very admirable.

[CK] 11:13:48

They would leverage on the American Chamber of Commerce. To get their people in.

[CK] 11:13:54

So before I started link sales guy this guy, good Ryan, good night to go in the US, before he shows up.

[CK] 11:14:03

To a country a lot of the things have been put in place.

[CK] 11:14:07

Because, you know, and we've seen Elon Musk even right now is very close to presidents and DTC. He will continue to leverage that relationship to push starlink.

[Wycliffe Bahati] 11:14:16

Of course.

[CK] 11:14:19

Into the different markets yeah But run it.

[Wycliffe Bahati] 11:14:21

Yeah, yeah. Okay. Yeah.

[Wycliffe Bahati] 11:14:27

Hello? Yeah, yeah, yeah, yeah. So actually, I will come to i think we'll come to that someday.

[CK] 11:14:28

Yes, I am Stacy.

[Wycliffe Bahati] 11:14:33

When you look at the sustainability factors. So I just joined us to divert a bit, discuss briefly from your perspective and what you've had How is this?

[CK] 11:14:37

Okay. Okay.

[Wycliffe Bahati] 11:14:46

For example now the service offered by Starlink compare with the fiber products we have and um and the mobile broadband we have in terms of speed, latency and reliability.

[CK] 11:15:00

Yeah. Okay. So I haven't tried one because I couldn't get one here.

[CK] 11:15:05

Not for any other reason, because I was also very curious. And I was like, this is the only way But I know people have tried this service and um Especially when the service began, everybody was very happy with it

[CK] 11:15:18

Because, you know, there's a selling doesn't give you an SLA. There's no This is when I know what they were talking about introducing SLAs early this year, but I haven't checked And so a lot of the enterprise customers were reluctant. So your MNOs, your AATLs, the big

[CK] 11:15:38

M&os. Most of them will be reluctant in the sense of there's no SLA. So if it works, it works. So you'll get a speed and etc But even that is not committed. And I saw a lot of tests that we did where

[CK] 11:15:56

When the service worked, it worked very well. However, we got feedback from like a number of the cruise customers who are SES customers at the time.

[CK] 11:16:09

And they also, they tested the service because what Starling could do is they'd come, they'll tell you have a service and they'll send you the kit and for the big customers that they were targeting it all here try it for free

[CK] 11:16:23

Walks. We start the service.

[CK] 11:16:27

So within the cruise sector, I know one of the things they did to improve the uptime was to set up what we're calling a daisy chain of dishes.

[CK] 11:16:40

So, you know, to improve the throughput and to enhance failover and whenever the different satellites are operating are passing overhead, their constellation of satellites if there's any lag during handover, whatever the issues could be, they were able to still

[CK] 11:16:57

Connection. This was in the early early days When they're still working out some of the kinks in their system.

[CK] 11:17:05

And so... We found that a lot of The cruise line has picked up Starlink.

[CK] 11:17:13

And what they ended up, and for us it was a big shock because at the time it was all the meal capacity for that segment we dominated meal dominated the the global cruise industry which is We're talking about multi-million dollars of revenue per annum

[CK] 11:17:31

But what they did is they segregated the traffic. So they said we'll take on Starlink as much as we can.

[CK] 11:17:39

And the starlink they will take will be for crew welfare.

[CK] 11:17:42

So people are able to communicate UTC because when it works, it works. And, you know, when they're at sea and it works, it is to see.

[CK] 11:17:50

But for their mission-critical applications, that navigation or communicating to the show, what are all those things?

[CK] 11:17:57

Things that they do, then they'd keep the medium out orbit satellites.

[CK] 11:18:01

And because of the limited space, real estate on a cruise ship is a big deal everything that is there. You don't have the luxury of space like you know you have a data farm and you can just put as many antennas.

[CK] 11:18:15

So it was those two. But we saw a few other industries take on that approach as well. So on the mobility space.

[CK] 11:18:25

We know that for a long time it has been geo and I've seen that a few airlines that have signed up with Starlink as well to use that for their in-flight connectivity, ETC.

[CK] 11:18:39

We've seen this also in the mining sector, mining oil and gas.

[CK] 11:18:45

That's also a very big sector in terms of satellite comms because typically your amayun is in the middle of nowhere when you're doing offshore drilling or even onshore drilling usually it's places where connectivity is limited and yet where it's most needed.

[CK] 11:19:00

So in those places we've seen that they picked up Starlink and because of the lack of SLA and the fact that you cannot put all your mission critical data there, so there's a segregation. So there's a bit of Leo.

[CK] 11:19:13

In this case, we've seen cases of even one web. And some of the mining places you'll have a bit of one web and then you'll have a bit of meal the middle mouth orbit because of the low still low latency high throughput

[CK] 11:19:28

And in some cases where the real estate allows, as long as it's not a rig or an FPO, then you'll have some sort of geo connectivity available.

[CK] 11:19:40

So i What was the question that I was answering?

[Wycliffe Bahati] 11:19:47

No, it was about the, okay, you've answered it. In terms of service reliability.

[CK] 11:19:51

Okay. Yeah. Okay. Yeah. So the quality of service. So the big issue still remains the SLA. And for us, for myself being in the MIO space and even the geo One of our key counter selling points against Starlink on one web was they don't give you an SLA. And so you mr customer are you going to risk

[Wycliffe Bahati] 11:19:55

Yeah.

[CK] 11:20:18

The operations for this But the reason all these guys are mentioned to con starlink What's because of their affordability?

[CK] 11:20:25

And some came and said, look. It's post-COVID. We are all concerned about our cost of doing business, our cost of operation.

[CK] 11:20:35

So the Starlink services maybe half your price to us.

[CK] 11:20:40

So we'll cut our connectivity by half and you know we'll we'll um will save money.

[CK] 11:20:49

So the fact that even without an SLA, because the commercials are so attractive.

[CK] 11:20:56

Styling still has space on the table because if they were as expensive as Mio or geo or fiber, not fiber, but I think just compare the satellite options. Therea is a the way it's expensive as that is the other technologies would have stayed.

[CK] 11:21:12

So that's what I'd say there.

[Wycliffe Bahati] 11:21:15

And having come from a circular background What limitations do you think you know you saw in that industry and if Can it affect the leo services.

[CK] 11:21:20

Yeah.

[Wycliffe Bahati] 11:21:34

Because I think Leo services mostly are operating on KA and K.

[Wycliffe Bahati] 11:21:38

Fund.

[CK] 11:21:39

Yes, yes. So when there's a time where the band, the battle of the bands really mattered. But we think because you know even i talk about ka band If I remember a bit of my science a little bit is that whole we have the whole discussion of the free space laws

[CK] 11:21:59

And the fact that the satellites are closer so you're still able to mitigate, whether you talk about the traditional challenges of geo the attenuation, is it the rain rain fade and all those things But when you look at now.

[CK] 11:22:18

And it was presented pictorially, as you're familiar with. So you have your geo here that is 7,000 kilometers. Then you put your meads, they're 8,000 kilometers from the earth and then you put your leos leo's about 100 about um

[Wycliffe Bahati] 11:22:34

552. 270.

[CK] 11:22:36

Yes, yes. Yes, yes. So they are much much closer. And that has its own there are pros and cons from the science point of view.

[CK] 11:22:46

So one being that the geos being because they're very far they give you more coverage. And me a little bit more Leo's a little bit less.

[CK] 11:22:59

But then, you know, with a constellation, because the closer you are to there, then you need a bigger constellation to provide a full coverage. So that is a challenge.

[CK] 11:23:06

But it's not a sustainable challenge because and these guys are launching satellites every day now we're having discussions around whatabouts space debris because space debris aside from the gen one of starlink, I saw something was it last week that a lot of that gen one

[CK] 11:23:23

A lot of those gen ones satellites are falling back. But that's a whole long conversation and we had this instant in Kenya recently, last year, where something fell from the sky.

[CK] 11:23:34

For one of those launches. I know, just a human being. But those are now other conversations that we need to have but i mean going back to your question in terms of the differences between the three so there's the coverage of course

[Wycliffe Bahati] 11:23:36

Like it didn't fall on a car. How?

[CK] 11:23:53

But that is being mitigated by the constellation so with one geosate you can do a lot then with me you need a few more with dtcs you need a few more.

[CK] 11:24:03

But then what does that mean in terms of cost? So... The interesting thing is launching one geosatellite compared to the cost of launching multiple LEO satellites.

[CK] 11:24:18

The economics on the leo satellites actually are cheaper. And this is because there's been a lot of change in the last decade or so.

[CK] 11:24:28

In terms of the launch cost And mostly thanks to SpaceX.

[CK] 11:24:32

Because you have your reusable launch vehicles, you have your electric propulsion the general, when you look across the three different uh satellite configurations.

[CK] 11:24:47

The cost has come lower. It's come significantly lower. I've seen an article, I think we just was talking about the cost in the last decade, satellite lunch costs have reduced almost tenfold.

[CK] 11:25:01

But then on the flip side, in terms of money that has come into the satellites uh sector money money has like I have those statistics and I hope I can find them.

[CK] 11:25:12

But there's a lot more money that has come into this general sector because now people are talking about the space race every nation on earth wants to go to the moon. I don't know what's in the moon.

[CK] 11:25:24

They want to go to mars we have satellite connectivity is becoming a thing A few years back, it was out of fashion and fiber came, but people were seeing the applications and they need to to invest there.

[CK] 11:25:40

Yeah, sorry, go ahead.

[Wycliffe Bahati] 11:25:41

Actually, you are just going into the next uh topic which is the sustainability and economics of this constellations in terms of long term long-term sustainability.

[CK] 11:25:55

Yeah. Yeah.

[CK] 11:26:00

Long time. Yeah, so that's a challenge because there are a challenge There are all these environmental concerns that have been raised.

[Wycliffe Bahati] 11:26:02

Yeah.

[CK] 11:26:08

In terms of even being able to see space because you have all these constellations I think you have, what, maybe about more than 3000 plus things in space. I think it's about 5,000 or so And all this time I'm talking about starling

[CK] 11:26:26

Then you need to add Kuiper, which needs to launch another few thousands of satellites then one web still needs to complete their launch of the other 100.

[CK] 11:26:35

And then you have the Chinese with a thousand sales Or is it what is it they're calling this new constellation that they're launching. So I think we're getting to a place whereby we haven't fully contemplated what the environmental impact is going to be with all these metallic objects

[CK] 11:26:55

Floating in space. And I know there've been a lot of conversations around

[CK] 11:27:05

Things they sound like science fiction, that there's even a potential to arm things in space and what does that mean for sovereignty and and uh

[Wycliffe Bahati] 11:27:14

Actually, yes, that is really understanding. The Russians and the Indians, I think they tested the and to circulate missile.

[Wycliffe Bahati] 11:27:27

Sometime back. So actually, the Russians were destroying one of them Is it one of the old set like I've forgotten the name, but they were just trying to destroy it and right now it's responsible for about 70% of the space debris.

[CK] 11:27:38

Yeah.

[Wycliffe Bahati] 11:27:43

In Leo. As a result of the destruction of that settlement. Most of the the the the dismantle objects have not fallen back into the atmosphere so they're still hanging around there so yeah the environmental factor is definitely a a really big concern.

[CK] 11:27:53

Yeah.

[CK] 11:28:03

Yeah. Yeah, it is.

[Wycliffe Bahati] 11:28:06

But now I want to draw you to something else. Because um This is... No, no, no, no. I don't know. Are you... you almost you have another one?

[CK] 11:28:09

Okay. The time.

[CK] 11:28:13

Because of time. That sounds about me.

[CK] 11:28:20

Yes, I have a call at 12. I need to, I have a hard stop in by 11 45 so that I have 15 minutes to prepare for that call.

[Wycliffe Bahati] 11:28:24

Okay.

[Wycliffe Bahati] 11:28:29

Okay, okay, okay, fine. No problem. So there is those first the telescope.

[CK] 11:28:31

Yeah. But yeah, we can carry on till then. Yeah.

[Wycliffe Bahati] 11:28:42

The leo's that were tried and they failed mostly because of they went bankrupt.

[CK] 11:28:50

Yes, yes.

[Wycliffe Bahati] 11:28:51

So now there's a lot of speculation of what is actually the cost of this constellations.

[Wycliffe Bahati] 11:28:59

Because there isn't actually much information shared by Specific, understanding.

[CK] 11:29:07

Hmm.

[Wycliffe Bahati] 11:29:08

And people, people are waiting for the Europeans to do their eyes They're waiting for that then then we can know exactly what is the cost of uh of doing these constellations. And if it's actually commercially viable on the internet.

[CK] 11:29:16

I race too. Yeah, it is.

[CK] 11:29:30

Hmm.

[Wycliffe Bahati] 11:29:34

I'm just trying to figure it out that comparing to the cost of doing terrestrial infrastructure.

[Wycliffe Bahati] 11:29:41

And given that the user also is demanding for more and more capacity.

[CK] 11:29:41

Mm-hmm.

[Wycliffe Bahati] 11:29:47

So I'd just like to know your view if actually this leo broadband will still be able to compete and provide the capacities And we mentioned earlier that actually they are not going to rural areas they just coming to the same market will they be able to deliver the same market

[Wycliffe Bahati] 11:30:05

Bigger capacities in the future and compete and sustain the existence

[CK] 11:30:12

Yeah. So that's a good question. So in terms of compared to terrestrial infrastructure.

[CK] 11:30:22

Of course, rolling out terrestrial is the cheapest thing to do.

[CK] 11:30:27

Because you cannot compare to the cost of satellite launch. And it always shows in the cost per megabit. And we've had that internal debate also within SAS, like some of the price points we'd see for Starlink.

[CK] 11:30:40

We're like, that cost doesn't make sense based on the way we know the math.

[CK] 11:30:45

Of satellite. But the one thing I'll tell you for sure is the mystery you're talking about, about not having full transparency on what it costs per satellite An army street has always existed within even the geo world.

[CK] 11:31:00

I know, I mean, asking those questions within essays and trying to figure out what are the absolute costs And unless you've seen the original business plan which always differs from what the end product is.

[CK] 11:31:17

There's still no very clear answers. The one thing I know for sure is that there's some rationale to believe that the prices have come down considerably this whole on the engineering side in terms of reusable This is with a variable launch vehicles. This was by SpaceX.

[CK] 11:31:38

The advancements basically has been doing on vessels like they call it the dragon that's a dragon their

workhorse When you look at that, the rationale is there, that you're not burning up everything and into space And then when you look at the facts that Amazon is very keen i think every

[CK] 11:31:59

Billionaire the world has tried to get into space. So maybe there's something there Some of the costs I know we've looked at.

[CK] 11:32:08

Like they're just estimates so it's maybe you know between 200 to half a million dollars for just manufacturing the satellite the Starlink satellite.

[CK] 11:32:21

And then you have to count the cost, but then the cost also varies depending on which vehicle is being used and how many things you're launching.

[CK] 11:32:33

So the math can be very, very challenging. But for sure, even without absolute numbers it is much cheaper to build terrestrially And to extend whatever you have is your fiber and actually be able to build a product that can rival

[CK] 11:32:55

What's the likes of Starlink provide. If that existed, if you had quality connectivity in all these capital cities globally.

[CK] 11:33:05

When as far as the London and so forth. Then Starlink wouldn't find a market in the urban cities, they would be forced by the virtue of There is no demand in those cities.

[CK] 11:33:19

To put this connectivity in remote areas. But I think... operators, all these big operators, we have the local operators, Safari comms, the DTC.

[CK] 11:33:32

They really need to hold a mirror up for themselves and ask, are they providing the best possible service at a price that makes sense.

[CK] 11:33:40

And how do they disrupt their own industry? So that when the lights of Starlink do what they're doing then they

have enough critical mass and people connected that they can you know market dynamics they can push starlink to whichever market segment they need to operate from

[CK] 11:33:59

But I think when I see Starlink selling out and all this demand coming from the capital cities.

[CK] 11:34:04

For me, I don't ask the question, why are they sailing there? Because they're not forcing these people to buy. They're not actually holding.

[CK] 11:34:11

Guns to people's heads saying you have to buy. I think that is a feeling of the local operators there's There's a market demand that is not being met.

[CK] 11:34:22

And also, you know, if it's a failing, it's an opportunity. So the supplier needs to think Why is it that Starlink can come into their own backyard?

[CK] 11:34:32

And disrupt the market in this way. What is it? Some of the things they may not be able to answer, maybe some of these people buying Starlink terminals don't want the government listening to them. And we know that's a

[CK] 11:34:43

For example, had that issue where people are like garb for them to give out information etc So when people want to maintain their individual sovereignty, that's problematic.

[Wycliffe Bahati] 11:34:47

True.

[Wycliffe Bahati] 11:34:54

And...

[CK] 11:34:54

Because that's also national security issues. All those things. So I think... A scholar such as yourself needs to sit and give us some answers.

[Wycliffe Bahati] 11:35:00

Yeah.

[Wycliffe Bahati] 11:35:05

So actually now you just actually the remaining time you can just delete the regulatory and policy challenges so What are the ABS regulatory challenges that you see this sleeve that's facing to gaining market entry.

[CK] 11:35:13

Yeah.

[Wycliffe Bahati] 11:35:20

Maintaining operations. And then if you have a an example of countries you have a unique, you can just give us

[CK] 11:35:23

Yeah.

[CK] 11:35:27

Yeah, yeah. Okay, so I'd maybe ask the question a little bit differently because The Leo, okay, well, let me answer it as it is then twist it So I think a lot of deli operators, by virtue of their strategy, specifically to Starlink, which we expect

[CK] 11:35:47

That Kaipur to do the same People are saying maybe Kaipo will behave differently. They'll come and hold the hands of the local ISPs. But people are saying, look at Jeff Bezos, look at his business model.

[CK] 11:35:59

It disrupted the whole entire ecosystem of delivering goods and services is not going to play by the old rules. There's a likelihood he won't.

[CK] 11:36:09

No one who's playing by these rules is Warren Webb. Because I feel like their products are a little bit inferior in the sense that it's not as advanced, doesn't have as much coverage, and they have limitations within their coverage, ETC to see.

[CK] 11:36:23

But I would say that the approach, the strategy that the Starlink has taken and Kuiper anticipated.

[CK] 11:36:34

They just show up in the market. They don't consult the local regulator.

[CK] 11:36:40

I know that they have done in some of these countries, but they come in with their rules.

[CK] 11:36:48

And they come in with a big brand of Starlink backed by SpaceX.

[CK] 11:36:53

Backed by Elon Musk. And, um.

[CK] 11:36:58

It's a different type of competitor And I know within SESA conversations we had, we're saying, look.

[CK] 11:37:05

When we think we think we're competing with Starlink or the traditional settler operators think they're competing in Starlink, but this is a different type of competitor.

[CK] 11:37:14

They don't play by the normal business rules. They have very deep pockets.

[CK] 11:37:18

They don't have the pressure of quarterly reporting to show what profits they made.

[CK] 11:37:23

Now, SES tells us that Intelsat. Cannot do that.

[CK] 11:37:29

They don't have... the ability, they are not big enough to compete.

[CK] 11:37:34

With Starlink and all the companies back. And also some of the things that we had to come into realization two years back and say.

[CK] 11:37:42

We have to be serious about this and we have to be serious about this you're making your own kombucha you know kombucha that drink in your backyard Trying to compete with Coca-Cola.

[CK] 11:37:56

It kind of feels like that. The traditional settler operators that those guys making kombucha and then you're competing with Coke.

[CK] 11:38:02

The market dynamics are very different. As a result, the big ISPs are able to show up in a very brazen manner in the different markets.

[CK] 11:38:12

They know that there's a natural affinity for the brand They don't do much marketing because Elon Musk is already a brand by himself. Starlink is a huge brand.

[CK] 11:38:23

And so someone sitting somewhere in South Sudan, if they want connectivity and they hear there's a Starlink kit and someone tells them, I bring you one across the border.

[CK] 11:38:31

And set it up by yourself. You don't need an engineer to come to your house and start doing things. Instructions are there. You get a very nice box.

[CK] 11:38:39

Very clear it's hard to point it and fire it up.

[CK] 11:38:43

So because of the ease of of deploying the solution, the ease of billing and collection um the fact that they're going direct that that has made some governments mistrust them. And that's why a lot of governments in Africa, especially I know Latin America as well, have said no.

[CK] 11:39:04

We are not very keen to work with you because we don't know what you're doing with our citizens.

[CK] 11:39:09

And that becomes the biggest regulatory challenge I think they have. It's just that the way the way they're entering the disrupted the long respected model where the government and the regulator set the rules of engagement and they come in the completely new business model

[CK] 11:39:27

On how to enter a market segment that has always been very sensitive because We know the history of

communication across Africa, ATC.

[CK] 11:39:38

But they are winning. Because our infrastructure, because I'm in Africa is not as mature. So when we have multiple fiber cuts like we did last year.

[CK] 11:39:47

And if you look at what happened after those cuts we had And the number of countries that quickly turned around and said no to Stalin overnight, they were saying yes.

[CK] 11:40:01

That was different. Kenya is a curious case, for example, because in Kenya, we welcome Starlink with open arms and it's a company called, I think, Caribou Net.

[CK] 11:40:10

That was set up and they did this big press release, but carbonace, all they do is they have a bunch They warehouse the equipment.

[CK] 11:40:20

But they're not in charge of the billing Or I mean, I can go to NIVAS and NIBAs are selling Starlink terminals.

[CK] 11:40:27

Not nervous about my styling time, you know? All I have to do is go online register with a Starlink and be able to bring the terminal up.

[CK] 11:40:39

So I like these leos what they're doing is they're rewriting the rules of regulation and policy.

[CK] 11:40:50

They're using a very abrasive market entry. And like before when it was cordial and there were rules of engagement and you had to get a license Starlink doesn't have licenses in all these countries.

[CK] 11:41:03

They have been awarded, in some cases yes they're awarded licenses or they're awarded their rights to operate But we've not seen this before. You've never seen... a big operator being given the green light by government. So they're building things as they go.

[CK] 11:41:19

What they are is they're very clear on their vision that all 248 countries on Earth they won't have access.

[CK] 11:41:27

And you know what that means because then they're able to hold the subscriber base And so now when they get into their fight with Kuiper then it's It's a different fight. And whoever owns the customer at the end of the day is the one who

[CK] 11:41:42

Controls the market outcome. So when you understand that strategy it tells it gives you an idea of why they're moving the way they are.

[Wycliffe Bahati] 11:41:45

Yeah.

[CK] 11:41:50

Now, different countries yeah sorry go ahead.

[Wycliffe Bahati] 11:41:52

And... Yes, sir. How would the country, sir. Just want to find out about the different for example your If, for example, you are to deploy your a CS service in a country like Kenya.

[CK] 11:42:06

Yeah.

[Wycliffe Bahati] 11:42:07

You normally have to the licensee is per terminal or blanket.

[CK] 11:42:12

So we don't even care about that. All we need to do, and the same with the Intel SAT, UTELSAT, all they need is we need to come and find a local guy who holds all the licenses, sits with all the operational risk.

[CK] 11:42:26

So like we look for an offset. That's what we did with AFSAD. So AFSAD had all the licenses And all we do a lot of these big operators, the top, whatever, all these isps service providers.

[Wycliffe Bahati] 11:42:31

Okay.

[CK] 11:42:41

Intel SAT. All they do is we tell them you're selling new megabits. How many megabits do you want or how many megahertz do you want?

[CK] 11:42:51

And that's the extent of our engagement will come and help you to sell LTC but regulatory

[Wycliffe Bahati] 11:42:56

So you don't have to like kind of apply for licensing for spectrum that is done by the law school employed Okay.

[CK] 11:43:00

No. Exactly. And because that's perceived by this listed European service providers, your utils at DTC.

[CK] 11:43:10

There's a level of risk. They perceive that as risk and they want to see it with as little risk as possible.

[CK] 11:43:15

Megahertz megabits equipment. Even equipment.

[Wycliffe Bahati] 11:43:17

Okay. Yeah, so now now compare that with the This is stalking.

[CK] 11:43:24

Starlink is happy to sell you equipment. Get you on an online portal.

[Wycliffe Bahati] 11:43:28

Yes.

[CK] 11:43:30

And they don't care what the government has. I mean, they can go I think they realized it's important to get some sort of green light handshake I got a handshake and the government will say, okay, fine, you can operate you send a

[CK] 11:43:45

And agreements in all those press releases. And we say, okay, they have been granted, but the regulator the rights to to operate, but they won't go and set up a Starlink office. I think this is the first year of actually seeing

[CK] 11:43:59

Staling quad MWC. Last year, they were also there But they don't operate like an ISP.

[CK] 11:44:07

Like a satellite operator. So sometimes when you try and compare them and they're the ones who are now putting together the blueprint of what LEO operators look like.

[CK] 11:44:17

Of the scale that they are. When you look at the way they function as an entity compared to your traditional satellite operators.

[CK] 11:44:26

Completely different. We don't see them in trade shows. Not the way you'll go and see a big SES stand there.

[CK] 11:44:34

Or a business that will stand there, no. So there are all those nuances, but their vision is very very clear And the rationale, we all have been told is Starlink is supposed to be raising money to finance the mission to Mars.

[CK] 11:44:48

So everybody is hard at work to make sure that there is cash flow coming to support this mass. And that's a whole other conversation.

[CK] 11:44:55

Because, you know, a bit... Yes.

[Wycliffe Bahati] 11:44:55

Okay, I think you're running out of time. You can conclude you are going to in terms of giving examples countries and then we can conclude there.

[CK] 11:45:05

Yeah, so example so uh regions where Neo Broadband differently okay So there are two key differences though. So OneWeb has gone the traditional way.

[CK] 11:45:17

And I know that they've gone like in South Africa. I think they have an office there and what they've done is set up published pricing, Gordon Local licensed operators to do that. I know they've set up teleports and likes of Namibia, Angola, ATC. They're working with local, they're doing local partnerships.

[CK] 11:45:39

And because, you know, they did this merger with util It feels very much like a traditional ISP, traditional satellite operator approach.

[CK] 11:45:49

And I think there's a question there. Is that the way to do it?

[CK] 11:45:53

We don't hear much about OneWeb as much as we hear about Starlink and what they're doing.

[CK] 11:46:00

So I think that's another whole conversation around what kind of model to use.

[CK] 11:46:07

So in time ground station. So that's an interesting one. So we haven't had much about Starlink. I've only heard from the grapevine like the industry says something up in Nairobi and i heard it from the guys that were talking to, I think they're trying to set something up in uganda as well because

[Wycliffe Bahati] 11:46:07

I'm big ground stations.

[CK] 11:46:24

All these multiple constellations, they do need ground stations. Within their areas of operation. So from the ground station side One web strategy has been very clear. They've been very strong like in the Southern African part of Africa.

[CK] 11:46:39

If I look at, if I follow the news, whether it's Angola, Namibia south africa Starlink haven't had much. I've only had whispers, corridor whispers.

[CK] 11:46:50

And I haven't quite frankly followed up much around that. But yeah.

[CK] 11:46:58

They definitely, they're setting up ground segments across across the globe just to be able to to learn the traffic locally.

Participant 6 TM

Name Of Participant: Tom Makau

Profession: Expert with over 20 years in Satellite and broadband services

Date of interview: 14th March 2025

[Wycliffe Bahati] 12:07:33

And then yeah you can So we can start with the first section competition market dynamics so from your experience how has the introduction of leo broadband affected the business strategies of ISPs and MNOs and With that, if you can identify some of the opportunities that

[Wycliffe Bahati] 12:07:55

And challenges that ISPs face in competing with the Leo broadband.

[Wycliffe Bahati] 12:08:01

Prudently does.

[Tom Makau (SNDBX)] 12:08:03

Yeah, so I think it depends on what level you want to look at it Because there are both benefits and drawbacks to Leo, especially to ISPs and others.

[Tom Makau (SNDBX)] 12:08:17

Starting with the benefits, first of all, is thanks to Leo and the low latency it provides.

[Tom Makau (SNDBX)] 12:08:24

Now leo is becoming a very good option or alternative to backhaul traffic.

[Tom Makau (SNDBX)] 12:08:32

Before when we had Jio. Using geo for backup was like the last option. There was the latency involved So now with leo I think from a buckle point of view.

[Tom Makau (SNDBX)] 12:08:50

It will provide an alternative to geo and also Even alternative to terrestrial background where there is no backup.

[Tom Makau (SNDBX)] 12:09:00

The Leo becomes like a good alternative. The drawbacks then mostly come when you look at it from the they're downstream and the market From... from the consumer point of view, I don't think it's a big threat to them because one the performance

[Tom Makau (SNDBX)] 12:09:23

In as much as there's low all latency in LEO compared to Gio.

[Tom Makau (SNDBX)] 12:09:28

But it's not been very low to the point of competing hand-to-hand with the with the rest of your solutions.

[Tom Makau (SNDBX)] 12:09:41

So in terms of directly competing with where the existing ISPs have infrastructure, it may not be a good alternative, but in terms of extending the ISPs or MNOs coverage Et a true... partnering with LEO providers. You saw the other day Safaricom looking at ways of partnering with the

[Tom Makau (SNDBX)] 12:10:05

For example starlink There have been... talks of For example, the utilisot service there have been partnerships with local providers and governments.

[Tom Makau (SNDBX)] 12:10:22

So I think from that point of view, there's an opportunity But the drawbacks are not really there because fiber is not is not really in direct competition with Leo.

[Tom Makau (SNDBX)] 12:10:37

If you compare oranges for oranges. But it offers an opportunity for In my nose and the ISPs to extend their coverage.

[Tom Makau (SNDBX)] 12:10:47

So the other thing why it's not really a threat for them is the price point. Now you can get with 2000, we can get unlimited connectivity for eight and bps But at the moment, in as much as LEO coverage has increased, the price point is still not

[Tom Makau (SNDBX)] 12:11:04

Especially in our market Yeah.

[Tom Makau (SNDBX)] 12:11:09

So that's how I'd frame it.

[Wycliffe Bahati] 12:11:11

So uh yeah so one of the the the okay the strategies for the leo providers was that they were going to connect to the unconnected.

[Wycliffe Bahati] 12:11:25

I mean, that's the first In fact, if you look at the funding of Starlink apart from the one that comes from Ministry

of Defense, they also have tapped into Congress funding that is used to fund providers to reach them connected.

[Wycliffe Bahati] 12:11:42

How do you think... broadband service.

[Wycliffe Bahati] 12:11:47

Is doing in that area.

[Tom Makau (SNDBX)] 12:11:51

I think in terms of, you know, you have to look at it from the point of view of are they extending coverage to the area that had no coverage.

[Tom Makau (SNDBX)] 12:12:00

Which is basically... covering the uncovered.

[Tom Makau (SNDBX)] 12:12:06

Or are they extending... affordable connectivity to those who could have been covered previously.

[Tom Makau (SNDBX)] 12:12:16

But do not afford. So in terms of extending coverage, yes, but in terms of extending connectivity.

[Wycliffe Bahati] 12:12:23

Mm-hmm.

[Tom Makau (SNDBX)] 12:12:23

Connecting the unconnected. It will happen, but not at the current price point. I think there needs to be a critical mass for that to happen.

[Tom Makau (SNDBX)] 12:12:35

At the moment, I don't think they have the critical mass to make that happen.

[Tom Makau (SNDBX)] 12:12:41

The way they envisage. But I think it's a good move because if you look at And some regions.

[Tom Makau (SNDBX)] 12:12:49

And like the Pacific Islands, for example, or even in Africa, if you look at some some remote islands.

[Tom Makau (SNDBX)] 12:12:59

There was very little or no coverage. Via fiber via basically low latency broadband service But now with the likes of leo services from Starlink and from maybe OneWeb and the likes those countries are now... coward. But in terms of

[Tom Makau (SNDBX)] 12:13:25

Most of Africa. I don't think that would be possible for now because of the price point.

[Tom Makau (SNDBX)] 12:13:32

And also the other thing that might be a challenge there is a

[Tom Makau (SNDBX)] 12:13:40

Regulatory frameworks. If you look at Kenya, for example. We have operators who basically have the regulator in their pockets not mentioning names, but in that case then in as much as their desire is to extend coverage and all that.

[Tom Makau (SNDBX)] 12:14:01

There is opposition. From incumbents.

[Tom Makau (SNDBX)] 12:14:05

Not incumbent from the point of them being government run and government, but the existing players who are licensed.

[Tom Makau (SNDBX)] 12:14:14

Are influencing their rate of growth. And... And you saw in Kenya where the Safaricom lodged a complaint to the regulator in terms of why They think Starling should be regulated.

[Wycliffe Bahati] 12:14:18

Yeah.

[Tom Makau (SNDBX)] 12:14:29

And there was even the talk of not even a talk even the regulator changed the licensing structure to accommodate or to now regulate the likes of Stalin.

[Wycliffe Bahati] 12:14:44

Mm-hmm. Okay.

[Tom Makau (SNDBX)] 12:14:46

Because the previous licensing framework didn't really cater for an entity coming to provide service remotely with an easy to assemble kit that can be bought from NIVAS you know before it was very easy to trace and track an operator.

[Wycliffe Bahati] 12:15:06

Yeah, yeah.

[Tom Makau (SNDBX)] 12:15:07

Because they would need technicians, they would need an office, they would need equipment.

[Tom Makau (SNDBX)] 12:15:13

Now you walk into NIVAS, pick a kit. Put it on your balcony and you're connected the regulation didn't really foresee such a situation.

[Wycliffe Bahati] 12:15:14

Hmm. Mm-hmm.

[Tom Makau (SNDBX)] 12:15:25

So there has been a change and a push to change the regulation and i'm talking with respect to Kenya So as to accommodate this new framework where we have providers who can provide a service end to end without requiring local presence or expertise.

[Wycliffe Bahati] 12:15:26

Hmm.

[Wycliffe Bahati] 12:15:30

Hmm. Mm-hmm.

[Wycliffe Bahati] 12:15:46

Okay.

[Tom Makau (SNDBX)] 12:15:47

So the fact that... And I'll go back to this word that people like me is using, digital transformation. If you look at the fact that an entity such as Starlink is able to digitally deliver value.

[Tom Makau (SNDBX)] 12:16:04

Across the globe. They do not need anyone on the ground other than shipping the kits to you.

[Tom Makau (SNDBX)] 12:16:10

The fact that organizations are now becoming digitally transformed to be able to deliver value over the internet.

[Wycliffe Bahati] 12:16:10

Yeah.

[Tom Makau (SNDBX)] 12:16:20

Means that all the regulatory approaches need to adapt to that.

[Wycliffe Bahati] 12:16:28

Yeah. So they... so it means that actually that you know they they one of the things that is highlighted is that And the... terrestrial broadband providers.

[Tom Makau (SNDBX)] 12:16:30

Yeah.

[Wycliffe Bahati] 12:16:52

Need to do almost justification to expand some areas, but some areas usually lack the the business numbers revenue that can warrant an extension of the networks either because of security or So from your point of viewpoint Do you think that

[Wycliffe Bahati] 12:17:15

I know you've mentioned it that it might not be. The only issue but do you think that by virtue of presence of standing these areas that don't have connectivity now will increase uptick of internet or broadband.

[Tom Makau (SNDBX)] 12:17:34

Yes, most definitely. And this is evidenced by, for example, in Kenya, Starlink grossly underestimated the consumption Especially over Nairobi and within three months after they launched the service They stopped offering additional setups.

[Wycliffe Bahati] 12:17:38

Hmm.

[Wycliffe Bahati] 12:17:45

Mm-hmm.

[Tom Makau (SNDBX)] 12:17:57

So that is already enough evidence that that these services will extend uptake of broadband in unconnected

areas.

[Wycliffe Bahati] 12:17:58

Okay.

[Tom Makau (SNDBX)] 12:18:09

Other than the unconnected areas, eh? There are very many pockets in the pockets with low qs And I think the the threat And the fact that the fact that the capacity for Starlink got consumed faster over Nairobi than any other region in Kenya is evidence that one

[Wycliffe Bahati] 12:18:12

Hmm.

[Wycliffe Bahati] 12:18:36

Mm-hmm.

[Tom Makau (SNDBX)] 12:18:37

The introduction of these services will not only extend coverage to those who are not covered.

[Tom Makau (SNDBX)] 12:18:43

But it will offer an alternative to those who are on terrestrial infrastructure but face severe QS.

[Tom Makau (SNDBX)] 12:18:54

Qs issues. And then... very many of the customers in Nairobi who are facing severe QS issues.

[Tom Makau (SNDBX)] 12:19:04

Shifted to or attempted to shift to stalin okay And the effect of that is that one cephardicom improved their quality of service. I think for some period they had even doubled the bandwidth for home users.

[Tom Makau (SNDBX)] 12:19:19

If you had 10 you could do 20. I think even here where I am now, we had 15, but now after today it's still 30 mbps.

[Tom Makau (SNDBX)] 12:19:28

Thanks to Starlink coming into the space. For the same amount.

[Wycliffe Bahati] 12:19:34

Okay.

[Tom Makau (SNDBX)] 12:19:34

Yeah, so we are now paying uh we are now paying for 15 mbps.

[Tom Makau (SNDBX)] 12:19:43

But we are getting up to 30 mbps or so. Yeah. So I think it's a good thing. It will extend the coverage and also offer offer an alternative to those who are facing QS issues even from a from players

[Tom Makau (SNDBX)] 12:20:04

Who are providing services now.

[Wycliffe Bahati] 12:20:07

And another area is... What do you think will be effect of this leo broadband on uh IoT connectivity support I know previously there's been attempts to make uh it's remote IoT devices connect via SIGFOX or LoRa1.

[Wycliffe Bahati] 12:20:34

Do you think there'll be a change your some way of it will improve connectivity to these remote devices.

[Wycliffe Bahati] 12:20:45

Especially IoT devices.

[Tom Makau (SNDBX)] 12:20:49

It is possible because of one And... the consumption of the consumption of of IoT devices in terms of data is extremely low.

[Tom Makau (SNDBX)] 12:21:05

So one, it is possible if the protocols that were used on IoT on terrestrial are also transferred to to satellites. I think there have been attempts around that But you see, for IoT IoT can work whether it's Geo or Leo.

[Tom Makau (SNDBX)] 12:21:27

The latency really doesn't count as such because it's not two-way real-time communication.

[Tom Makau (SNDBX)] 12:21:35

So whether it's Leo or Jio, but I think the effect of leo on that is because that is because of lowering the price point.

[Tom Makau (SNDBX)] 12:21:46

Remember, before all these leo um uh uptakes started. Like about 10 years ago, it used to cost Slightly

\$10,000 per kilo.

[Tom Makau (SNDBX)] 12:22:05

To space. Right now, as you speak.

[Tom Makau (SNDBX)] 12:22:11

We are talking of slightly below \$200. Of a kilo to space.

[Tom Makau (SNDBX)] 12:22:18

And because of... that significant reduction in cost of launch The price point will come very low and because the price point will be low It will give an opportunity for the players now to even experiment with other protocols because as it is.

[Tom Makau (SNDBX)] 12:22:41

Tcbip is very wasteful when it comes to IoT. IoT needs to send to send four or five bytes.

[Wycliffe Bahati] 12:22:46

Uh huh.

[Tom Makau (SNDBX)] 12:22:49

Four or five bytes. But IoT, but the current protocols like IP the minimum they're concerned is 64 bits.

[Tom Makau (SNDBX)] 12:23:03

So you see there's a lot of wastage there. You only need four but the minimum you can do is 64 bits So there needs to be new protocols that are efficient on the channel for IoT to happen. Those protocols are there. I'm not saying there needs to be new.

[Wycliffe Bahati] 12:23:04

Me.

[Tom Makau (SNDBX)] 12:23:19

They just need to be adapted for satellite. And there's been a lot of talk around using satellite for IoT. And the reason why it hasn't really taken off.

[Wycliffe Bahati] 12:23:23

Uh...

[Tom Makau (SNDBX)] 12:23:33

Wasn't because the technology couldn't be adapted for satellite. That's because of the price point.

[Wycliffe Bahati] 12:23:40

Okay. Okay.

[Tom Makau (SNDBX)] 12:23:41

Yeah, because you'd imagine if I'm in a very, if I have a sense in a very remote location, let's say at an oil rig.

[Tom Makau (SNDBX)] 12:23:50

It will be cheaper for me to use private wireless radio even shortwave to send four bits Because you can send for business with your own private shortage.

[Tom Makau (SNDBX)] 12:24:00

As opposed to subscribing to satellite. So the price point has to come to a point where it is attractive to the consumers.

[Wycliffe Bahati] 12:24:01

Yeah.

[Wycliffe Bahati] 12:24:10

And maybe... Do you think that now with the availability of the availability Leo that maybe the IoT producers will start making bandwidth hungry devices

[Tom Makau (SNDBX)] 12:24:25

And... I think the issue hasn't been really in terms of the IoT was designed that way because there was no capacity.

[Tom Makau (SNDBX)] 12:24:37

The IoT is designed that way because of that power the battery consumption you want a sensor to there are some sensors today that can hold power for for very long.

[Tom Makau (SNDBX)] 12:24:52

Because they are very low demand for power in terms of transmission and receiving.

[Wycliffe Bahati] 12:25:04

Hello? Okay.

[Tom Makau (SNDBX)] 12:25:05

Yeah, sorry, I was drinking water. So in terms of um them adapting like creating new bandwidth hungry options Those protocols are already there.

[Tom Makau (SNDBX)] 12:25:19

And the reason why they've not been adopted is not because there was no bandwidth, because on 4G, even on 5G, I think on 4G you can do NBIOT.

[Tom Makau (SNDBX)] 12:25:33

Which is an adaptation of the GSM signal to IoT. 5g is being touted as the IoT ready network But 5G was mostly addressing the number of devices. The challenge that will come with satellite is not really the bandwidth, but in terms of the number of devices one satellite can support. Because remember, if you have one satellite over Nairobi

[Wycliffe Bahati] 12:26:02

Okay.

[Tom Makau (SNDBX)] 12:26:05

It will not be able to do a million devices per square kilometer like 5G.

[Wycliffe Bahati] 12:26:11

Mm-hmm.

[Tom Makau (SNDBX)] 12:26:12

But unless something drastic happens in how they design it. But... it will still offer an alternative.

[Tom Makau (SNDBX)] 12:26:24

Especially in regions where connectivity is erratic or not available. The technologies for IoT or the protocols for IoT, some of them have already been adopted for satellite.

[Wycliffe Bahati] 12:26:31

Okay.

[Tom Makau (SNDBX)] 12:26:39

But the issue has been the price point.

[Wycliffe Bahati] 12:26:42

Okay. Alright, I think... That's okay for that. Now we can move to this performance and technical challenges So from your perspective, how does Lio Broadband compare to fiber or the other fixed services mobile in

terms of speed, latency and reliability and based on your experience

[Wycliffe Bahati] 12:27:10

What are the most significant technical limitations of Leo broadband.

[Wycliffe Bahati] 12:27:16

And how they would debunk the performance and scalability.

[Tom Makau (SNDBX)] 12:27:22

Okay. When you compare speed and reliability, I think we need to be cognizant of something.

[Tom Makau (SNDBX)] 12:27:33

That the internet has changed a lot. And that a lot of content is now hosted within the regions where it's consumed Like if you look at if you look at YouTube.

[Tom Makau (SNDBX)] 12:27:51

Or even Facebook or Netflix a lot of that content is cached locally here in Kenya or yeah And that has made it. And also the fact that these local ISPs are using Kenyan registered IP addresses it improves the user experience because content can be can be localized.

[Tom Makau (SNDBX)] 12:28:15

The challenge with LEO as it is right now, for example, Kenya.

[Tom Makau (SNDBX)] 12:28:20

Is that you get connectivity, but because you have an American IP address or you have an IP address of the art station where the service is landing.

[Tom Makau (SNDBX)] 12:28:31

There's a lot of user experience that is lost. But before I go there, in terms of speed.

[Tom Makau (SNDBX)] 12:28:40

I think at the moment, Leo is offering better speeds because of the way maybe simply because of simply because of the number of users on those networks even when... Starlink had said that they are no longer offering additional services over Nairobi.

[Wycliffe Bahati] 12:28:54

Okay.

[Tom Makau (SNDBX)] 12:29:03

The users were already at the network were still getting better speeds than terrestrial people because initially when they started they would hit highs of 200 mbps when the congestion became too much, they would hit 40, 60... mbps which is still even better than maybe some terrestrial service, eh?

[Tom Makau (SNDBX)] 12:29:30

The only challenge that comes in is in terms of

[Tom Makau (SNDBX)] 12:29:37

The service reliability from the point of view that There has been a lot of pushback in many countries for LEO services.

[Tom Makau (SNDBX)] 12:29:48

Especially those that are those coming in from other countries Most of this pushback would be political. As you've seen, what happened in South Africa where Starlink was uh of Starlink's operating license was withdrawn, I think, two weeks ago because of...

[Tom Makau (SNDBX)] 12:30:07

Elon yeah doubling in politics yeah So there's also a lot of pushback in some countries i think ukraine And I think some other countries that have expressed dismay Of course, the new American way of politics So the challenges that are coming in there is that, especially for the largest LEO provider, Starlink.

[Tom Makau (SNDBX)] 12:30:36

Than satanity. That comes with the political baggage.

[Tom Makau (SNDBX)] 12:30:41

Yeah. And when it comes to latency, I think for countries for For areas where the areas where the Leo does not have a local art station.

[Tom Makau (SNDBX)] 12:30:55

Or a local landing station, then the quality of service and latency will be poorer And... what's happening now is that many of these LEO providers are building local art station like Starlink is building one in karen which is nearly which i think is ready, but hadn't been commissioned

[Tom Makau (SNDBX)] 12:31:26

For users, again, due to political reasons. So I think... latency wise for real-time applications.

[Tom Makau (SNDBX)] 12:31:37

It works. Though it's not as low as as a... as terrestrial, but it works.

[Wycliffe Bahati] 12:31:45

I'm interested.

[Tom Makau (SNDBX)] 12:31:50

Yeah. In terms of reliability for the people who have used the service, I think They... have mentioned that it's definitely more reliable than a terrestrial, especially in Kenya.

[Wycliffe Bahati] 12:31:51

Okay. Okay.

[Tom Makau (SNDBX)] 12:32:08

But I've not experienced this myself. So there's an image I'm sharing with you.

[Wycliffe Bahati] 12:32:15

And the... Okay.

[Tom Makau (SNDBX)] 12:32:19

For the art station in current for Starlink. That.

[Tom Makau (SNDBX)] 12:32:25

That is under construction. So once that is complete, then you'll see an improvement in latency locally.

[Wycliffe Bahati] 12:32:36

Okay. So, and I know you've been you're from a satellite background a bit so i mean what is the effect that i understand that they use KU band for user connectivity and ka band for

[Tom Makau (SNDBX)] 12:32:38

Yeah.

[Wycliffe Bahati] 12:32:59

For grounds for downloading to the Yeah.

[Tom Makau (SNDBX)] 12:33:02

For the grounds ground link

[Wycliffe Bahati] 12:33:06

Would that fit the service?

[Tom Makau (SNDBX)] 12:33:10

I think that they use for KA band is because of the high capacity. K-band is at a higher frequency.

[Tom Makau (SNDBX)] 12:33:20

So under normal circumstances, K1 would be able to push K-band would provide a hotter beam And also therefore be able to push more traffic.

[Tom Makau (SNDBX)] 12:33:35

On cabin. However, the antennas need to have a very high gain for them to act. So for at stations where the antenna is pretty is big then that's okay. So it's good to use K. But the use of KU is from the fact that one, it's a lower frequency.

[Tom Makau (SNDBX)] 12:33:57

Therefore more reliable on most occasions compared to KA. Like KU, would be compared to K would be last to be impacted by changing weather. For example, when it's very heavily raining.

[Tom Makau (SNDBX)] 12:34:17

They also the use of KU makes the equipment easier to install because you just need a place where there's clear line of sight.

[Tom Makau (SNDBX)] 12:34:27

On average, whereas if it was ka bundle then the demands for where the dish would be located would be more strenuous for an average user to set up that maybe in their balcony or in their compound Yeah. And also the use of Ka would have made the a

[Tom Makau (SNDBX)] 12:34:49

The cp is slightly more pricier. Because at those high frequencies, you'd spend more to create a phased array antenna.

[Tom Makau (SNDBX)] 12:35:02

Compared to ku banking because remember that the advantage that guys uh the leo are offering is by the fact that now you can have a phased array antenna

[Tom Makau (SNDBX)] 12:35:18

With no moving parts, that the beam is the one that is moving and not the antenna reflect itself.

[Tom Makau (SNDBX)] 12:35:26

So it offers, at KUBand, it offers a cheaper CPE.

[Wycliffe Bahati] 12:35:36

And what about the scalability.

[Wycliffe Bahati] 12:35:43

I mean, spectrum Is there any limitations on that?

[Tom Makau (SNDBX)] 12:35:50

I think the limitations when it comes to using K and KU I don't think they're really a problem because One, in terms of the available spectrum right now, most of that KU had been utilized in geo And that is now being given up to be consumed at

[Tom Makau (SNDBX)] 12:36:20

For Leo satellites. Also.

[Tom Makau (SNDBX)] 12:36:26

In whether that can scale or not. I think... still spectrum is a finite resource.

[Tom Makau (SNDBX)] 12:36:36

So if you look at the United States, there are now talks of utilizing previous C-band capacity For 5G. I think if I'm not wrong last year, the FCC authorized the use of a C-band for 5G.

[Wycliffe Bahati] 12:36:57

Thank you.

[Tom Makau (SNDBX)] 12:36:57

And it immediately brought a challenge in that the I mean, the FCC authorized that but the faa the Federal Aviation Authority.

[Tom Makau (SNDBX)] 12:37:10

Pushed back on that because some bits of the CBA and the spectrum actually being used for guided landing and takeoffs in most airports. So there was an issue there.

[Tom Makau (SNDBX)] 12:37:23

So with that, I think the capacity for C-band can be availed.

[Wycliffe Bahati] 12:37:23

Okay.

[Tom Makau (SNDBX)] 12:37:30

To to extend the coverage of LEO services.

[Tom Makau (SNDBX)] 12:37:38

Yeah. So I think with time that moved to CBAN, but of course with CBAN now.

[Tom Makau (SNDBX)] 12:37:43

There needs to be new modulation techniques to make it carry as much capacity as as much traffic as the ku

[Wycliffe Bahati] 12:37:52

So then... So now we can look at the sustainability meaning sustainability Okay.

[Tom Makau (SNDBX)] 12:37:58

And I think maybe just before I leave that point. Is that there's a lot of advancements in AI, especially on the radio access networks.

[Tom Makau (SNDBX)] 12:38:16

And with what is happening right now, we are seeing significant improvements on on spectrum efficiency, even on terrestrial networks.

[Tom Makau (SNDBX)] 12:38:27

And through the adoption of AI. On the radio access network. And I think if that is also done it could really help scale the capacity of the existing KU band spectrum.

[Wycliffe Bahati] 12:38:40

Yeah, actually there's talk of using machine learning algorithms in No.

[Wycliffe Bahati] 12:38:49

In... in terms of the handover and you know allocation i think there's some some development in that area.

[Tom Makau (SNDBX)] 12:39:01

Actually, the biggest development is in the area of being farming. Yeah, so there's quite a lot there.

[Wycliffe Bahati] 12:39:02

Okay.

[Wycliffe Bahati] 12:39:06

Yeah.

[Tom Makau (SNDBX)] 12:39:09

Last year, I was at the Mobile World Congress. And there are crazy things happening when it comes to AI and radio.

[Tom Makau (SNDBX)] 12:39:16

So I think that that will be a very big boost also to satellite.

[Wycliffe Bahati] 12:39:23

Okay. Yeah. And how... How do you see... this long-term sustainability of broadband I mean, the space is finite.

[Wycliffe Bahati] 12:39:36

Stick to any cyanide. I mean, how do you see... them getting to a point where there's almost nothing more you can do.

[Wycliffe Bahati] 12:39:47

To add capacity. When you're talking about that

[Wycliffe Bahati] 12:39:54

Do you think that... it's a sustainable profitable business in the long run.

[Tom Makau (SNDBX)] 12:40:06

Yes, and the reason I'm saying so is because um the sustainability comes in terms of Previously, it was very difficult because once you launch something up there It was fixed.

[Tom Makau (SNDBX)] 12:40:22

But now with the software-defined networks and software-defined radios. What that means is that now as we move from For example, 4G to 5G to 6G or whatever it is possible to upgrade the service or the satellite.

[Tom Makau (SNDBX)] 12:40:48

Without having to like bring it down. And also the fact that a satellite is now significantly cheaper to build and launch.

[Tom Makau (SNDBX)] 12:41:00

The rapid change in technology would also be experienced on the satellite front.

[Tom Makau (SNDBX)] 12:41:07

Because now it doesn't it's not a very big sunk cost if you have a satellite up and halfway through its life 10-year life cycle, the technology changes significantly.

[Tom Makau (SNDBX)] 12:41:19

You're left with a technology that is obsolete or cannot scale.

[Tom Makau (SNDBX)] 12:41:25

Due to advancements in technology. So the fact that it is becoming significantly cheaper would mean that it is now possible to assure the long-term sustainability.

[Tom Makau (SNDBX)] 12:41:39

And the reason why I see this happening is this happening If you look at what has been happening in the last 10 years, especially in that space.

[Tom Makau (SNDBX)] 12:41:49

Who have been the big players? There's been a lot of interest in LEO or high altitude platforms.

[Tom Makau (SNDBX)] 12:41:59

By Facebook, you saw what they were trying to do recently.

[Tom Makau (SNDBX)] 12:42:04

Currently. We also had the Google Loons which is a similar approach to offering connectivity from above from the sky We also have... Blue Origin by Amazon, which is also trying the same thing And now at least Starlink has been

[Tom Makau (SNDBX)] 12:42:35

Successfully. And we are seeing that there's a lot of and talk around the fact that most of these people who are managing and running this service see the traditional isp as a stumbling block to their business model.

[Tom Makau (SNDBX)] 12:42:56

It is in the interest of Facebook for users to access Facebook for free.

[Tom Makau (SNDBX)] 12:43:00

It is the interest of Netflix to access Netflix for free.

[Tom Makau (SNDBX)] 12:43:04

And that is the reason why, for example, if you are an ISP in Kenya to peer to Netflix or to Facebook is free of charge. You just pay for the interconnect.

[Tom Makau (SNDBX)] 12:43:13

And because it is in their interest So an entity like uh twitter sees Sephardicom not as a partner, but as an impediment.

[Tom Makau (SNDBX)] 12:43:24

To the service. Reaching a wide uh a wide audience.

[Tom Makau (SNDBX)] 12:43:35

So what's going to happen is that because there's a lot of interest by these large players.

[Tom Makau (SNDBX)] 12:43:40

What's going to happen is that the investment in these networks is going to improve.

[Tom Makau (SNDBX)] 12:43:46

To increase as we go on. And I could be wrong, but I foresee a time where if you subscribe to Netflix.

[Tom Makau (SNDBX)] 12:43:55

And you have... 4G capable device that can can see a Starlink satellite instead of paying, let's say \$11 a month for Netflix you pay like \$14 and that includes the connectivity to Netflix itself to watch the content for free.

[Tom Makau (SNDBX)] 12:44:18

We have also seen the attempts by Google to try out similar services.

[Tom Makau (SNDBX)] 12:44:25

And there is a lot of interest in getting that to work. I think already in Kenya there is a, I'm trying to look for it on the map.

[Tom Makau (SNDBX)] 12:44:33

There's a... there's a trial going on with Airbus. Let me look for the image.

[Tom Makau (SNDBX)] 12:44:40

Where they are trying out a high altitude vehicle over over Kenya.

[Tom Makau (SNDBX)] 12:44:47

And that shows that there's a lot of interest in making that work.

[Tom Makau (SNDBX)] 12:44:53

Because that then cuts off the middleman, in this case the local isps from these large players. So Netflix can reach you directly Facebook can reach you directly.

[Wycliffe Bahati] 12:44:53

No kidding.

[Tom Makau (SNDBX)] 12:45:07

They are currently doing that because if you buy a data bundle from Airtel or wherever.

[Tom Makau (SNDBX)] 12:45:13

You get... through Facebook.

[Tom Makau (SNDBX)] 12:45:18

Yeah, because... Because Facebook has essentially bid the mobile operator.

[Wycliffe Bahati] 12:45:19

Yeah.

[Tom Makau (SNDBX)] 12:45:26

To allow you to connect for free. So they are paying on behalf of the consumer for some time.

[Wycliffe Bahati] 12:45:27

To allow. Okay.

[Tom Makau (SNDBX)] 12:45:32

So if they have their own networks then it means that uh it is going to grow their market share.

[Tom Makau (SNDBX)] 12:45:46

Yeah. So that's how I see. The sustainability comes from the reason why these guys are setting up these services.

[Wycliffe Bahati] 12:45:47

All right.

[Tom Makau (SNDBX)] 12:45:53

If you look at, for example, the traditional players like Intel SAT, Utelsat.

[Tom Makau (SNDBX)] 12:45:58

They were selling the connectivity itself as an end. But if you look at what's going to happen now, the connectivity will be a means to an end and the end is the number of people and the number of eyes

[Tom Makau (SNDBX)] 12:46:13

That are looking at adverts and all that.

[Wycliffe Bahati] 12:46:17

That's where the money will be. So yeah and

[Tom Makau (SNDBX)] 12:46:19

That's where the money is because even now traditional mobile operators or isps they are no longer they no longer feature very high in the value chain. They are just like a conduit to the value itself.

[Tom Makau (SNDBX)] 12:46:36

Meanwhile, I'm subscribing to an isp All I'm looking for is whether they are stable or not.

[Tom Makau (SNDBX)] 12:46:42

And I'm not really looking for like any other experience.

[Tom Makau (SNDBX)] 12:46:49

Most of my experience, the reason why I subscribe to, for example, Safari comics because I need to access netflix I need to access YouTube.

[Tom Makau (SNDBX)] 12:46:57

I need to access Microsoft. I need to access Facebook and all these.

[Tom Makau (SNDBX)] 12:47:02

That is the reason why I got the link. I didn't get the link because it is a superior link or there is a good service on it.

[Tom Makau (SNDBX)] 12:47:10

So for... for the traditional players, eh?

[Wycliffe Bahati] 12:47:10

Yeah.

[Tom Makau (SNDBX)] 12:47:15

They're trying to reinvent themselves by trying to offer all these additional services, calling them themselves take calls instead of 10 calls But the challenge now comes in in that these larger players who are offering these OTT services want to own the network end to end.

[Tom Makau (SNDBX)] 12:47:33

Yeah so now you'll be in a situation where you are on an android phone that whose operating system is by You're accessing a youtube app via maybe a 4G service that is offered through Starlink via Google.

[Tom Makau (SNDBX)] 12:47:48

Because Google can say tomorrow, YouTube is free on Starlink. Or yeah netflix Netflix, you can pay an additional two dollars and get connectivity.

[Wycliffe Bahati] 12:47:54

The most yeah most likely.

[Tom Makau (SNDBX)] 12:48:04

As part of the package. Yeah. So the sustainability of that will be there because for these entities to provide services digitally.

[Wycliffe Bahati] 12:48:08

No.

[Tom Makau (SNDBX)] 12:48:15

And without local presence they will need to offer services from the sky, basically.

[Tom Makau (SNDBX)] 12:48:21

Not from the ground.

[Wycliffe Bahati] 12:48:24

And because you mentioned many players Do you think that standardization in terms of between Leo providers will come at some point.

[Wycliffe Bahati] 12:48:38

And we connect benefits. In the future.

[Wycliffe Bahati] 12:48:42

Benefit the legal providers in the future.

[Tom Makau (SNDBX)] 12:48:45

Yes, standardization, I think, has already happened. It's only that it's not happened on the correct platform.

[Tom Makau (SNDBX)] 12:48:53

There is a lot of talk now and a lot of tests going on with, for example.

[Tom Makau (SNDBX)] 12:48:59

Starlink to provide a traditional 4G coverage from from their satellites.

[Tom Makau (SNDBX)] 12:49:07

And that's essentially a standardized technology. Yeah.

[Tom Makau (SNDBX)] 12:49:16

I'm not very optimistic about 5G, but come 6G. Where 6G will now be a unified standard for both mobile and Wi-Fi in a way.

[Tom Makau (SNDBX)] 12:49:30

Yeah there's that push to mulch the two together in a way into one standard day.

[Tom Makau (SNDBX)] 12:49:35

What that would mean is that whether you are on satellite, whether you are on your home network, whether you are on on the mobile network You will be on 4G.

[Tom Makau (SNDBX)] 12:49:51

And what will happen now is when I talked about these providers maybe taking advantage of these LEO services.

[Tom Makau (SNDBX)] 12:49:58

Safaricom, for example, could lose a slice because now with this new technology, there's network slicing.

[Tom Makau (SNDBX)] 12:50:05

So Falcom could lose a slice of the entire LEO capacity, maybe for Starlink or one maybe.

[Tom Makau (SNDBX)] 12:50:12

And use that as an alternative backup to to their network in places with erratic coverage.

[Tom Makau (SNDBX)] 12:50:20

So you'll be on the same network when you're in your house you are on the same network. When you're on the road, you're on the same network. If you're on the road in a place where there is poor coverage.

[Tom Makau (SNDBX)] 12:50:29

It switches to satellite but on the same network. So that standardization has to happen for the value to be delivered to the end user.

[Tom Makau (SNDBX)] 12:50:37

Any entity that does not standardize their their technologies may not really be able to scale Yeah, yeah.

[Wycliffe Bahati] 12:50:50

Okay, and so... Okay, now I think we can It will come back to user adoption. We can talk about regulatory and policy challenges.

[Wycliffe Bahati] 12:51:03

So I don't know what you think are the biggest regulatory challenges that Leo broadband providers are facing to gain method entry.

[Wycliffe Bahati] 12:51:12

And maintaining operations. And if you know of different countries where Maybe let us.

[Wycliffe Bahati] 12:51:22

Viewed LEO services differently. Maybe you can just highlight.

[Tom Makau (SNDBX)] 12:51:27

So I think the biggest regulatory challenge now is the fact that the regulation are not kept pissed with the changing technology.

[Tom Makau (SNDBX)] 12:51:41

Most of the regulations are not fit for purpose. And that is the biggest impediment.

[Tom Makau (SNDBX)] 12:51:48

Yeah, like if you look at the licensing requirements for Kenya They include a lot of things like type approval registration of your technicians with the regulator registration of the area where you need to operate the type of licenses, they're also very prohibitive.

[Tom Makau (SNDBX)] 12:52:18

If you look at Kenya, for example, to be a Taiwan provider.

[Tom Makau (SNDBX)] 12:52:24

The license is very prohibitive in most countries. And the return on investment is still not very clear for most of these players, especially in a continent such as africa where the regulatory scene is very fragmented due to very many countries having very different

[Wycliffe Bahati] 12:52:29

Yeah.

[Tom Makau (SNDBX)] 12:52:49

And varied regulatory frameworks. And also the protectionist nature of most of these countries.

[Tom Makau (SNDBX)] 12:53:00

For example, entities like the incumbent like Safariko Masame. We paid a lot of license for us to be licensed.

[Tom Makau (SNDBX)] 12:53:10

We've spent a lot of capex in infrastructure we needed to be we need to be protected so that we can do copla investments.

[Tom Makau (SNDBX)] 12:53:20

I think that's a very old way of looking at regulating the telecom framework. It used to work maybe when the market was being liberalized.

[Tom Makau (SNDBX)] 12:53:30

So the biggest impact yeah the biggest impediment is the regulatory frameworks for most countries.

[Wycliffe Bahati] 12:53:30

Yeah, no police.

[Tom Makau (SNDBX)] 12:53:38

Are not in tandem with the changes in technology. And the digital transformation that has happened in the telecom space.

[Tom Makau (SNDBX)] 12:53:48

Yeah. Also, the regulation is very regulation is very technology specific.

[Wycliffe Bahati] 12:53:50

So...

[Wycliffe Bahati] 12:53:57

Mm-hmm.

[Tom Makau (SNDBX)] 12:53:58

And it's not really outcome specific. Yeah um and uh and and and That's a big issue, yeah.

[Tom Makau (SNDBX)] 12:54:10

So... that is one challenge. If you look at, for example, the example I've given south africa

[Tom Makau (SNDBX)] 12:54:19

Recently banned Starlink. And you can look at that from a regulatory point of view because the banning has to come from a regulator.

[Tom Makau (SNDBX)] 12:54:28

Yeah so the regular current regulators under a lot of political influence.

[Tom Makau (SNDBX)] 12:54:36

They are not really independent regulators as such. Yeah. The second thing in terms of regulatory challenges comes in terms of the fact that there was a lot of focus from the regulator.

[Tom Makau (SNDBX)] 12:54:59

In being... fair and utilizing public scarce resources such as spectrum And with the advances in technology today.

[Tom Makau (SNDBX)] 12:55:15

It's not a big issue today because if you have a very good algorithms and the use of AI today.

[Tom Makau (SNDBX)] 12:55:24

Most of these players would not need to pay so much for Spectrum locally.

[Tom Makau (SNDBX)] 12:55:32

And that, of course, would make the rollout of these services faster and more affordable.

[Tom Makau (SNDBX)] 12:55:38

So that's the challenge that I see from a regulatory point of view.

[Tom Makau (SNDBX)] 12:55:46

Yeah.

[Wycliffe Bahati] 12:55:47

And... in terms of Okay, I know we've touched it a bit your one result due to about One is the regulation of internet in countries and second is the geopolitical restrictions for example you'll find... South Africa.

[Wycliffe Bahati] 12:56:15

The bounce to link and now they've allowed the chinese to operate.

[Wycliffe Bahati] 12:56:24

So how does this work because like you mentioned it's very it's it's not very cheap to launch these constellations and most of these

[Wycliffe Bahati] 12:56:40

Companies launching them even though they seem to be independent it looks like there's quite a lot of interest in with the governments of the hosting nations there seem to be a lot of uh Because I think also the licensing of these vehicles and all that.

[Wycliffe Bahati] 12:56:59

So, uh. How do you see this play out?

[Wycliffe Bahati] 12:57:04

With the geopolitics and second is We've seen one like if I give an example in Kenya.

[Wycliffe Bahati] 12:57:15

In the events of 25th and the distrust and then you have place like Uganda where this book is, I don't know if it's open now, but we'll just banned.

[Wycliffe Bahati] 12:57:28

How does that play out? In terms of

[Tom Makau (SNDBX)] 12:57:31

Yeah, I think the, if you framed it correctly, because currently the regulation of these players, especially Leo.

[Tom Makau (SNDBX)] 12:57:45

Is more about balancing interest of like national security interests balancing protectionism measures because

the original regulatory frameworks offered some form of inbuilt protectionism for the licensed operators, right?

[Tom Makau (SNDBX)] 12:58:09

So you'd see Safaricom tamping their chest and saying that uh Stalin cannot operate in Kenya because they are not licensed here.

[Tom Makau (SNDBX)] 12:58:20

Or Starlink can operate in Kenya

[Tom Makau (SNDBX)] 12:58:25

We paid a lot and we were forced to extend our networks beyond our capital capability And now we need to recoup our investments. So that is factored in in the regulatory framework.

[Tom Makau (SNDBX)] 12:58:42

But the biggest thing that is impeding these players from from a regulatory point of view is the fact that most countries today or most governments today Unlike... a while back where they had a very good grasp of the communication mechanism, radio stations

[Tom Makau (SNDBX)] 12:59:10

Newspapers yeah the government control the narrative. But now with this new technology such as the internet.

[Tom Makau (SNDBX)] 12:59:19

The government is no longer in control of the narrative. And the only way to control is through regulating ISPs.

[Tom Makau (SNDBX)] 12:59:30

So with the coming of Leo operators who are not domiciled in that country.

[Tom Makau (SNDBX)] 12:59:35

Regulation becomes tricky, especially if this leopreditors are coming from powerful countries like the US.

[Tom Makau (SNDBX)] 12:59:42

So regulation becomes very tricky.

[Tom Makau (SNDBX)] 12:59:47

And you've seen what is happening now with the USA putting sanctions on any country that tries to impede the growth of american goods and services yeah So there is that bit.

[Tom Makau (SNDBX)] 13:00:04

The fact that very many countries now do not have control of the internet They had gained that control through regulating the local terrestrial providers.

[Tom Makau (SNDBX)] 13:00:14

But they will lose that control now. With the advent of Leo satellites.

[Tom Makau (SNDBX)] 13:00:21

So the regulation needs to change. To one.

[Tom Makau (SNDBX)] 13:00:26

Accommodate the new realities Yeah. And also two recognize that leo satellites can actually be used as complementary to the existing services.

[Tom Makau (SNDBX)] 13:00:42

Yeah, as opposed to them being independent services. So from a national security or political expediency point of view.

[Tom Makau (SNDBX)] 13:00:53

It's an impediment. The current regulations are an impediment. However, these regulations could also be a plus because unlike terrestrial providers.

[Tom Makau (SNDBX)] 13:01:04

That maybe have very many network points or nodes in a country.

[Tom Makau (SNDBX)] 13:01:11

Switching them off is not as easy. Yeah but now with the leo providers, for example, the image I've shown you for the current station switching off Starlink just means switching off that equipment I've showed you there.

[Wycliffe Bahati] 13:01:26

Mm-hmm.

[Tom Makau (SNDBX)] 13:01:26

At one spot. But switching off Safarico needs a little bit more work it's possible.

[Tom Makau (SNDBX)] 13:01:34

But needs a little bit more work to achieve it. So it depends on how the regulator looks at it.

[Tom Makau (SNDBX)] 13:01:41

Yeah. Yeah.

[Wycliffe Bahati] 13:01:42

And not about that gpdr and this uh data privacy laws how do you see it play out in terms of the regulatory and politic framework now that most countries are trying to enforce the data privacy rules to the data how you access it how will it affect how does it

[Tom Makau (SNDBX)] 13:01:47

Gdpr and all that.

[Wycliffe Bahati] 13:02:04

Work with this.

[Tom Makau (SNDBX)] 13:02:05

It will have no effect because the same laws have not had an effect on google microsoft and uh and all these other players with what they do with our data

[Tom Makau (SNDBX)] 13:02:18

The When you look at it from a data privacy and protection point of view.

[Tom Makau (SNDBX)] 13:02:27

I don't think... Leo introduction of Leo, for example, in Kenya, we will have any significant uh changing how we look at our data protection laws because current live in the current laws do not really work.

[Tom Makau (SNDBX)] 13:02:43

Yeah, but the only challenge that will come there is in terms of uh It's not really in terms of data protection or data privacy.

[Wycliffe Bahati] 13:02:43

Mm-hmm.

[Tom Makau (SNDBX)] 13:02:56

Is the fact that...

[Tom Makau (SNDBX)] 13:03:00

Our data in the grand scheme of things, data, especially from third world countries.

[Tom Makau (SNDBX)] 13:03:05

Especially for users. Is not that valuable.

[Tom Makau (SNDBX)] 13:03:12

Yeah. And it's the truth. It's not that valuable uh to speak the truth.

[Tom Makau (SNDBX)] 13:03:22

Yeah. So I don't think there's an effect on that in terms of data privacy and Because currently, those laws do not seem to apply on OTT services.

[Wycliffe Bahati] 13:03:36

Okay. So now I just want you to summarize in terms of What do you think... would be factors that would drive customer choosing Globroadband of a traditional internet services at least she wants.

[Wycliffe Bahati] 13:03:56

And do you think there's an affinity of certain types of customers to choose Leo?

[Wycliffe Bahati] 13:04:01

Rather than out of.

[Tom Makau (SNDBX)] 13:04:06

I think, yes, there are factors. For example, in Kenya, the big drive towards the adoption of Leo was more of a it was more of a protest and and the fact that people felt the current provider In Kenya embedded government in as far as giving them information.

[Tom Makau (SNDBX)] 13:04:27

Yeah. And there has been a lot of talk of mobile operators being used to locate people who have said something against the government.

[Tom Makau (SNDBX)] 13:04:38

And I think that is a factor that is pushing people to adopt Leo satellites.

[Tom Makau (SNDBX)] 13:04:42

Not because the Leo satellites are any superior in terms of QSA, but because people feel they are more secure and more private when dealing with LEO satellites.

[Tom Makau (SNDBX)] 13:04:57

Yeah, that is one. One user.

[Tom Makau (SNDBX)] 13:05:05

The other thing that could push adoption is price.

[Tom Makau (SNDBX)] 13:05:15

If... if you get to a point where

[Tom Makau (SNDBX)] 13:05:22

Okay, let me take that back a bit. Let me not call it price.

[Tom Makau (SNDBX)] 13:05:27

But the proponents of leo uh of leo network, sir. So.

[Tom Makau (SNDBX)] 13:05:37

So I've come to something that's going to just wait a chair.

[Tom Makau (SNDBX)] 13:05:40

Okay, okay, okay. Sorry. So I was saying that the proponents or the organizations investing into into these uh network, sir. Their motive is different.

[Tom Makau (SNDBX)] 13:06:01

From the current income pencil. And... The end game is to provide connectivity for free as a means to an end.

[Tom Makau (SNDBX)] 13:06:12

Where that's the current incumbents providing the connectivity is their bread and butter.

[Tom Makau (SNDBX)] 13:06:18

So the adoption of Leo will not be because of of any other reason other than the fact that the connectivity in Leo will be more of a means to an end than the ending itself. Safaricom would want or another local SPA would want to make money from selling the bandwidth.

[Tom Makau (SNDBX)] 13:06:39

But Netflix or google is not interested in making money from the bandwidth.

[Tom Makau (SNDBX)] 13:06:44

But making money from the services that they they provide all this bandwidth.

[Tom Makau (SNDBX)] 13:06:49

So that is the second thing that will push adoption of Leo because the proponents of leo that that them they are being driven by different drivers compared to terrestrial networks.

[Tom Makau (SNDBX)] 13:07:05

Then the third thing also is... consistency of service. Even today in Kenya. Nairobi to 10 five kilometers away. Service is unbelievably poor.

[Tom Makau (SNDBX)] 13:07:19

Yeah. And if mobile operators do not scale up to provide uniform coverage. The challenge with the current terrestrial networks the lack of uniform QoS across large regions. Qs varies drastically within a very short distance.

[Tom Makau (SNDBX)] 13:07:39

So that's going to be a challenge, especially as people come to adopt network slicing and private networks.

[Tom Makau (SNDBX)] 13:07:49

Because remember, as we move on from 5G to 6G and onwards.

[Tom Makau (SNDBX)] 13:07:55

We'll move on to unified networks so that if I'm here in the house the network slice that i have that connects my office. So I'm on the same line as the office but on the mobile network.

[Tom Makau (SNDBX)] 13:08:08

If I'm on the road, my phone connected to 5g is on the same quote unquote LAN or network as my office and my house.

[Tom Makau (SNDBX)] 13:08:18

So the provision of one unified network across the globe is one of the things that is making Leo very attractive.

[Tom Makau (SNDBX)] 13:08:27

Because you're going to imagine now if for some reason, not even for some reason.

[Tom Makau (SNDBX)] 13:08:33

When Starlink launches 4G services across the globe Things like roaming.

[Tom Makau (SNDBX)] 13:08:41

Will be history, yeah. Things of spotty coverage, things of uncertainty on the quality of service will be history yeah So that is another push towards people adopting layo services.

[Tom Makau (SNDBX)] 13:08:59

The third push, of course, then I think even when you look at most of the advancements that have been made in terrestrial communications.

[Tom Makau (SNDBX)] 13:09:10

The initial application was in satellite. Yeah so the the fact that a lot of R&D will now or has gone into these technologies means that And the fact that the go-to-market is very fast.

[Tom Makau (SNDBX)] 13:09:30

Means that uh means that I mean, Leo networks will expand very rapidly.

[Tom Makau (SNDBX)] 13:09:38

Because if you look at the last two years. No mobile network would have expanded the way Starlink has expanded across the globe.

[Tom Makau (SNDBX)] 13:09:46

It would have been very difficult. So that will be the attractive point. So even players who want to provide global coverage are not looking at setting up terrestrial networks. They're looking at setting up Leo or high altitude vehicle, balloons coverage.

[Tom Makau (SNDBX)] 13:10:05

Yeah.

[Wycliffe Bahati] 13:10:06

Okay, and... They want us to touch on.

[Wycliffe Bahati] 13:10:14

There is...

[Wycliffe Bahati] 13:10:22

Just a moment. What about the

[Wycliffe Bahati] 13:10:30

Somebody mentioned that somebody mentioned that I don't know, Kenya Power has over a million customers, right?

[Wycliffe Bahati] 13:10:39

Okay, they were trying to compare in terms of easily power and internet subscription So there's still low uptake of broadband ma'am population.

[Wycliffe Bahati] 13:10:54

Not only in Kenya but in currently living in some parts of the US. So I just wanted to discuss what could be the reason and Does digital literacy play a part in this?

[Tom Makau (SNDBX)] 13:11:14

Starting with the last bit in terms of digital literacy, I think for some time, yes, it has played a part because most of the internet was in English.

[Tom Makau (SNDBX)] 13:11:28

And most of the internet to so many people was a platform for entertainment and and amazement. It wasn't really a serious tool But with the internet becoming a critical and fundamental part of ASEAN.

[Tom Makau (SNDBX)] 13:11:51

And the fact that the fact that AI is now able, for example, to translate there's a lot of translation of languages that's happening So digital literacy or the utility of digital services will improve because now most of these services will be available in the local languages.

[Tom Makau (SNDBX)] 13:12:13

And digital literacy has also improved because now even the devices that people are using will be intuitive and in their own local languages for them to use.

[Tom Makau (SNDBX)] 13:12:23

Yeah. This one. The other bit where somebody is trying to compare power and internet uptake The main reason is because currently if you look at it.

[Tom Makau (SNDBX)] 13:12:39

The second largest producer of producer of power in terms of capacity.

[Tom Makau (SNDBX)] 13:12:46

In the country Sappharico. Why? Because of all the generators that they have in the base stations.

[Tom Makau (SNDBX)] 13:12:55

If Safaricon was to pop up all their generators, eh? They would generate a lot of power across the country.

[Tom Makau (SNDBX)] 13:13:01

And the reason why they have all these backup generators everywhere at nearly each station.

[Tom Makau (SNDBX)] 13:13:07

Is because of the unreliability of power. For very long that is number one number two And... that is changing now because of the adoption of solar.

[Tom Makau (SNDBX)] 13:13:19

Even some of these best stations are being powered by solar.

[Tom Makau (SNDBX)] 13:13:24

Most homesteads today that are not on the grid are on solar.

[Tom Makau (SNDBX)] 13:13:28

Because solar has become affordable. So that will be able to extend the consumption of digital services.

[Tom Makau (SNDBX)] 13:13:36

The other thing also is uh with a lot of consciousness around carbon.

[Tom Makau (SNDBX)] 13:13:46

Carbon footprint. The carbon footprint for 1 GB of data over a mobile network in Kenya is very, very high.

[Tom Makau (SNDBX)] 13:13:56

I can't remember the figures. But the carbon footprint of, for example, a leo satellite is extremely small.

[Tom Makau (SNDBX)] 13:14:06

In fact, it's in fact much, much smaller compared to terrestrial networks.

[Tom Makau (SNDBX)] 13:14:15

So that push is making people live in question. Yeah. And their consumption like if you have an entity that is very conscious about their ESG reporting.

[Tom Makau (SNDBX)] 13:14:30

And they would switch to a provider that has the lowest carbon footprint.

[Tom Makau (SNDBX)] 13:14:37

If that impacts them, yeah? Even today some organizations like if you're in Deloitte now and you're booking a flight Other than looking at the cost, you are also supposed to look at the number of kilos of carbon you'll put in the air by using that airline.

[Tom Makau (SNDBX)] 13:14:56

So those two come into the decision.

[Tom Makau (SNDBX)] 13:15:02

So as... unless the mobile networks or the current ISP get to a point where the carbon footprint matches the carbon footprint of of leo uh there'll be a lot of uptake of leo being pushed by uh environmental consciousness.

[Wycliffe Bahati] 13:15:02

Okay.

[Wycliffe Bahati] 13:15:25

Okay. All right so and um maybe you can just as we conclude, give me your future outlook and as you do this just there's a lot of talk of people putting data centers in the skies putting routers in the skies.

[Wycliffe Bahati] 13:15:53

I mean, where do you think that we'll go with Leo in the future?

[Wycliffe Bahati] 13:15:59

And... Would...

[Tom Makau (SNDBX)] 13:16:03

Yeah.

[Tom Makau (SNDBX)] 13:16:07

Let me approach it from two points of view. From a technical point of view.

[Tom Makau (SNDBX)] 13:16:15

That talk is valid now because one.

[Tom Makau (SNDBX)] 13:16:22

With everything now becoming softer. For example, 10, 15 years ago, if you talk to Varuta in the sky.

[Tom Makau (SNDBX)] 13:16:29

You could even visualize a Cisco device yeah physical hardware tool.

[Wycliffe Bahati] 13:16:33

Mm-hmm.

[Tom Makau (SNDBX)] 13:16:34

Yeah but now when you talk of a router, it is software based. It will be running on a Windows machine.

[Tom Makau (SNDBX)] 13:16:41

Yeah so the fact that most of these functions have been virtualized through what is known as network function virtualization nfv It means that it is now possible to to run these, for example, in the sky.

[Tom Makau (SNDBX)] 13:17:00

Because they can be upgraded. Through software that is a number one then number two the changing nature of our consumption of internet, a lot of localized content it will be very crucial to have for leo operators to have operators

[Tom Makau (SNDBX)] 13:17:23

Caching or content delivery systems that are on the satellite itself.

[Tom Makau (SNDBX)] 13:17:31

So yes, Starlink can provide you the bandwidth and connectivity But you as Netflix would find it worth your while to lease some content delivery capacity from Starling.

[Tom Makau (SNDBX)] 13:17:45

To host the Nigerian movies on the satellite that crosses over Africa.

[Tom Makau (SNDBX)] 13:17:52

To host the Korean movies of satellites that cross Korea. Yeah, and so on and so forth. So that will happen due to NFV.

[Tom Makau (SNDBX)] 13:18:02

And also the other thing that will make that happen is because for a very long time, the limitation has been the size of the launch vehicle.

[Tom Makau (SNDBX)] 13:18:10

Now we are seeing Elon Musk testing this Starship. Other than his dreams of taking people to Mars, what Starship has done is that it's made it's possible to create much, much larger satellites.

[Tom Makau (SNDBX)] 13:18:29

And therefore carry more equipment on the satellite and carry more batteries, carry more solar panels.

[Tom Makau (SNDBX)] 13:18:36

And provide more power for any compute that might be put on those satellites.

[Tom Makau (SNDBX)] 13:18:43

So that's one approach. The second approach is when the need to have data centers and caching on satellites.

[Tom Makau (SNDBX)] 13:18:55

Is driven by the user experience. Currently, if you are using, for example.

[Tom Makau (SNDBX)] 13:19:03

Starlink. The content is not really, really, really, really customized to the location you are in yeah So there'll be a need to be a need replicate what terrestrial providers have done.

[Tom Makau (SNDBX)] 13:19:21

By localizing content, local hosting caching, and so on and so forth.

[Wycliffe Bahati] 13:19:31

Yeah, so do you see um Leo taking over broadband completely or it will just be one of the services

[Tom Makau (SNDBX)] 13:19:44

The two will work together. And what's going to happen is that currently even when you look at Nairobi, for example.

[Wycliffe Bahati] 13:19:46

Okay.

[Tom Makau (SNDBX)] 13:19:54

There's very spotty coverage in most buildings. You get into a building second third floor or basement uh service drastically drops yeah So, uh.

[Tom Makau (SNDBX)] 13:20:09

What's going to happen is that the two will work together.

[Tom Makau (SNDBX)] 13:20:16

In tandem, not necessarily because the two services are provided by different operators.

[Tom Makau (SNDBX)] 13:20:23

But... the fact that it will be worth they are wild for Safaricom, for example.

[Tom Makau (SNDBX)] 13:20:31

To utilize leo to provide coverage over very large areas of, for example, the city and then utilize terrestrial GSM coverage, 4G or whatever to supplement that coverage using small cells.

[Tom Makau (SNDBX)] 13:20:51

The biggest challenge currently with 5G rollout is the number of base stations that are required, especially at the millimeter, however.

[Tom Makau (SNDBX)] 13:20:59

That can be overcome using Leo. That's number one. Number two.

[Tom Makau (SNDBX)] 13:21:04

The fact that the OTTs who are now the main value drivers for the broadband ecosystem.

[Tom Makau (SNDBX)] 13:21:16

Are now trying to look at Leo. It means that um With time, Leo will be dominant.

[Tom Makau (SNDBX)] 13:21:22

From the fact that The Leo service will be uh quote unquote free compared to terrestrial services.

[Tom Makau (SNDBX)] 13:21:34

And that is what will drive the adoption of LEO satellites.

[Tom Makau (SNDBX)] 13:21:40

Then the other thing also in terms of the future

[Tom Makau (SNDBX)] 13:21:49

The merger of Wi-Fi and on the mobile GSM, eh?

[Wycliffe Bahati] 13:21:55

It's G.

[Tom Makau (SNDBX)] 13:21:59

With the launch of 6G, because 5G is dead by the way 5g we just forget about 5G. We are now moving to 6G.

[Tom Makau (SNDBX)] 13:22:06

With the launch of 6G, Leo will have an advantage of being able to provide one network across very large geographical regions.

[Tom Makau (SNDBX)] 13:22:18

Which uh which which the rest of providers cannot provide at that price point.

[Tom Makau (SNDBX)] 13:22:26

For example, if I'm traveling, if somebody is traveling from the US to Kenya.

[Tom Makau (SNDBX)] 13:22:31

They will be on the same office network when they're in Kenya.

[Tom Makau (SNDBX)] 13:22:35

And security managing security of that network is much easier. So the adoption of leo will come in from the point of the fact that Because there are very few Managing security will be will be will be easier and simpler.

[Tom Makau (SNDBX)] 13:22:54

And quality assurance will be easier and simpler. So very many organizations will start adopting Leo services across the globe.

[Tom Makau (SNDBX)] 13:23:03

Which

[Tom Makau (SNDBX)] 13:23:07

Many international organizations, for example, struggle with it. We find like an entity We love to work with very many local operators.

[Tom Makau (SNDBX)] 13:23:20

To provide connectivity to their offices in those countries. With varied quality of service, varied level of support.

[Tom Makau (SNDBX)] 13:23:29

Verde... without a common shared understanding.

[Tom Makau (SNDBX)] 13:23:34

But Leo will be very attractive because this will be one unified network.

[Tom Makau (SNDBX)] 13:23:42

So the future is in such a way that Leo will make internet Free.

[Wycliffe Bahati] 13:23:43

Okay.

[Wycliffe Bahati] 13:23:48

Homogeneous offering.

[Tom Makau (SNDBX)] 13:23:49

As in that internet will not be the main product as such it will be it will be the service we are after.

[Tom Makau (SNDBX)] 13:23:57

Currently, when I'm consuming water like in my house, I'm more concerned about the quality of the water.

[Tom Makau (SNDBX)] 13:24:04

Than how big that pipe is. To my housing or whether that pipe is plastic or whether it is lead.

[Wycliffe Bahati] 13:24:08

Okay. The quality vote.

[Tom Makau (SNDBX)] 13:24:12

Is the quality of the water the the the current broadband market as it is We love to change.

[Tom Makau (SNDBX)] 13:24:22

The value proposition has to change.

[Wycliffe Bahati] 13:24:23

Okay. All right. Thank you. I think I've covered almost everything.

Participant 7 HN

Name Of Participant: Henrich Nothnagel

Profession: Expert with over 35 years in Satellite services

Date of interview: 14th March 2025

[Heinrich Nothnagel] 16:07:48

It should have said that this can be used in a court of law against you or something like that.

[Wycliffe Bahati] 16:07:57

No, no, no. This is not for good.

[Wycliffe Bahati] 16:08:01

So I effectively... I mean, they preview Yeah, because I've managed to at least come up with the right thing.

[Heinrich Nothnagel] 16:08:03

Oh, yeah.

[Heinrich Nothnagel] 16:08:07

Yes.

[Wycliffe Bahati] 16:08:14

So I'll just start with the first section, which is the competition and market dynamics.

[Wycliffe Bahati] 16:08:22

So like from your experience, how has the introduction of Lio Broadband affected the business strategies of ISPs and the main nodes?

[Wycliffe Bahati] 16:08:31

And while discussing that if you can identify any challenges opportunities space in competing with your broadband providers.

[Heinrich Nothnagel] 16:08:45

I think it's different for ISPs and for M&Os. M&Os is probably easier to answer.

[Heinrich Nothnagel] 16:08:51

In the sense that M&Os are always looking for lowering the cost of the backhaul.

[Wycliffe Bahati] 16:08:52

Mm-hmm.

[Heinrich Nothnagel] 16:09:01

And apart from that okay okay so they're looking for the lowering of cost of backhaul because of course The further they go out to the rural areas, the more expensive it becomes.

[Heinrich Nothnagel] 16:09:14

Both in terms of deployment of infrastructure and tobacco And in many cases, those sites are just not sustainable.

[Heinrich Nothnagel] 16:09:28

Was much of the back hole technology suites were available before Liu came.

[Heinrich Nothnagel] 16:09:37

So you can say in some of those technologies, there was going to be a natural progression to a lower price.

[Heinrich Nothnagel] 16:09:43

But there was nothing that really forced them. So Liu came and it offered that lower price. I think we had this discussion before that even though these guys keep saying that they want to help the guys who doesn't have access

[Heinrich Nothnagel] 16:09:55

They don't do so directly, but they do so indirectly by pushing the price down.

[Heinrich Nothnagel] 16:10:01

Which makes it possible to have sites that go further and which are self-sustainable.

[Heinrich Nothnagel] 16:10:08

So I think for M&Os, this is just great. In terms of getting into rural areas.

[Wycliffe Bahati] 16:10:17

Okay, so they can easily use them to backhaul their services to the remote areas.

[Heinrich Nothnagel] 16:10:25

Yes, they do. I know that a number of them tested on Starlink.

[Wycliffe Bahati] 16:10:27

Okay.

[Heinrich Nothnagel] 16:10:31

Safari Chrome being one of them. We implemented, I think, for two mnos.

[Heinrich Nothnagel] 16:10:39

And for our own, for ISET's own Tao.

[Heinrich Nothnagel] 16:10:46

They're using it or they are going to use it. And then we also Yeah, we actually at the moment.

[Wycliffe Bahati] 16:10:50

Okay.

[Heinrich Nothnagel] 16:10:57

Trying to convince the M&O to use one web for this as well but that's a different issue.

[Heinrich Nothnagel] 16:11:04

Well, ISPs, I think it's different because for ISPs The way I understand what ISB mean, Internet Service Provider.

[Wycliffe Bahati] 16:11:13

The traditional one with the fiber and

[Heinrich Nothnagel] 16:11:16

Yeah, that business became a utility business. It is all about price.

[Wycliffe Bahati] 16:11:22

Mm-hmm.

[Heinrich Nothnagel] 16:11:23

And for them it basically should just offer another technology at a lower price point.

[Heinrich Nothnagel] 16:11:35

Which again means that they can probably also reach more people in a rural area, which they couldn't have done so before with the prices which were available.

[Heinrich Nothnagel] 16:11:46

I understand. That actually in some cases, people migrate from existing infrastructure to even in urban areas from existing infrastructure to Starlink.

[Heinrich Nothnagel] 16:11:58

Starlink is probably The one that is quiet is used. I understand there are more than 5 million subscribers worldwide.

[Heinrich Nothnagel] 16:12:07

Yeah, so for me for them it is just a matter of they get an opportunity to get to more customers.

[Heinrich Nothnagel] 16:12:19

At the same time, I should say that the likes of Starlink You might recall from our days in iWay that One of the big things we struggled with was these layers which are between the servers and the end customer.

[Heinrich Nothnagel] 16:12:39

And to make it worth the while for each of those layers to be there, because each one of them had an important function to fulfill.

[Heinrich Nothnagel] 16:12:48

I think the thing that Starlink has done very successfully is They have just completely removed those layers, some of those layers.

[Heinrich Nothnagel] 16:12:58

There's nobody coming to you doing a marketing job for Starlink.

[Heinrich Nothnagel] 16:13:01

You'll find it in the shock. Install it. Boom. Off you go Whereas we had to go through a distributor and then a sales guy and blah, blah, blah.

[Heinrich Nothnagel] 16:13:12

So for that reason, the price can be much better. Amongst other they just cut out a lot of the overhead I'm not exactly sure why I say that but yeah It's just that it's just I think that is a risk to the ISPs that these guys are actually becoming an international ISP.

[Wycliffe Bahati] 16:13:20

Magoon.

[Heinrich Nothnagel] 16:13:33

Which competes with them on their turf. So for them, it's difficult.

[Heinrich Nothnagel] 16:13:39

Paymentos is easy. They need that service. They can buy it from anyone and now they don't have to buy it locally they can buy it from Starlink if regulation allow For the ISPs is difficult because this guy is going to compete with him directly

[Heinrich Nothnagel] 16:13:54

So a guy that's already in the utility business is going to find that this utility has lost even more price.

[Wycliffe Bahati] 16:14:03

So they... The other... The thing I was looking at is because of this frequent Do you think that the leo providers can act as disaster recovery and we're back up to the rest of the infrastructure where there's for your buckets.

[Heinrich Nothnagel] 16:14:34

I don't think in its current form. I don't think that the capacity which they have on these satellites, which they can deliver reliably.

[Heinrich Nothnagel] 16:14:42

In the current form, the likes of Empower can, but the likes of OneWeap and Starlink, they cannot.

[Heinrich Nothnagel] 16:14:49

At this yeah i mean unless they i i understand that Starlink is working on a more.

[Heinrich Nothnagel] 16:14:58

Business-like model. And if Elon Musk or Starlink thinks there's an opportunity, they will for sure go for it but I mean, I... I wonder if they will want to be in that person. I think they will want to be in the business just cut the fiber guy out completely and just go directly to the internet. Why bother?

[Heinrich Nothnagel] 16:15:19

So instead of restoring the fiber I just saw you from your home.

[Heinrich Nothnagel] 16:15:29

Under your own control. Because regulation makes that difficult or a challenge but yeah i think that's what they're really after.

[Wycliffe Bahati] 16:15:29

Okay. Yep. And so like

[Wycliffe Bahati] 16:15:41

So the these comparison in terms of how much this providers are backhauling now on fiber, for example.

[Heinrich Nothnagel] 16:15:54

Mm-hmm.

[Wycliffe Bahati] 16:15:55

Some of those that capacities in in terabits Is there a chance that Leo will get there?

[Wycliffe Bahati] 16:16:06

And be able to transfer to of Apple service in the framework.

[Heinrich Nothnagel] 16:16:13

I don't see that. And the limitation for me is spectrum.

[Heinrich Nothnagel] 16:16:21

They cannot concentrate that much spectrum in one location. I just saw a new a new constellation, which I think European Space Agency are involved with their total capacity for the whole world is two terabits per second.

[Wycliffe Bahati] 16:16:41

Well, that's one flavor.

[Heinrich Nothnagel] 16:16:45

Yeah, so I mean, you cannot the spectrum is just not there.

[Heinrich Nothnagel] 16:16:50

So what Empower does is they just put a very big terminal and a very big gateway.

[Wycliffe Bahati] 16:16:51

Okay.

[Heinrich Nothnagel] 16:16:57

And they just use a very high and a very efficient mod code.

[Heinrich Nothnagel] 16:17:02

And they go in a frequency band where there's a bit more spectrum than traditionally for satellite.

[Wycliffe Bahati] 16:17:10

So what about the the the airline industry the maritime industry I mean, that's almost... Because I was talking to commander the other day and she was telling me that They immediately felt the impact of sterling.

[Wycliffe Bahati] 16:17:33

The marine time traffic, which is usually prime for them.

[Heinrich Nothnagel] 16:17:41

Yeah, I think, was it you who told me about my asset's troubles because of... One whip.

[Wycliffe Bahati] 16:17:49

Mm-hmm.

[Heinrich Nothnagel] 16:17:49

Because of Starlink. Yeah, I think, you know.

[Heinrich Nothnagel] 16:17:57

Compared to a geo service you know you don't have the latency and all of that crap.

[Heinrich Nothnagel] 16:18:03

Involved. And it's just much cheaper per bit per second. So yeah, I mean, if it can work it surely is going to have an impact on that market.

[Heinrich Nothnagel] 16:18:19

But those... I'm not sure if those are traditional ISP markets. That's sort of another it's it's almost the service on its own, right? Or do you consider the guys who provide services to those airlines also as ISPs.

[Wycliffe Bahati] 16:18:19

So, and...

[Wycliffe Bahati] 16:18:38

I mean, you see the same geo satellites that were providing And with service raw soul.

[Wycliffe Bahati] 16:18:48

Provides broadband to parts of the world if they were banking I mean, removing that part of business from them I think would affect them.

[Heinrich Nothnagel] 16:19:09

Yeah, I mean, there's no doubt about it.

[Wycliffe Bahati] 16:19:13

Okay, so and uh like from What about... I know you've talked about the latency but i mean in terms of if you compare the services that we're getting on fiber and and mobile broadband in terms of speed latency and reliability

[Wycliffe Bahati] 16:19:38

Compared to what you are providing. Is there... a big difference is the is it is it a big difference is it Are there any challenges that Leo might face to provide the same quality of service as fiber

[Heinrich Nothnagel] 16:19:59

I have to be very honest with you. I have not used Leo Broadband myself.

[Heinrich Nothnagel] 16:20:04

I can only tell you what I've heard from prospective customer of ours. Well, it's not a prospective customer. It's the guy for whom I'm going to do the installation in the Congo.

[Heinrich Nothnagel] 16:20:15

They use Starlink. All over Africa, but in DRC was not allowed. So they go they're going to put one web terminal And I have seen some of the tests which we did this week on OneWeb.

[Wycliffe Bahati] 16:20:17

Mm-hmm.

[Wycliffe Bahati] 16:20:24

Mm-hmm.

[Heinrich Nothnagel] 16:20:30

So that's basically my experience. And the thing that struck me about what I've seen thus far, and this guy has actually said the same is that the performance is great.

[Heinrich Nothnagel] 16:20:46

The latency in our case to the internet with one web was 30 milliseconds something like that.

[Heinrich Nothnagel] 16:20:55

Now, I don't think I would think that in most cases you don't even get that with fiber.

[Wycliffe Bahati] 16:20:56

Okay.

[Heinrich Nothnagel] 16:21:02

And you know that it's your environment.

[Wycliffe Bahati] 16:21:03

True. Yeah, it depends on... It depends on... where the content is So where are we doing this test?

[Heinrich Nothnagel] 16:21:14

Yes. The test was done from Zambia and the link landed in Johannesburg.

[Wycliffe Bahati] 16:21:22

Okay, so most of the time maybe you're accessing content interns back that would make it 50. But if you were

to just get out of there and go to europe to be more than fame.

[Heinrich Nothnagel] 16:21:27

Very possible.

[Heinrich Nothnagel] 16:21:36

Yeah, then if they do that, then they have to use the fiber to get to Europe. So that's added.

[Heinrich Nothnagel] 16:21:42

In that case. Yeah.

[Wycliffe Bahati] 16:21:42

Correct. Yeah, so actually that's what I also found out. I was trying to sort of investigate because you see the the way the internet was designed some time back it's not the same way it is now.

[Wycliffe Bahati] 16:21:58

Because you find... that a lot of content has tried to move close to the user just like you pointed out you are getting 50 milliseconds Originally, just a few years ago.

[Heinrich Nothnagel] 16:22:05

Hmm.

[Wycliffe Bahati] 16:22:14

A lot of content was based in Europe. And uh so uh it was almost impossible to to get any latency of 50 milliseconds to anywhere so One thing that I'm not sure.

[Heinrich Nothnagel] 16:22:18

Hmm.

[Heinrich Nothnagel] 16:22:28

Yeah.

[Wycliffe Bahati] 16:22:34

Leo Broadband. Providers will do is, for example, we are talking about Zambia If you are using the terrestrial cable in Zambia to reach South Africa, the latency will be much lower.

[Wycliffe Bahati] 16:22:50

I don't think you'll even hit 20 milliseconds.

[Heinrich Nothnagel] 16:22:55

If it's a good cable. I wonder if there's a link between South Africa and a large part of Africa.

[Wycliffe Bahati] 16:22:57

Yes.

[Wycliffe Bahati] 16:23:04

Yes. Yes, that's liquid and I think there's work some cable in between there.

[Wycliffe Bahati] 16:23:12

Through the border. So it will be much lower although not reliable It's

[Heinrich Nothnagel] 16:23:12

Yeah. But...

[Heinrich Nothnagel] 16:23:21

But there's a point where I think it doesn't really make a difference. I know for the guys who trade on the stock exchange, I'm not sure if 10 milliseconds will make a difference. I don't know not in that business.

[Wycliffe Bahati] 16:23:32

Mm-hmm.

[Heinrich Nothnagel] 16:23:32

The gamers, I don't know if it makes a big difference to them.

[Heinrich Nothnagel] 16:23:36

Those are the guys that are really chasing latency but in terms of data transfer, cloud business, VPNs, blah, blah, blah.

[Heinrich Nothnagel] 16:23:46

I would think that, you know, a few tens of milliseconds difference and perhaps a more reliable link is better.

[Heinrich Nothnagel] 16:23:54

Then having a fast link which is not that reliable. Because these protocols which you use they really suffer for bad links.

[Wycliffe Bahati] 16:24:05

So so it yeah it makes the trade-off between the tool especially in a country like Zambia makes a difference
So I was just saying that now you find a lot of content, for example, in Africa, it's in South Africa, it's in Kenya.

[Heinrich Nothnagel] 16:24:15

Yeah.

[Wycliffe Bahati] 16:24:26

At least wherever there's a lot of users i was surprised the other day Google and Facebook really enclopiat
they have their crashes in Ethiopia So, so.

[Heinrich Nothnagel] 16:24:38

Jeez, okay.

[Wycliffe Bahati] 16:24:42

The design that of course over time I don't know what will happen with rest infrastructure.

[Wycliffe Bahati] 16:24:52

But... And again.

[Wycliffe Bahati] 16:24:58

Not verified because the figures are actually these are reverse engineered reverse engineered figures But
they're seeing that one ground station for Starlink would go to most 5 million?

[Wycliffe Bahati] 16:25:12

Usd or something like that. So if it actually is that expensive Right. Now, I'm just wondering how will they deal
with the fact that you have to have Looks like the only way to bring down latency is by putting down ground
stations in specific locations.

[Heinrich Nothnagel] 16:25:16

Mm-hmm.

[Wycliffe Bahati] 16:25:34

Because often in kenya i think latency improved from 80 milliseconds to 2030 after they turned on the ground
station.

[Heinrich Nothnagel] 16:25:44

Mm-hmm.

[Wycliffe Bahati] 16:25:44

So, so... My thinking is the design architecture of of the Leo broadband.

[Wycliffe Bahati] 16:25:56

Maybe will not eventually scale to the satisfaction of users if they trust infrastructure update.

[Wycliffe Bahati] 16:26:07

I would note the opinion could be.

[Heinrich Nothnagel] 16:26:12

I hope I understand the question correctly. See, I don't think that they are using inter-satellite links widely at the moment.

[Heinrich Nothnagel] 16:26:25

So intersatellite links makes it possible to if you see that a given ground station is is congested you can go somewhere else but then of course you'll add a little bit of latency because you have to come back with the fiber.

[Heinrich Nothnagel] 16:26:37

But then I'm also sure that Starlink, for example, has their own international network like one web to link the ground stations.

[Wycliffe Bahati] 16:26:49

Yeah, I'll try to link them, I think. There have to be i think around 1000 kilometers apart minimum and then they link them with Thank you.

[Heinrich Nothnagel] 16:26:52

Yeah.

[Heinrich Nothnagel] 16:26:58

Yeah. And so the bigger problem I see for them is that, again, it's spectrum.

[Heinrich Nothnagel] 16:27:10

If the spectrum you use for landing at that gateway is no longer available to a user.

[Heinrich Nothnagel] 16:27:16

It's as simple as that. Laws of physics. So the only way to improve that is to improve the mod code.

[Heinrich Nothnagel] 16:27:27

And you can do that by putting a bigger antenna. So I don't know if you've seen, I'm sure you've seen the Starlink ground stations. They're not very big.

[Wycliffe Bahati] 16:27:35

Yeah, yeah.

[Heinrich Nothnagel] 16:27:38

The small dome. And the way they increase capacity is by just adding.

[Wycliffe Bahati] 16:27:40

Yeah.

[Heinrich Nothnagel] 16:27:46

Those things i understand that Nicola every now and then they are there, they're adding.

[Heinrich Nothnagel] 16:27:54

But I still think that there's going to be a point where that big antenna where the size of the antenna doesn't justify the gain which you get from the mod code. I don't know where that number is.

[Heinrich Nothnagel] 16:28:08

But I foresee that spectrum is a limitation unless they start building gateways in the middle of the ocean where there's no one And then they can intercept that link them there And they can use the spectrum as they wish because they're no users.

[Heinrich Nothnagel] 16:28:25

Oh, and the bowls or something like that.

[Wycliffe Bahati] 16:28:28

Yeah, but then they have to come back to where users are.

[Heinrich Nothnagel] 16:28:30

Yeah, but you guys put fiber everywhere. So that's not a problem.

[Heinrich Nothnagel] 16:28:35

Just send a vote.

[Wycliffe Bahati] 16:28:37

Yeah. So coming to that, I don't know if you realize that They seem to be running out of bandwidth again so fast.

[Wycliffe Bahati] 16:28:50

We talked about the idea that they don't actually basically like styling you know The hope was that the most places it will be used will be in the rural areas or areas where we don't have connectivity but from... what people are observing is that these people are coming right where they

[Wycliffe Bahati] 16:29:21

Kick is in the urban centers and they seem to be running out of capacity almost immediately.

[Heinrich Nothnagel] 16:29:28

Hmm.

[Wycliffe Bahati] 16:29:29

So then you have this It's a furnace in the furnace.

[Wycliffe Bahati] 16:29:38

In the obit i mean okay you could try to add as much as pump as much as you want.

[Wycliffe Bahati] 16:29:44

I don't know how i don't know. Because it seems as though right now it's not something that can actually scale.

[Wycliffe Bahati] 16:29:55

I have no idea. I don't know what you think.

[Heinrich Nothnagel] 16:30:01

I think your feeling is correct that i mean maybe my technical knowledge is also too limited But there is certainly no shortage of new ideas. I mean, China is working on a Starlink-like service.

[Wycliffe Bahati] 16:30:20

When they're deploying in South Africa.

[Heinrich Nothnagel] 16:30:24

Oh, is it up? I didn't know.

[Wycliffe Bahati] 16:30:27

Not this yeah they said growing it so perfect. I think they loved it.

[Heinrich Nothnagel] 16:30:34

Oh, okay.

[Wycliffe Bahati] 16:30:35

After the... back and forth with elon musk So the Chinese were allowed to come in.

[Heinrich Nothnagel] 16:30:41

Yeah. I'm sure, well, Cooper or whatever they're called they are coming

[Heinrich Nothnagel] 16:30:51

So it's almost like scalability comes in the form of new networks and then you are correct. You are eventually limited but then limited it then Wagliff when we worked at iowa and certainly 10 years ago, 15 years ago This all low earth over thing was still a dream. I remember when I came to East Africa in 1997, there was a big movement

[Heinrich Nothnagel] 16:31:22

For Liu. Satellites. There was Ico from inmarsat. There was Global Star, there was iridium and blah, blah, blah.

[Heinrich Nothnagel] 16:31:31

None of them made it. Some of them are still alive, but alive in another form.

[Wycliffe Bahati] 16:31:33

Yeah, two.

[Wycliffe Bahati] 16:31:37

Yeah, and... actually yeah and uh That brings me to the next thing is the next thing a lot of people feel that The real cost of this thing is not That's poignant.

[Heinrich Nothnagel] 16:31:56

Is not.

[Wycliffe Bahati] 16:31:57

Transparent in terms of the real cost Yeah.

[Heinrich Nothnagel] 16:32:01

Oh. I think that's true.

[Wycliffe Bahati] 16:32:05

So... Because think about it, the teledasics and the iridiums They were purely commercial, right?

[Wycliffe Bahati] 16:32:16

And they really didn't really do very well of course technology yes change spacecraft is now It's cheaper now. The thing is about \$1,500 per kilo.

[Wycliffe Bahati] 16:32:29

To go to to medium but to the numbers you know They think that they think But behind the scenes, there's a lot that is going on that we are not being shown.

[Wycliffe Bahati] 16:32:47

That this internet service side is just a face the real business is something else.

[Wycliffe Bahati] 16:32:56

I don't know what your opinion is on that. That actually that actually the deployment costs of higher money.

[Heinrich Nothnagel] 16:33:00

I have the same opinion.

[Wycliffe Bahati] 16:33:06

And we see but they make money by other means.

[Heinrich Nothnagel] 16:33:14

I think that Elon Musk with Starling is building an ecosystem.

[Heinrich Nothnagel] 16:33:21

These cars are going to be self-drive. So he wants to provide the internet to them other things which he does is his launch vehicles The only thing that I found strange, which I now I think I understand.

[Heinrich Nothnagel] 16:33:41

Is why he bought X? For the rest, I sort of saw that this guy is building this self-reliance

[Heinrich Nothnagel] 16:33:53

And then while X is now become something else is a tool to be politically active. But in X, there lies a in my view, in my view a warning.

[Heinrich Nothnagel] 16:34:06

In that i personally don't think that these guys are making money.

[Heinrich Nothnagel] 16:34:11

And Elon Musk doesn't have to make money. He just goes to the stock exchange tell a big story and the price of something goes up so you can pay for it or he gets um Trump to put his truth

[Heinrich Nothnagel] 16:34:24

Through the social or whatever on starlink and or buy a Tesla and blah, blah, blah, the price goes up So money is not the limiting factor here. But what has happened with axes I understand that people thought I don't know enough about it but that with eggs he's never going to get his money back.

[Heinrich Nothnagel] 16:34:45

Apparently x is at the point of turning its first profit.

[Wycliffe Bahati] 16:34:53

Yeah, I...

[Heinrich Nothnagel] 16:34:54

So don't underestimate this guy.

[Wycliffe Bahati] 16:34:59

Yeah, they couldn't mean but Boom.

[Wycliffe Bahati] 16:35:06

Is it possible that he's making money more from the military contract.

[Heinrich Nothnagel] 16:35:15

It's possible, for sure, it is possible. I'm not sure that was his original idea, but of course I cannot It might be that this thing was a bit early because he's self-drive cars are not there yet.

[Heinrich Nothnagel] 16:35:30

So he needs to make money in another place like the military and the FAA and those guys and he's now getting that.

[Heinrich Nothnagel] 16:35:37

Some of it. You know, it reminds me of O3B, which had a certain objective But when they launched it, it found a completely different market, which nobody thought about.

[Heinrich Nothnagel] 16:35:52

So the same thing might have happened here. And maybe the war in Ukraine gave him a lot of advertising.

[Wycliffe Bahati] 16:35:52

Yeah, yeah.

[Wycliffe Bahati] 16:36:02

And so, I mean... I think maybe when the Europeans launched their a squared will be able to know exactly what all the books look like.

[Wycliffe Bahati] 16:36:16

Because maybe I think we won't be able to hide the books the way Elon is able to do.

[Heinrich Nothnagel] 16:36:23

Yeah.

[Wycliffe Bahati] 16:36:24

Yeah so because a lot of information that we've got about this deal is actually done by reverse engineering.

[Wycliffe Bahati] 16:36:33

Regard to the failings and try to work backwards and say okay how much is this guy spending?

[Heinrich Nothnagel] 16:36:34

Yeah.

[Heinrich Nothnagel] 16:36:40

Yeah.

[Wycliffe Bahati] 16:36:40

So, yeah. So the other issue was about regulatory and policy challenges So, I mean, how did Do you think these guys are facing challenges?

[Wycliffe Bahati] 16:36:55

Gaining market entries. Maintaining the operations.

[Heinrich Nothnagel] 16:37:01

Yeah, I think they do. I mean, Elon Musk in South Africa is a good example. There were several terminals

here and eventually it was blocked.

[Heinrich Nothnagel] 16:37:10

In the DRC, I understood that there was a certain regulatory framework under which these things can be used as a VSAT and you buy one license and off you go.

[Heinrich Nothnagel] 16:37:22

But understand, the licensing regime has changed completely Because these guys see an opportunity to make a lot of money.

[Heinrich Nothnagel] 16:37:31

And... that I don't think... should have been unexpected.

[Heinrich Nothnagel] 16:37:40

And then, of course, African politicians doesn't like people to have access to the world.

[Heinrich Nothnagel] 16:37:48

So yeah, I see politically, I see a challenge in places like Africa and some developing parts of the world And I see that the regulators are seeing a good chance to make good money because this is the flavor of the day.

[Heinrich Nothnagel] 16:38:05

And that they are putting obstacles.

[Wycliffe Bahati] 16:38:06

And what? What about the licensing model the way you know the way the the old uh the way regulators are licensed in the normal satellite service vis-a-vis the business model for legal.

[Wycliffe Bahati] 16:38:23

How does it compare?

[Heinrich Nothnagel] 16:38:27

Michael, I'm not really in a position to talk about that. I can only say what I've heard and i've heard from Rakesh that the regulators have as many as the traditional satellite model of regulation they don't believe that they make enough money from it.

[Heinrich Nothnagel] 16:38:49

So they're changing the model. In order to make more money from the likes of Starling.

[Heinrich Nothnagel] 16:38:57

But the exact details I'm not that glued up on.

[Wycliffe Bahati] 16:39:03

So there's... one week later in Wanda who was saying that um

[Wycliffe Bahati] 16:39:11

Then... The used license circulates per terminal.

[Heinrich Nothnagel] 16:39:18

Yeah.

[Wycliffe Bahati] 16:39:20

No, all of us didn't have like a thousand of these things It became challenging it became challenging to license the empathino so they They have to change their lessons and model to a blanket.

[Heinrich Nothnagel] 16:39:25

The money's good.

[Heinrich Nothnagel] 16:39:35

Yeah.

[Wycliffe Bahati] 16:39:35

But again. Some countries don't even give license. They just make agreements to operate we just give guys say okay we are allowed to come in and operate there's no I don't know if there's something that is done at that level

[Wycliffe Bahati] 16:39:54

Exempt them from the normal process.

[Heinrich Nothnagel] 16:39:57

I can imagine if your view is that the internet contributes to the economy.

[Heinrich Nothnagel] 16:40:04

Which i think there's plenty of studies to say that that is the case.

[Heinrich Nothnagel] 16:40:10

And you don't have this political fear. That people will know too much.

[Heinrich Nothnagel] 16:40:17

Then... I think that it's a good thing to open it.

[Heinrich Nothnagel] 16:40:26

But in Africa. Like to be fearful of openness.

[Heinrich Nothnagel] 16:40:34

And we don't care too much about the economy. We care about sharing. We don't care about growing.

[Heinrich Nothnagel] 16:40:43

But yeah, I looked... I'm not good at regulatory. I mean, I don't understand. I'm not really For me, it's just frustrating.

[Wycliffe Bahati] 16:40:54

What about the geo i mean you you think But do you see geopolitics?

[Wycliffe Bahati] 16:41:02

Becoming a factor becoming a fact We're seeing almost these companies that are launching the levels in the orbit some of them look private but there seem to be a lot of the hosting governments stick in it.

[Wycliffe Bahati] 16:41:20

So it's becoming almost like um as a risk.

[Wycliffe Bahati] 16:41:26

As this race. And, uh.

[Wycliffe Bahati] 16:41:33

You know these um I think... you know where sometimes you find some countries they impose they don't like something so they put sanctions I mean, I mean with this particular technology, unlike the intel certs and i mean i think those ones are owned mostly by governments and there wasn't much of interference in terms of operation.

[Wycliffe Bahati] 16:42:02

For these particular ones do you see geopolitics playing out?

[Wycliffe Bahati] 16:42:08

A big factor in acceptance.

[Heinrich Nothnagel] 16:42:14

Well, perhaps the case of South Africa and Starlink is a good one.

[Heinrich Nothnagel] 16:42:19

Perhaps the fight between the Russians and Starlink about what's happening in Ukraine and that the fact that the services provided there is kind of an example.

[Heinrich Nothnagel] 16:42:32

But as you know, these satellites don't only do communication services they do they're almost like a global positioning system.

[Wycliffe Bahati] 16:42:32

What about?

[Heinrich Nothnagel] 16:42:41

A very accurate one. I'll not be surprised if we fly some cameras on it.

[Wycliffe Bahati] 16:42:44

Mm-hmm.

[Wycliffe Bahati] 16:42:50

Yeah, definitely. You can't tell. And you can't stop it from flying above you Absolutely.

[Heinrich Nothnagel] 16:42:53

Yeah. No. And there's a... There's a military strategy of the americans which came i think i i don't know when it came, but there's a military doctrine saying he who controls space controls the world That's why everybody's fighting to get this spy satellites in space.

[Heinrich Nothnagel] 16:43:18

But again, you know, it's all gossip and it's all, I think there's a lot of truth in it.

[Heinrich Nothnagel] 16:43:24

It makes sense to me. But...

[Wycliffe Bahati] 16:43:27

It must be true if they are fighting about the ice in the North Pole.

[Wycliffe Bahati] 16:43:33

No, they didn't.

[Heinrich Nothnagel] 16:43:34

Yes, which in an earth which might not even exist then

[Wycliffe Bahati] 16:43:42

So it's definitely true. Yeah, these guys it's it's crazy what people are fighting for.

[Heinrich Nothnagel] 16:43:42

These guys are going to swoop.

[Heinrich Nothnagel] 16:43:50

But yeah, I think white of it is fair to say a general statement to say that geopolitics is going to be a challenge for these guys.

[Wycliffe Bahati] 16:43:50

And then

[Heinrich Nothnagel] 16:43:58

The likes of Intelsat when they came into being. And all the countries of the world as part of it.

[Heinrich Nothnagel] 16:44:04

They were all stakeholders. And that is now no longer the case. These things are now in private hands.

[Heinrich Nothnagel] 16:44:12

And some very funny guys ants So, yeah.

[Wycliffe Bahati] 16:44:17

Yeah.

[Heinrich Nothnagel] 16:44:17

Was he but then you know we cliff at the same time we are saying this thing is not That's scalable and we're saying that There's limits to what nature will allow you to do.

[Heinrich Nothnagel] 16:44:30

Maybe they don't care too much about losing some countries.

[Wycliffe Bahati] 16:44:36

Actually, the funniest thing is uh Somebody did an analysis.

[Wycliffe Bahati] 16:44:45

This and try to place the size of Europe in Africa. I think it can fit twice.

[Wycliffe Bahati] 16:44:50

In africa but in terms of revenue in terms of if you were to sell something It's it.

[Heinrich Nothnagel] 16:44:51

Mm-hmm.

[Wycliffe Bahati] 16:45:04

The... the two is totally incomparable in terms of how much money we'll make.

[Heinrich Nothnagel] 16:45:13

Yeah.

[Wycliffe Bahati] 16:45:14

So... Like you say that's why it's possible just to say, okay, we can non-league just concentrate on and give service here and we still make our amendment.

[Heinrich Nothnagel] 16:45:27

Yeah.

[Wycliffe Bahati] 16:45:27

Yeah, I think I'm not about the other thing that seen this little constellations because they are being driven by private hands and

[Wycliffe Bahati] 16:45:40

I mean, do you... think they'll open up to standardization the way the geo and traditional circulates have now international bodies that almost standardized the operations um Do you think they'll open up to each other and say, okay, fine, we can collaborate we can work

[Wycliffe Bahati] 16:46:01

Design things in a standard width.

[Heinrich Nothnagel] 16:46:06

That is a very good question. You know, the guys in the VSAT industry could never reach that point.

[Heinrich Nothnagel] 16:46:16

They all use very similar way of doing things, but they could never standardize on in a way that you could put any reset on any hub.

[Heinrich Nothnagel] 16:46:29

I am not too convinced. I think that there's still a lot of cutting edge technology going into this stuff.

[Heinrich Nothnagel] 16:46:39

And they are not too keen to share that. I don't think that Liu Even though internet access is a commodity.

[Heinrich Nothnagel] 16:46:48

I don't think this leo technology is a commodity yet. I think they're still building some secrets in there.

[Heinrich Nothnagel] 16:46:56

Which will give them a competitive advantage.

[Wycliffe Bahati] 16:47:01

Okay, let's finish.

[Heinrich Nothnagel] 16:47:04

If you just think of... the antenna.

[Heinrich Nothnagel] 16:47:08

For Starlink that the antenna was the big problem for low earth all the satellites.

[Heinrich Nothnagel] 16:47:14

And many people said that this will never happen. I was always concerned. I didn't say it will never happen but I could almost say that.

[Heinrich Nothnagel] 16:47:23

Until Elon Musk said, okay, in that case, I'll just subsidize it until I sell so many that I can make them at the price which I sell them So that technology was a major obstacle.

[Heinrich Nothnagel] 16:47:38

But they overcame it. That gave them a competitive advantage in the ability to be first. But now that technology is out, now they can be caper, there can be others.

[Heinrich Nothnagel] 16:47:48

So I think there are other things there which they don't necessarily want to share yet.

[Heinrich Nothnagel] 16:47:56

If they will ever, like I said, the VSAT guys in When was the first visa installed? Probably in the 70s.

[Heinrich Nothnagel] 16:48:02

In more than 30, 50 years now. They cannot there's no standard

[Heinrich Nothnagel] 16:48:12

So no, I don't see that.

[Wycliffe Bahati] 16:48:12

And... Okay, now if they don't do that, then it means that everybody has just to launch their own satellites.

[Wycliffe Bahati] 16:48:22

And be independent of the other So that comes to the environmental factors of space delivery.

[Wycliffe Bahati] 16:48:29

You know everybody is saying banking on that the other guy is nice and he's going to make sure that he's sick like you do a bit.

[Wycliffe Bahati] 16:48:42

After five years. So I think it looks like now it's five years for the five Leo, five to nine years, it depends So there's a case i think i can't remember one of the satellites russian satellites was in uh leo

[Heinrich Nothnagel] 16:48:48

Mm-hmm.

[Wycliffe Bahati] 16:48:58

And I think they needed to think bring it down something. So these Russians developed an anti-syillidate missile, I think.

[Wycliffe Bahati] 16:49:13

And blew it out of the... deliver them.

[Wycliffe Bahati] 16:49:18

Of course it now that satellite accounts for about 70% of the debris that is in that outer space.

[Wycliffe Bahati] 16:49:27

Of course it's coming down and burning but not as fast as as opposed to so this guy is, I mean, if they don't collaborate, they don't standardize Who stops?

[Wycliffe Bahati] 16:49:43

I mean, the future who stops china or india I mean, to... make this the leo atmosphere you know I mean, unusable.

[Wycliffe Bahati] 16:49:56

That's important.

[Heinrich Nothnagel] 16:50:01

Wycliffe, that answer I don't have for you again. It's something I guess all of us wonder about. I'm not clever enough to know.

[Heinrich Nothnagel] 16:50:09

But I have a feeling that we as a think that because we're so clever and we developed all these technologies, we will find technology answers to all our questions. Like, for example, the the environmental issue.

[Heinrich Nothnagel] 16:50:24

I mean, the guy, the likes of Trump and co mushered the only way I can imagine they think about this is that we will find the technology to solve this problem.

[Heinrich Nothnagel] 16:50:34

But technology cannot solve all problems. It might be able to.

[Heinrich Nothnagel] 16:50:38

But it might be too late. So then there was over years, there's been talk about how are we going to clean up the debris in space.

[Heinrich Nothnagel] 16:50:50

And then the damn russians go. Test the missile to blow up a satellite.

[Heinrich Nothnagel] 16:50:54

Just to show the Americans we can. And now you say, I didn't know this, 70% of debris comes from there.

[Wycliffe Bahati] 16:51:04

Yeah, it's a step late.

[Heinrich Nothnagel] 16:51:05

So... So I remember one of the talks were that they're going to put this launched this big magnet but i think If I understand it correctly.

[Heinrich Nothnagel] 16:51:20

This debray moves at a speed that'll probably go straight through the magnet

[Wycliffe Bahati] 16:51:26

Oh, yeah. I'm sure you heard about the incident in Kenya the 500 kilogram

[Heinrich Nothnagel] 16:51:32

Yes, yes.

[Wycliffe Bahati] 16:51:35

Yeah, and that's actually I mean, we... It's something that people need to worry about Unlike... there isn't one we know about the un space you know But it's toothless.

[Wycliffe Bahati] 16:51:55

It doesn't stop anybody from I mean, the only thing that is limiting me to launch something in space I don't have money to build that rocket.

[Heinrich Nothnagel] 16:51:56

Hmm.

[Heinrich Nothnagel] 16:52:06

Hmm.

[Wycliffe Bahati] 16:52:06

If I went somewhere in a very well and built that thing I'll launch it.

[Heinrich Nothnagel] 16:52:12

It's true. Yeah, it's true.

[Wycliffe Bahati] 16:52:13

I don't know why it will go, but... Yeah, so... So I think there's quite a bit of concern on that and unless These fellas can work together because they have the monies to do this thing We'll keep on having surprises.

[Wycliffe Bahati] 16:52:30

Because if that 500 kilogram thing landed on a building or something increase or a cut Yeah, I think then, so, I mean, I finally, I think just take your view of how you think.

[Heinrich Nothnagel] 16:52:39

Yeah. Yeah.

[Wycliffe Bahati] 16:52:53

You've seen the industries evolve what do you think this How will this

[Wycliffe Bahati] 16:53:02

Play out.

[Heinrich Nothnagel] 16:53:07

You know uh Wycliffe, one thing... I have observed And which sort of makes sense to me is that whichever technology you bring to provide internet access.

[Heinrich Nothnagel] 16:53:24

Or communication. For data networks.

[Heinrich Nothnagel] 16:53:29

If the price is fair, it doesn't even have to be great you will find a market simply because there's such a demand for bandwidth.

[Heinrich Nothnagel] 16:53:39

And we're saying these guys will not be able to fulfill the need of the world.

[Heinrich Nothnagel] 16:53:45

I mean, I also have to say that I think a lot of the data which is being used in the world is just waste. It's like we wasted water many years ago. Now he's trying to save it.

[Heinrich Nothnagel] 16:53:54

It's the same kind of thing.

[Wycliffe Bahati] 16:53:57

No.

[Heinrich Nothnagel] 16:53:59

So I think, you know, when in in 2010, the run about everybody said that geo is dead I think there's today, I'm pretty sure there's today more geo capacity than what there was then.

[Heinrich Nothnagel] 16:54:16

So the arbitrary was written a bit too soon. And it's for two reasons one people have this demand for bandwidth and they're in places where that's the only option they have

[Heinrich Nothnagel] 16:54:32

People have devised other ways to use because people are smart, they find ways to do things.

[Heinrich Nothnagel] 16:54:39

The geo guys, the Leo guys will do the same. They will find clever ways to use what they have better.

[Heinrich Nothnagel] 16:54:45

They will find clever ways to get to other applications, some of which we might not even know today.

[Heinrich Nothnagel] 16:54:54

So I don't think, I think they will find a slot Like Jew found a slot, which is always also always evolving Like Fiverr has a big slot like copper has some place like microwave, I have some place And I do think that other technologies will also come. Laser, I think, is one which there's also a lot of talk about.

[Heinrich Nothnagel] 16:55:19

But yeah, like I said, I'm not clever enough, but that should promise a lot of capacity.

[Heinrich Nothnagel] 16:55:26

So what will happen to Leo? I think it will be there. I think it will grow. I think it will find its slot.

[Heinrich Nothnagel] 16:55:32

And it will probably go through the same life cycle as you, grow, grow, grow, and then start declining.

[Heinrich Nothnagel] 16:55:38

Depending on how people find other applications for it and depending on how new technologies come about to rich areas which today rely to reach areas to provide reliable comms

[Heinrich Nothnagel] 16:55:55

Yeah.

[Wycliffe Bahati] 16:55:56

And these astronauts are complaining that those things I'm blocking them from seeing the stars.

[Wycliffe Bahati] 16:56:05

And they're not yet. Then there are only 7,000.

[Wycliffe Bahati] 16:56:09

They're supposed to launch 42,000. No.

[Heinrich Nothnagel] 16:56:12

Yeah, that's what's on the list now.

[Wycliffe Bahati] 16:56:15

What will happen to geo I mean, would it affect the the leo spectrum coming.

[Heinrich Nothnagel] 16:56:29

Block the signals. I guess there has to be something like that because even an airplane flying through a beam give you a blip.

[Wycliffe Bahati] 16:56:30

Yeah.

[Heinrich Nothnagel] 16:56:44

You know, what might well happen at some stage is that the guys are saying, listen, you guys just switch off give us that here's a couple of billion dollars. Give us that spectrum.

[Heinrich Nothnagel] 16:56:55

That might also happen. I don't know, but I don't It's too easy it's too soon to say that this is not going to really have that much of an impact on geo People will also find clever ways to come around that, I'm pretty sure.

[Heinrich Nothnagel] 16:57:13

They do wonderful thing in communications, as you know.

[Wycliffe Bahati] 16:57:13

Yeah. I forgot one thing there.

[Wycliffe Bahati] 16:57:20

Your take on... So for now, I think they're trying, they're using KU band for standing alone they use the terminals and use KA for the ground station.

[Heinrich Nothnagel] 16:57:38

Mm-hmm.

[Wycliffe Bahati] 16:57:40

What's your take with I mean, with the weather and do you think this possibility in some areas you might not be able to deliver the cell day okay this they don't have SLA actually but usability of the service.

[Heinrich Nothnagel] 16:57:59

No, I don't think so. I mean, when we launched the highway in what was it, 2000, somewhere there KU Ben, people said forget it. You guys, this is never going to fly

[Heinrich Nothnagel] 16:58:14

Today, nobody even asks the question. Then came and it was the same thing. This thing is not going to work.

[Heinrich Nothnagel] 16:58:24

It works. It has its limitations, but people also found clever ways to work with it.

[Heinrich Nothnagel] 16:58:32

And these things, the low Earth orbit satellites are so close to the earth that I don't really see too much of an issue for it.

[Heinrich Nothnagel] 16:58:42

And... How reliable is... I've got an internet I've got a fiber link here at home Is it 100% here? No.

[Heinrich Nothnagel] 16:58:53

Is it good yeah Is it okay for what I do? Absolutely.

[Heinrich Nothnagel] 16:58:59

More than okay. So... guys who want to do serious communication like you die you guys doing cloud servers and what what you'll book the five before it.

[Heinrich Nothnagel] 16:59:15

And you will build a reliable network for it. That's not where these guys are going.

[Heinrich Nothnagel] 16:59:21

I don't see that. And we've already said they don't have the capacity for it.

[Wycliffe Bahati] 16:59:28

And one thing that's has been observed the guy who did... you know who did a test and Because I think styling.

[Wycliffe Bahati] 16:59:42

The first array antenna i mean he was trying to see you know the I think that... they do handoff between the set rates and they do so basically because they don't have the they don't rotate they have basically they try to do the beam

[Heinrich Nothnagel] 16:59:58

Mm-hmm.

[Wycliffe Bahati] 17:00:02

Beamforming or something. Okay. Yeah. So he realized that he realized the handoff every time i think it was 15 seconds

[Heinrich Nothnagel] 17:00:04

Lymphoma, yeah.

[Wycliffe Bahati] 17:00:16

There was a dip in terms of the latency goes high or you get bucket loss so apparently they they've really not perfected the it's not seamless steel.

[Wycliffe Bahati] 17:00:31

This you can notice the handoff when you're doing it.

[Wycliffe Bahati] 17:00:36

Australian IP tests. Between the acetylites as they move around.

[Wycliffe Bahati] 17:00:44

So whether... the increase of satellites will change that or they'll overcome that because apparently when they launched it was worse.

[Wycliffe Bahati] 17:00:53

Now, I think they've approved it it's manageable but it's still noticeable

[Heinrich Nothnagel] 17:01:01

See, the position of the satellite is predictable. And they must load a file in that in that remote site.

[Heinrich Nothnagel] 17:01:12

I think they call it the euphemiris file or something like that. It's got a name.

[Heinrich Nothnagel] 17:01:17

Which tells the terminal where the next satellite is. I am not sure. I'm not clever enough to know how accurate that information is all the time.

[Wycliffe Bahati] 17:01:22

Okay.

[Heinrich Nothnagel] 17:01:32

And the part, I also don't know i also don't know exactly how uh this um phase the rate lock onto the satellite.

[Heinrich Nothnagel] 17:01:41

So it has to do two things. It has to log onto the satellite and then it has to lock onto the carrier.

[Heinrich Nothnagel] 17:01:48

And then, of course, the gateway also have to lock onto the carrier which this thing transmits.

[Wycliffe Bahati] 17:01:55

So I think... So I think the observation was Apparently, I think that's what I was trying to i think we were discussing it some other time is the Doppler effect or something So the distance from the distance from the distance from

[Heinrich Nothnagel] 17:01:55

All of that can happen.

[Wycliffe Bahati] 17:02:10

Come in to the satellite. As it moves.

[Wycliffe Bahati] 17:02:15

Yeah and then yeah it picks up the next one i think Yeah, this stuff.

[Wycliffe Bahati] 17:02:26

I think we've tried to perfect it but I think it still doesn't ip doesn't

[Wycliffe Bahati] 17:02:36

They're not affected.

[Wycliffe Bahati] 17:02:42

Maybe that's the best they could they can do with that technology

[Heinrich Nothnagel] 17:02:47

No, I'm not sure about that. The retrace of phased array.

[Heinrich Nothnagel] 17:02:59

Is... It's electronic.

[Heinrich Nothnagel] 17:03:04

It's the next bit. So that is not the issue the issue Not that you're saying that's the issue, but in my view, that is not the issue. The issue in my view is that that VSAT can steer that or the phase array can steer that beam anywhere it wants, right? Within its range. It can steer it anywhere.

[Heinrich Nothnagel] 17:03:27

In fractions of a second. But still, it must be accurate on the satellite.

[Heinrich Nothnagel] 17:03:34

And that's the part I'm not 100% sure about. In due, we lock on to if we have big antennas with narrow beams, we lock onto a beacon.

[Heinrich Nothnagel] 17:03:45

And then you have fancy electronics which track the beacon.

[Heinrich Nothnagel] 17:03:52

Now... there must be something similar going on here so that your antenna Otherwise, how does this thing track the satellite through the phase that I track the satellite through space. There must be some form of there must be a way in which it

[Heinrich Nothnagel] 17:04:08

Logs onto it. I don't think it can be a predetermined algorithm of this is where the satellite is.

[Wycliffe Bahati] 17:04:13

It's it's it's um So from what I get, it's not actually Okay, I think the there's supposed to be many antennas, not one.

[Wycliffe Bahati] 17:04:28

One in the in that dish?

[Heinrich Nothnagel] 17:04:34

Mm-hmm.

[Wycliffe Bahati] 17:04:34

So... I think by position of where the small antennas are, they're able to cover I think a bigger area, right?

[Wycliffe Bahati] 17:04:46

And be able to refocus the the signal as in when they transmit like that by the time it gets to the satellite it forms one.

[Wycliffe Bahati] 17:04:58

One wave or something like that, I think.

[Heinrich Nothnagel] 17:05:02

Yeah, that's how phase theory works. But they have a phase three in that phase the rate in the panel which you have.

[Heinrich Nothnagel] 17:05:08

Which is just a set of dipoles which you steer electronically by just changing the face.

[Wycliffe Bahati] 17:05:16

Oh, yeah, yeah, yeah, yeah. I've seen that. It's an interesting so do you think still that Because I think in Kenya they're selling it at \$400 or something.

[Wycliffe Bahati] 17:05:31

So you think they're still subsidized for that type of equipment.

[Heinrich Nothnagel] 17:05:33

Mm-hmm. No, I think by now he's probably producing those things at cost i yeah i know i think You know, face the rays are now everywhere.

[Heinrich Nothnagel] 17:05:44

One way abuses them use as several Well, they have a different technology but the phased array flat widely used, widely produced.

[Heinrich Nothnagel] 17:06:00

So I think it's sorted in that respect.

[Wycliffe Bahati] 17:06:01

Yes.

[Wycliffe Bahati] 17:06:06

Okay, but now the default i think i there's a guy in... Malawi. He has an ISP there.

[Wycliffe Bahati] 17:06:17

And this is the same Actually, when Stalin came in with he didn't lose his normal customers. He only lost his premium customers Because... feeling is that the cost of the 400 and \$50 or something is quite a lot a normal person too.

[Heinrich Nothnagel] 17:06:30

Mm-hmm.

[Heinrich Nothnagel] 17:06:41

It's true.

[Wycliffe Bahati] 17:06:46

Goodbye just for internet. Yeah. So the observation was that it's still prestige.

[Heinrich Nothnagel] 17:06:48

It's true.

[Wycliffe Bahati] 17:07:00

Products.

[Heinrich Nothnagel] 17:07:01

Yeah, it is, but we're a long way off from the thousands of dollars we used to charge.

[Wycliffe Bahati] 17:07:09

Actually, when you go to heaven, God will ask you that question.

[Heinrich Nothnagel] 17:07:13

No, no, no, no, no, no, no, no. But then, you know, Salim always had a very good analogy. He said to the guys would complain to him say this V set of yours is too expensive. And he would ask them.

[Wycliffe Bahati] 17:07:23

Okay.

[Heinrich Nothnagel] 17:07:26

How much does that laptop you're using cost and it was the same cost So that mobile phone you are using there, how much does that cost?

[Wycliffe Bahati] 17:07:35

Oh yeah, it's too expensive.

[Heinrich Nothnagel] 17:07:37

So then it's \$450 expensive? It's expensive. It's a lot of money, sure.

[Heinrich Nothnagel] 17:07:42

But then compared to other technologies and gadgets and whatever and what it does for you is it

[Wycliffe Bahati] 17:07:50

Let's see. I think you try to compare it with the prices for their modules so 5.

[Heinrich Nothnagel] 17:07:57

Yeah. Look, man, we as humans will always complain about the price.

[Wycliffe Bahati] 17:08:00

Okay.

[Heinrich Nothnagel] 17:08:03

Otherwise, prices will never come down.

Participant 8 AK

Name Of Participant: Arthur Kinyanjui

Profession: Expert with over 35 years in Satellite services

Date of interview: 14th March 2025

[Wycliffe Bahati] 13:32:26

I'll quote you i'll let you know it before. Yeah, so I shared with you the questions Oh.

[Arthur] 13:32:32

Okay.

[Wycliffe Bahati] 13:32:38

I think the first one is your strong point. The competition and market dynamics. So how do you think the Leo

Broadband services is affecting the business strategies of ISPs and mobile network operators.

[Wycliffe Bahati] 13:32:54

And then when you're doing that just try to focus on what are the opportunities that are there and what are the challenges.

[Arthur] 13:32:55

Mm-hmm.

[Arthur] 13:33:03

Give me one minute. I'm trying to settle in a quiet room

[Wycliffe Bahati] 13:33:06

Okay, no, no, take your time. No problem.

[Arthur] 13:33:19

Let me try and open the questions that you answered you will send to me where it can go.

[Wycliffe Bahati] 13:33:26

No.

[Arthur] 13:33:47

These questions I've been encountering on a day-to-day basis. So I'm able to quickly go through and through them.

[Arthur] 13:33:56

I mean, the first one is quite, of course. Very straightforward i mean Some of the hindrances that has been there with Legas, etc. Service is the cost And then the cost from an acquisition perspective and then also post from the recurring one

[Arthur] 13:34:18

And looking at the introduction of leo That has quickly gone down well making satellite service available and affordable for sectors that initially couldn't afford them then number two is that user experience has greatly improved.

[Arthur] 13:34:47

With the LEO services, noting that on the early access, you could either on geo mayo or on hts you would be doing between anywhere between 700 to 1500 milliseconds so mission critical applications that require low

latency and were dependent on that business is dependent on

[Arthur] 13:35:13

Could not operate. With the Leo that had been taken care of because with the inward teleports within countries you're able to do anywhere between 30 to 80 milliseconds which Most applications ERPs would work.

[Arthur] 13:35:28

That's one. And then the second one is which is a positive contribution to ISPs is that now they have a technology they are able to deploy to be able to satisfy the SLAs that they are giving their customers in different

[Arthur] 13:35:43

The verticals and also MNOs. But on the other side is that this has come to heat into their market.

[Arthur] 13:35:53

Which is really affecting them and if you take case in point that the introduction of Starlink in kenya within one year it was already ticking about 5% of the broadband market which if you look at traditionally

[Arthur] 13:36:15

The ISPs and MNO struggled to actually get that i mean it's a record time getting to 5% of the market share in Kenya was quite something where But it took the traditional ISPs and MNOs somewhere between two to five years to achieve.

[Arthur] 13:36:36

Yep.

[Wycliffe Bahati] 13:36:37

Yeah, so the apart from um One of the things that uh Leo Burn Band was started to do was to connect areas where traditional isps could not connect.

[Wycliffe Bahati] 13:36:55

Is that the reality? Is that what's happening?

[Arthur] 13:36:58

Yes, it's a priority that areas that were initially not connected because satellites usually have a universal coverage of a country. There is coverage in those particular places. That's why. Number two.

[Arthur] 13:37:12

That not only did they connect, they are also able to give affordable and reliable service to those particular

areas.

[Arthur] 13:37:21

Yes, they have been an influence on connecting the green fields But also, we have seen an upside in connectivity using the leo in traditional strongholds that these ISPs and mnos holds.

[Wycliffe Bahati] 13:37:38

So basically. They kind of come to share the market.

[Arthur] 13:37:45

Yes. Simply because everyone, I mean, people are very industrious and they like testing new things.

[Wycliffe Bahati] 13:37:47

Okay and

[Arthur] 13:37:54

I mean, we saw some people trying to trying to change their traditional fiber links to styling of course at that time they had to go back to their family but it is something that we saw in the market where

[Arthur] 13:38:08

I mean, the statement that curiosity killed the cat was best exemplified.

[Wycliffe Bahati] 13:38:14

Yeah. And where do you think they came back?

[Arthur] 13:38:17

One is that The Leo failing capacity of service is a mid-type service.

[Arthur] 13:38:29

In the traditional fiber services is an unlimited service. So one of the things is that it was started to be cheap, but when you exhaust your data and you have to buy a new bundle and then that that means the service expensive over the traditional fiber services in the traditional markets that these ISPs and mnos operate.

[Wycliffe Bahati] 13:38:54

And do you see it? The leo service being let come back up to terrestrial link failures.

[Wycliffe Bahati] 13:39:08

Especially in Africa.

[Arthur] 13:39:11

Yes, from a technological perspective, EM, it is a true backup.

[Arthur] 13:39:20

To the traditional paper but that will be dependent on analysis of the data on the usage of that service to better package that service to complement the existing PIPAA and terrestrial services.

[Wycliffe Bahati] 13:39:39

And so we're going to number two where What have you heard from the consumers in terms of the latency the speed and reliability of this service leo service.

[Arthur] 13:39:55

Oh, um... Even before we go to consumers, the usage patterns have shown that after Starlink launched.

[Arthur] 13:40:03

Within one year, they were already facing charges of clogging in the network because people had oversubscribed on the service.

[Arthur] 13:40:17

On the capacity available to use them. So is there an opportunity for them yeah that was We saw that. Now, from a user experience, I mean, they came in because they could get proper speeds i mean I did several speech tests and you could clearly go beyond 150 megabits per second on the downlink.

[Wycliffe Bahati] 13:40:24

Mm-hmm.

[Arthur] 13:40:40

On styling in various locations. And you could do well between 10 to 15 megabits per second on the app link.

[Arthur] 13:40:48

So from user experience, I guess, before the challenge of a subscription came in, it was good.

[Wycliffe Bahati] 13:40:49

Mm-hmm.

[Arthur] 13:40:59

Then that is from a challenge perspective is that it came with upfront acquisition of the kit at no cost so that

that became a barrier to entry.

[Arthur] 13:41:12

Because, I mean. Many consumers wouldn't afford the \$500 for outright purchase of the kit.

[Arthur] 13:41:19

And they are used to the traditional ISPs giving them a free non-recurrent cost on the acquisition of technology.

[Wycliffe Bahati] 13:41:28

Have you had over any complaints of, for example, down times or unreliable service or anything like that?

[Arthur] 13:41:35

Yes, after a subscription they have been reliability issues And new user experience. Yes, those ones have been there.

[Arthur] 13:41:46

Especially in there that was very rampant in Rwanda and the part of the the central parts of Kenya and Nairobi.

[Wycliffe Bahati] 13:42:00

Yeah, and now from your experience from I set that background. How do you think this leo constellations.

[Wycliffe Bahati] 13:42:12

Are sustainable in the long run. They bring return of investment

[Arthur] 13:42:20

Well, I mean... From where I sit, if we are able to cure the cost of acquisition of It's sustainable that's why two the model of how they intend to come into the market should they follow what Starlink did where it was to come

[Arthur] 13:42:40

And compete directly with mobile network operators and local ISPs. I don't think that will be sustainable or they will be able to actually be sustainable into the market i guess they we need to work on the cost of acquisition of technology that's a service. And then number two, probably look at reviewing their market model

[Arthur] 13:43:03

For my direct format, probably utilizing the local infrastructure variable with mnos and the ISPs in the country.

[Wycliffe Bahati] 13:43:13

This can collaborate. Okay.

[Arthur] 13:43:14

Corroborate, yes.

[Wycliffe Bahati] 13:43:18

I don't know if they're going to change that. And what about...

[Arthur] 13:43:23

I mean, you see that is the styling but we have the quipa coming and we have the Chinese constellation. I mean, we look forward to see how they're going to come into the market.

[Wycliffe Bahati] 13:43:28

Yeah.

[Wycliffe Bahati] 13:43:37

Okay. And... What do you think is a challenge in regards of regulatory and licensing?

[Arthur] 13:43:46

Maybe that is a big big and good question. I don't think many regulators understand

[Arthur] 13:43:56

What is the satellite service? That site has always eased.

[Arthur] 13:44:03

Steve resistance especially from the regulatory perspective. Because many regulators do not.

[Arthur] 13:44:11

Or have not understood that other technologies are evolving. Also, certified services are evolving.

[Arthur] 13:44:17

So they still want to treat satellite as that old legacy CPAN, KU band model which was meant to be used by businesses and reach high net worth people But it's moving to become a mass market product. And from a regulatory perspective, it meant to be treated as such. The regulator needs to start looking at

[Arthur] 13:44:46

The tight note is a satellite technology, but as any other technology that is bridging the connectivity divide are more or less like LTE, FTTB, FTTH, they need to be seen as another model of bridging the digital divide but

not as a technology that needs to be penalized or being there.

[Wycliffe Bahati] 13:45:12

And so I just want to take you back a bit on the user uptake.

[Wycliffe Bahati] 13:45:18

Apart from affordability. I mean... What do you think is the problem with uptake in areas which current providers have not followed.

[Wycliffe Bahati] 13:45:33

Appetite to deploy Yes, apart from the affordability.

[Arthur] 13:45:39

One of them is the model that is used to reach tech consumers we use it's wrong.

[Arthur] 13:45:46

I mean, Kenyan, Kenya and east africa nearing 80 70 percent of of the transactions are done on The model and in this case the constellation and in this case satellite to use cards. I am not sure that is going to make it easier for people to access it that's one.

[Arthur] 13:46:11

Number two, after it is acquired, how do you support those particular clients for those particular customers without local support architecture.

[Arthur] 13:46:25

And then number three is they tried doing the mass market but doing it through the hypermarkets and supermarkets which are only available in the urban centers and they are not available in the green fields New.

[Wycliffe Bahati] 13:46:41

Who leads uh still doesn't treat the remote areas.

[Arthur] 13:46:45

No. And again, capacity. I mean, it's supposed to be a free deployment do it yourself type of service.

[Wycliffe Bahati] 13:46:47

Okay. And...

[Arthur] 13:46:56

Do people know how to install it? Without help, no.

[Wycliffe Bahati] 13:47:01

It will still require help to install. Okay. And what about... the issue of governments and regulation of internet.

[Arthur] 13:47:01

Yep. Yes.

[Wycliffe Bahati] 13:47:14

How does it apply? Study or the the regulation for now you find some governments have restricted Facebook, some governments have want to be able to turn off the internet.

[Arthur] 13:47:17

How is it?

[Wycliffe Bahati] 13:47:31

The entrance of leo how does it vary With that.

[Arthur] 13:47:39

And that's a very good question. I would want to wear a political hats when I wear them.

[Arthur] 13:47:44

When i'm answering you that question but i will opt not to.

[Arthur] 13:47:49

The constellations are putting locally reports in each country. So more or less like a fiber landing point which is prone to interference by the However, it may be so i guess from a compliance perspective on that they are doing way more better than they are supposed to be doing.

[Arthur] 13:48:11

I mean, and the ends I would also say that it offers a different route than the available roof which the government more or less would interfere with the connectivity in country

[Wycliffe Bahati] 13:48:28

Yeah, actually... We're just thinking of this thing.

[Wycliffe Bahati] 13:48:33

Each ground station is about \$5 million or something. Will they be able to deploy this everywhere, really, in

every country?

[Arthur] 13:48:39

Yeah.

[Wycliffe Bahati] 13:48:42

And some countries, of course, they're not even getting the license

[Arthur] 13:48:46

I've seen them struggling in that one in getting the type approvals for equipment number two getting expertise to deploy and maintain But I don't think that um the neo providers have any challenge with capital if they want to be given a free and i believe they have studied

[Arthur] 13:49:09

The architecture of what they want to to deploy and probably They have... they have factored that in their in the business plan to be able to deploy this but having said that we have seen them when like Starlink when they launched in kenya

[Arthur] 13:49:29

And in Rwanda, they used an external teleport to do that from Nigeria and back hold on fiber rush traffic But hey, we are deploying three stations currently in Kenya so i i I guess if there is the business, there's appetite for the service. I believe they also have

[Arthur] 13:49:53

The required resources to do some use by particular customers.

[Wycliffe Bahati] 13:50:00

And so... like from where I sit now um What do you think is the future will look for like for the Leo broadband providers.

[Arthur] 13:50:13

I mean, they got to change their business model and they got to change their use cooperation mode and going direct to the market that way it will be sustainable They are also going to also bridge the connectivity divide by opening up areas where cost of acquisition of technology to deploy those areas would

[Arthur] 13:50:35

Would have not would have been inconceivable but with this one now we are able to bridge that gap And with the work in technology, we will be looking at Other variables that can be ramped onto the onto the leo to make

them relevant into the market case in point we are seeing the test for direct to mobile happening

[Arthur] 13:51:03

In the United States, and I want to believe When you go to any African countries and I want to use Africa because that's where I know best.

[Arthur] 13:51:12

You go and find, I guess, the most connected countries, Kenya sitting at 68%.

[Arthur] 13:51:17

Meaning 22% of the country is dark So if you want to even talk about providing those particular places with mobile connectivity you will probably want to explore how to use leo to bridge that instead of going to build the stations that would provide that.

[Arthur] 13:51:36

So yeah, there would be a user case of loading more variants onto the service to make it relevant to people or to answer what those people currently want done.

[Arthur] 13:51:51

Data from a data perspective or from a voice perspective.

[Wycliffe Bahati] 13:51:58

I think... I think we've covered most of the items.

[Wycliffe Bahati] 13:52:04

Okay, one more thing. Have you seen that Is it... for the leo services Is it generally all uses rainbow to use rainbow get it or is only specific type of users who are acquiring this?

[Wycliffe Bahati] 13:52:26

Legal services.

[Arthur] 13:52:30

I would say it is some specific type of user One of the biggest users for Leo are the existing people is existing people and businesses with existing service where we are using Leo to complement their current services, either as primary or redundancy

[Arthur] 13:52:52

Then the second cluster of people I've seen are people who require connectivity and And they have redefined

their businesses or their way of life is dependent on them being online. I haven't seen an uptick of really your services by the common warranty

[Arthur] 13:53:15

In relief use. I haven't seen that.

[Arthur] 13:53:22

But I wasn't seen the developer of tapping into leo to start scaling the variables or the development that they are making outside their urban center.

[Arthur] 13:53:36

Where there is no per seal coverage. But I haven't seen a normal woman whom you are just going to whom to the shop and procuring a styling kit then we are seeing them doing the Jeep on fiber or the metro fiber networks.

[Arthur] 13:53:59

Sparring across different villages. Up and centers and shopping centers in the country.

Declaration of Authenticity



Declaration of Authenticity

I hereby declare that I have completed this Master's thesis on my own and without any additional external assistance. I have made use of only those sources and aids specified and I have listed all the sources from which I have extracted text and content. This thesis or parts thereof have never been presented to another examination board. I agree to a plagiarism check of my thesis via a plagiarism detection service.

A handwritten signature in blue ink, appearing to be 'W. B.', is written above a horizontal line.

Nairobi, 25/05/2025

Place, Date

Student signature

